

27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session

Prague, 16. – 18. September 2026

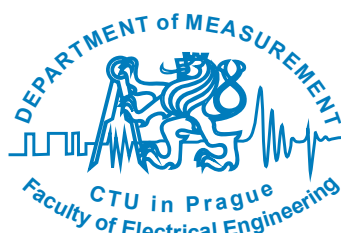


Measuring the Future

**IMEKO TC-4 International Symposium
with TC-11 Technical Session**

PROCEEDINGS

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Prague, 16. – 18. September 2026



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Welcome Message from the Organizing Committee

On behalf of the Organizing Committee, we extend a warm welcome to you for the 27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session.

The conference commences with a Welcome Ceremony and Get-Together Reception on September 16th, 2026, held in the Penthouse on the 10th floor of the CIIRC CTU building. Perched above Prague with its own terrace and a view across the rooftops to Prague Castle, the Penthouse — part of the Czech Institute of Informatics, Robotics and Cybernetics — offers an informal and relaxed setting in which to reconnect with old friends and meet new colleagues before the technical programme begins.

The Scientific Conference itself takes place on September 17th and 18th at the Czech Technical University in Prague, Faculty of Electrical Engineering, on the university's Dejvice campus. On the evening of the first conference day, delegates are invited to the IMEKO TC4 Gala Dinner at Restaurace Mlýnec, a restaurant occupying a distinctive spot on Novotného lávka with a terrace sitting directly beneath Charles Bridge and a view across the Vltava river toward Prague Castle. Part of the Zátíší Group and Michelin Selected, Mlýnec pairs a contemporary, water-inspired interior with a kitchen that blends traditional Czech technique and seasonal, local ingredients — a fitting way to celebrate a conference themed "Measuring the Future" in one of Europe's oldest university cities.

This edition also features a Special TC11 Technical Session dedicated to "Testing, Inspection and Certification in Electrical Sector," bringing the digital transformation of metrology, blockchain-based traceability and secure data infrastructures alongside TC4's traditional programme.

The main objective of the IMEKO TC4 conference is to foster collaboration between academic researchers and industry engineers, with a specific focus on measurement and instrumentation for electrical and magnetic quantities, ADC and DAC modelling, characterization and testing, calibration, traceability and metrological standards, advanced sensors and signal processing, and applications in energy systems, e-mobility, industry automation and emerging technologies.

IMEKO TC4's primary interest lies in the theoretical and practical aspects of measuring electrical quantities through electronic instruments, encompassing measurement accuracy, instrumentation design, calibration principles, uncertainty analysis and digital metrology.

We would like to take this opportunity to express our gratitude to all the members of the Technical Programme Committee, the Special Session Organizers, and the anonymous Reviewers, whose contributions have made this outstanding programme possible. We also extend our appreciation to the Authors for submitting their papers and sharing their scientific findings with the IMEKO TC4 & TC11 community. Over the course of three days, the technical programme features six invited keynote lectures and about forty oral contributions from authors around the world, organized into eight thematic sessions.

Special thanks are due to all the local Organizing Committee members, whose significant efforts were devoted to the successful organization of this event, and to our sponsors and partners — Teste, HTEST, and Termovize TMV SS — for their generous support, which made this event possible.

The Organizing Committee:

General Conference Chair – Assoc. Prof. Jakub Svatos, Ph.D., CTU in Prague, Czechia

Scientific Chairs – Dr. Dragana Popovic Renella, SENIS AG, Switzerland;

Prof. Luca Callegaro, INRIM, Italy;

Prof. Carlos Roberto Hall Barbosa, PUC-Rio, Brazil

Technical Program Overview

The technical programme of the 27th IMEKO TC4 Conference 2026 with Special TC11 Technical Session evolves over three days (September 16-18, 2026) and includes six keynote speeches and eight oral sessions, including a Special TC11 Technical Session on "Testing, Inspection and Certification in Electrical Sector."

The keynote speeches are:

- One Bit to Useful Knowledge: Rethinking Data Acquisition for Low-Power and High-Speed Measurement Systems by Prof. Paolo Carbone, University of Perugia, Italy
- Major European Research Projects in Electrical Metrology by Prof. Jiří Tesař, Director of the Czech Metrology Institute (CMI), Czechia
- Measuring the Performance of C-UAS Systems: How Hard Can It Be? by Dr. Cristian Coman, NATO Communications and Information Agency (NCIA), The Hague, Netherlands
- Artificially indeterminate: measuring quantity, cost and impact of Artificial Intelligence by Dr. Michael Street, NATO Communications and Information Agency (NCIA), The Hague, Netherlands
- Trust by Design in Digital Metrology: From Machine-Readable Documents to Privacy-Preserving Computation by Prof. Kruno Miličević, J. J. Strossmayer University of Osijek, Croatia
- EUROLAB: Empowering Digital and Sustainable Labs and other CABs by Laura Martin, EUROLAB

The eight technical sessions cover the whole spectrum of electrical and magnetic measurements, from calibration and traceability to advanced sensors and emerging applications, alongside the Special TC11 Technical Session.

The oral sessions titles are:

- Session 1: Calibration, Traceability & Metrological Standards I
- Session 2: Calibration, Traceability & Metrological Standards II
- Session 3: Uncertainty Evaluation & Electrical/Magnetic Instrumentation
- Session 4: Electrical & Magnetic Instrumentation
- Session 5: ADC/DAC, Waveform & Data Conversion Metrology
- Session 6: Advanced Sensors & Signal Processing I
- Session 7: Advanced Sensors & Measurement Systems II
- Session 8 - Special TC11 Technical Session: Applications in Energy Systems, Emerging Technologies & Materials Characterization (incl. Testing, Inspection and Certification in Electrical Sector)

As Technical Program Chairs, we would like to thank all the members of the Technical Program Committee and the Reviewers, who allowed to improve the level of the contributions.

The Committee

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IMEKO TC4 members: <https://www.imeko.org/tc4-homepage/tc4-members>

Program IMEKO-TC4 & TC11 Technical Session 2026

Day 1 - Wednesday,, September 16, 2026		
17:00-19:00	Welcome Ceremony & Get-Together Reception	Penthouse - CIIRC, CTU
Day 2 - Thursday, September 17, 2026		
08:30-08:55	Conference Opening & Welcome Address	
Session 1: Calibration, Traceability & Metrological Standards I (Chair: Marija Cundeva-Blajer)		
08:55-09:30	KEYNOTE — One Bit to Useful Knowledge: Rethinking Data Acquisition for Low-Power and High-Speed Measurement Systems	<i>Prof. Paolo Carbone</i>
09:30-09:45	* Development of System for Disseminating AC/DC Voltage Difference Using Ranges of Transfer Standard and Investigating Stability	Valentyn Isaiev
09:45-10:00	* MetroMag: A European Infrastructure for Low Magnetic Field Metrology	Marco Coisson
10:00-10:15	o Franco Cabiati, electrical impedance metrology, and IMEKO	Luca Callegaro
10:15-10:30	* Exploring calibration schemes for electric impedance measurements beyond the short-open-load	Alessandro Cultrera
10:30-11:00	<i>Coffee Break</i>	
Session 2: Calibration, Traceability & Metrological Standards II (Chair: Luca Callegaro)		
11:00-11:30	KEYNOTE — Major European Research Projects in Electrical Metrology	<i>Prof. Jiri Tesar</i>
11:30-11:45	o High-Temperature Superconductor Josephson Junction Arrays Coupled to Microwave of Different Polarizations	Changhao Zhao
11:45-12:00	* MEMS-based Long-haul Optical Link Asymmetry Estimation for Atomic Clock Comparison	Michal Spacek
12:00-12:15	* A Model for Secure Exchange of Digital Calibration Certificates	Ahmed Khan Leghari
12:15-12:30	* Metrological Traceability in IoT Sensor Networks: A Microservice-based Framework for Automated Calibration Management	Marco Sento
12:30-12:45	* Implementation of Monte Carlo Simulation for measurement uncertainty estimation in electrical power and energy reference standards calibration	Kiril Demerdziev
12:45-13:00	TMV SS - Use of Thermography for Non-Destructive Testing	Antonin Krnoul
13:00-14:00	<i>Lunch Break</i>	

Session 3: Uncertainty Evaluation & Electrical/Magnetic Instrumentation (Chair: Paolo Carbone)

14:00-14:30	KEYNOTE — Measuring the Performance of C-UAS Systems: How Hard Can It Be?	<i>Dr. Cristian Coman</i>
14:30-14:45	* GUM-compliant uncertainty evaluation in measurements based on ML regression or classification models	Leopoldo Angrisani
14:45-15:00	* Quality Infrastructure for Resilient Electrical Sector - Ensuring Confident Measurement Results in Calibration of High Currents Instruments	Marija Cundeva-Blajer
15:00-15:15	o Development of the Digital Twin for a Digital Impedance Ratio Bridge	Ana Lucia Medina Rodriguez
15:15-15:30	* Accuracy Analysis of THD Measurement of Power Signals by Means of a Low-cost Oscilloscope	Gabriel Gasparesc
15:30-16:00	<i>Coffee Break</i>	

Session 4: Electrical & Magnetic Instrumentation (Chair: Cesare Svelto)

16:00-16:15	o The primary Tesla Standard at PTB	Franziska Weickert
16:15-16:30	o A traceable calibration of low-field magnetic coil systems in noisy industrial environments	Michal Ulvr
16:30-16:45	* Model-based magnetization measurement in Fluxgate sensors	Jiri Maier
16:45-17:00	* Estimation of torquer rod magnetic moment with fluxmetric method and compact sensing coils	Michal Janosek
17:00-17:15	* Compact modular cryogenic dielectric resonator for surface impedance measurements in high magnetic fields	Kostiantyn Torokhtii
19:00-21:00	Gala Dinner	

Day 3 - Friday, September 18, 2026

Session 5: ADC/DAC, Waveform & Data Conversion Metrology (Chair: Pedro M. Ramos)

08:30-09:00	KEYNOTE — Artificially indeterminate: measuring quantity, cost and impact of Artificial Intelligence	<i>Dr. Michael Street</i>
09:00-09:15	* A systematic aliasing suppression architecture for real-time FI-DAC based on linear-phase complementary filter and oversampling	Yang Shen
09:15-09:30	* Parameter estimation of complex-valued sinewaves by a frequency-domain linearized sine-fit algorithm	Dominique Dallet
09:30-09:45	* Benchmarks for IIR Model Identification Based on Binary Measurements	Paolo Carbone
09:45-10:00	o A Digital Voltage Ramp Generator with Integrating Compensation	Jianting Zhao
10:00-10:30	<i>Coffee Break</i>	

Session 6: Advanced Sensors & Signal Processing I (Chair: Jan Holub)		
10:30-11:00	KEYNOTE — Trust by Design in Digital Metrology: From Machine-Readable Documents to Privacy-Preserving Computation	Prof. Kruno Milicevic
11:00-11:15	o Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds	Martin Maxa
11:15-11:30	o Lock-in detection algorithm for signals modulated with irregular intervals	Shingo Tamaru
11:30-11:45	o A Wireless Acoustic Monitoring System for Drone Localization Based on MEMS Microphones and LoRa Communication	Kamil Herman
11:45-12:00	* Comparative Analysis of Search Algorithms for Automatic L-type Matching Network	Joao Pedro Torres S. Andrade
12:00-12:15	* Airborne Navigation During GNSS Outages Using Tightly Coupled IMU/DVB-T2 Fusion	Michal Koldinsky
12:15-13:15	Lunch Break	
Session 7: Advanced Sensors & Measurement Systems II (Chair: Luca De Vito)		
13:15-13:30	* Microcontroller Based Software Defined Instruments as an Economical Substitute of Virtual Measurement Instruments	Jakub Svatos
13:30-13:45	* Exploring a new indirect approach to latency measurement	Jakub Turinsky
13:45-14:00	* Detection of Mercury Ion in Water Samples Using a Low-Tech EPS-Coated Interdigitated Capacitive Sensor through S-Parameter Analysis	Henrique Dantas Silva
14:00-14:15	* Development and Experimental Evaluation of a Three-Lead ECG Holter System	Ondrej Kovac
14:15-14:30	* Multiparametric metrological characterization of an inductive sensing element for mechanical wear estimation	Davide Ceschini
14:30-15:00	Coffee Break	
Session 8: Applications in Energy Systems, Emerging Technologies & Materials Characterization (Chair: Gabriel Gasparesc)		
15:00-15:15	KEYNOTE — EUROLAB: Empowering Digital and Sustainable Labs and other CABs	Laura Martin
15:15-15:30	* Detection of Physical Layer Anomalies and Eavesdropping in Real World Data Cables	Amit Duwal
15:30-15:45	* Sustainability-oriented optimization of a sensing board for applications as a cyber-physical measurement system (CPMS)	Leopoldo Angrisani
15:45-16:00	* Vector Fitting-Based Feature Extraction from Electrochemical Impedance Spectroscopy for SOC and SOH Estimation	Pedro Ramos
16:00-16:15	* Measurement of the characteristics of EDL and hybrid supercapacitors versus temperature	Ornella Conti
16:15-16:30	o Non-Destructive Microwave Dielectric Characterization of Leather for Automotive Applications	Kostiantyn Torokhtii
16:30-16:45	* Measurements of Wetsuits Thermal Properties at 100 kPa and 600 kPa Absolute Pressure	Cesare Svelto
16:45-17:00	Closing Remarks & Best Paper Awards	

* paper included in the proceedings volume (multi-page) o one-page abstract only (not included)

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Development of System for Disseminating AC/DC Voltage Difference Using Ranges of Transfer Standard and Investigating Stability

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Abstract – The paper describes a system for disseminating a reference value of AC/DC voltage transfer difference, starting with that attributed to a planar multi-junction thermal converter and, further, distributed across ranges of an AC/DC transfer standard. The second purpose of the developed system is to monitor the stability of the quantity under consideration. A key feature of the design is the integration of five double elements as the basis for conceptual realization, involving goals, standards, their peculiarities, and the methods used. The span between two AC/DC differences is a characteristic parameter for assessing the actual internal state and stability of the standard under study, as well as for disseminating the quantity by adding it to the initial reference value. Along with the methodology depicted, the calibration data analysis and experimental verification results are provided.

Keywords: AC/DC transfer; voltage; stability; dissemination.

I. INTRODUCTION

The metrological traceability of low AC voltage measurement results up to 1000 V at a frequency of up to 1 MHz is established through combining a DC voltage standard with an AC/DC transfer standard. The main condition for international recognition of measurement results is a verified link with the SI system. Among the approaches to ensuring metrological traceability is calibration against a primary standard.

AC/DC transfer in the specified voltage and frequency ranges can be implemented using a thermocouple-based voltage thermal converter or using a voltage converter based on a root-mean-square (RMS) voltage sensor, such as the Fluke 792A. Calibration of such a reference instrument relative to the primary AC/DC transfer standard may sometimes cause legal and logistical problems associated with the lack of harmonization of national laws and the peculiarities of transport and customs procedures. These difficulties may disrupt the regular

calibration schedule, lower metrological reliability, and raise doubt on the established traceability.

The authors propose a specific system for stability assessment based on data from a series of calibrations combined with internal observations of the change over time of two precision voltage converters. The introduction of the assessing parameter ($\Delta_{kj}^{int} - \Delta_{kj}^{cal}$) is the main difference from the others. In addition, the presented methodology allows us to address the issue of traceability establishment in low-frequency voltage measurements. The proposed approach seems feasible for assessing the internal state and deterioration of metrological characteristics of multi-range precision instruments.

II. BACKGROUND AND METHODOLOGY

A. General notes

The procedure for comparing the input voltage conversion characteristics of two standards (Fluke 792A and PMJTC) allows us to record the span of their AC/DC differences. The basic elements of the common measurement setup, in a somewhat simplified form, are shown in Fig. 1.

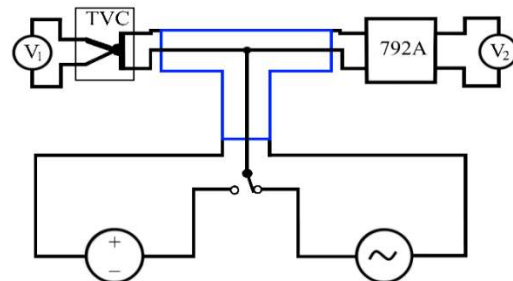


Fig. 1. Basic measurement setup to compare AC/DC differences of two voltage converters

The T-connector (marked in blue) allows parallel supply of voltage from either a DC or AC source, alternately using switch S. The six selected frequencies (10 and 45 Hz, 1, 10, and 100 kHz, 1 MHz) are the base range for

characterizing the PMJTC reference thermal converter at a voltage of 1.5 V, which is the nominal value. To compare readings of the multimeters V_1 and V_2 , the Fluke 792A instrument must be set to voltage conversion mode on the 2.2 V range.

It is worth noting the long-term study of the stability of the Fluke 792A standard [1], which has been ongoing since 2004 and includes accumulated measurement data at 1.5 V and 1 V. Statistical analysis [2] showed no discernible trend in the AC/DC difference over 20 years of observation. Confidence in the results is bolstered by the fact that the voltage converters being compared have fundamentally different design features. The Fluke 792A voltage converter contains electronic components that require power from the included battery pack. The RMS sensor performs precision conversion of the input voltage using a differential amplifier and transistors, with the resistor's temperature affected by the current [3]. In contrast, the PMJTC thermal converter is based on Seebeck's effect, with several thermocouples connected in series, creating a total thermo-electromotive force (EMF) output of 100 mV, which is a convenient value for measurement [4].

Assuming the invariance (or minimal shift over the observation period) of the AC/DC transfer characteristic of the PMJTC reference voltage converter, a conclusion has been made about the satisfactory condition of the internal elements of the Fluke 792A under study, as well as the high stability of its characteristics. The high stability over the years of PMJTCs, incorporating a series of thermocouples, has been confirmed in research [5] led by the German National Metrology Institute. Also, a comparative analysis of AC/DC voltage transfer standards [6] showed the relative advantage of PMJTC converters over other types. This is the basis of the developed system for disseminating the reference AC/DC voltage difference value and monitoring the stability of the investigated characteristic alongside the spotlights [7] and [8] that highlight propagation across different measurement levels.

B. Steps of dissemination

Having considered the design features of the 792A instrument [9], presented in Fig. 2, we can distinguish three groups of input circuits (a circuit for conditionally direct supply of input voltage to the RMS sensor; 4 circuits with voltage dividers for inputs higher than 2 V; 3 circuits with input voltage amplification for inputs lower than 0.6 V). It can be seen that the input impedance contains both resistive and reactive capacitive components. The minimum resistance is 420 Ω and corresponds to the input range with direct input voltage up to 2.2 V, while the largest resistance of 10 M Ω is at the three amplification ranges (22, 220, and 700 mV). The reactive components in the lower three ranges do not exceed 40 pF, while those in the others do not exceed 20 pF. The details of the element base for the amplification ranges are not shown in Fig. 2

because they are more complex blocks [9]. The mechanical switch (Range Switch) is designed to connect the input circuit of the selected range to the RMS sensor.

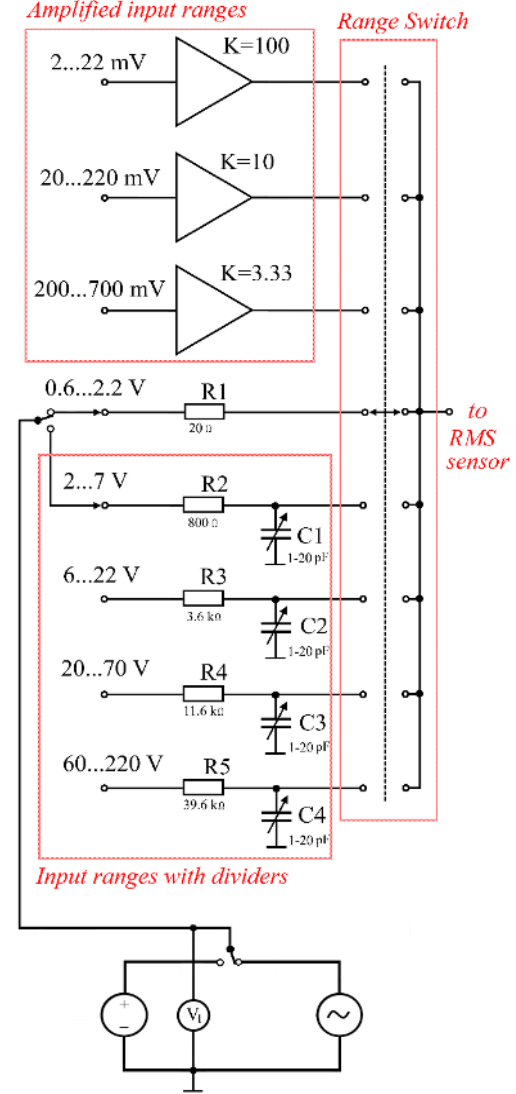


Fig. 2. Setup for disseminating the AC/DC difference value across ranges of the Fluke 792A (indicating 3 groups of input circuits)

After comparing the readings of the two converters (Fluke 792A and PMJTC), the following propagation of the reference AC/DC difference is made by comparing the output of the multi-range AC/DC transfer standard under study on two adjacent ranges at the same voltage. High-stable AC and DC voltage sources (e.g., Fluke 5720A) alternately feed the input circuits, and a precision multimeter V_1 (e.g., Keysight 3458A) in the appropriate mode monitors the invariance of the applied signal. The additional increase of uncertainty due to the multimeter's resolution in AC voltage measurement mode is from 0.5 to 1.5 μ V/V with each transition to the next range.

C. Stability assessment

As noted above, the previous study on the stability of the Fluke 792A aimed to collect data comparing the readings of the two converters at 1 V and 1.5 V [2]. Given that the nominal voltage of the PMJTC is 1.5 V, this value is suitable for studying the input circuits of the 2.2 V and 7 V measurement ranges. If we also consider 0.6 V (at which the PMJTC output thermo-EMF is approximately 14 mV), we can investigate the stability of the Fluke 792A at the adjacent 2.2 V and 700 mV ranges.

According to the data in the Fluke 792A calibration certificate, the characteristics can be compared at the same voltage (for example, 600 mV for the range up to 700 mV and 0.6 V for the range up to 2.2 V). The observation points are selected so that, for all input voltage ranges of the device under study, there is one or two such positions. Since the Fluke 792A standard has been calibrated 4 times over its 30 years of existence, the accumulated dataset has enabled analysis to assess the relationships between the AC/DC differences Δ_{kj} (where k represents the input voltage and j is the AC frequency) and to compare them across different years. Moreover, using the setup in Fig. 2,

one can estimate the actual value of this characteristic. The results of the analysis of the accumulated data and the experimental comparison using the proposed setup are presented in the next section. Fig. 3 shows a visual interpretation of the proposed approach to monitoring the long-term stability of the Fluke 792A standard.

In Fig. 3, a list of points for determining the AC/DC differences δ_{kj} during calibration, as well as points for observing the stability of the characteristic, is shown, with the points enclosed by curly brackets. Here, the higher input voltage range is denoted by H, and the lower voltage range by L. Blue arrows indicate the imaginary input-output step sequence in the data analysis procedure. The difference Δ_{kj} between two δ_{kj} of adjacent ranges is a unique parameter (at frequency j and voltage k) for assessing stability at the time of evaluation. This allowed us to evaluate the stability and technical state of the device under study between calibrations and to use it for comparing the obtained experimental data. The graph on the right in Fig. 3 is the outcome of the stability analysis by the proposed system.

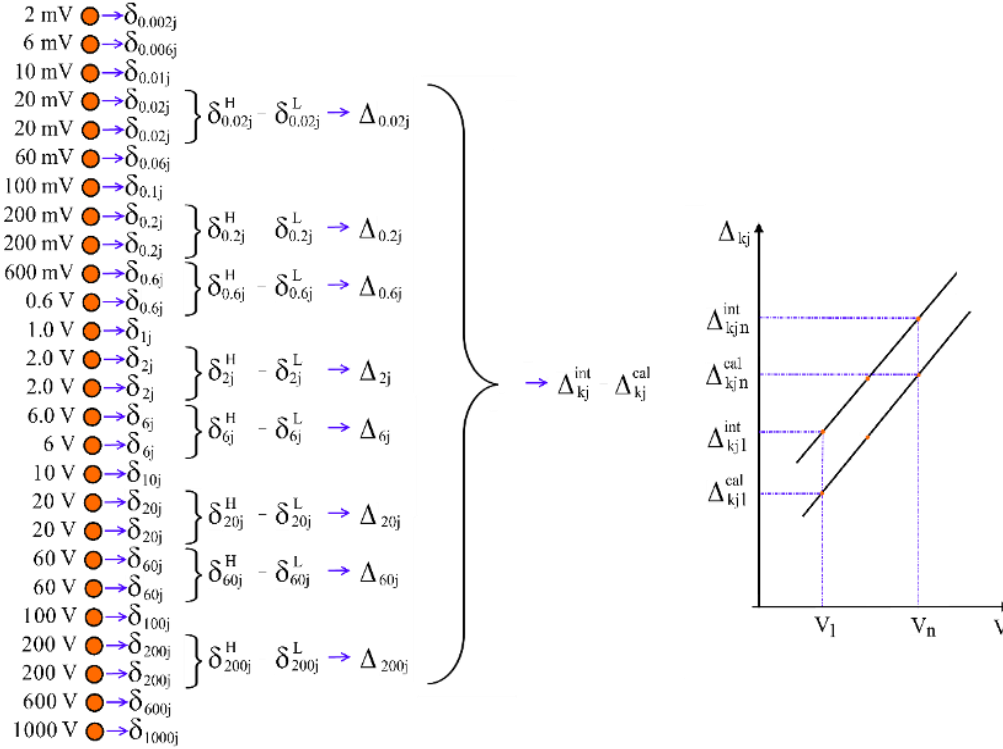


Fig. 3. Visual representation of the idea to derive information on actual Δ -parameter for further implementation to disseminate AC/DC difference and to assess its stability

The difference between Δ_{kj}^{int} and Δ_{kj}^{cal} reflects the quality of the measurements, the refinement of the setup, and the perfection of the procedure, as well as a probable shift in the controlled quantity.

D. Uncertainty analysis

Several sources of uncertainty were assessed in measurements according to the outlined methodology. Over 20 years of observation of the Fluke 792A, there was some degradation and deterioration of the cables, contacts,

and internal components and characteristics of the instruments, including short-term stability of the voltage sources, which in some way affected the repeatability [2].

The measurement model has the form:

$$\delta_{ST} = \delta_{DIF} + \delta_{792A} + \delta_{MS} + \delta_C + \delta_{PMJTC} + \delta_{SH} + \delta_T + \delta_{EMI}, \quad (1)$$

where δ_{DIF} is the AC/DC difference between two converters; δ_{792A} is the AC/DC difference of the Fluke 792A; δ_{MS} characterizes the impact of the measurement setup; δ_C represents the effect of cables and connectors; δ_{PMJTC} reflects a drift of the PMJTC; δ_{SH} is a parameter related to self-heating of the Fluke 792A; δ_T characterizes the impact of the temperature; δ_{EMI} is the effect of electromagnetic interference.

According to model (1), the uncertainty budget includes two contributions from the Fluke 792A: the type B standard uncertainty and the drift due to self-heating. The contribution of the measuring setup includes the reversal of the DC voltage, the commutation of the special switch, the mutual influence of its elements, the stability of the voltage sources and the output signal meters during the observation cycle, etc. The influence of threaded connectors, conductors, and cables due to contact resistance and contact thermo-EMFs is also taken into account. The drift of the PMJTC (instability of output thermo-EMF) is another source of uncertainty. To minimize the influence of electromagnetic interference, coaxial connectors, the shortest possible wires, shielding, and a grounded room were used. The ambient temperature was kept as stable as possible, and its contribution was also considered in the combined uncertainty. The list of estimated components for characteristic frequencies is given in Table 1.

Table 1. Sources of uncertainty

Uncertainty component	Estimate ($\mu\text{V/V}$) at frequency (kHz)		
	0.01	1	1000
Standard uncertainty of Fluke 792A (δ_{792A})	6	2	14
Measurement setup (δ_{MS})	4	0.5	7
Connectors and cables (δ_C)	1	0.5	6
Drift of PMJTC (δ_{PMJTC})	1.5	1	4
Self-heating of Fluke 792A (δ_{SH})	1.5	0.5	3
Temperature (δ_T)	0.5	0.5	0.5
Electromagnetic interference (δ_{EMI})	1	1	1
Standard deviation (δ_{DIF})	1	0.5	1.5

Recall that the idea behind the work is to use PMJTC as the initial link for dissemination of AC/DC difference. However, during the long-term stability study and verification of the presented results, data from the Fluke 792A calibration certificates were used, as PMJTC has not been calibrated for 20 years. Regarding the further steps of the reference AC/DC difference propagation, the same

components should be added to the initial stage uncertainty, except for the Fluke 792A's uncertainty and the PMJTC's drift. The measurement setup, connectors, and cables were slightly changed.

III. OBTAINED RESULTS

A. Calibration and measurement

According to the procedural logic of Fig. 3, the Fluke 792A instability parameters were determined to compare the averaged values of the difference Δ_{kj}^{cal} with the values Δ_{kj}^{int} obtained from the measurement results using the setup in Fig. 2. Several specific measurement points were selected for observation, as listed in Table 1.

Table 2. Observation points of AC/DC voltage difference of PMJTC and Fluke 792A

Voltage (V)	Frequency (kHz)
0.2	0.01; 1000
0.6	0.01; 0.02; 1; 20
20	0.01
200	0.02; 100

The metrological characteristics of the Fluke 792A voltage converter in the range from 2 mV to 1000 V were previously determined several times through calibration using the primary AC/DC voltage transfer standard, i.e., its AC/DC differences and corresponding expanded uncertainties were added to the accumulated data array. Table 3 presents the results of experimental evaluation of the relative difference δ_{kj} for those characteristic observation points as in Table 2.

Table 3. Experimental comparison results of AC/DC voltage difference between adjacent ranges of Fluke 792A

Frequency (Hz)	AC/DC difference ($\mu\text{V/V}$) depending on year of calibration			
	Lower range		Upper range	
	Mean ($\mu\text{V/V}$)	SD ($\mu\text{V/V}$)	Mean ($\mu\text{V/V}$)	SD ($\mu\text{V/V}$)
<i>At a voltage of 0.2 V</i>				
0.01	240.7	1.8	56.0	2.1
1000	2848.7	1.5	2809.2	2.2
<i>At a voltage of 0.6 V</i>				
0.01	-9.5	2.4	72.6	1.1
0.02	-6.6	1.3	-8.2	0.7
1	-9.8	0.5	-13.9	0.6
20	-8.9	0.3	-0.7	1.2
<i>At a voltage of 20 V</i>				
0.01	-15.5	1.6	-163.7	1.5
<i>At a voltage of 200 V</i>				
0.02	-35.7	1.7	-66.0	2.0
100	-126.4	1.3	-136.8	1.2

Furthermore, for the observation points indicated in Table 2, Table 4 lists the values of the studied characteristic δ_{kj} , obtained in 1996, 2002, 2008, and 2023, which are extracted from calibration certificates.

Table 4. AC/DC difference assigned to Fluke 792 in calibration

Frequency (Hz)	AC-DC difference ($\mu\text{V/V}$) depending on year of calibration			
	1996	2002	2008	2023
<i>At 0.2 V for the range of 200 mV</i>				
0.01	-12	-10	-12	-19
1000	296	316	283	566
<i>At 0.2 V for the range of 700 mV</i>				
0.01	-202	-198	-196	-189
1000	381	404	392	425
<i>At 0.6 V for the range of 700 mV</i>				
0.01	-56	-52	-53	-46.4
0.02	-2.4	1.3	2.3	3.8
1	-11	-2.2	0.3	0.1
20	-44	-5.3	-4.7	-3.8
<i>At 0.6 V for the range of 2.2 V</i>				
0.01	41	47	49	45.1
0.02	7.7	4.9	7.4	6.5
1	0.6	-2.4	-2.4	-1.3
20	3.7	0.5	0	2.5
<i>At 20 V for the range of 22 V</i>				
0,01	201	198	180	201.8
<i>At 20 V for the range of 70 V</i>				
0,01	53	54	59	54.4
<i>At 200 V for the range of 220 V</i>				
0,02	51	50	51	52
100	10	5	9	10
<i>At 200 V for the range of 2.2 V</i>				
0,02	8.9	4.9	4	5
100	-18	-18	-15	-19

B. Data analysis

The analysis begins with Table 5, which presents comparative data for the Fluke 792A instrument. These results compare the AC/DC differences between adjacent ranges at a specified voltage over the years when the standard was calibrated, and the maximum differences between any two calibrations recorded.

Recall that the instability of the characteristic in this work is defined as the shift of the parameter Δ_{kj} at a certain observation point over some period. From Table 5, it can be seen that the maximum shifts $\Delta_{\max}$ of the Δ_{kj}^{cal} parameters between calibrations were smaller for frequencies from 20 Hz to 100 kHz and for passive (those that do not require voltage amplification) measurement ranges. The 20 V and 200 V voltages were compared without amplification, and the values did not exceed 10 $\mu\text{V/V}$, except at 10 Hz.

On the other hand, 0.2 and 0.6 V should be compared within measurement ranges that include input signal amplifiers. Table 5 shows that only the 20 Hz point received an estimate of less than 10 $\mu\text{V/V}$. The worst comparison points, characterized by the strongest influence of external electromagnetic interference and the largest reactive components, are the amplified voltages at 1 MHz. At 0.6 V, one of the adjacent ranges is passive, and the other is active, but at 0.2 V, both compared ranges are active.

Table 5. Calibration data processing results

Voltage (V)	Frequency (kHz)	Parameter Δ_{kj}^{cal} ($\mu\text{V/V}$) depending on year				$\Delta_{\max}$
		1996	2002	2008	2023	
0.2	0.01	190	188	184	170	20
	1000	-85	-88	-109	141	250
0.6	0.01	-97	-99	-102	-91.5	10.5
	0.02	-10.1	-3.6	-5.1	-2.7	7.4
	1	-11.6	0.2	5.4	1.4	11.8
	20	-47.7	-5.8	-4.7	-6.3	43
20	0.01	148	144	121	147	27
200	0.02	42.1	45.1	47	47	4.9
	100	28	23	24	29	6

As a result, the largest difference in comparison errors ($\Delta_{\max}=250 \mu\text{V/V}$) was for adjacent ranges (220 mV and 700 mV) at 0.2 V and a frequency of 1 MHz. The identified problem point requires further observation and deeper analysis. The last step, according to the procedural logic of Fig. 3, is the comparison of the parameters Δ_{kj}^{int} and Δ_{kj}^{cal} .

Based on this, Table 6 summarizes comparative data for the Fluke 792A instrument on its application: expanded uncertainty according to the 2023 calibration certificate, expanded measurement uncertainty by [10] when determining Δ_{kj}^{int} , and the results of comparing the latter parameter with the averaged parameter Δ_{kj}^{cal} over four calibrations.

Table 6. Comparison results of calibration data and internal experiment

Voltage (V)	Frequency (kHz)	$\Delta_{kj}^{\text{int}} - \Delta_{kj}^{\text{cal}}$ ($\mu\text{V/V}$)	$U_{\text{exp}}^{\text{cal}}$ 2023 p., ($\mu\text{V/V}$)	$U_{\text{exp}}^{\text{int}}$ by Fig.3, ($\mu\text{V/V}$)
0.2	0.01	-1.7	4	19
	1000	-74.8	10	38
0.6	0.01	-15.3	2.5	17
	0.02	-7.0	2.5	14
	1	-5.9	1.6	7
	20	-7.9	1.6	8
20	0.01	-8.1	2.1	18
200	0.02	15.0	6	18
	100	15.6	12	36

Note: the difference ($\Delta_{kj}^{\text{int}} - \Delta_{kj}^{\text{cal}}$) should not exceed the root sum of squares of $U_{\text{exp}}^{\text{cal}}$ and $U_{\text{exp}}^{\text{int}}$.

According to the comparative data on the difference $\Delta_{kj}^{\text{int}} - \Delta_{kj}^{\text{cal}}$, an excess over the expanded uncertainty was fixed at 0.2 V and 1000 kHz. This result is consistent with the conclusion from Table 5. For a more in-depth investigation, trends in the AC/DC difference over the years were analyzed and presented in Fig. 4. Since this problematic point was fixed in the amplified ranges, all available calibration data from 2 mV to 600 mV at 1000 kHz were analyzed more carefully. Fig. 4 shows that the AC/DC differences for voltages of 2, 6, and 10 mV follow similar decreasing patterns from the first to the third point.

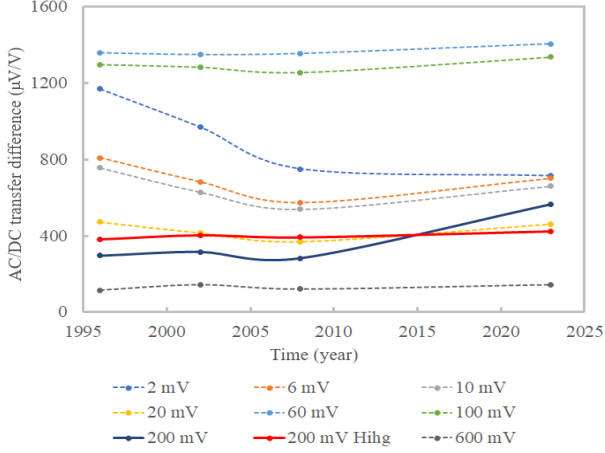


Fig. 4. Long-term change in the voltage conversion characteristics of the standard under study

However, the fourth point returns to or approaches the value obtained in 2002 for 6 and 10 mV. Note that the best expanded uncertainty for these values is at least 100 $\mu\text{V/V}$, as per the calibration yielded in 2023. The rest of the calibration result sequences have a smoother, more uniform appearance, except for 200 mV (dark blue bold curve) in the 220-mV measurement range.

Further testing of pairwise convergence of the available data using the E-criterion allowed us to isolate inconsistent results for 2008 & 2023 at 100 mV and the pairs 2008 & 2023 and 2002 & 2023 at 200 mV (range 220 mV). For the 200-mV search point, the E-criterion was near 5, which indicated a significant discrepancy in the analyzed data.

The analyzed metrological challenge (0.2 V/1 MHz) for dissemination from the 700 mV to 200 mV range may be explained by the operating feature of the 792A instrument at 1 MHz (a noticeable discrepancy due to differences in the dynamic characteristics of the amplifier paths). The 200-mV range operates in almost full-scale mode with maximum output signal, whereas in the 700-mV range (the beginning of the scale, where the signal-to-noise ratio is worse), the input impedance changes significantly due to parasitic capacitances, the skin effect, etc. To overcome this issue and cover the difference ($\Delta_{kj}^{\text{int}} - \Delta_{kj}^{\text{cal}}$), this circumstance prompts an uncertainty extension from 38 to 76 $\mu\text{V/V}$ at the 1 MHz observation point.

IV. CONCLUSION

The presented system is realized by the quintuple-two approach that integrates five double-components, namely:

- two objects (Fluke 792A and PMJTC instruments) are observed in the study;
- two setups are used to compare AC/DC differences;
- two adjacent ranges of the AC/DC transfer standard are investigated;
- two methods are applied to derive information to evaluate Δ -parameters (calibration data analysis and

measurement);

- two goals are pursued, which are the dissemination and stability assessment of the quantity under study.

The evidence of conceptual correctness is experimental and calibration data that substantiate the feasibility of implementation of the developed system.

The presented data analysis allowed us to localize a problematic point (0.2 V / 1 MHz) that prompted a more in-depth investigation of amplified input circuits. Although currently, uncertainty extension is a solution, further data accumulation is needed.

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MetroMag: A European Infrastructure for Low Magnetic Field Metrology

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Abstract – Measurements in the low magnetic field range [10 μ T – 10 mT] are increasingly required by industry due to strong demand and recent developments in key areas such as electric mobility, the medical sector, industry, and magnetic field sensing applied to the detection and localisation of ferromagnetic and conducting objects in safety applications and prospecting for natural mineral resources. However, only very few European National Metrology Institutes have the required capabilities to perform traceable measurements in the low magnetic field range. Consequently, the adoption of novel technologies and materials is hindered by the lack of pan-European metrological expertise in this area. The 24RPT02 MetroMag project will establish standards to measure weak magnetic fields, develop a transfer standard to enable easier and faster comparisons, develop novel methods to cancel environmental magnetic fields, and establish a European infrastructure in the low magnetic field ranges to address the stakeholders' needs.

Keywords: DC magnetic field metrology; interlaboratory

comparison; magnetic field sensing; environment magnetic field cancellation.

I. INTRODUCTION

Measurements in the low magnetic field range [10 μ T – 10 mT] are increasingly required by industry and other sectors due to strong demand and recent developments. In particular, low-intensity magnetic fields measurements are needed: in healthcare and environmental monitoring [1–3], where low exposure levels need to be assessed for long periods with suitably calibrated sensors (for example in support of Directive 2013/35/EU related to the exposure of workers to the risks arising from physical agents (electromagnetic fields)); in biomedicine [4–8], where the weak magnetic signals generated by the neuronal activity of the human body need to be reliably measured for diagnostic applications; in environmental and industrial detection [9–14], where low-intensity static magnetic fields are a useful tool for non-invasive and non-destructive testing; in quantum computing and sensing [15, 16], which is pushing the current metrological limits to unprecedented levels both at extreme high and extreme

low magnetic field strengths; in new classes of magnetic sensors [17–19], which are exploiting new effects of magnetic materials and are finding application in ICT and related industries; in the electrical mobility and energy transformation sectors [20–22], where magnetic field measurements are of the utmost importance for assessing power consumption and the efficiency of components and processes, in support of the European Green Deal; in space applications [23, 24], both for scientific investigation and for ensuring the magnetic field cleanliness of components and devices.

Very few European countries currently have measurement and calibration capabilities for magnetic parameters. However, the current offerings from National Metrology Institutes (NMIs) and Designated Institutes (DIs) are not sufficient to meet industrial demand, and a wider network of laboratories having the capability to perform magnetic measurements and calibrations is needed.

Within this framework, the project 24RPT02 MetroMag [25], funded by EURAMET within the Horizon Europe Programme, aims at the following objectives:

1. To develop optimal and effective traceable measurement standards for magnetic fields in the range [10 μ T – 10 mT] with uncertainty (at $k = 2$) between 2 mT/T and 0.2 mT/T.
2. To develop a robust and feasible transfer standard and cross-calibration method to provide traceability to the metrological laboratories with an expected accuracy of 10^{-4} . This transfer standard will be used in an interlaboratory comparison in this project and to validate calibration and measurement capacities (CMCs).
3. To develop new methods for magnetic field cancellation in the presence of local gradient fields, compensating for DC power noise and other sources of noise due to urban development, that hinder the calibration of new, low magnetic field systems.
4. To develop a metrological infrastructure for low-intensity magnetic fields to address the main needs of the stakeholders.

In this paper we will discuss the activities performed in the first months of the 24RPT02 MetroMag project to address these objectives.

II. LOW INTENSITY MAGNETIC FIELD SENSING

Nowadays, primary standards of magnetic field are based on the nuclear magnetic resonance (NMR) technique in water samples, which ensures traceability through the frequency standard. This technique can only be used in DC magnetic field measurements and in a limited magnetic field range, with a lower limit of 10 μ T (free precession technique) or 30 mT to 40 mT (forced precession technique). Below 10 μ T the NMR signal vanishes due to a reduced difference in thermal population of spin-up and spin-down nuclear spin states. Even in

magnetic fields between 10 μ T and 10 mT, the NMR method imposes strict constraints on the experimental conditions, such as high homogeneity of the field over a large volume, e.g. a sphere of 4 cm diameter. Whilst some other potential measurement techniques are already available at a research laboratory level, e.g. optical magnetometers, fluxgates or SQUIDs, the necessary studies and developments to bring them into a metrological chain to provide the realisation of the unit of magnetic field are still lacking. For this reason, alternative and easy to use calibration capabilities are highly desired, especially in new developing calibration laboratories with limited infrastructure.

During the first quarter, the MetroMag project has established a characterisation protocol that will help the participating laboratories in assessing the performance of their standard magnetometers used for magnetic field metrology. Even the laboratories exploiting coils as standard will benefit from a characterisation of transfer standards like Hall probes teslameters. The aim of this activity is to build a public dataset reporting the performance of the different type of sensors and magnetometers, for different magnetic field ranges and for different applications.

III. INTERLABORATORY COMPARISON

To ensure confidence in the measurements and calibrations provided by NMIs/DIs, it is necessary that the measurements are traceable, robust and comparable. Therefore, for an extended network of NMIs/DIs providing magnetic measurements and calibrations to be effective, timely and reliable laboratory intercomparisons are required, covering all necessary magnetic field ranges. Up to now, interlaboratory comparisons on magnetic field measurements have been very scarce [26], and new protocols to enable comparison and mutual recognition are needed.

In the first quarter of the project, the consortium has identified a Garrett solenoid (Fig. 1) that will act as travelling coil among the participants, to perform an interlaboratory comparison. The coil was characterised in the laboratories of CMI, its stability over time was checked against historical data, and a numerical representation



Fig. 1. The Garrett solenoid chosen for the interlaboratory comparison, CMI and CTU.

through a digital twin is currently being developed. The coil is compatible with the experimental setups of all participants, who will measure the coil constant (in mT/A) in two subranges ($20 - 50 \mu\text{T}$ and $1 - 5 \text{ mT}$). Equivalence of the participants in the comparison will be established according to [27], and will ensure the metrological robustness of their approaches.

IV. MAGNETIC FIELD CANCELLATION

The Earth produces its own magnetic field, typically $40\text{--}50 \mu\text{T}$, and this is of sufficient strength to be a perturbing factor in low-intensity magnetic field measurements. Current environmental magnetic field cancellation techniques rely on shielding or on passive or active feedback on compensation coils (Fig. 2) to cancel the Earth's and environmental magnetic fields in a given volume of space. However, due to constant urban development and local DC power noise, cancellation systems are starting to be insufficient to maintain their initially designed performance. This issue is particularly important for low range magnetic fields, whose measurement is even more affected by uncompensated, time-varying background noise. A typical example of background magnetic noise is shown in Fig. 3, which reports the time dependence, over 5 days, of the environment magnetic field along one direction (approximately East-West). Daily cycles are clearly visible, due to human activities in the laboratories and in the surrounding neighbourhood, but also fluctuations at higher frequency are clearly visible, which negatively affect calibration tasks that can last a few minutes each.

In the first quarter of the project, the participants have agreed on a protocol to benchmark the environment magnetic field compensation systems present in their laboratories. The protocol assumes that each partner compensates the environment magnetic field at the best of their capabilities, and then, with feedback turned off, the magnetic field fluctuations will be recorded for a time interval of one minute. Then, the same procedure will be repeated, but with the feedback for active compensation



Fig. 2. A typical 3-axial Helmholtz coil system for compensating the environment magnetic field, at INRIM laboratories.

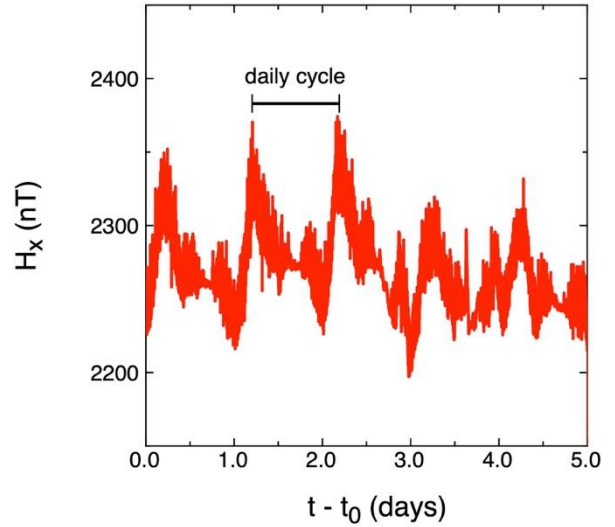


Fig. 3. A representative time dependence over a few days of one component of the environment magnetic field at INRIM laboratories.

turned on. An example of what is obtained by following such a procedure is shown in Fig. 4, for the same component of the environment magnetic field shown in Fig. 3. Please note the very different time scale, that allows to appreciate the undesired fluctuations over the duration of a typical calibration procedure. While passive compensation (second inset of Fig. 4) preserves the same amplitude of fluctuations in the background magnetic field, active compensation (third inset) may induce additional oscillations due to improper tuning of the feedback loop or to the presence of fluctuating gradients. Subsequent data analysis will allow benchmarking of the different compensations systems, highlighting their strong points and the areas of improvement. The benchmarking results will constitute a dataset that will be published online.

Moreover, a few partners will implement novel strategies to try to improve over existing magnetic field cancellation techniques, by exploiting the measurement of carefully selected components of local magnetic field gradients, or by exploiting forecast techniques based on statistical analysis and machine learning to help improving the performance of a magnetic field cancellation system.

V. IMPACT

The increasing industrial demand for reliable measurement of low magnetic field strengths is driving the need for development of a new metrological infrastructure at the European level, including new standards and new ways to disseminate the unit tesla to NMIs and DIs, universities, research institutes, test laboratories and manufacturing companies that rely on magnetic measurements and calibrations for their research and development, quality control, production and technology.

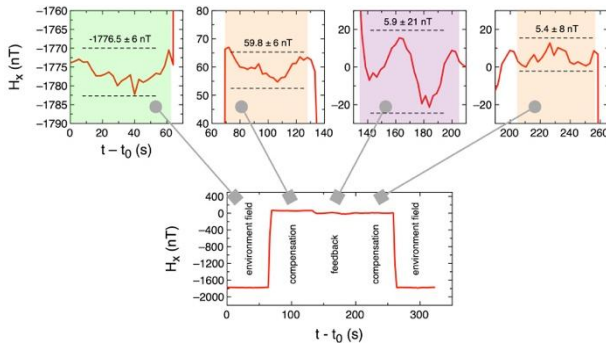


Fig. 4. A representative example of the fluctuations over time of the environment magnetic field without compensation (first inset), with passive compensation (second inset), with active compensation (third inset), and again without compensation (fourth inset).

These organisations would benefit from a proper metrological infrastructure for low strength magnetic fields to support new technologies, markets and applications oriented to new materials, environment monitoring, the prospecting for natural resources, healthcare, electrical mobility, etc.

The project is actively seeking for stakeholders, which may include academia, research institutes, research groups, and companies, who might be interested in the development and outcomes of the project (see the website [25] for contact points).

VI. CONCLUSIONS

The project addresses the challenge of developing low range magnetic field metrology focussing on three interdependent issues, each requiring progress beyond the state of the art: i) traceable standards; ii) a transfer standard enabling more frequent intercomparisons; iii) improved cancellation techniques for environment magnetic fields.

The new standards and methods developed in the project for realising low-intensity magnetic fields and for active compensation of environment magnetic fields will allow additional laboratories to calibrate and characterise probes in the sub 10 mT range, and thus facilitate new or improved services to end users.

VII. ACKNOWLEDGMENTS

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Exploring calibration schemes for electric impedance measurements beyond the short-open-load

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Abstract – This work introduces a generalized three-load (LLL) calibration scheme for impedance meters, aimed at enhancing the measurement accuracy over the complex impedance domain. A Möbius transformation, with parameters identified from measurements of three reference impedance standards, is employed to model the meter error. The associated uncertainty is evaluated following the Supplement 2 of the *Guide on Expression of Uncertainty in measurements (GUM) Supplement 2*. A practical implementation of the proposed calibration on a commercial handheld meter is presented.

Keywords: Impedance measurement; calibration; measurement uncertainty; linear fractional transformations.

I. INTRODUCTION

Nowadays, electrical impedance measurements are performed with an electronic meter, an LCR meter or an impedance analyzer. When the impedance under measurement (the device under test, or DUT) is connected to the instrument, it completes a circuit mesh that includes a sinewave generator and a reference impedance standard (typically a resistor). The meter act as a voltage ratio bridge [1, Sec. 4.4]: the complex-valued ratio of the voltage phasors on the DUT and on the standard is measured, and numerically processed to provide the displayed impedance reading.

To improve the meter accuracy, the user can perform measurements of both the DUT and on additional calibrated impedance standards, in a process called in literature *calibration*, but also *compensation* or *correction*.

A calibration scheme typically employed, and often available in the instrument firmware, is the *short-open-load* (SOL) [1, Sec. 4.8.2.2], [2, App. C]. In SOL, the short S is an impedance whose value is zero (that is, zero voltage drop for any current), the open O is an impedance having an infinite value (zero current flows for any voltage applied), the load L a resistive (real-valued) impedance in the same range of impedance magnitude of the DUT. A four terminal-pair definition of the standards allows to build “perfect” open and short standards; in simpler impedance

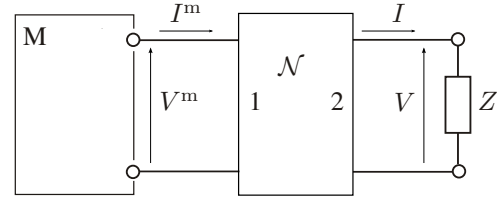


Fig. 1. Measurement network schematic representation. M is an ideal LCR meter, N is a two-port linear network representing the measurement errors and Z is the DUT.

definitions, the open and/or short behaviour are approximated. Two-load scheme has also been proposed [3]. The standard SOL provides a limited error correction, in particular for the imaginary part of impedance measurements.

In this work we consider a three-load (*load-load-load*, LLL) calibration scheme. The scheme was previously proposed, although not developed, by one of the authors [1, Sec. 4.8.2.1]. The scheme computes the DUT impedance from the readings of the meter on the DUT and on three calibrated impedance standards (loads) having a known value. The uncertainty evaluation asks for a propagation in a complex-valued nonlinear model.

II. THEORY

Fig. 1 gives a circuit representation of the meter error model, which is the same employed for the development of the SOL calibration.

The impedance Z is a two-terminal element, with current I developing voltage V . The meter is modeled as a cascade of a two-port error network N , which maps I to I^m and V to V^m , and an ideal meter M that provides the reading $Z^m = I^m / V^m$.

If N is a two-port linear network with ABCD parameters,

$$\begin{bmatrix} V^m \\ I^m \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \cdot \begin{bmatrix} V \\ I \end{bmatrix}. \quad (1)$$

From (1),

$$Z^m = \frac{V^m}{I^m} = \frac{AV + BI}{CV + DI} = \frac{AZ + B}{CZ + D}. \quad (2)$$

The error model is a *Möbius transformation*, a linear fractional transformation of the impedance value Z in the meter reading Z^m . If $D \neq 0$, as occurs in all cases of practical interest, Eq. (2) can be rewritten as

$$Z^m = \frac{\alpha Z + \beta}{\gamma Z + 1}, \quad (3)$$

and its inverse as

$$Z = \frac{Z^m - \beta}{-\gamma Z^m + \alpha}, \quad (4)$$

Equation (3) and (4) make evident the dependence of the transformation on *three* complex-valued parameters α , β , γ . These can be determined by measuring (at least) three known impedance standards Z_1 , Z_2 , and Z_3 , connected in sequence to M, and by solving the system of equations

$$\begin{cases} Z_1 \alpha + \beta - Z_1 Z_1^m \gamma = Z_1^m \\ Z_2 \alpha + \beta - Z_2 Z_2^m \gamma = Z_2^m \\ Z_3 \alpha + \beta - Z_3 Z_3^m \gamma = Z_3^m. \end{cases} \quad (5)$$

SOL assumes $Z_1 \equiv 0$, $Z_2 \equiv \infty$, and $Z_3 = Z_L$ resistive. Here we generalize to arbitrary Z_1 , Z_2 , Z_3 in the complex plane. The system (5) has an unique solution in α , β and γ which, when substituted in (4), gives

$$\begin{aligned} Z &= \frac{N}{D}, \quad \text{where} \\ N &= Z_1 Z_2 Z_2^m Z_3^m - Z_1 Z_2 Z_1^m Z_3^m + Z_1 Z_2 Z_1^m Z_2^m \\ &\quad - Z_1 Z_2 Z_2^m Z_1^m + Z_1 Z_3 Z_1^m Z_2^m - Z_1 Z_3 Z_2^m Z_3^m \\ &\quad - Z_1 Z_3 Z_1^m Z_2^m + Z_1 Z_3 Z_3^m Z_2^m - Z_2 Z_3 Z_1^m Z_2^m \\ &\quad + Z_2 Z_3 Z_1^m Z_3^m + Z_2 Z_3 Z_2^m Z_1^m - Z_2 Z_3 Z_3^m Z_1^m, \\ D &= Z_1 Z_1^m Z_2^m - Z_1 Z_1^m Z_3^m - Z_1 Z_2^m Z_1^m + Z_1 Z_3^m Z_1^m \\ &\quad - Z_2 Z_1^m Z_2^m + Z_2 Z_2^m Z_3^m + Z_2 Z_1^m Z_1^m - Z_2 Z_3^m Z_1^m \\ &\quad + Z_3 Z_1^m Z_3^m - Z_3 Z_2^m Z_3^m - Z_3 Z_1^m Z_1^m + Z_3 Z_2^m Z_1^m. \end{aligned} \quad (6)$$

The limit of (6) for $Z_1 \rightarrow 0$, $Z_2 \rightarrow \infty$ retrieves the equations for the SOL calibration scheme.

III. UNCERTAINTY

The standard impedances Z_1 , Z_2 , Z_3 values are known up to an uncertainty; the readings Z_1^m , Z_2^m , Z_3^m , Z^m can also be affected by an uncertainty due to the meter noise and resolution. These uncertainties propagate through the model (6) to give the uncertainty of the mesaurand Z .

The propagation of complex quantities through a nonlinear model can be performed in the framework of the *Guide on the Expression of Uncertainty in measurement* Supplement 2 [4].

An analytic expression of the sensitivity coefficients is cumbersome. We therefore employed for the calculations the package `Metas.UncLib` [5, 6], using the `LinProp`

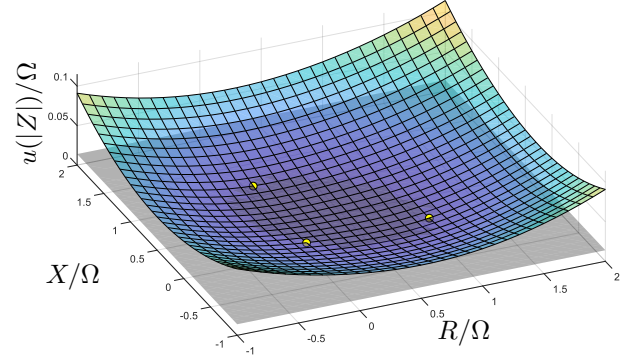


Fig. 2. A toy example of the uncertainty propagation through the three-load calibration scheme. Color map of the uncertainty $u(|Z|)$ of the magnitude of the Z , the model outcome. Z_1 , Z_2 and Z_3 are plotted as yellow markers. The light red plane intersecting the color map represents the uncertainty on the on the magnitude of the three standards.

module, which performs automated algorithmic differentiation [7].

Fig. 2 provides a toy example of the outcome of uncertainty propagation through the model (5). In the example, the standards values are $Z_1 = (1 + 0j) \Omega$, $Z_2 = (0 + j) \Omega$, $Z_3 = (0 + 0j) \Omega$; the uncertainties of all the three standards are set to $1 \times 10^{-2} \Omega$ for both the real and imaginary part (uncorrelated). The uncertainty of the readings is set to zero.

As expected, Fig. 2 shows that the uncertainty is the smallest when Z lies within the triangular area defined by the three standards, and is therefore interpolated from the known values. The uncertainty grows quickly outside that area, where the model solution becomes an extrapolation.

IV. EXPERIMENTAL

To test the calibration scheme, we set up an experiment using commercial meters and a set of impedance standards.

A. Instruments and standards

- The meter M on which the calibration scheme is applied is a Keysight U1732C handheld impedance meter. It can measure three-terminal impedances with a $4\frac{1}{2}$ digit resolution in magnitude, 0.1° in phase. The specified relative base accuracy is 0.2 %.
- The reference impedance values are provided by measurements of a Keysight E4980B bench-top impedance bridge, having specified relative base accuracy of the instrument is 500×10^{-6} . The meter natively employs a four terminal-pair (4P) impedance definition. To ensure a impedance definition consis-



Fig. 3. Experimental setup. The bench-top Keysight E4980B LCR bridge is measuring the parallel of a 10 nF and of a 10 k Ω resistor, both defined as two-terminal pair impedance standards.

tent with M, a 4P to two terminal-pair (2P) adapter was employed.

- The standards Z_1 , Z_2 , Z_3 and the DUT impedance Z are realized with standard resistors belonging to an Agilent 42030A set, and standard capacitors from a custom-made thermostated set [8]. The impure standards are obtained from the pure ones with parallel or series connections. The impedance definition includes the connecting cables.

Figure 3 shows the E4980A meter, the capacitance standard set, and one of the standard resistors during a measurement.

B. An example of results

In the following, we provide an example of application of the scheme.

Four impedance standards are employed: the calibration standard nominal values are $Z_1^{\text{nom}} = 1 \text{ k}\Omega$, $Z_2^{\text{nom}} = 100 \text{ k}\Omega$ and $Z_3^{\text{nom}} = 1 \text{ nF}$.

The DUT is the RC parallel $Z^{\text{nom}} = 10 \text{ nF} // 10 \text{ k}\Omega$.



Fig. 4. Experimental setup. The three-terminal connection of the meter M, Keysight U1732C, to the two coaxial cables leading to the impedance under measurement.

The measurement frequency is set to 1 kHz. All measurements are performed in a thermostated laboratory at 23.0(5) $^{\circ}\text{C}$.

As reference values Z_1 , Z_2 , Z_3 and Z of the standards, the measurements of the E4980B are considered, with uncertainties set to the meter specifications.

The readings Z_1^m , Z_2^m , Z_3^m , Z^m are stable within the resolution of M, which is therefore considered as the only contribution to the corresponding uncertainty.

Figure 5 gives a graphical representation of the calibration scheme effectiveness:

- Z and its 95 % elliptical region of confidence is plotted in **red**;
- Z^m and its region of confidence is plotted in **green**;
- The outcome of the calibration Z_{3L} is plotted with its region of confidence in **blue**.

It can be easily recognized that Z_{3L} is much closer than Z^m to the reference value Z , and that the region of uncertainty of the former is smaller. The effectiveness of the

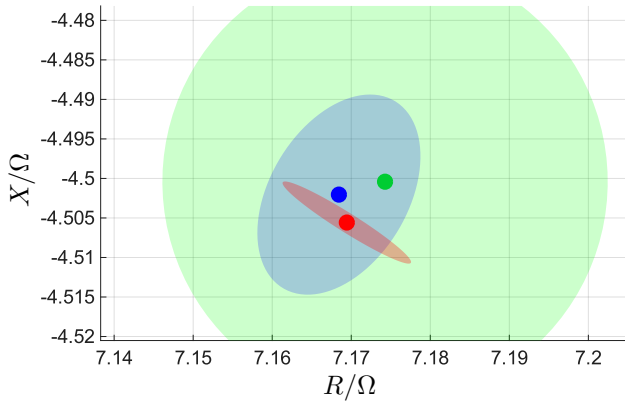


Fig. 5. Z (red), Z^m (green) and Z^{3L} (blue) estimations of Z^{nom} and their corresponding regions of confidence.

calibration scheme in improving the meter accuracy is confirmed.

V. DISCUSSION AND OUTLOOK

The LLL calibration scheme here proposed is of simple application in the laboratory environment, comparable to the SOL scheme. The required calibrated standards are often already available in a calibration laboratory. The automation of the uncertainty evaluation provided by the `Metas.UncLib` package allows to quickly simulate the effectiveness of the scheme given the standards available.

In Sec. ii. the LLL calibration equations are derived from a model where Z is defined as a two-terminal impedance. Suzuki et al. [9] have analyzed an extended model where Z is defined as a 4P impedance and have shown that the SOL equations maintain their validity. Work is ongoing to show that the LLL equations also satisfy this extended model.

Recent research on primary impedance metrology focuses on extending impedance traceability over the entire complex plane, thus allowing the calibration of DUTs having arbitrary phase angles [10, 11, 12]. The method proposed constitutes a tool in this direction.

An application to more accurate meters and measurements, using reference impedance standards calibrated with primary methods, is presently under development.

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MEMS-based Long-haul Optical Link Asymmetry Estimation for Atomic Clock Comparison

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Abstract – Precise atomic clock comparison over long-haul optical fibers requires rigorous compensation of link asymmetry. While White Rabbit (WR) technology provides sub-nanosecond synchronization, chromatic dispersion and physical path differences in Wavelength Division Multiplexing (WDM) systems introduce non-negligible timing offset. Classical calibration techniques typically rely on manual fiber reconnection to toggle signal direction, a process prone to human error and known to cause rapid degradation of optical connectors. This paper proposes an automated *in-situ* asymmetry estimation technique utilizing an auxiliary optical channel and MEMS-based switching. By replacing manual intervention with high-reliability MEMS technology, the proposed architecture ensures a more robust measurement environment and establishes a framework for future continuous, long-term monitoring of synchronization stability without physical link disturbance. The methodology was validated on a 50 km laboratory spool, demonstrating agreement within 100 ps of theoretical dispersion models. Field deployment on a 20 km urban link connecting CTU FEE and IPE CAS successfully determined link asymmetry (t_{asm}) and compensated for an $11,248.0 \text{ ps} \pm 41.7 \text{ ps}$ synchronization offset, effectively reducing link-induced synchronization errors to sub-nanosecond levels and enabling high-precision comparison of remote atomic timescales.

Keywords: Optical fiber links, White Rabbit, Asymmetry calibration, Atomic clock comparison, Chromatic dispersion, MEMS switches

I. INTRODUCTION

The precise comparison of remote atomic timescales is a fundamental requirement in contemporary metrology, supporting applications ranging from the generation of Coordinated Universal Time (UTC) to relativistic geodesy. When physical proximity between timing nodes

is not feasible, optical fiber links represent the state-of-the-art medium for high-fidelity time and frequency transfer. However, achieving sub-nanosecond synchronization accuracy requires rigorous characterization of the propagation delays within the optical path.

A primary challenge in long-haul optical links is optical path asymmetry. In systems utilizing Wavelength Division Multiplexing (WDM), such as White Rabbit (WR) technology [1, 2], the transmit (T_x) and receive (R_x) signals propagate at different wavelengths over a single fiber. This introduces a timing offset due to the chromatic dispersion of the optical fiber, approximately $17 \text{ ps nm}^{-1} \text{ km}^{-1}$ for standard G.652 fiber. As demonstrated in [3], this dispersion-induced asymmetry is the limiting factor for sub-nanosecond accuracy in long optical links. Alternatively, links utilizing separate fibers for T_x and R_x at identical wavelengths suffer from physical path length differences. In both scenarios, accurate estimation of the link asymmetry t_{asm} is essential, as the resulting synchronization error is equivalent to half of this total round-trip asymmetry.

While local calibration is possible for short-range installations, it becomes increasingly difficult for deployed fiber links that traverse multiple reconnection points or utilize Dense Wavelength Division Multiplexing (DWDM) to share the medium with high-speed data traffic. This is the case for the link connecting the time laboratories at CTU FEE (Czech Technical University in Prague, Faculty of Electrical Engineering) and IPE CAS (Institute of Photonics and Electronics of the Czech Academy of Sciences). Both facilities are equipped with Cesium atomic clocks, but complex urban fiber routing necessitates a robust, *in-situ* calibration methodology that does not rely on manual intervention, aligning with the broader strategy for establishing reliable clock network services over existing fiber footprints [4].

This paper proposes an automated asymmetry estimation technique utilizing a White Rabbit link integrated with an auxiliary optical reference channel. By employing optical MEMS switches to interchange the direction of a co-

herent 10 MHz reference signal, we can isolate and measure the link asymmetry independently of the chromatic dispersion on the auxiliary optical channel. The main contributions of this work are:

- The development of a hardware architecture capable of autonomous, active asymmetry monitoring using high-repeatability MEMS-based optical switching.
- A two-stage calibration methodology involving initial laboratory characterization of hardware delays followed by automated field deployment.
- Experimental verification of the technique on a 50 km laboratory spool and a real-world installed urban link, demonstrating the capability to resolve picosecond-level synchronization offset.

II. MEASUREMENT PRINCIPLE AND SYSTEM ARCHITECTURE

This section details the theoretical framework for asymmetry estimation, which is built upon the established White Rabbit calibration model [5], and the integrated hardware architecture developed to facilitate autonomous sub-nanosecond synchronization monitoring.

Figure 1 shows the overall measurement architecture including the White Rabbit nodes, auxiliary channel, and MEMS switching layer.

A. Theoretical Framework

To estimate t_{asm} *in-situ*, we utilize an auxiliary DWDM channel at wavelength λ_{aux} . By employing a common-path approach with alternating measurement directions, we define the measured time-skews at Node A (skew_A) and Node B (skew_B) as follows:

$$\text{skew}_A = t_{1\text{PPS-A}} - (t_{10\text{MHz-B}} + \delta_{\text{aux}}) \quad (1)$$

$$\text{skew}_B = t_{1\text{PPS-B}} - (t_{10\text{MHz-A}} + \delta_{\text{aux}}) \quad (2)$$

where δ_{aux} is the fiber propagation delay specific to the auxiliary channel at wavelength λ_{aux} , $t_{1\text{PPS}}$ represents the rising edge of the local 1PPS signals, and $t_{10\text{MHz}}$ denotes the rising edge of the 10 MHz reference signal received from the remote node. **Crucially, the 1 PPS and 10 MHz signals at each node are phase-coherent**, as they are derived from a common internal clock source within the WR switch.

By subtracting these reciprocal measurements, the common-mode fiber delay δ_{aux} is canceled. The relationship between the measured skews and the node-to-node offset (skew) is derived as:

$$\text{skew}_A = \text{skew} - \delta_{\text{aux}} \quad (3)$$

$$\text{skew}_B = -\text{skew} - \delta_{\text{aux}} \quad (4)$$

Consequently, the synchronization offset and the total link asymmetry are isolated as:

$$\text{skew} = \frac{\text{skew}_A - \text{skew}_B}{2} \quad (5)$$

$$t_{\text{asm}} = \text{skew}_A - \text{skew}_B \quad (6)$$

B. Hardware Implementation

The physical layer is built upon White Rabbit Low Jitter (WRLJ) switches capable of picosecond-level resolution and sub-nanosecond synchronization accuracy. The architecture consists of three functional layers:

- **Timing Layer:** WRLJ switches serving as Grandmaster (GM) and Slave, disciplined by local Cesium atomic standards. These devices ensure the required phase-coherence between the 1 PPS and the 10 MHz auxiliary output.
- **Acquisition Layer:** High-resolution Time Interval Counters (TIC), specifically Pendulum CNT-104s, which perform the skew measurements with a resolution of 7 ps.
- **Switching Layer:** Optical MEMS switches integrated with Optical-to-Electrical (OE) and Electrical-to-Optical (EO) converters. Unlike manual patching, which is susceptible to connector wear and insertion loss variability [6], MEMS switches provide the high repeatability and operational longevity required for autonomous field monitoring [7].

C. Reciprocity Considerations

The validity of this methodology rests on the reciprocity of the auxiliary channel. By utilizing the same DWDM channel and optical fiber for both skew_A and skew_B measurements, chromatic dispersion effects are identical in both directions. The MEMS switches ensure that internal hardware delays remain constant. Provided the switching cycle is performed within the thermal coherence time of the environment, this reciprocity allows the system to accurately distinguish between link-induced asymmetry and internal equipment offsets.

III. LABORATORY CALIBRATION AND METHODOLOGY

To isolate the link-induced asymmetry from the internal delays of the measurement hardware, a multi-stage calibration procedure was implemented. This methodology ensures that the final estimation reflects only the optical path characteristics and is independent of equipment-specific latencies.

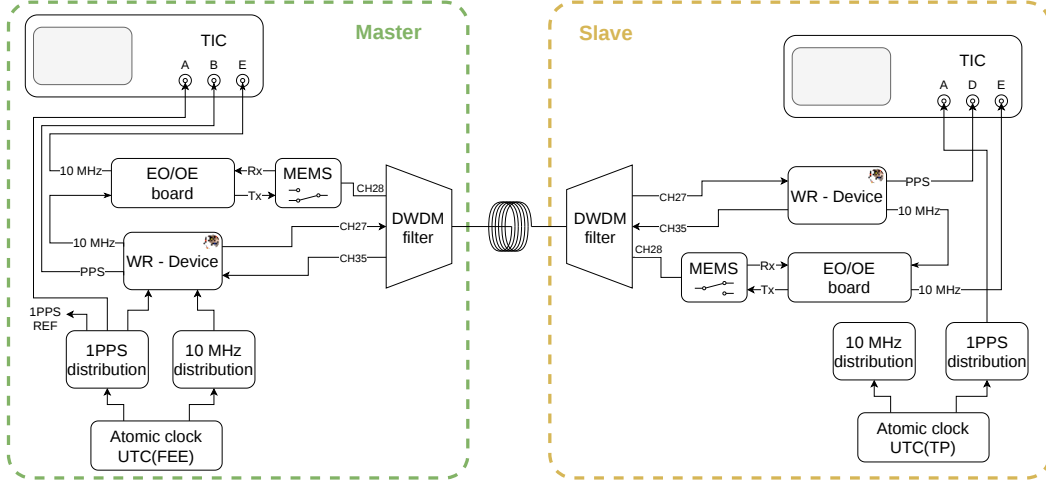


Fig. 1. Overall measurement setup for in-situ asymmetry estimation using an auxiliary optical channel and MEMS switching.

A. Internal Delay Characterization

The hardware components, including SFP modules, PCB routing on the custom electro-optical/optical-electrical (EO/OE) experimental boards, and coaxial interconnects, introduce fixed time offsets. These internal delays must be characterized before field deployment. The primary sources of these delays include:

- **SFP Transceiver Latency:** Delays originating from internal electrical routing and laser driver response times.
- **Interconnect Asymmetry:** Small length differences in the coaxial cables connecting the White Rabbit switch (WRS) outputs to the Time Interval Counters (TIC).
- **MEMS-Induced Delays:** The propagation delay within the optical MEMS switch, which may vary slightly between the different switching states.

B. Reference Setup and Baseline Measurement

A "short-fiber" configuration was established using a 1 m patch cord to connect the two White Rabbit nodes. In this controlled environment, chromatic dispersion is negligible. The initial offset between WR nodes, measured as the time difference between the respective 1PPS signals, was 1350 ps. This reflects a total reference setup asymmetry of 2700 ps.

A specific calibration correction was configured in the WR Slave node to compensate for this asymmetry, which reduced the residual offset to -7.7 ± 4.2 ps. Following this internal synchronization, the time difference between the local 1PPS signal and the 10 MHz reference received via the auxiliary channel was recorded for both nodes. These

values, denoted as $\text{skew}_{A,ref}$ and $\text{skew}_{B,ref}$, serve as the baseline calibration for all subsequent measurements. The results of the reference calibration procedure are summarized in Table 1.

To isolate the true link characteristics from internal hardware latencies in subsequent tests, the calibration-corrected synchronization offset (skew) is formally defined as:

$$\text{skew} = \frac{(\text{skew}_A - \text{skew}_{A,ref}) - (\text{skew}_B - \text{skew}_{B,ref})}{2} \quad (7)$$

Table 1. Reference calibration results (1 m fiber)

1PPS offset	$\text{skew}_{A,ref}$	$\text{skew}_{B,ref}$
-7.7 ± 4.2 ps	$57,338.0 \pm 12.0$ ps	$57,309.0 \pm 11.5$ ps

IV. EXPERIMENTAL VALIDATION (SPOOL TESTING)

Following the laboratory calibration, the short reference fiber was replaced with a 50 km optical fiber spool to emulate the characteristics of a real-world deployed link. This stage serves to validate the measurement principle against theoretical dispersion models.

A. 50 km Fiber Emulation

The fiber spool used for validation is a standard G.652 single-mode fiber (SMF) with a manufacturer-specified length of 50.47 km, nominally referred to as a 50 km spool. With the White Rabbit link operating at DWDM channels 27 ($\lambda_{Tx} = 1555.75$ nm) and 35 ($\lambda_{Rx} = 1549.32$ nm), the wavelength separation is $\Delta\lambda = 6.43$ nm. Using the nominal chromatic dispersion coefficient $D \approx 17$ ps nm⁻¹ km⁻¹,

for standard G.652 fiber [8], the expected delay asymmetry t_{asm} is calculated as:

$$t_{\text{asm}} = D \cdot \Delta\lambda \cdot L \approx 17 \cdot 6.43 \cdot 50.47 \approx 5516.9 \text{ ps} \quad (8)$$

The resulting synchronization offset observed in the 1PPS signal is expected to be half of this value, approximately 2758.4 ps. The goal of the validation is to observe a matching estimated asymmetry using the auxiliary channel, corrected by the baseline skew_{A,ref} and skew_{B,ref} values established during calibration.

B. Comparison with Theoretical Models

The measurement sequence was initiated by toggling the MEMS switches every 300 s to allow the TIC to reach a stable average. The experimental results, summarized in Table 2, demonstrate a high degree of correlation with the theoretical model.

Table 2. Fiber spool validation results (50 km)

1PPS offset	skew _A	skew _B
2718.5 ± 3.9 ps	40,447.0 ± 11.0 ps	35,081.0 ± 13.3 ps

C. Calculation Breakdown

The synchronization offset caused by the validation fiber spool was 2668.5 ± 17.4 ps after applying the reference calibration. The theoretical value was computed using the widely adopted nominal chromatic dispersion coefficient $D \approx 17 \text{ ps nm}^{-1} \text{ km}^{-1}$ for standard G.652 single-mode fiber, whereas the actual dispersion coefficient of the specific fiber spool may differ from this nominal value within the manufacturer specification of $D \leq 18 \text{ ps nm}^{-1} \text{ km}^{-1}$ [9]. The observed deviation of -89.9 ps is attributed to this discrepancy between the nominal and actual dispersion coefficient of the validation spool. These results demonstrate that the MEMS-based switching architecture can reliably resolve and characterize real-world, optical link asymmetry.

V. FIELD DEPLOYMENT AND RESULTS

Following the successful laboratory validation, the measurement system was deployed on a real fiber link connecting two metrological laboratories. This section describes the field infrastructure and the results of the *in-situ* asymmetry estimation.

Figure 2 illustrates the deployed optical path between the CTU FEE and IPE CAS laboratories, including DWDM/CWDM multiplexing.

A. Infrastructure Overview

The experimental passive link consists of an approximately 20 km span of single-mode fiber. Unlike the laboratory spool, this link is routed through multiple urban

reconnection points and utilizes Dense Wavelength Division Multiplexing (DWDM) and Coarse Wavelength Division Multiplexing (CWDM) to coexist with other network traffic.

The reciprocity assumption regarding one-way delay stability, discussed in Section II.C, remains valid over the selected switching period. Buried optical fiber responds to surface temperature excursions as a strongly damped low-pass filter. Field measurements reported in [10] characterized a thermal time constant of approximately 12.7 days for a buried single-mode optical fiber cable. Over a single switching period, this implies that the fiber temperature changes by at most a fraction $1 - e^{-t_{\text{sw}}/\tau_{\text{th}}} \approx 5.5 \times 10^{-4}$ of any surface excursion. Under a conservative worst-case assumption of a 10°C step change applied uniformly across the entire link length, a scenario far exceeding any realistic condition, the resulting fiber temperature change remains below 0.006°C . Applying the thermal coefficient of delay $\text{TCD} = 40 \text{ ps km}^{-1} \text{ }^\circ\text{C}^{-1}$ from [11] over the 20 km link yields a worst-case one-way delay variation of:

$$\Delta\delta_{\text{fiber}} = \text{TCD} \cdot L \cdot \Delta T_{\text{fiber}} \approx 4.4 \text{ ps} \quad (9)$$

This bound is negligible relative to the synchronization performance targeted by the system, confirming that the reciprocity assumption in (5) holds robustly over the selected switching period. This conclusion is further supported by the soil thermal diffusion model of [12], which shows that even the diurnal temperature cycle, the fastest realistic ambient excursion, arrives at burial depth with strongly attenuated amplitude and a phase delay of several hours, rendering thermal fluctuations occurring within the switching period negligible for buried fiber installations.

The hardware configuration remained consistent with the laboratory setup:

- **Node A (Grandmaster):** Situated at the CTU FEE Time laboratory, synchronized to a local Cesium atomic clock.
- **Node B (Slave):** Situated at the remote IPE CAS laboratory, also equipped with a Cesium reference for independent verification.
- **Optical Layer:** DWDM filters were used to multiplex the White Rabbit traffic (Channels 27 and 35) and the auxiliary reference channel 28 (λ_{aux}) onto the shared fiber.

B. Initial Measurement on Deployed Link

The laboratory setup was integrated into the real installation to validate the connections and demonstrate expected behavior. The measurement sequence followed the same steps as the laboratory testing, using the "short-fiber" calibration as the baseline. The initial measurements are summarized in Table 3, where the final calibrated synchronization offset (skew) is evaluated according to (7).

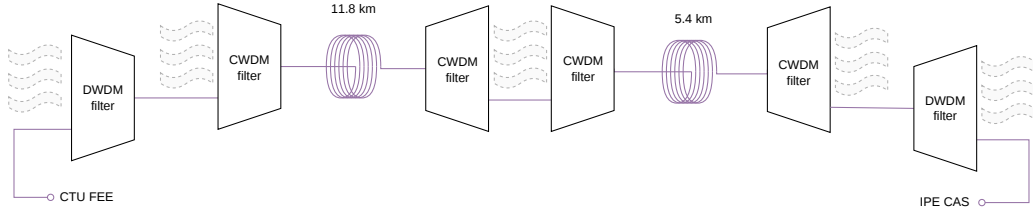


Fig. 2. Optical link deployment between CTU FEE and IPE CAS showing fiber routing and wavelength multiplexing.

Table 3. Installed link measurements

skew _A	skew _B	skew
$-61,217.4 \pm 58.7$ ps	$-83,742.4 \pm 59.1$ ps	$11,248.0 \pm 41.7$ ps

These measurements allow for the compensation of link asymmetry, which significantly improves synchronization accuracy. Based on (7), the WR Slave node was configured to compensate for the observed synchronization offset, and the calibration was verified through subsequent measurement cycles.

C. Comparison with Atomic Scales

The 1PPS outputs of the WR nodes and the local Cesium clocks were compared using the multichannel TIC. The time difference between the CTU FEE atomic clock and the WR GM is denoted as skew_{UTC(FEE)}, and the difference between the IPE CAS atomic clock and the WR Slave is labeled skew_{UTC(TP)}. The total difference between the atomic timescales skew_{TOT}, accounting for the WR-induced skew, is shown in Table 4.

Table 4. Timescale measurements and total synchronization offset

skew _{UTC(FEE)}	skew _{UTC(TP)}	skew _{TOT}
$19,183.9 \pm 13.5$ ps	$-273,904.8 \pm 85.4$ ps	$293,103.5 \pm 86.4$ ps

After applying the estimated asymmetry t_{asm} , the residual synchronization error was reduced and final measured offset was $293,103.5 \pm 86.4$ ps. A significant portion of this absolute offset is caused by the cabling at the IPE CAS laboratory. Due to structural constraints, the connection to the WR system requires an extended coaxial path, which introduces a substantial constant delay. The total delay attributed to the calibrated cabling on both nodes was characterized as $193,202.8 \pm 23.6$ ps. After subtracting these local instrumental delays, the final estimated offset between the remote cesium atomic timescales was determined to be $99,900.7 \pm 89.6$ ps.

Even this limited initial field deployment confirms that the auxiliary channel and MEMS switching methodology effectively track and eliminate deterministic link asymme-

tries, providing a stable foundation for remote clock comparison.

VI. DISCUSSION

The experimental results from both the laboratory spool and the field deployment demonstrate that the MEMS-based auxiliary channel effectively isolates link-induced asymmetry. The use of MEMS technology proved superior to traditional mechanical optical switches by providing high switching repeatability and minimal insertion loss variability. This stability is a critical threshold for maintaining sub-100 ps synchronization. While the current DWDM approach is naturally constrained by available spectral gaps in the fiber, the requirement of only a single narrow-band channel makes the system highly compatible with existing telecommunication infrastructures without the need for dedicated "dark" fibers. Furthermore, the automated switching cycle allows for the compensation of temperature-induced timing link fluctuations, which can significantly impact synchronization stability over long fiber spans [13]

VII. CONCLUSION

This paper presented an automated methodology for the *in-situ* estimation and compensation of link asymmetry in long-haul optical fiber networks. By integrating White Rabbit nodes with an automated MEMS-based optical switching layer, we successfully resolved synchronization offsets induced by chromatic dispersion and physical path differences. The laboratory validation on a 50 km fiber spool demonstrated a high correlation between the theoretical dispersion models and experimental measurements, with results agreeing within 100 ps.

Furthermore, the field deployment on a 20 km urban link confirmed the system's effectiveness in a real-world environment, identifying and compensating for a link-induced synchronization offset of $11,248.0 \pm 41.7$ ps. Future work will focus on full software automation and the investigation of long-term asymmetry stability to account for seasonal environmental fluctuations in fiber infrastructure.

VIII. ACKNOWLEDGMENTS

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A Model for Secure Exchange of Digital Calibration Certificates

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Abstract—The exponential growth of IoT applications and related hardware has dramatically changed the manual way of doing things. Metrology is one such field that is witnessing widespread changes caused by digitalization and connectivity of metrological hardware. Metrological devices need calibration at regular intervals to maintain their metrological accuracy. After the calibration, each device gets a calibration certificate that documents the findings of the calibration and is valid until the next calibration. These calibration certificates are issued on a physical paper or as a PDF document to be human readable. However, efforts are under way to come up with a globally acceptable machine readable calibration certificate called DCC (Digital Calibration Certificate). Growing demand of digitalization and connectivity of metrological devices means that the paper and PDF based calibration certificates will be soon thing of the past, and will eventually be replaced by machine readable Digital Calibration Certificates (DCCs). DCCs when transferred over the network could be exposed to security threats such as man-in-the-middle (MITM) attack. This paper presents a high level architectural model for secure DCC exchange, that is based on existing tried and tested techniques, and ensures integrity, confidentiality, non-repudiation and authenticity of DCCs, sent over network, from a calibration lab to the organization that requested for the equipment calibration.

Index Terms—Digital Calibration Certificate, man-in-the-middle (MITM), Calibration, IoT, Security, Integrity, Confidentiality

I. INTRODUCTION

The science of measurement is known as Metrology [1]–[4]. Accuracy, precision and correctness of the equipment used for metrological purpose play a critical role to establish trust and faith in metrological processes and measurements. The accuracy, precision and correctness of the devices and equipment can be maintained by a process known as “calibration”, performed at certain intervals. The process of calibration involves tuning and configuring a device or equipment to ensure its accuracy matches against a known standard.

The frequency and intervals of calibration differ from device to device. However, it is important for a metrological device before being used for measurements to retain a certain accuracy, thus, metrological devices and equipment usually go through multiple calibrations during their life span. At the end of calibration process, a certificate is issued that contains the device information, properties and findings of the process, physical environment that was maintained during the calibration such as temperature,

pressure, humidity etc, as well as the parameters used for the calibration. This certificate is valid until next calibration is required. Traditionally calibration certificates were issued on paper. However, nowadays companies also issue calibration certificates in digital formats such as Word and PDF. Due to its platform independent nature that ensures the same look and feel across all platforms, PDF is widely used for sharing calibration certificates. However, PDF is just a digital representation of the same paper based calibration certificate.

Like every scientific field, Metrology is also moving towards digitization and automation. One focus of research and development in Metrology is to develop a global standard for calibration certificates. A standard that can be used to represent all the calibration related information as an XML document [5]–[7]. A transition to Extensible Markup Language widely known as XML from PDF has enormous benefits, as it is a popular format for data storage and transfer. XML documents are extensible, so, users can add their own attributes, and fields in an XML document. This feature allows an XML document to contain lots of relative information, such as metadata, that if added to a PDF certificate or traditional paper based certificates will grow their size to many number of pages. Flexibility, clarity and simplicity of XML not only makes it human readable, but also ideal for machine readability. A calibration certificate in XML format is known as a Digital Calibration Certificate (DCC). A DCC would not only ensure global compatibility, but also makes it easier for machines to parse, interpret and store it on the device or in a database. A DCC in XML format would allow companies to exchange, transfer and store it in their Enterprise Resource Planning(ERP) systems, Quality management systems (QMS) and Supply chain tracking systems. Calibration certificates in PDF are great for human readability like their paper based counterparts, however, they can not offer the benefits of XML format. There are other popular machine readable data transfer formats such as JavaScript Object Notation (JSON), that is widely used for RESTful services [8]–[13]. Though, JSON is lighter, faster and easier to work with, however, XML due to its powerful features that allow it to represent complex descriptive attributes, namespace, comments, and tags in hierarchical fashion makes it better suited to represent digital calibration certificates.

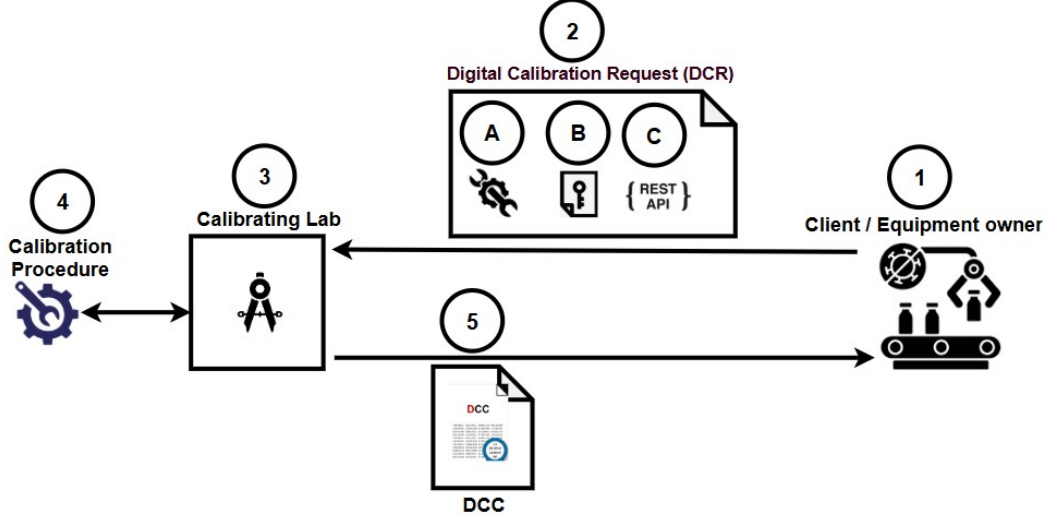


Fig. 1. Insecure exchange of DCC

Calibration certificates in machine readable formats known as Digital Calibration Certificates (DCCs) can be transferred over the network. It means that once calibration process for an equipment is complete, a DCC can be issued, and transferred to the equipment owner or the company that requested for the calibration. A calibrated equipment can be used in sectors, such as pharma, healthcare and defense, thus, information provided in a DCC can be very sensitive and needs proper security and safety measures to maintain the privacy, confidentiality and integrity of the information. Therefore, a DCC transferred from the calibration lab to the organization that needed calibration must ensure that information exchanged is safe against cyber threats, such as man-in-the-middle (MITM) attack, eavesdropping and information tempering [14]–[16]. This paper presents a simple, but effective approach to ensure that a machine readable DCC issued and transferred over the network is secure against cyber threats. A trusted and safe exchange of DCC ensures that it can be directly parsed and stored in a database or can be processed according to the internal requirements of the organization.

The paper is organized as follows. Section-I, introduces DCC and relevant security threats, Section-II highlights contribution and purpose of the work, Section-III presents related work, Section-IV provides background details, Section-V provides methodology and implementation details and Section-VI presents conclusion of the proposed framework.

II. RESEARCH CONTRIBUTION

The work presented here does not claim to introduce any groundbreaking or revolutionary approaches to secure exchange of Digital Calibration Certificates. It rather proposes a security framework that can be implemented using existing tried and tested approaches. The novelty of this work is to

answer a fundamental question that companies frequently ask that how to secure exchange of a DCC without investing in expensive hardware and software resources. This paper aims to prepare organizations adopt DCCs once their use is widespread by satisfying following important considerations.

1. Ensuring authenticity of calibration certificates ,
2. DCC confidentiality during inter-organizational exchange,
3. Traceability of potential certificate modifications, and
4. Compliance with national and international security standards.

III. RELATED WORK

Research and development in digital calibration certificates is still in its early stages. However, the exponential growth of IoT and related ecosystems is driving major industry players and knowledge producing institutions to focus on the DCCs. Some of the notable work is cited here. Gadelrab et al. in their work carry out analysis of calibration process and digital calibration certificates [17]. They identify several use cases and also discuss drawbacks and benefits of various DCC formats. Mustapää et al. in their work propose how to ensure and validate trustworthiness and data quality using digital calibration certificates, Digital SI (D-SI) data model and cryptographic based device identifiers [18]. Hackel et al. in their work comprehensively introduce the need for global standard in Metrology and calibration and aim to create a globally recognized DCC format [5]. Nummiliuikki et al. in their work [19] present a study based on a digitized proof-of-concept environment. The environment is then used for calibration related data generation, data transfer, and data usage. Hackel et al. in their work emphasis on the importance of DCC to improve the quality of entire digital infrastructure [7]. Their study suggests that DCCs can be used in the industry for cost saving and error prevention caused by manual intervention. Nummiliuikki et al. present advantage of standardization in the field of DCC and Metrology, and

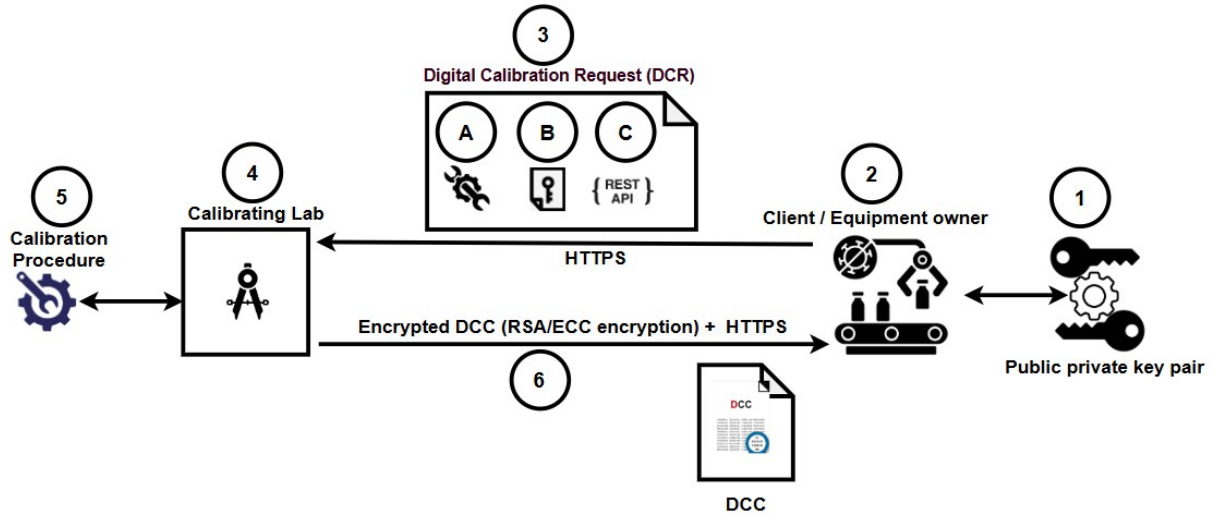


Fig. 2. Encrypted exchange of DCC

propose an interoperable platform that can be beneficial for all the Metrology stockholders [20]. Neyezhnikov et al. in their work present benefits of DCC in improving the traceability, and a scheme to implement digital calibration certificates in organizational setup [21]

IV. BACKGROUND

A digital calibration certificate (DCC) is an XML document that consist of four main sections as follows:

- 1) Administrative data:** This section is regulated and contains important administrative information such as, client, device and manufacturer details.
- 2) Calibration results:** This section contains the unit used for calibration and the resultant calibration details.
- 3) Comments:** This is a non-regulated section, and provides useful information and data that is agreed between client and the calibration lab.
- 4) Document:** This section contains necessary calibration details specially aimed for human readability.

The aforementioned structure of DCC provides important information about the sections which could possibly contain calibration details, client and manufacturer related sensitive information. As XML is a text based document, structured as a tree or in hierarchical order, thus, it is possible to just protect the elements containing the sensitive data from potential threats. However, for the sake of simplicity and easier implementation it is assumed in this text that the calibration lab and the equipment owner would like to protect the entire DCC i.e. the XML document from potential threats. The following section provides the implementation details aimed at protecting the entire DCC during the transfer.

V. METHODOLOGY

A DCC is an XML document, and an XML document is a piece of structured data. Thus, like any other piece of data, an XML document or DCC can also be protected

using modern cryptography techniques. Refer to Fig-1 that depicts exchange of DCC without any security mechanism in place. In this figure an equipment owner sends a DCR to a calibration lab and calibration lab after necessary calibration dispatches the calibration certificate using similar insecure mechanism. However, various sensitive fields such as pharma and defence need a secure and properly safeguarded means of DCC exchange.

In contrast to Fig-1, Fig-2 depicts a secure transfer of DCC from the calibration lab to the equipment owner or the client. The entire process takes place in 6 steps, as shown encircled in Fig-2. The steps are as follows.

Step-1: In the very first step, the equipment owner generates a pair of public-private keys using an asymmetric cryptography algorithm such as RSA (Ron Rivest, Adi Shamir, and Leonard Adleman) or ECC (Elliptic Curve Cryptography) [22]–[25]. There are various online services and tools that can be used to generate an asymmetric key pair. It is also possible to initiate the process using symmetric key cryptography, such as AES-256, which is one of many quantum-resistant algorithms. However, using AES-256 or any other symmetric key cryptography requires secure transfer of keys to the party that performs encryption, and in our case it is the calibration lab. Thus, to ensure simplicity this text uses asymmetric key cryptography.

Step-2: In step two, the client after obtaining cryptography public private key pair assembles Digital Calibration Request (DCR) or Request for Calibration (RFC). The DCR in the proposed approach has three components, A) Information that is required by the calibration lab to perform calibration and subsequently issue a DCC, B) A public key that will be used by the calibration lab to encrypt the DCC and C) A RESTful service or API endpoint including the token or authentication

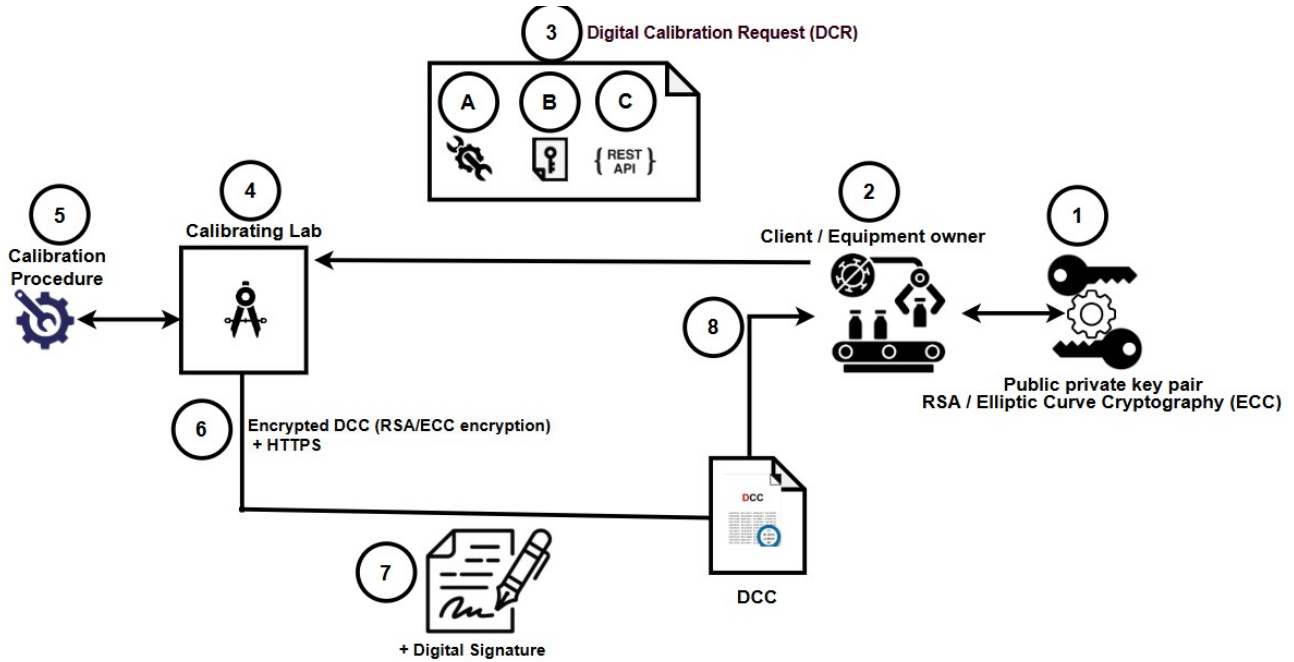


Fig. 3. Encrypted and digitally signed exchange of DCC

information that will receive the DCC at the client side.

Step-3: In step three, the client will send DCR containing all three components to a calibration lab over HTTPS. For the sake of simplicity, it is assumed that the calibration lab has an application running that receives DCRs from clients, and that each client gets a unique id, and authorization details from a calibration lab to securely interact with the online application.

Step-4: In step-4, a calibration lab that receives a DCR ensures that all the resources required to calibrate an equipment are present at the time of calibration.

Step-5: In step-5 once calibration process is complete and all the required information to create a DCC is present in a database, the actual XML document is generated. An automated generation of DCC from the database contents are possible using a batch job, that runs at certain intervals and generates DCC from the raw data stored in a database.

Step-6: In step-6, the DCC is encrypted using RSA or ECC by the public key shared by the client. The encrypted DCC is then sent using HTTPS by calling the API endpoint provided by the client. After receiving the encrypted DCC, client decrypts the cipher text, and then it's up to the client to process, update, or perform intermediate operation before storing it into the database or any other system.

The aforementioned process ensures that DCC during the transfer is safe, and secrecy, confidentiality and privacy are preserved. However, it doesn't guarantee non-repudiation. If client's DCR is compromised then it's possible for anyone to generate a fake DCC and then call client's API to send the bogus DCC. It is also possible that a calibration lab

performs a calibration that is erroneous and could then cause serious issues and monetary losses for clients when the equipment is used for various metrological projects in sectors, such as in pharma, healthcare and defense. Without any proper mechanism that ensures that a calibration process was properly performed and a valid DCC was later sent to the client, it is possible for the lab to deny that it carried out erroneous calibration and issued a flawed DCC. Thus, to ensure that no one else can claim as the calibration lab, and calibration lab can be traced back as the issuer of the DCC a non-repudiation mechanism is required.

There are many tools to ensure non-repudiation, such as verification and validation of identity using third party services, using audit logs, and adding timestamps to the messages exchanged between client and a calibration lab. However, one widely popular method to ensure non-repudiation is digital signatures as shown in Fig-3. Digital signatures provide cryptographic proof of a sender's identity and the integrity of a message. Fig-3 illustrates that the entire process will take place as mentioned in Fig-2 and explained earlier, with just one significant addition of digital signatures applied at step-7. Digital signatures are also based on public-private key pairs. The calibration authority generates a public-private key pairs with algorithm such as RSA, and then creates a hash of the DCC. The DCC and the hash are then encrypted using the public key provided by the client. The encrypted DCC is then sent over HTTPS to the client. The client then deciphers or decrypts the message received by the calibration lab and then hashes the message with calibration lab's corresponding public key. If the hash of

the DCC and hash sent by the calibration lab are equal, that means that message is authentic, integrity is preserved, and is sent by the genuine calibration lab and not from anyone else, as no one else except the calibration lab has access to the private key used to hash the DCC. Though, digital signatures and encryption both use cryptographic algorithms, one major difference between hashing and encryption is that, hashing is an irreversible one way function.

So, hash of a message can not be converted back to the original message. While on the other hand, encryption is a reversible function, and a cipher or encrypted text can always be converted back to the original text using the corresponding private key, if asymmetric encryption is used or secret key if symmetric cryptography has been used to encrypt the message.

VI. CONCLUSION

This paper presents popular cryptography techniques and applies them to the Digital Calibration Certificates (DCCs). The text in this paper not only provide detailed introduction of XML based Digital Calibration Certificates, but also explains how to protect them in various steps using practical techniques. These techniques not only secure a DCC from man-in-the-middle (MITM) attack, but also ensures non-repudiation. Exchanging DCC over HTTPS and adding digital signatures and applying a 256-bits encryption to the DCC adds multiple layers of security.

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Metrological Traceability in IoT Sensor Networks: A Microservice-based Framework for Automated Calibration Management

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Abstract – The widespread use of wireless sensor networks has made traditional calibration procedures logistically unfeasible in many contexts, thereby risking the compromise of the metrological traceability of the collected data. To address this challenge, this paper proposes a novel microservice-based framework for automated calibration management. The architecture abstracts sensor hardware into JSON-based digital models by integrating the Digital Unit System (D-SI) to ensure correct data interpretation and by establishing its interoperability on the concept of interchangeable units of measurement (MU). A field-deployed Travelling Standard executes automated in-situ calibration routines, communicating securely with a cloud backend to automatically evaluate measurement deviations and generates machine-actionable, PKI-signed Digital Calibration Certificates (DCCs). By natively integrating "Traceability by Design," this framework minimizes network downtime, eliminates human transcription errors, and bridges the gap between low-cost edge sensors and robust digital quality infrastructures.

Keywords: Calibration; DCC; Travelling standard; sensor networks.

I. INTRODUCTION

The global testing, inspection and certification (TIC) sector relies on a robust metrological infrastructure to ensure the reliability of measurements, with National Metrology Institutes (NMIs) maintaining the primary standards of the International System of Units (SI), whilst accredited calibration laboratories (CAL-LABs) ensure traceability down to industrial instruments. Traditionally, this requires a documented chain of physical calibrations however, the continuous growth of the Internet of Things (IoT) and massive Wireless Sensor Networks (WSNs) renders conventional, paper-based calibration procedures unsustainable for industries. Field-deployed sensors are subject to physical wear and environmental stresses, and subsequent mea-

surement drift therefore, without periodic calibration, the measurements provided by these sensors, which are often low-cost, will eventually deviate from the stated confidence intervals. Traditionally, devices are removed from the field and calibrated in highly controlled laboratory environments before they can return to service. In the context of WSN networks, which comprise a large number of sensors spread across wide geographical areas, this static approach is logistically unsustainable, and ignores non-deterministic network variables, such as transmission latency. Furthermore, the calibration process typically takes several days; during this period, the units are unavailable, which compromises the full functionality of the network [1, 2]. Although collaborative sensing networks often rely on mathematical self-calibration to handle large volumes of data, this approach has a fundamental limitation if it is not constantly anchored to calibrated reference nodes, leaving the network completely unaware of collective systematic drifts [3]. In the past, various solutions have been proposed to address these issues; in particular, the concept of Travelling Standards was introduced, which refers to remotely controlled, precise reference devices deployed directly at the measurement site. Some approaches exploit the surrounding ambient as a natural comparison medium: by placing a calibrated wireless reference node in close proximity to the field sensor, the calibration can be performed by relying on natural environmental conditions [4]. Although this approach avoids the need to use sophisticated instruments other than the reference, the dependence on environmental conditions limits the applicable calibration ranges. Consequently, in several other cases, in addition to incorporating reference sensors to carry out calibration procedures by comparison, the instruments used also included actuators to generate the stimuli to be measured. These techniques have been used in various sectors, such as the calibration of data acquisition (DAQ) boards, multifunction calibrators, as well as pressure and chemical sensors for detecting pollutants [1, 2, 5, 6].

Although this makes it possible to replicate real-world operating conditions and eliminates discrepancies between the laboratory and field environments, this approach, be-

ing inherently customised, lacks universality and requires specialised hardware interfaces and proprietary communication protocols. Consequently, it is unable to scale and support the thousands of heterogeneous nodes found in Industrial IoT (IIoT) ecosystems. The industrial automation sector offers a solution to the limitations associated with the sectorisation of reference devices: the IO-Link protocol and IO Device Description (IODD) files. An IODD file is a standardised, machine-readable XML file that provides a detailed description of a sensor's parameters, enabling 'plug-and-work' interoperability regardless of the manufacturer [7]. Another central aspect in the digitisation and standardisation of metrology is the Digital Calibration Certificate (DCC), based on the universal Digital SI (D-SI) data model. Unlike simple automation payloads, a DCC provides machine-actionable XML tags for rigorous metrological data, ensuring every value is inextricably linked to its SI unit, expanded uncertainty, and coverage factor [8, 9]. To overcome these issues, therefore, it is necessary to adopt a 'Traceability by Design' approach, by natively integrating metrological capabilities into network architectures.

II. THE PROPOSED ARCHITECTURE

To ensure the integrity of data collected on-site, a unified framework is required that bridges the gap between low-cost edge nodes and the digital infrastructure. The proposed architecture achieves this through three key concepts: the usage of JSON-based Digital Twin of the nodes, interchangeable Measuring Units (MU) and the automated in-situ calibration. The server's microservice architecture ensures the traceability of data acquired from network nodes through Digital Templates, which map the physical and metrological characteristics of the nodes into a data structure based on the Digital System of Units (D-SI) vector, enabling computational dimensional analysis and a high degree of interoperability. Furthermore, the development of edge nodes with hot-swappable Measurement Units (MUs) featuring unique 32-bit identifiers enables the automated association of physical hardware with valid DCCs in the cloud, in accordance with the DX digital document template [10]. Finally, the deployment of a mains-powered Travelling Standard that executes automated calibration algorithms directly in the field enables the generation of cryptographically signed PTB-compliant XML DCCs.

A. Seamless assurance of metrological traceability

The core of the framework's interoperability lies in the abstraction of sensor hardware into Digital Templates (DT), which consist of JSON files that define all the technical and metrological aspects of each device in the network. These templates act as the foundational *primitives* for all device instances, whereas a physical

device is associated with an instance of the template itself via a unique identifier, known as a 32-bit Extended ID.

These templates comprehensively outline the device's technical specifications, physical and electronic limitations, and primary metrological parameters such as resolution, calibration constant and main uncertainty contributions. All values reported within the template must ensure the correct interpretation of measurements and are therefore accompanied by the corresponding measurement unit in accordance with the notation adopted by D-SI.

The algorithmic advantage of D-SI becomes evident when automated operations need to be performed on input data (for example, when applying calibration constants, combining quantities or propagating uncertainty). The algorithm is based on Computational Dimensional Analysis (CDA): the standardised strings from D-SI are translated into corresponding seven-dimensional vectors representing the base International System of units.

In this way, metrological calculation is abstracted into vector algebra, as a matter of fact, the determination of a derived quantity is not performed by searching text-based dictionaries, but by performing mathematical operations on the exponents of the vector. This approach ensures that every calculation performed on the data is intrinsically validated from a dimensional perspective. Subsequently, a deterministic match is performed between the resulting matrix and the hash tables of the D-SI ontology; in this way, the exact semantic representation corresponding to the vector is extracted and the resulting unit of measurement from the operations performed is obtained.

The result is a digital measurement chain where uncertainty and traceability are automatically calculated and propagated through each subsequent operation, ensuring absolute dimensional verification without the need for human intervention in the parsing of the measurement units. Furthermore, to ensure the metrological traceability of all measurements obtained from the sensor network, the hardware architecture of the edge devices is strictly divided into two modular components: a primary Control Unit and one or more interchangeable Measurement Units (MUs) [11]. The Control Unit manages the communication stack; specifically, the network employs the MAC layer of LoRaWAN. On the other hand, the MU represents the sensing element of the edge node: each physical MU has its digital twin, an instance of the DT of the Measurement Unit model, tracked within the cloud database via a unique 32-bit hardware identifier. This identifier links the physical MU directly to its specific characteristics, including its calibration constants and the association with its latest valid DCC. This separation of the unit managing the MAC and transport layers from the sensing unit allows the metrological management of the nodes to be abstracted from any heterogeneous communication protocol used between the network nodes,

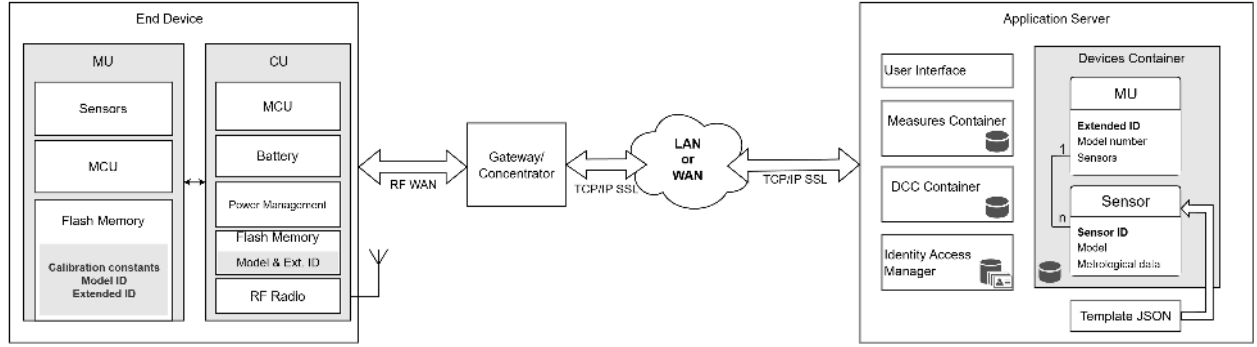


Fig. 1. Architecture of the Wireless Sensors Network and client-server application.

resulting in a significant reduction in the complexity of the devices verification. Fig. 1 shows the architectural diagram of the proposed framework.

B. Calibration management

In a standard IoT network, replacing a faulty, drifting, or uncalibrated sensor requires network downtime and manual updates to the backend enterprise database to link the new sensor's calibration data. In the proposed framework, an operator can replace a degraded MU with a newly calibrated one of the same model, directly in the field without decommissioning the LoRaWAN Control Unit. Upon connection, the Control Unit instantly detects the hardware change via the MU's 32-bit identifier and transmits a hardware-change payload over the LoRaWAN network. The cloud framework receives this payload, queries the digital twin database, identifies the newly mounted MU, and automatically associates the new device with the control unit (CU) to which it is physically connected, displaying all information relating to the new MU, such as its valid DCC, if available. Consequently, the metrological traceability chain remains completely unbroken, and the physical maintenance substitution occurs without any significant downtime for the entire sensor network. While this approach solves the immediate need to replace out of validity or faulty sensors, the operation of removing MUs and returning them to a laboratory for recalibration remains a significant logistical burden. To eliminate this bottleneck, the framework introduces the use of travelling standards directly to the field. A travelling standard operates as a localized, high resolution, transportable reference node that executes automated calibration routines directly on the deployed MUs, enabling to maintain the traceability in distributed sensing networks.

Usually, IoT devices are subjected to strict duty cycles, fair-access policies, and narrow bandwidth and thus, they are inherently unsuitable for supporting a continuous exchange of data, as required during the calibration process [12].

This has led to the development of a travelling standard that interfaces with the sensing units (MU) of the nodes through a shared UART interface, enabling to maintain continuous, secure, high-bandwidth communication with the central server (during the calibration procedure) by means of the Wi-Fi connectivity of the reference unit. Analogous to the monitoring nodes, it comprises a central processing unit and multiple high-precision reference, the travelling standard incorporates actuators capable of generating targeted physical stimuli to create a stable, controlled environment for the calibration. Unlike the low-power nodes, the travelling standard is based on a 64-bit quad-core ARM Cortex-A72 System on Chip (SoC) operating at 1.5 GHz and equipped with 8 GB of LPDDR4 SDRAM; it features a PCI Express bus dedicated to connecting the USB 3.0 controller and a dedicated chip for hardware decoding used to manage the operating UI. Moreover it embeds a Ethernet controller and a radio module for dual-band 802.11ac Wi-Fi (2.4 GHz and 5.0 GHz) network connectivity and therefore, it requires substantial electrical power to operate, drive the actuators, process real-time metrological algorithms and maintain connectivity to the server; for this reason, it is powered by the mains supply.

C. Certificate management

In highly regulated industrial sectors, the authenticity and non-repudiation of calibration data are of critical importance; the framework addresses this issue by integrating a robust digital signature mechanism directly into the DCC generation microservice. The system ensures data integrity through a standard asymmetric cryptographic workflow based on a Public Key Infrastructure (PKI). When a DCC is generated, the microservice computes a unique cryptographic hash of the XML payload using the SHA-256 algorithm. This hash, or "digital fingerprint," is then encrypted using the framework's private key-maintained within a secure cryptographic vault, to produce the digital signature. By utilizing this dedicated PKI layer, the system ensures that the signature represents the institutional au-

thority of the issuing laboratory, providing a formal guarantee of the document's provenance [8]. End-users, enterprise quality managers, and third-party auditors can utilize the platform's verification tools to automatically verify the digital signature of any DCC in their possession. This process involves decrypting the signature with the corresponding public key to retrieve the original hash and comparing it with a new one calculated from the received document. Any discrepancy between the two hashes indicates that the DCC has been compromised. This level of encryption ensures a greater reliability of the automated metrology infrastructure for edge computing, similar to that offered by blockchain solutions, but implemented within a highly efficient, standards-compliant microservices architecture.

D. Conformity assessment

The system performs out regular checks between active MU units and their assigned DCCs to assess their validity periods and current compliance status. Since all historical calibration data is stored in deeply structured, machine-readable XML, the framework facilitates advanced, automated data analysis that was previously difficult with static PDF certificates. The microservices can parse the historical XML DCCs of a specific MU, or perform comparative analytics across a broader batch of MUs to monitor, as an example, measurement drift over extended periods for devices of the same model. By indexing the deviation values and uncertainty budgets across multiple verifications, the system can implement predictive analytics to estimate the time remaining since the last calibration before the measurements provided by the measuring unit (MU) exceed the critical operational tolerance thresholds. This predictive capability transitions the management of the sensor network from a reactive maintenance model to a proactive, predictive compliance strategy. The framework scales further by allowing users to input specific regulatory (or process) requirements into the system; the backend algorithms will then automatically infer, based on the stored DCC XML files, whether the devices currently in operation remain compliant with these customised thresholds, immediately flagging any node requiring urgent calibration verification.

III. DCC GENERATION

The travelling standard automates the entire calibration lifecycle: once a certain period of time has elapsed since the last verification, or if there are any doubts that a device is not operating in accordance with its stated specifications, an operator connects the target sensing unit (MU) to the travelling standard, by means of a suitable hardware connector; thereby, a fully automated, closed-loop algorithm for the calibration is initiated, involving the parties represented in Fig. 2. The workflow of the automated in-situ calibration process is executed as follow:

1. The operator physically connects the target MU to the travelling standard. The standard queries the MU unique 32-bit identifier and fetches the device's JSON Digital Template from the cloud server.
2. By parsing the D-SI vectors within the template, the travelling standard identifies the physical quantities measured by the MU. It dynamically populates the local User Interface (UI), offering only applicable calibration routines to the operator.
3. Based on the selected calibration function, the MU's physical limits defined in the template, and the standard's actuator capabilities, the system automatically determines the optimal distribution of calibration set-points.
4. The travelling standard drives its actuators to generate the physical stimuli, holding stable conditions in order to carry out a verification by comparison. It simultaneously records high-frequency telemetry from both its internal reference sensor and the target MU.
5. The service of the server interprets the measurement errors by comparing them against the manufacturer specifications defined in the device's Digital Template to evaluate whether the MU is compliant. If the MU is non-compliant, the system proceeds with a software adjustment: utilizing regression algorithms, it calculates the new calibration constants required to correct the MU's output curve. Conversely, if the MU is compliant, the software adjustment can be avoided. However, a warning threshold can be implemented (e.g. 70% of the specification limit); if even a single measurement error exceeds this threshold, the system prompts the user to decide whether to proceed with the software adjustment anyway.
6. If a software adjustment is performed, the calibration constants are sent back to the travelling standard and written directly into the MU's physical flash memory. Simultaneously, the updated parameters are synchronized inside the instantiated Digital Twin of the MU.

To ensure data integrity and end-to-end security during the transmission to the server over potentially vulnerable network infrastructures, a Virtual Private Network (VPN) tunnel is established between the travelling standard and the central server. Upon completion of each calibration step, the collected measurements are encapsulated into a JSON payload and transmitted via the VPN to a central Mosquitto MQTT broker. To ensure scalability and robust data handling, a dedicated bridge service routes the incoming MQTT messages to a Kafka topic.

1. Persistent Storage: the measurements and associated metadata are stored in a central database to maintain a calibration history for the specific Measurement Unit.

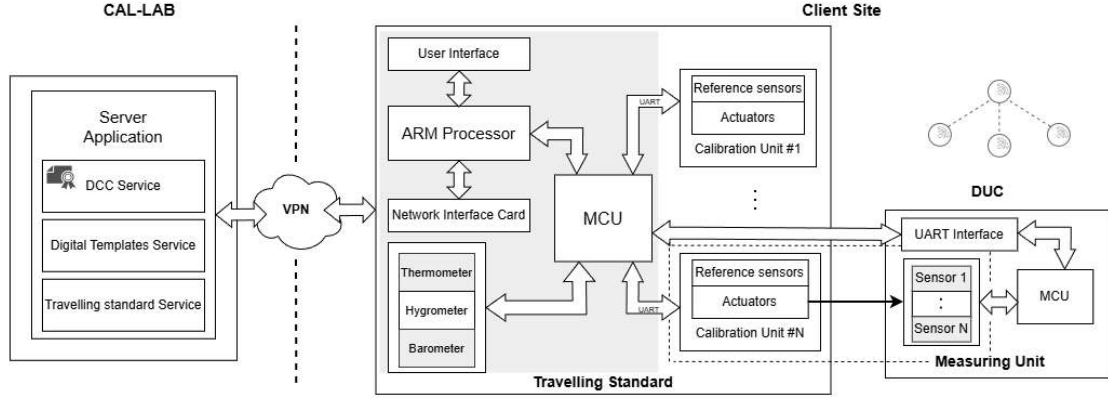


Fig. 2. Architecture of the remote calibration system and block diagram of the travelling standard.

2. Automated Computation: the service initiates the metrological evaluation of the readings obtained, as previously mentioned, based on the units of measurement associated with the measurements in accordance with the Digital System of Units (D-SI), as specified in the sensor's Digital Template stored on the server. By applying Computational Dimensional Analysis (CDA) based on the dimensional vectors that define the semantic representations of units according to the D-SI, the algorithm calculates the errors, evaluates the measurement and error uncertainty, as well as the new calibration constants, ensuring the mathematical integrity of the results and their units of measurement without any human intervention.

Once the calibration has been completed, a draft Digital Calibration Certificate (DCC) is compiled according to the PTB-compliant XML schema, it is placed in a *Pending* state, as shown in Fig. 3. This introduces a crucial quality control process: the certificate remains suspended until a qualified operator reviews the results and provides formal validation. Once approved, the DCC is cryptographically signed using a public key infrastructure (PKI) and published on the platform's web dashboard for the end user to access. This innovative approach closes the traceability loop to minimise manual intervention and, as a result, reduce human error almost entirely.

Following the successful execution of an in-situ calibration routine by the travelling standard, the GEMIMEG microservice is used in order to fill the official PTB XML schema and to generate the corresponding Digital Calibration Certificate draft. This machine-actionable XML file embeds all requisite administrative data, the measured errors, the calculated measurement uncertainty (incorporating both the variance of the measures provided by the device under calibration, its resolution and the uncertainty of the travelling standard), and the newly identified calibration constants. Moreover, the framework generates also a human-readable PDF version of the certificate. In this way,

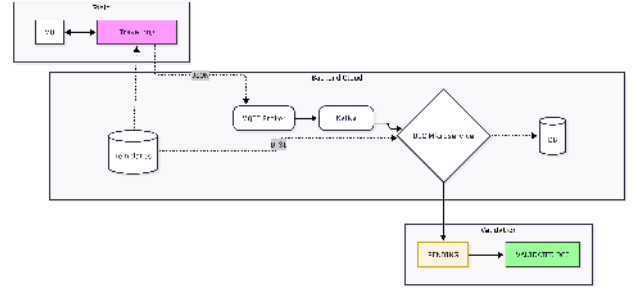


Fig. 3. Automatic Calibration Workflow

the platform provides client companies with a comprehensive, web-based dashboard to archive, manage, and verify their DCCs. Moreover, clients can upload external XML DCCs for legacy devices not natively part of the network, or rely entirely on the dynamically generated certificates from the travelling standards.

IV. CONCLUSIONS

This paper presented a novel microservice-based framework designed to ensure metrological traceability within large-scale IoT sensor networks. By abstracting sensor hardware into Digital Templates based on the Digital System of Units (D-SI) and decoupling the Measuring Units (MUs) from the communication interface, the proposed architecture achieves robust semantic interoperability. The deployment of a Travelling Standard enables fully automated, in-situ calibration routines, drastically reducing network downtime and eliminating the human errors associated with manual data handling. The subsequent automated generation of machine-actionable Digital Calibration Certificates (DCCs) aligns with the metrological community's transition toward an end-to-end Digital Quality Infrastructure [13]. Furthermore, the integration of a PKI-based digital signature mechanism within the framework ensures the cryptographic integrity and non-repudiation of the distributed metrological data, successfully addressing

the security vulnerabilities inherent in communicating calibration data over public networks. Future work will focus on the experimental validation of the proposed architecture, exploring and expanding the use case of in-situ thermometer and barometer calibration using the travelling standard, to thoroughly evaluate the complete framework performances in a real-world scenario.

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Implementation of Monte Carlo Simulation for measurement uncertainty estimation in electrical power and energy reference standards calibration

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Abstract – In the paper a step-by-step methodology for measurement uncertainty calculation, in a protocol for electrical power and energy reference standards calibration, will be presented. Initially, all predictable influencing factors that may be mathematically expressed as single uncertainty components will be identified, as well as the adopted matching probability distributions. Sequentially, two methodologies for evaluation of the measurement uncertainty, attributed to the calibration result, will be conducted. The straightforward principle of uncertainty propagation, according to the Guide to the Expression of Uncertainty in Measurement (GUM), will be complemented by the implementation of the stochastic Monte Carlo Simulation approach of distribution propagation. Both concepts' outcome will be analyzed from the perspective of real time measurements, conducted in an accredited calibration laboratory, by using reference standards of the highest accuracy class available, which are traceable to the intrinsic primary reference standards of BIPM.

Keywords: measurement uncertainty, reference standards, Monte Carlo Simulation.

I. INTRODUCTION

The accurate measurement of electrical power and energy plays an essential role in the operation of modern power systems, regarding the fact that the power flow monitoring, system's stability and commercial energy transactions heavily dependent on it [1-2]. For providing confidence in the measurement data obtained from the applied instrumentation, these measurement units are subjected to periodical calibration and verification with reference standards (RSs) of higher accuracy class [3-7]. At the top of the metrological traceability chain are the primary reference standards for the aforementioned electrical quantities, which serve as a benchmark for further performance evaluation of the working standards and field instruments [8-10].

The establishment of an examination protocol, for electrical power and energy RSs calibration, is a rather complex process [5]. For calibration of high accuracy class standards, a suitable measurement configuration is required, consisting of both a RS of level up accuracy and a signal source for providing the desired voltage, current and phase shift values [11-12]. Both the Unit Under Test (UUT) and the RS in such a configuration are supposed to be synchronized, for valid calibration data to be obtained. Complementary, a proper connection between the instruments is supposed to be ensured [13]. The electrical power and energy RSs usually possess several voltage and current measurement ranges and may rely on different measurement modes (e.g. different techniques for reactive power/energy measurement) [14]. As no standardized test points are available [5], as is the case of multimeters [15] or oscilloscopes [16] calibrations, the laboratory itself is required to provide a tailor made examination protocol. An essential part of the protocol establishment is the determination and mathematical evaluation of all the influencing factors that affect the measurement process, in the form of measurement uncertainties [12, 17].

The uncertainty analysis in calibration laboratories usually follows the framework provided in the Guide to the Expression of Uncertainty in Measurement (GUM) [18]. However, scenarios exist where the GUM [18] based approach is not suitable, due to the non-linearity of the model for uncertainty propagation, or the correlation existence between the single sources of uncertainty, or the fact that the dominant influencing factors cannot be regarded as random variables with Gaussian (normal) distributions [19]. These limitations are addressed by implementation of the stochastic Monte Carlo method, based on distribution propagation [20].

The implementation of the Monte Carlo method [20] in the domain of instruments for electrical quantities calibrations has been covered in several research papers [21-24]. In this paper, an analytical evaluation of single influencing factors as uncertainty components, in a calibration procedure for electrical power and energy RSs,

will be provided. The measurement uncertainty, attributed to the calibration result, will be further calculated by implementing both the straightforward GUM [18] based approach, as well as the Monte Carlo distribution propagation method [20]. By analyzing real-time data, obtained from testing conducted with high accuracy class instrumentation, traceable to the intrinsic primary RSs of BIPM [25], the eventual variations in the obtained uncertainties will be discussed.

II. CALIBRATION CONFIGURATION

The experimental part of the work is conducted in the Laboratory for Electrical Measurements (LEM), at the Faculty of Electrical Engineering and Information Technologies (FEEIT), at Ss. Cyril and Methodius University in Skopje (UKIM). It is an accredited laboratory for calibration of instruments for electrical quantities, according to the international standard ISO/IEC 17025:2017 [26]. In the disposal of the laboratory, there are two RSs for electrical power/energy. A three phase energy comparator of accuracy class 0.01, with voltage measurement ranges up to 480 V and current measurement ranges up to 160 A, ZERA COM 3003 [27], acts as the primary RS of LEM, for the electrical quantities of interest. A three phase voltage and current source, i.e. a three phase power calibrator, of accuracy class 0.02, constructed for generation of voltages up to 560 V per phase and currents up to 120 A, CALMET C300 [28], acts as laboratory secondary (working) RS.

When a metering RS of high accuracy class is subject of calibration, both standards of LEM [27-28] are deployed for realization of the testing protocol. Such a configuration will be regarded in the practical part of the work. The three phase calibration connection is illustrated in Fig. 1. The current terminals of the UUT, the RS and the signal source are connected in series, while the voltage terminals of the three units are connected in parallel. The BNC frequency output of the UUT is connected to the BNC frequency input of the primary RS, thus enabling the measured power/energy by the examined unit to be automatically compared with the reference quantity. The number of pulses with particular frequency are directly proportional to the measured quantity, for both instruments, the RS and the UUT. The result of the calibration, in the form of measurement error, is obtained as:

$$\varepsilon_{\%} = \frac{N_{UUT} - N_{RS}}{N_{RS}} \cdot 100, \quad (1)$$

where N_{UUT} is the number of pulses proportional to the measured power/energy by the UUT, and N_{RS} is the number of pulses proportional to the same quantity, measured by the RS. The setting of both the primary RS [27] and the signal source [28] is done remotely, by using a computer and appropriate connection interface.

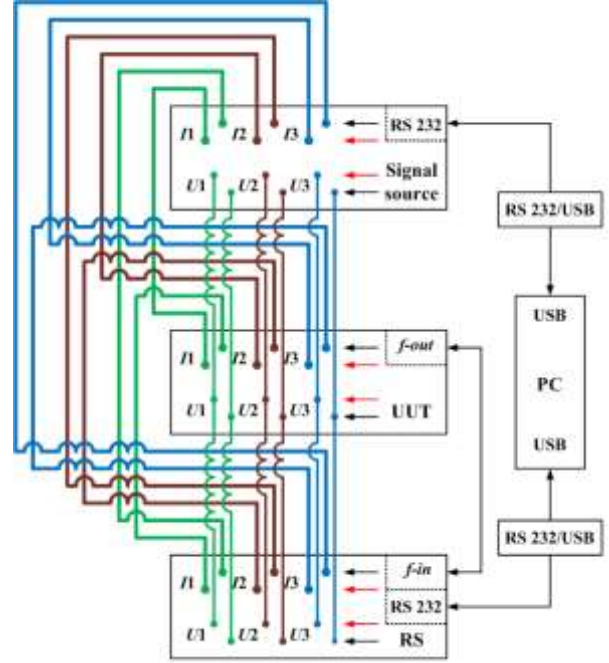


Fig. 1. Three phase electrical power/energy RS calibration test set-up

III. MEASUREMENT UNCERTAINTY BUDGET EVALUATION

In the following chapter, the influential factors that affect the calibration of an electrical power/energy RS will be evaluated as standard uncertainty components. According to the calibration working instructions of LEM [29], for this particular procedure, total of five influential factors are regarded in the overall budget, and they are treated as mutually uncorrelated, which is a usual approach in domain of electrical measurements [30].

The first standard uncertainty is related to the statistical scattering of the measured data, around the mean value. It takes into consideration the random variations of the measured quantity both by the UUT and by the primary RS [27]. In a single measurement point, usually several error recordings, according to (1), are performed, and the mean error, $\overline{\varepsilon}_{\%}$, is taken as the most appropriate result of calibration:

$$\overline{\varepsilon}_{\%} = \frac{1}{n} \sum_{i=1}^n \varepsilon_{\%,i}, \quad (2)$$

where $\varepsilon_{\%,i}$ are the single error recordings, obtained according to (1), and n is the number of repetitive measurements in the particular measurement point. The measurement uncertainty component arising from the random variations of the single recordings, is then calculated as standard deviation of the mean, $\sigma_{\%}$, i.e. according to GUM [18], it is evaluated as Type A uncertainty $u_{A,\%}$:

$$u_{A,\%} = \sigma_{\%} = \sqrt{\frac{1}{n(n-1)} \sum_{i=1}^n (\varepsilon_{\%,i} - \bar{\varepsilon}_{\%})^2}, \quad (3)$$

by assuming Student t-distribution, regarding the fact that only finite amount of single recordings are performed.

The other four uncertainty components are related to the performance of the reference unit, for this calibration configuration that is the primary RS of LEM [27]. The first one is related to the accuracy of the RS, i.e. it is obtained from its technical datasheet [27]:

$$u_{ACC,\%} = \frac{U_{ACC,\%}}{\sqrt{3}}, \quad (4)$$

and it is calculated by assuming rectangular (uniform) distribution, regarding the fact that no additional data is available in [27], except for the accuracy limits, $U_{ACC,\%}$, hence the division by $\sqrt{3}$ [18-19]. Next, the long-term stability of the RS is regarded as an influence factor, the standard uncertainty component is calculated as follows:

$$u_{ST,\%} = \frac{U_{ST,\%}}{\sqrt{3}} \cdot y, \quad (5)$$

where the $U_{ST,\%}$ is the long-term stability data obtained from RS's datasheet [27], expressed in (%/year), and y is the number of years that have passed since the last calibration and adjustment of the RS. This uncertainty is once again evaluated by assuming uniform distribution, the division by $\sqrt{3}$ is used for obtaining the standard component from the distribution half-width [18-19]. The third uncertainty component that emerges from the specifications of the RS [27] is related to the temperature influence of the standard's performance. The analytical expression for calculation of the standard uncertainty is given as:

$$u_{T,\%} = \frac{U_{T,\%}}{\sqrt{3}} \cdot \Delta T, \quad (6)$$

where $U_{T,\%}$ is the temperature related influence presented in [27], expressed in (%/K) and ΔT is the temperature variation interval, in which the calibration is performed. As was the case with the previous two influencing factors, rectangular distribution and accordingly division by factor $\sqrt{3}$ are adopted for calculation of the standard uncertainty from the distribution half-width.

The last uncertainty component, attributed to the calibration result, is related to the traceability of the primary RS of LEM, i.e. its level-up calibration. Since the calibration uncertainty is usually obtained from several components and in a calibration certificate it is presented as expanded combined uncertainty [18-19], with statistical probability of approximately 95 %, the standard component may be evaluated as:

$$u_{CAL,\%} = \frac{U_{CAL,\%}}{2}, \quad (7)$$

while the division by factor of 2, is due to the adopted Gaussian (normal) distribution.

If the GUM [18-19] based approach is adopted, the standard combined uncertainty attributed to the calibration result, calculated from the five previously listed components, will be calculated as follows:

$$u_{C,\%} = \sqrt{u_{A,\%}^2 + u_{ACC,\%}^2 + u_{ST,\%}^2 + u_{T,\%}^2 + u_{CAL,\%}^2}. \quad (8)$$

In a calibration certificate, the uncertainty is presented as expanded combined uncertainty, corresponding to a coverage interval of approximately 95 %, by assuming Gaussian distribution, due to the multiple influencing factors considered [18-19]:

$$U_{C,\%} = k \cdot u_{C,\%} = 2 \cdot u_{C,\%}, \quad (9)$$

hence the multiplication by coverage factor $k=2$.

Taking into consideration that the standard uncertainties are not modelled by assuming normal distributions only, i.e. rectangular distributions are used as well, the resultant distribution prescribed to the calibration result may actually differ from Gaussian [19]. In such a case, the Monte Carlo method [20] is found to be more adequate for providing the resultant distribution. The number of simulations, M , required for the coverage interval, corresponding to the desired statistical probability, p , to be determined, is given as follows:

$$M > \frac{10^4}{1-p}. \quad (10)$$

As the measurement uncertainty, calculated according to GUM [18], is presented for statistical probability of approximately 95 % ($p=0.95$), the number of simulations required for obtaining the same information by using Monte Carlo method [20], exceeds 200000.

IV. CASE STUDY RESULTS

The 95 % confidence interval measurement uncertainty, attributed to the calibration result, is going to be evaluated, by implementing both GUM [18] and Monte Carlo [20] methods, on a real case dataset, obtained from calibration protocol conducted on a three phase electrical power and energy RS of accuracy class 0.05. The calibration procedure is conducted in multiple measurement points, corresponding to a fixed test voltage (240 V), test currents that vary between 20 mA and 50 A, and three power factor values of 1, 0.5 L and 0.8 C. The results are presented as mean relative errors, (2), from $n=5$ single recordings, according to (1), of the three phase active power/energy in symmetrical conditions.

In Table 1 the standard uncertainties' magnitudes are presented, alongside with the adopted distributions. The long-term stability, temperature and level-up calibration

related components are with fixed values, in every measurement point. The distribution half-width of the accuracy related component equals either $\pm 0.008\%$ or $\pm 0.01\%$, depending on whether the measured current is higher or lower than 50 mA [27]. The Type A uncertainty is dependent on the recorded dispersion of the measured errors, according to (3), hence its magnitude is denoted as *var.* The measured standard deviations vary between 0 and 0.00217 %.

Table 1. Standard uncertainty components' magnitudes and distributions.

Uncertainty component	Expanded uncertainty/ Distribution half-width (%)	Standard uncertainty (%)	Distribution
$u_{A,\%}$	<i>var.</i>	<i>var.</i>	Student (t)
$u_{ACC,\%}$	0.008/ 0.01	0.00462/ 0.00577	Rectangular
$u_{ST,\%}$	0.012	0.00693	Rectangular
$u_{T,\%}$	0.0005	0.00029	Rectangular
$u_{CAL,\%}$	0.012	0.00612	Gaussian

The RS's accuracy, long-term stability and traceability, as influencing factors, dominantly shape the overall budget. They are all of an equal order of magnitude value, as shown in Table 1. Both the accuracy and the long-term stability related components are obtained by adopting distribution different than Gaussian. By taking this into consideration, the implementation of the Monte Carlo method [20] for uncertainty evaluation is justified.

The expanded measurement uncertainties, calculated according to GUM [18], in all the measurement points, except for those that correspond to test currents lower than 50 mA, equal $\pm 0.021\%$. The expanded uncertainties, when the UUT is examined with currents of 20 mA, for the three aforementioned power factor values, equal $\pm 0.022\%$. It is obvious that this difference is a result of the alteration of the accuracy related component, as was highlighted earlier. The relatively constant measurement uncertainty values indicate that the random variations of the single error recordings around the mean, have little influence on the overall calibration performance, if the GUM [18] approach is adopted.

The Monte Carlo Simulation [20] results are presented for several measurement points, considering the alterations of the Type A and accuracy related uncertainties discussed earlier. The equivalent distributions, obtained from the conducted simulations using MATLAB®, are depicted in Fig. 2. The evaluation according to Monte Carlo [20] is realized by performing $M=1000000$ (one million) simulations, which is in accordance with [20] and (10) for

obtaining coverage interval of approximately 95 %.

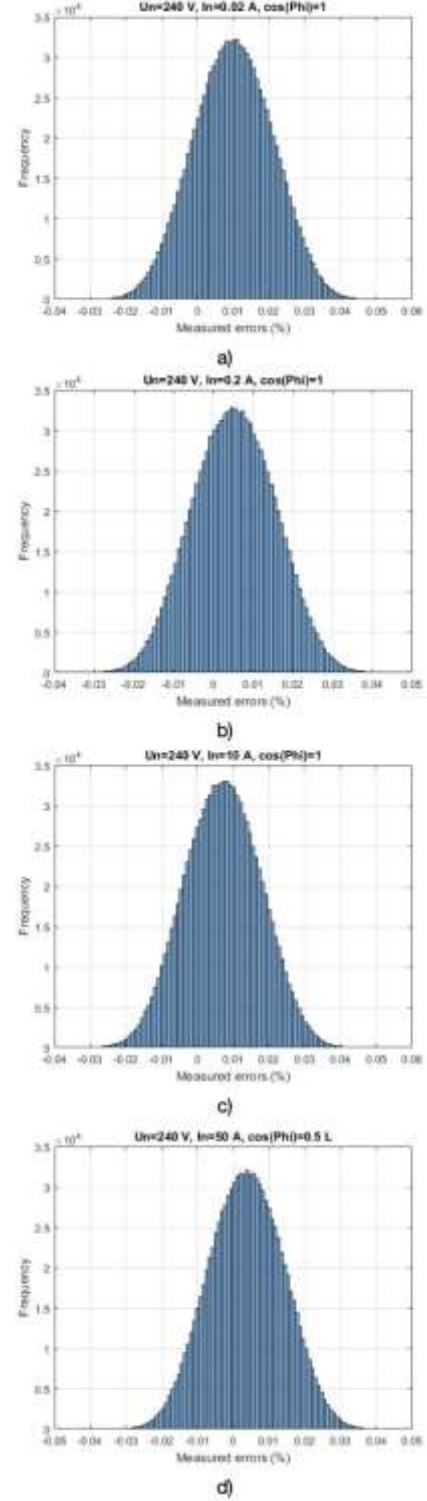


Fig. 2. Distribution of measurement results according to Monte Carlo Simulation at four measurement points labelled as a) MP1, b) MP2, c) MP3, d) MP4.

In Fig. 2 a) the equivalent distribution of the calibration result regarding a measurement point MP1, corresponding to a test current of 20 mA and power factor of 1 is illustrated, where a higher accuracy related uncertainty is present. Fig. 2 b) depicts the distribution of simulation results for MP2, at test current of 200 mA and power factor 1, in which no recorded dispersion of measurement data is detected, i.e. $u_{A,\%}=0\%$. Fig. 2 c) presents the simulated results for a measurement point MP3, corresponding to a test current of 10 A and power factor of 1, in which the highest standard deviation is recorded. In Fig. 2 d) a scenario with the highest test current, equaling 50 A, and a power factor of 0.5 inductive, labelled as MP4, is regarded. As can be seen from the figures, despite the fact that two out of three dominant influential factors are obtained by assuming rectangular distributions, the resultant distribution of measurement data tends toward Gaussian, mainly due to the contribution of the RS's traceability component.

In Table 2, the summarized results for the four measurement points are presented. The uncertainty corresponding to a coverage interval of approximately 95 % is almost identical if either GUM [18] or Monte Carlo [20] methods are applied. The Monte Carlo Simulation [20] provides slightly lower measurement uncertainty, which on the other hand encourages the further usage of the more simple, uncertainty propagation principle according to GUM [18], as it may be regarded as a safe side approach.

Table 2. Comparison between GUM and Monte Carlo evaluated uncertainties.

Measurement point	Mean error recorded, $\bar{\epsilon}_{\%}$ (%)	Uncertainty, U_C (%)	
		GUM	Monte Carlo
MP1	0.010	± 0.022	± 0.021
MP2	0.0052	± 0.021	± 0.020
MP3	0.0071	± 0.021	± 0.020
MP4	0.0037	± 0.021	± 0.020

V. CONCLUSION

In the paper, an analysis of the components that make up the measurement uncertainty budget, when performing a calibration of electrical power/energy reference standards, in an accredited laboratory, was conducted. The influential factors that may be mathematically evaluated as standard uncertainties are highlighted, altogether with the distributions adopted for their contribution presentation. For the evaluation of the uncertainty that corresponds to a 95 % coverage interval, two approaches are regarded, the straightforward uncertainty propagation principle, according to GUM, as well as stochastic Monte Carlo Simulation method, based on distribution propagation.

The implementation of both methods for uncertainty

evaluation was practically presented, by taking into consideration real-time measurement data, obtained from a calibration of high accuracy class reference standard. By performing both the calculation according to GUM, as well as the Monte Carlo Simulation, by using MATLAB®, similar magnitudes for the 95 % coverage interval uncertainty, were obtained. Even though the outcome of non-normal resultant distributions was expected, due to the fact that two out of three dominant influential factors were evaluated by assuming uniform distributions, the remaining one, evaluated by adopting Gaussian distribution, contributes to the dispersion of overall calibration results, according to the same statistical principles implied in the GUM approach. It is important to be emphasized that the variations of the input influential factors were taken into consideration, when Monte Carlo Simulation was performed. In all these measurement points slightly lower measurement uncertainties were obtained according to the distribution propagation method, which justifies the usage of the undemanding, GUM based, methodology and puts it as a safe side approach in conformity assessment decision making processes.

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GUM-compliant uncertainty evaluation in measurements based on ML regression or classification models

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Abstract – The integration of Machine Learning (ML) into measurement processes requires a rigorous approach to ensure metrological traceability and compliance with international standards. This paper presents the first steps toward a unified framework for the evaluation of measurement uncertainty when ML is used within measurement processes. In such applications, ML may introduce an additional uncertainty component due to its data-driven nature, which can be non-negligible and should therefore be included in the overall uncertainty budget to properly express the final measurement result. The proposed approach applies to both continuous and ordinal output scales, thus covering regression and classification problems. Its effectiveness is shown through two case studies, namely power plant output estimation and battery State-of-Health assessment.

Keywords: GUM-compliant measurement; ML-based measurement models; Measurement model uncertainty; Ordinal and continuous scales.

I. INTRODUCTION

The widespread adoption of data-driven techniques is establishing Machine Learning (ML) as an integral component of modern metrology [1, 2]. Specifically, ML models can be used to infer from data the relationship between the relevant input quantities and the measurand, making them suitable to operate as measurement models when no explicit functional relationship is available or adequate [3]. According to the International Vocabulary of Metrology (VIM) [4], this role is formally accommodated for both continuous-scale measurands, addressed via regression-based models, and ordinal-scale measurands, addressed via classification-based models.

This development, while offering significant opportunities for metrology, also raises important questions concerning trustworthiness, traceability, and accountability, especially in safety-critical and high-stakes measurement applications. As a consequence, the focus is shifting from purely performance-oriented metrics toward a broader perspective that also includes the reliability and demonstrability of ML-based results. This evolution is reflected in the emerging regulatory landscape. In particular, the European Artificial Intelligence (AI) Act [5] requires AI systems, including those based on ML, to ensure appropriate levels of accuracy, robustness, and transparency throughout their lifecycle, together with adequate technical documentation, conformity assessment procedures, and non-misleading declarations of performance. These requirements become especially challenging when the measuring systems incorporate a learned, data-driven component. Within this context, a metrological approach for ML is increasingly recognized as a natural means to ensure that ML-derived measurement results are traceable, comparable, and consistently interpretable across applications and jurisdictions.

A critical challenge arises from the fact that, unlike conventional measurement models, which are explicitly defined by the practitioner on the basis of known physical laws, ML models are data-driven and obtained through learning processes whose outcome depends on several factors, including the quantity and quality of the training data, the selected hypothesis space, the hyperparameter configuration, and the training strategy [6, 7]. As a result, in ML-based measurement applications, the uncertainty associated with the model itself cannot in general be assumed negligible. This marks a significant departure from the conventional framework of the Guide to the Expression of Uncertainty in Measurement (GUM) [8], in which the measurement model is assumed to be known and the uncer-

tainty associated with it is considered negligible. Such an assumption is generally justified for analytical or physics-based models, where the relationship between the input quantities and the output quantity is explicitly defined. This challenge necessitates the formal evaluation of an additional component to the uncertainty budget, referred to in previous work as *measurement model uncertainty*¹ [10]. Accordingly, the measurement result shall be expressed in a way that jointly accounts for the measurement model uncertainty component and for the conventional one, which is propagated through the measurement model from the physical input quantities, referred to in previous work as *physical quantity uncertainty* component [10].

Building on these premises, this work takes the first steps toward a unified GUM-compliant approach for evaluating measurement uncertainty in ML-assisted measurement applications. The proposed methodology focuses on the evaluation of the measurement model uncertainty component, and on its joint consideration with the physical quantity uncertainty one. If the former is negligible, or can be made negligible through suitable corrective actions, the conventional GUM-based uncertainty evaluation remains applicable. Otherwise, the two components should be properly combined [11]. To show the generality of the approach, the analysis is developed for both continuous and ordinal output scales, thus covering regression- and classification-based measurement applications. Two case studies are considered for this purpose: a regression-based application involving a Combined Cycle Power Plant (CCPP), where an ML model estimates the full-load electrical power output, and a classification-based application concerning battery State-of-Health (SOH) assessment, where ordinal categories (Healthy, Warning, and Critical) are predicted. While the general rationale remains the same, the work also highlights the differences in the metrics required to express the best estimate of the measurand and the associated dispersion for continuous and ordinal scales.

The remainder of the manuscript is organized as follows. Section ii. presents the proposed methodology, Section iii. discusses the results obtained from the two case studies, and Section iv. summarizes the main findings and outlines future developments.

II. METHOD

The starting point of the proposed approach is an ML model that has already been developed, trained on an appropriate dataset, and made available for use in the intended measurement application. The design, selection,

¹According to ML terminology, the measurement model uncertainty component encompasses both model and approximation uncertainty, and can be further decomposed into epistemic and aleatoric contributions [9]. The epistemic contribution reflects a lack of knowledge regarding the data-generating process or limitations in the model structure, while the aleatoric one originates from the intrinsic variability of the data.

and training of the model are therefore assumed to have been addressed beforehand according to established ML best practices. On this basis, the methodology is applied to the trained model within the measurement process and is structured into two main steps.

A. Physical quantity uncertainty evaluation

The first step consists in evaluating the physical quantity uncertainty component, denoted as u_{phys} . This can be done by propagating the probability distributions assigned to the input quantities $(X_1, X_2, \dots, X_N)$ through the ML-based measurement model $\hat{h}$ by means of the Monte Carlo procedure described in Supplement 1 to the GUM (GUM-S1) [12]. In this way, the distribution of the measurand Y is obtained starting from the distributions assigned to the input quantities.

The way the measurement result is expressed depends on the measurement scale:

- In *regression-based* measurements, where the measurand is defined on a continuous scale, Y is described by a Probability Density Function (PDF). Following GUM-S1 [12], the best estimate of the measurand is identified as the mean of the distribution, while u_{phys} is quantified by its standard deviation.
- In *classification-based* measurements involving ordinal quantities, Y is described by a Probability Mass Function (PMF) over the admissible classes. In this case, the best estimate of the measurand is identified with the mode, *i.e.*, the most probable class. This choice is consistent with the GUM framework, which explicitly recognizes the mode as a valid estimator of a measurand value (Definition C.2.19 in [8]). To quantify dispersion, a suitable index must then be adopted. Although several metrics are available in the literature, such as Shannon entropy, Gini index, and Probabilistic Margin, this work adopts the Standard Deviation of the Mode (SDM) [13]. Unlike information-theoretic or inequality-based metrics, the SDM expresses dispersion directly in the category space, thus quantifying the spread of the values reasonably attributable to the measurand.

However, in ML-assisted measurement applications, this conventional evaluation is not sufficient by itself. Indeed, unlike conventional analytical models, an ML-based measurement model may introduce an additional uncertainty component due to its data-driven nature, namely the measurement model uncertainty, denoted as u_{mod} . Considering only u_{phys} would therefore understate the overall measurement uncertainty. For this reason, u_{mod} must be evaluated separately and then combined with u_{phys} within the overall uncertainty budget.

B. Measurement model uncertainty evaluation

The second step consists of evaluating the measurement model uncertainty component u_{mod} . To this end, for each model output, it is necessary to evaluate the variability reasonably attributable to the measurand by determining the corresponding distribution around its best estimate and deriving from it a suitable index of dispersion [4]. Notably, this contribution must be assessed separately from the uncertainty associated with the input quantities, so that the two sources can be kept distinct within the overall uncertainty budget. For this reason, in the present step, the input quantities are considered without uncertainty and the analysis is focused on the model output alone.

The way this model's output distribution is obtained depends on whether the adopted ML model is stochastic or deterministic. In stochastic models, such as deep ensembles or Monte Carlo dropout schemes [14, 15], multiple outputs are generated for the same input directly by construction, so that the corresponding distribution can be estimated at inference time. In deterministic models, instead, such a distribution is not directly available and must be reconstructed through a calibration procedure carried out on an independent and representative dataset. This procedure relates model predictions to the corresponding reference values, thereby enabling the estimation of both the measurement error and the uncertainty contribution associated with the model output [10]. For consistency, the best estimate and the associated dispersion are then determined according to the same criteria introduced above for continuous and ordinal scales. Accordingly, in regression-based measurements, u_{mod} is quantified by the standard deviation of the output PDF, whereas in classification-based measurements involving ordinal quantities, it is quantified through the SDM of the output PMF. At the end of this step, each output of the ML-based measurement model can be expressed in terms of a best estimate of the measurand and the corresponding uncertainty contribution associated with the measurement model.

C. Overall measurement uncertainty evaluation

Once the measurement model uncertainty contribution, u_{mod} , has been evaluated, it must be combined with the contribution arising from the uncertainty of the physical input quantities, u_{phys} . Since the two components are evaluated separately and originate from distinct sources, they can be treated as statistically independent: u_{phys} reflects the uncertainty affecting the physical input quantities, whereas u_{mod} accounts for the dispersion associated with the measurement model itself. Under this assumption, the overall standard uncertainty associated with the measurement result Y is obtained through variance decomposition:

$$u(Y) = \sqrt{u_{phys}(Y)^2 + u_{mod}(Y)^2} \quad (1)$$

III. RESULTS

The proposed methodology is validated through two industrial case studies, representative of ML-assisted measurement applications on continuous and ordinal scales, respectively.

A. ML-based regression measurement

The first case study concerns an ML-assisted regression measurement task in the field of electric power monitoring, involving a Combined Cycle Power Plant (CCPP) [16]. The aim is to estimate the full-load electrical power output (P_E) from four measurable input quantities, namely ambient temperature (AT), atmospheric pressure (AP), relative humidity (RH), and condenser vacuum (V).

To address this problem, a feedforward Artificial Neural Network (ANN) with Rectified Linear Unit (ReLU) activation functions was adopted as the ML-based measurement model. The publicly available dataset [17] was partitioned by allocating 70 % of the data to training and validation, and the remaining 30 % to calibration and testing. The optimal model architecture was then selected through a grid search over the hyperparameter space reported in Table 1.

Table 1. Search Space Adopted for the Hyperparameter Fine-Tuning

Tuned Hyperparameters	Search Space
number of hidden layers	{1, 2, 3, 4}
number of neurons in each layer	{10, 20, 50, 100}
learning rate	{ 10^{-2} , 10^{-3} , 10^{-4} }
batch size	{32, 64, 100}

1. *Physical quantity uncertainty evaluation*: First, the uncertainty contribution arising from the input quantities was evaluated on a representative test instance, whose values are reported in Table 2.

Table 2. Input Quantities for the Employment of the ML-Based Measurement Model in Case Study I

Input Quantity	Best Estimate [NU]	Standard Uncertainty	Distribution
AT	27.17 °C	0.29 °C	Uniform
AP	1011.4 mbar	0.2 mbar	Uniform
RH	65.32 %	0.29 %	Uniform
V	61.02 cm Hg	0.43 cm Hg	Uniform

These quantities were assumed to be measured by off-the-shelf sensors (STTS22H, LPS22DF, HDC3120, and DVR2pro, respectively), and their probability distributions were propagated through the trained model by means of 10^6 Monte Carlo simulations. The resulting output distribution was described by a PDF with mean equal to 451 MW, taken as the best estimate of

the measurand, and relative standard deviation corresponding to $u_{phys} = 0.77\%$.

2. *Measurement model uncertainty evaluation:* Since the selected ANN is a deterministic model, the output distribution required for the evaluation of u_{mod} cannot be obtained directly at inference time. For this reason, u_{mod} was evaluated through the calibration procedure described in Section B.. The resulting calibration diagram is reported in Fig. 1. For the selected operating point (represented by the best estimate obtained in the previous step), the calibration procedure returned a relative measurement model uncertainty $u_{mod} = 0.14\%$.

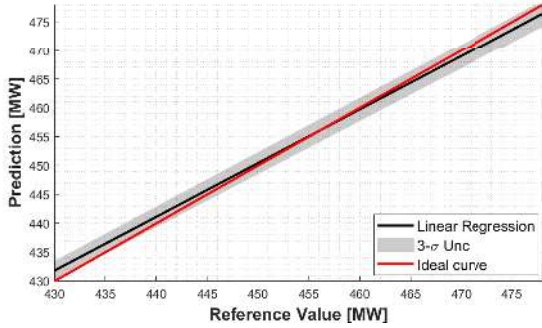


Fig. 1. Calibration diagram of the ML-based measurement. The plots compare the model output P_E with reference values, including the linear regression (black line), ideal trend (red line), and 3σ uncertainty band (gray area).

3. *Overall measurement uncertainty evaluation:* Since u_{mod} is not negligible with respect to u_{phys} , the two contributions were combined according to Eq. (1). This yielded a combined standard uncertainty of $u(Y) = 0.78\%$, which is higher than the uncertainty contribution associated with the physical input quantities alone. This confirms that considering only u_{phys} would have understated the overall measurement uncertainty.

B. ML-based classification measurement

The second case study concerns an ML-assisted measurement task on an ordinal scale, focused on the assessment of the State of Health (*SOH*) of lithium-ion batteries. The *SOH* is a key indicator of the remaining useful life of a battery, but it is difficult to measure directly because of the complexity of the underlying degradation mechanisms. In this context, an ML-based measurement model was used to assess *SOH* from seven measurable input quantities, namely charging/discharging current (I_{ch}/I_{dis}), voltage (V_{ch}/V_{dis}), temperature (T_{ch}/T_{dis}), and battery capacity (BC). To support maintenance decisions, the measurand was expressed on an ordinal scale with three classes:

- *Class #C0* (Healthy, $SOH \geq 90\%$);
- *Class #C1* (Warning, $70\% < SOH < 90\%$); and
- *Class #C2* (Critical, $SOH \leq 70\%$).

The dataset was obtained from the NASA Prognostics Data Repository [18]. A feedforward ANN was adopted as the classification-based measurement model, with two hidden layers (20 and 10 neurons, ReLU activation functions) and a Softmax output layer. The model was trained with the Adam optimizer and balanced class weights to ensure comparable behavior across the three classes.

1. *Physical quantity uncertainty evaluation:* First, the uncertainty contribution arising from the input quantities was evaluated on a representative test instance, whose normalized values are reported in Table 3.

Table 3. Input Quantities for the Employment of the ML-Based Measurement Model in Case Study II

Input Quantity	Best Estimate [NU]	Standard Uncertainty [%]	Distribution
I_{ch} [A]	0.6385	0.02	Uniform
V_{ch} [V]	0.6414	0.05	Uniform
T_{ch} [°C]	0.7899	0.05	Uniform
I_{dis} [A]	-0.0108	0.02	Uniform
V_{dis} [V]	0.8408	0.05	Uniform
T_{dis} [°C]	0.4316	0.05	Uniform
BC [Ah]	0.5455	0.02	Uniform

These quantities were assumed to be acquired by off-the-shelf sensors (Kaiweets multimeter and DHT11 sensors), and their probability distributions were propagated through the trained model by means of 10^6 Monte Carlo simulations. The resulting output was described by a PMF over the three admissible classes. The best estimate was identified with the mode of this PMF, corresponding to Class #C1, while the associated SDM yielded $u_{phys} = 0.04$.

2. *Measurement model uncertainty evaluation:* Since the selected ANN is a deterministic model, the output distribution required for the evaluation of u_{mod} cannot be obtained directly at inference time. For this reason, u_{mod} was evaluated through the calibration procedure described in Section B.. A balanced set of 90 instances ($M = 30$ for each class) was used to reconstruct the empirical PMFs reported in Fig. 2. For each class, the mode was identified as the best estimate of the measurand, while the associated u_{mod} was quantified through the SDM. The calibration results showed different levels of dispersion across the ordinal scale: $SDM = 0$ for *Class #C2*, $SDM = 0.05$ for *Class #C1*, and $SDM = 0.34$ for *Class #C0*. For the selected operating point (represented by the best estimate obtained in the previous step), the calibration procedure returned $u_{mod} = 0.05$.

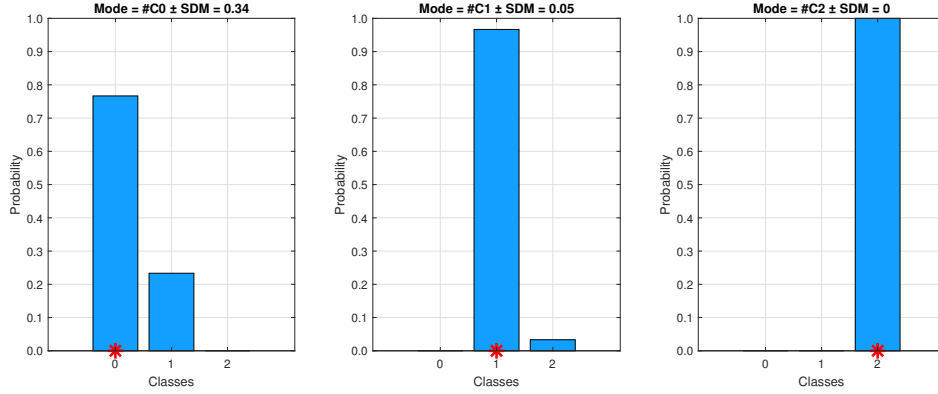


Fig. 2. Empirical PMFs obtained during the calibration phase for the three target classes. The red asterisks (*) indicate the identified mode for each distribution, representing the best estimate, while the model uncertainty (u_{mod}) is quantified through the SDM reported above each plot.

3. *Overall measurement uncertainty evaluation:* Since u_{mod} is not negligible with respect to u_{phys} , the two contributions were combined according to Eq. (1). This yielded a combined standard uncertainty of $u(Y) = 0.07$, which is higher than the uncertainty contribution associated with the physical input quantities alone. This confirms again that considering only u_{phys} would have understated the overall measurement uncertainty.

IV. CONCLUSIONS

This work presented the first steps toward a unified, GUM-compliant approach for the evaluation of measurement uncertainty in ML-assisted measurement applications. The proposed methodology addressed the additional uncertainty contribution introduced by the data-driven nature of ML-based measurement models, referred to as measurement model uncertainty (u_{mod}), and showed how it can be evaluated separately from the uncertainty contribution arising from the physical input quantities (u_{phys}). The two components were then combined within a common uncertainty budget, thus enabling ML-based results to be expressed in a form consistent with established metrological principles. The two case studies confirmed the applicability of the proposed approach to both continuous and ordinal scales. These results support the view that ML outputs can be treated as measurement results, provided that the associated uncertainty contributions are properly identified and combined. Future developments will extend the proposed approach to nominal quantities, thus addressing classification problems in which no natural ordering exists among categories. Further research will also investigate stochastic ML models, such as Deep Ensembles and Bayesian approaches, for which the output distribution can be obtained directly at inference time. These developments will contribute to a broader and more systematic metrolog-

ical treatment of ML-based measurement systems.

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Quality Infrastructure for Resilient Electrical Sector-Ensuring Confident Measurement Results in Calibration of High Currents Instruments

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Abstract – The electrical power systems resilience is strongly dependent on confident measurement results from the quality infrastructure (inspection/certification bodies, testing/calibration laboratories). The reliability of the conformity assessments is deeply rooted in the results traceability, enabled by calibration of used testing equipment. One challenging metrology issue is the calibration of instruments for extreme (very high or low) electrical currents, due to the complex measurement procedures, unestablished traceability chain, and high number of uncertainty influential factors. After thorough analysis of the international CMCs in the area of very high electrical currents, the developed calibration procedure for extreme electrical currents instruments by the Laboratory for Electrical Measurements (LEM) will be presented. The uncertainty will be analyzed according to the GUM and the advanced Monte Carlo methodology. Further conformity assessment of a calibration artefact will be conducted against prescribed decision making rules, using an original software MonteCalc Uncertainty Toolkit, developed in LabView™ by LEM.

Keywords: high currents calibration, Monte Carlo uncertainty evaluation, quality infrastructure, electrical metrology, LabVIEW.

I. INTRODUCTION

The resilience of some critical infrastructure systems, like the power supply systems, is strongly dependent on the confident conformity assessments conducted by the testing, inspection and certification bodies (TIC), [1]. Electrical equipment and installations, such as induction motors, transformers, and others have parameters of voltage, current, active power, apparent power, power factor, that constantly change at any time along with changes in load, disturbance, or other abnormal conditions. This critical electrical equipment and installations need to be monitored to identify their condition in order to prevent eventual serious problems with the system. Lately the

integration of various power resources, intensifies new challenges regarding grids monitoring and supervision, [3]. For instance, more challenges exist in front of accurately measuring, monitoring, and controlling electrical currents in transmission lines. Reliable current measurement is essential for ensuring the stability, efficiency, and safety of modern smart grids. A transparent power grid represents a new development form under the energy transition trend, integrating the power grid with new generation information and communication technologies, [3]. The use of micro smart sensors allows real-time monitoring of many important parameters of power systems via current, which is crucial for ensuring safe, stable, intelligent, and green power grid operation.

The establishment of confident metrology infrastructure for reliable conformity assessments in electrical and energy sector, in particular in the calibration of extreme (very high or very low) electrical currents is the backbone of delivering correct decision rules against various prescribed acceptance criteria for the components of the power supply systems. The calibration of instruments for extreme, both low and high electrical currents, is a metrologically very challenging task, due to factors like:

- a complex measurement procedure to be validated,
- very often unestablished measurement traceability chain for extreme current values to the SI units, and
- high number of influential factors in the uncertainty budget, in many cases not sufficiently predictable.

Various contributions are published for enhancement of the labs infrastructure for calibration of high electrical currents instruments, [4]-[10], with improvement in different aspects of the calibration procedures.

In this paper, after conducting a thorough analysis of the current calibration and measurement capabilities in the area of high electrical currents at international level, the developed calibration procedure for instruments for high electrical currents by the Laboratory for Electrical Measurements (LEM) at the Ss. Cyril and Methodius University (UKIM) will be presented. The uncertainty of the calibration results will be analysed by applying the

GUM approach as well as the advanced Monte Carlo methodology. For that purposes an original software MonteCalc Uncertainty Toolkit, [11], [12], developed in LabVIEW™, [13], by LEM will be applied. The results derived by the two methods will be compared and discussed by using experimental data. Based on the uncertainty propagation distribution gained by the two approaches, assessment of the conformity compliance of calibration artefacts in particular measurement points will be conducted against prescribed decision making rules embedded in the MonteCalc Uncertainty Toolkit.

II. CURRENT STATE OF THE ART IN THE FIELD OF HIGH ELECTRICAL CURRENT METROLOGY

The establishment of an unbroken chain of international measurement traceability in the area of high currents, both AC and DC is rather challenging, [14], [15]. For appropriate needs assessment for expanding the metrology infrastructure, an analysis of the current-state-of-the-art is conducted. This survey is based on the published best calibration and measurement capabilities (CMCs) of some of the most relevant National Metrology Institutes (NMIs) in the Key Comparison Database (KCDB) of the International Bureau of Measures and Weights (BIPM), in the field of high currents, [16].

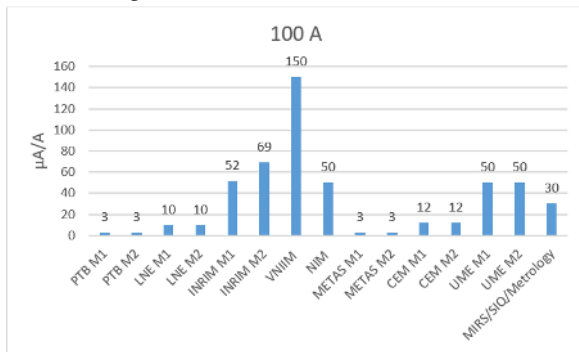


Figure 1. Expanded measurement uncertainties of DC electrical current of 100 A at international level

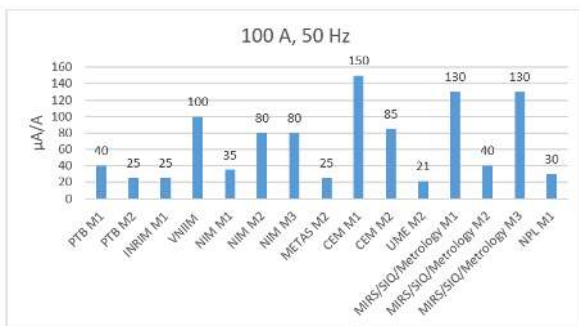


Figure 2. Expanded measurement uncertainties of AC electrical current of 100 A, 50 Hz at international level

The best CMCs of 100 A electrical current calibration of the most relevant international NMIs are displayed in Figure 1 for DC current and in Figure 2 for AC current.

The LEM at UKIM, provides calibration services mostly for the electrical and energy sector in South Eastern Europe. Therefore, a comparative analysis of the best CMCs of the NMIs established in the region of South Eastern Europe is conducted, which is restricted up to 20 A for DC electrical currents as in Figure 3, and up to 100 A, at 50 Hz for AC electrical currents shown in Figure 4.

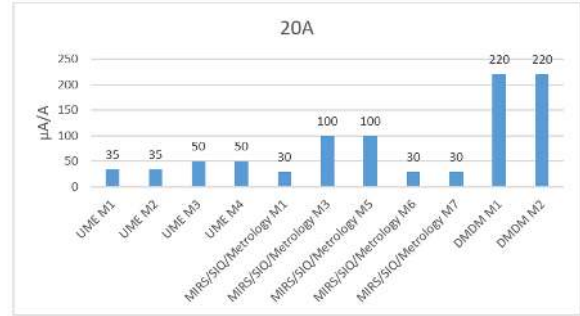


Figure 3. Expanded measurement uncertainties of DC electrical current of 20 A in South Eastern Europe

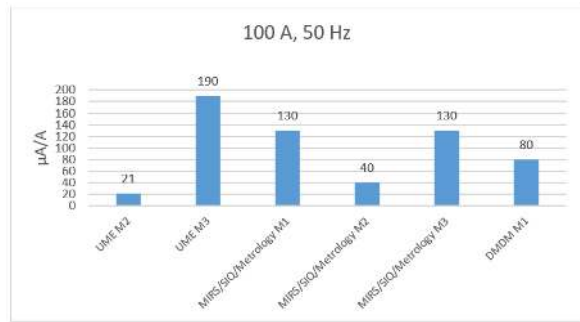


Figure 4. Expanded measurement uncertainties of DC electrical current of 100 A, 50 Hz in South Eastern Europe

LEM has been an accredited calibration laboratory for electrical quantities instruments according to ISO 17025 since 2015, as presented in [17] and [18]. Lately, LEM has expanded its accredited scope for calibration of high current instruments up to 1500 A, both DC and AC. A reference standard Transmille 4015 Multifunction calibrator, with an option for high current calibration coil EA002, has recently been acquired in LEM, with technical specification provided in [19], and established measurement traceability through the producer's accredited calibration laboratory traceable to national (NPL) and international primary reference standards (BIPM).

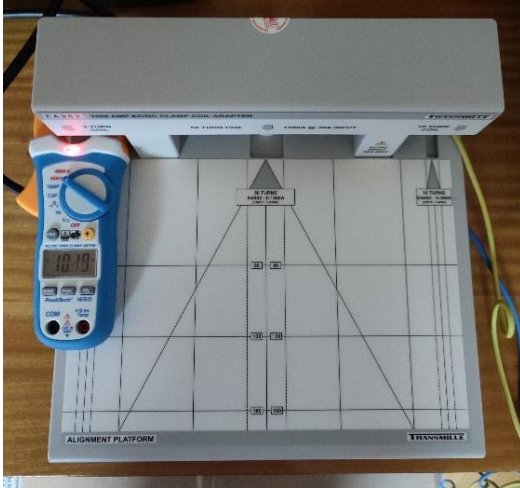
III. CALIBRATION PROCEDURE OF INSTRUMENTS FOR HIGH DC AND AC ELECTRICAL CURRENT DEVELOPED IN LEM

The NMIs and the calibration laboratories in the field of electrical quantities develop diverse calibration methods for instruments for high DC or AC electrical current, [4]-[10]. In this case study, the LEM, develops a calibration

procedure for high DC and AC current clamps. For validation of the established procedure, the unit under test (UUT) is PeakTech 1650, with technical specification in [20]. The test set-up for calibration of the high DC and AC current clamp connected to the LEM high current reference standard is shown in Fig. 5, a) the reference standard and b) the connected UUT.



a) LEM reference standard multifunctional calibrator



b) Calibration of UUT-DC and AC clamp meter with the high-current reference standard calibration coil connected to the multifunctional calibrator

Figure 5. DC and AC high current calibration test set-up

IV. UNCERTAINTY ESTIMATION IN CALIBRATION OF HIGH CURRENT INSTRUMENTS BY DEPLOYMENT OF DIFFERENT APPROACHES

In LEM, two procedures for measurement uncertainty estimation are developed: 1) traditional evaluation according to the Guide to the expression of uncertainty in measurement (GUM), [21] and 2) estimation by propagation of distributions using a Monte Carlo method, according to the Supplement 1 to the GUM, [22]. The calibration of high current instruments belongs to extreme metrology and the selection of the appropriate approach for uncertainty estimation is a complex task, [23].

A. GUM data fusion approach for uncertainty estimation

The initial uncertainty analysis begins with the generally most adopted approach, the GUM methodology. To obtain the uncertainty budget of the calibration results, data from a component of type A, as denoted in the GUM, and components of type B are fused, following the guidelines of GUM, by using the original LEM data fusion concept as in [24]. The type A uncertainty, u_A , is obtained through statistical processing of the experimental data, by the mean value, X_{mean} , and the measurements standard deviation, as:

$$u_A = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_{icor} - X_{mean})^2} \quad (1)$$

where

$$X_{mean} = \frac{1}{n} \sum_{i=1}^n X_{icor} \quad (2)$$

$$X_{icor} = X_i - X_{ref} \quad (3)$$

X_i is the measured value in the specific point and X_{ref} is the value generated by the reference standard. The combined uncertainty of type B stands for:

$$u_B = \sqrt{u_{res_inst}^2 + u_{res_refst}^2 + u_{d_refst}^2 + u_{cal_refst}^2} \quad (4)$$

where the particular uncertainty input components are:

u_{res_instr} - from the resolution of the calibrated instrument,

u_{res_refst} - from the resolution of the reference standard,

u_{d_refst} - component corresponding to the accuracy and the drift of the reference standard, and

u_{cal_refst} - from the calibration of the reference standard.

The total combined uncertainty is:

$$u_c = \sqrt{u_A^2 + u_B^2} \quad (5)$$

The expanded uncertainty, used in conformity statement decision process is:

$$U = 2 \cdot u_c \quad (6)$$

B. Monte Carlo approach for uncertainty estimation

The GUM, enables harmonization in an extensive scope of practical measurement problems, and its traditional methodology is grounded in the Law of Propagation of Uncertainty (LPU). However, GUM's preassumption on linearization and the approximations of the Student's t-distribution can be with restricted applicability. Therefore, JCGM 101:2008, [22] proposes the Monte Carlo approach, propagating entire probability distributions, and not just some parameters. Various published case studies highlight the effectiveness of Monte Carlo simulations in assessing uncertainty in diverse testing and calibration processes, [25]. The Monte Carlo based on random selections, provides compensation of the GUM constraints, such as when nonlinearities appear that cannot be represented by the assumptions of the GUM, or when a normal distribution is attributed to the probability density of the output quantity. However, if there are significant deviations from the normal distribution, which happens in certain cases of calibration of high current instruments, GUM will not provide an adequate representation, which is particularly evident in

cases of extreme high current values. The Welch-Satterwhite method, [25] allows calculation of the effective degrees of freedom, defining the output distribution with a t-distribution, thus enabling influence reduction of deviations from an ideal distribution; however, in real case scenarios evidently greater errors can be generated. Therefore, when one of the uncertainty contributions is significantly larger than the others, or when the order of magnitude of the output quantity is the same or close order to the uncertainty, the only reliable option is the Monte Carlo approach. The GUM Supplement 1 [22], suggests an algorithm on how to perform a Monte Carlo simulation, as in the following sequence: i) defining the measured and input variables, ii) modeling, iii) propagation of the input distributions estimation, iv) simulation/calculation and v) results presentation. When executing a Monte Carlo simulation, a certain number of trials need to be conducted, as each trial produces a random variable depending on the distribution of the input variable. The number of necessary trials to be performed is determined as:

$$M > \frac{10^4}{1-p} \quad (7)$$

where, M is the calculated number of trials, and p is the required probability. For coverage factor $k=2$, at the probability of 95%, i.e. $p=0.95$, the minimal required number of trials is 200000. With sufficient processing capacity, most of the Monte Carlo applications perform 10^6 simulations. The higher the number of trials, more confident the result; however, the laboratory should optimize.

C. Description of the MonteCalc Uncertainty Toolkit

The originally developed software package in LEM for Monte Carlo simulation entitled MonteCalc Uncertainty Toolkit, features the following functionalities described in [11]. The MonteCalc Uncertainty Toolkit is universal and can be applied to any case of testing/calibration uncertainty estimation, with no restrictions on the type of instrument, the physical quantity measured, the range, the type of the probability distribution etc. The Monte Carlo simulation is performed using a virtual instrument palette that allows the generation of random numbers with an appropriate distribution. The generation of the distribution in this case is done depending on the Type B parameters that are entered. For each Type B component, a distribution of random numbers is generated and summed together with the Type A component. From the resultant distribution, the uncertainty both by GUM and the Monte Carlo approach is calculated. The Toolkit has an incorporated complex evaluation module for conformity assessment against different prescribed decision-making rules in line with the Guideline ILAC G8:09/2019, [26]. The built-in decision-making rules in MonteCalc are the binary rule,

the binary rule with guard band, the non-binary rule with guard band and with measurement uncertainty, and other.

V. RESULTS AND DISCUSSION

The results presented in this contribution are only selected specific calibration points from the overall conducted calibration protocol, derived by the MonteCalc Uncertainty Toolkit. In Table 1 are the measurement uncertainties derived from the Toolkit, by applying both the GUM and the Monte Carlo calculations, at the points of 100 A, both DC and AC at 50 Hz, respectively.

Table 1. Comparison of extended calibration uncertainty calculated according to GUM and Monte Carlo at points of 100 A, DC (0 Hz) and AC (50 Hz)

Measurement point		Expanded measurement uncertainty	
Current	Range	GUM	Monte Carlo
(A)		(mA/A)	(mA/A)
100	DC	9,4	7,0
100	AC	9,3	6,7

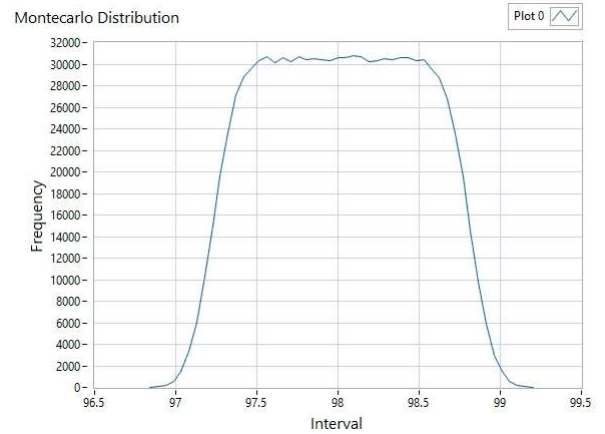


Fig. 6 Probability distribution from MonteCalc Uncertainty Toolkit at point of 100 A in DC range

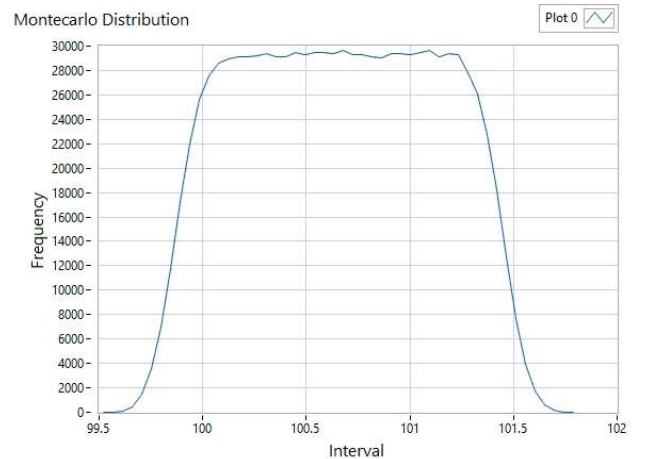


Fig. 7 Probability distribution from MonteCalc Uncertainty Toolkit at point of 100 A AC range at 50 Hz

In Figure 6 is the probability distribution gained at the calibration point of 100 A in DC range, while in Figure 7 is the distribution at the point of 100 A at 50 Hz in AC range, both obtained by the MonteCalc Uncertainty Toolkit. In both figures, the x-axis (abscissa) presents the range of values that may be attributed to the measured quantity, therefore it is labeled as “Interval” in Amps, while the y-axis (ordinate) corresponds to the number of repetitive results, regarding the performed Monte Carlo simulation, and hence it is titled “Frequency”. The derived calibration results with the accompanying uncertainty are subjected to conformity assessment by the built-in option of the MonteCalc Uncertainty Toolkit, against different decision making rules, as in [26]. The compliance graphs obtained by the Monte Carlo simulation method depending on the limit value set with an appropriate control for the particular decision making rule, i.e. guard band of three expanded uncertainties $w=3U$, i.e. the Six-Sigma rule, at

the calibration point of 100 A DC in Figure 8, and AC at 50 Hz in Figure 9 of the UUT, are displayed. The values along the vertical (ordinate) axis in Figures 8 and 9, are the measured values of the current in Amps A.

The two methods yielded noticeably different uncertainty values. This is expected as the GUM assumes a normal distribution, an assumption that is not fully true in case of calibration at high currents values and particularly in cases of low-resolution artefacts of calibration. Here it is this case, as the UUT-current clamp is with dominant uncertainty component due to its resolution with rectangular probability distribution. In contrast, the Monte Carlo simulation accounts for the actual distribution by randomly generating variable values, providing more reliable conformity assessment decision. In both cases the results obtained by the two approaches are acceptable according to the applied Six-Sigma decision rule.

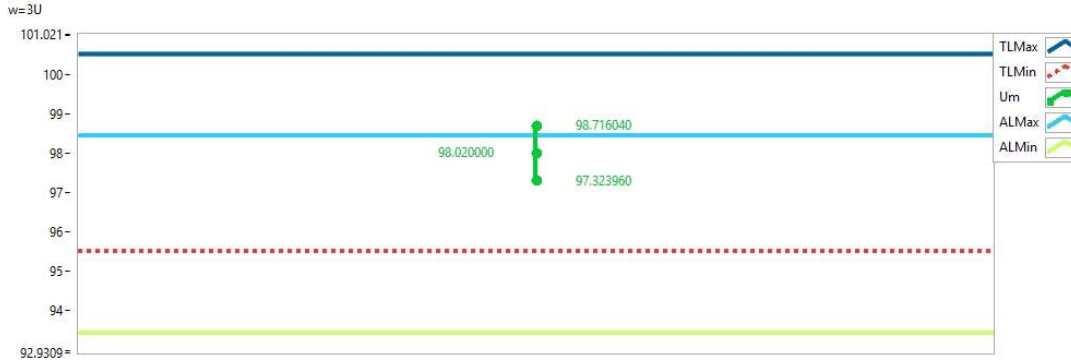


Fig. 8 Result at a calibration point of 100 A at DC range with guard band $w=3U$, Six-Sigma decision rule

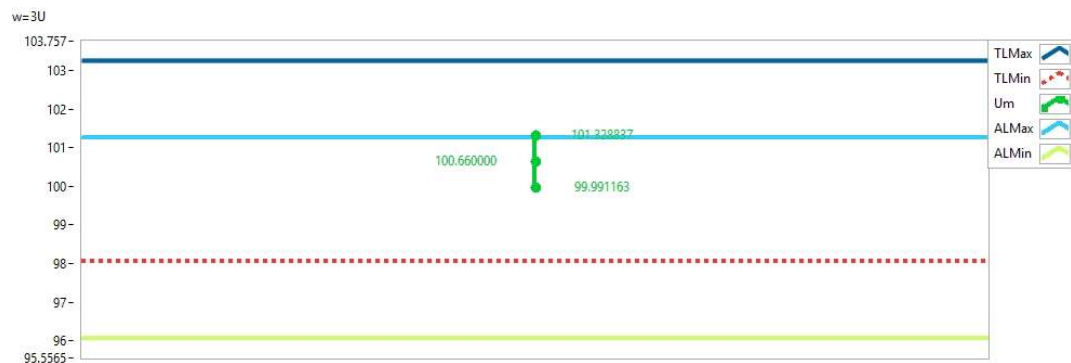


Fig. 9 Result at a calibration point of 100 A, 50 Hz at AC range with guard band $w=3U$, Six-Sigma decision rule

VI. CONCLUSIONS

After a thorough analysis of the up-to-date CMCs in the area of high electrical current at international level and rationale confirmation for development of high current calibration infrastructure in South Eastern Europe, a calibration procedure for instruments for extreme electrical currents in the LEM at UKIM is developed and accredited. The uncertainty of the calibration results are analyzed by applying the GUM and the Monte Carlo methodology, incorporated an original software

MonteCalc Uncertainty Toolkit, developed by LEM. The results derived by the two methods are compared and discussed by using experimental calibration data of a high current instrument-current clamp in DC and AC range. Based on the uncertainty propagation distribution gained by the two methods, conformity compliance assessment of the calibration artefact in particular points is conducted against prescribed decision making rules embedded in the Toolkit. The satisfactory results confirm the effectiveness of the developed approaches.

ACKNOWLEDGEMENT

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Accuracy analysis of THD measurement of power signals by means of a low-cost oscilloscope

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Abstract – In this paper the accuracy of the Total Harmonic Distortion ratio (*THD*) measurement of power grid voltages achieved by using a low-cost USB oscilloscope is investigated. The *THD* measurements provided by the USB oscilloscope are analyzed in the case of synthesized signals for different *THD* and signal frequency values. That comparison allows us to determine the accuracy of the *THD* measurements provided by the USB oscilloscope. Moreover, the *THDs* of the power grid voltages are measured by the USB oscilloscope and compared with those achieved by using a very accurate power quality logger.

Keywords: error analysis; power signals; statistical analysis; Total Harmonic Distortion ratio (*THD*); USB oscilloscope.

I. INTRODUCTION

One of the most important parameters used to characterize a power signal is the Total Harmonic Distortion ratio (*THD*) since it indicates the amount of harmonic distortion into the analyzed signal. For proper operation of the equipment and instruments connected to the power systems the *THDs* of the power signals should be smaller than the limits specified in the IEEE 519:2022 Standard - Recommended Practice and Requirements for Harmonic Control in Electric Power Systems [1] corresponding to the used power system. Therefore, the *THD* should be accurately measured in real-time. To this aim the Discrete Fourier Transform (DFT)-based algorithms are often used. When the DFT is employed the IEEE 519:2022 Standard recommends a measurement window equal to 10 (12) cycles for nominal frequency equal to 50 (60) Hz for estimating the magnitude (or rms value) of the signal harmonics [1]. In practice *THD* is often measured by means of power analyzers which, in most situations, use for this purpose a DFT-based algorithm. The prices of these instruments are relatively high. In this work a low-cost solution for relative accurate and fast *THD* measurements of power grid voltages is presented. That solution consists in the utilization of a low-cost USB

oscilloscope, which is the PicoScope 2205A. That USB oscilloscope is characterized by a vertical resolution of 8 bits, a bandwidth of 25 MHz, a maximum sampling rate of 200 MSa/s, a memory capacity of 16 kSa, a harmonic distortion smaller than -50 dB (0.32%), and a noise smaller than 150 μ V rms [2].

The *THD* measurements provided by the PicoScope 2205A oscilloscope are analyzed for different *THD* and signal frequency values. Their accuracy has been determined in the case of synthesized harmonically distorted signals by using as reference values those achieved by means of the Interpolated DFT (IpDFT) algorithm based on the Hann window [3]-[5]. Also, the *THDs* of the power grid voltages are measured by the PicoScope 2205A oscilloscope and compared with those provided by a very accurate power quality logger.

The rest of the paper is organized as follows. In Section II the accuracies of the *THD* measurements achieved by means of two DFT-based algorithms, which are the Energy-Based (EB) algorithm [6]-[9] and the two-point IpDFT algorithm [3]-[5] are investigated through simulations. The *THD* measurement accuracy achieved by the PicoScope 2205A oscilloscope is investigated in Section III by using synthesized signals. Then, it is used to measure the *THDs* of the power grid voltages. The most important conclusions of this work are given in Section IV.

II. ACCURACY OF THE THD MEASUREMENT BY THE DFT-BASED ALGORITHMS

The expression for the *THD* is given by:

$$THD = \sqrt{\frac{\sum_{k=2}^{K_H} A_k^2}{A_1^2}}, \quad (1)$$

where A_1 is amplitude of the fundamental component, A_k is the amplitude of the k th harmonic, and K_H is the number of the significant harmonics of the analyzed signal ($K_H = 51$ according to the IEEE 519:2022 Standard [1]).

From (1) it follows that the accuracy of the *THD* measurement depends on the fundamental and harmonic amplitudes estimation accuracies. In the case of

noncoherent sampling, as often occurs in practice, two DFT-based algorithms are often used for the amplitude estimation. They are the so-called EB algorithm [6]-[9] and the two-point Interpolated DFT (IpDFT) algorithm [3]-[5]. In the case of the EB algorithm the rms of the k th signal component is computed by using the highest DFT sample corresponding to that component and a number of DFT samples situated on the left and right sides of the above highest DFT sample. The total number of the used DFT samples is denoted by L in the following. When the analyzed signal is *a-priori* weighted by a H -term cosine window, L is often chosen as $L = 2H + 1$ [6]-[8] (H DFT samples are used in both sides of the related highest DFT sample). In the case of the two-point IpDFT algorithm the inter-bin frequency of the k th component is firstly estimated. After then, the amplitude is estimated by means of the Discrete-Time Fourier Transform (DTFT) of the used window at the estimated inter-bin frequency and the corresponding highest DFT sample [4].

Fig. 1 shows the mean $\pm$ the standard deviation of the $THDs$ achieved using the EB and the two-point IpDFT algorithms based on the Hann window [10] as a function of the waveform frequency f in the case of a waveform affected by the first 15 harmonics and an additive white noise of zero mean and a Signal-to-Noise Ratio (SNR) equal to 50 dB. The waveform frequency f varies in the range [48, 52] Hz with a step of 0.2 Hz. The nominal frequency is $f_n = 50$ Hz and 10 nominal cycles are acquired with a sampling rate of $f_s = 20.48$ kSa/s. Thus, the acquisition length is $M = 4096$ samples. In the EB algorithm $L = 2H + 1 = 5$ DFT samples are considered. The harmonic amplitudes are chosen as follows: A_2 for all even harmonics, $5A_2$ for the third harmonic, $4A_2$ for the 5th and 7th harmonics, and $2A_2$ for the 9th, 11th, 13th, and 15th harmonics. The amplitude A_2 is chosen to achieve a $THD = 4.50\%$, that is $A_2 = 5 \cdot 10^{-3}$ p.u. For each waveform frequency 1000 runs of $M = 4096$ samples each are used with signal tone phases randomly chosen in the range $[0, 2\pi)$ rad.

In Fig. 1 it can be observed that both IpDFT and EB algorithms provide very accurate THD measurements. The bias of the THD measurements achieved by the EB algorithm slightly increases as the distance of the waveform frequency from the nominal frequency increases. Conversely, for the considered waveform frequency, the bias of the THD measurements achieved by the IpDFT algorithm is almost the same. Moreover, in Fig. 1 it can be observed that the standard deviations of the THD measurements achieved by both algorithms are very close.

Fig. 2 shows the bias and the standard deviation of the THD measurements achieved by the EB algorithm (Fig. 2(a)) and the two-point IpDFT algorithm (Fig. 2(b)) as a function of THD when the waveform frequency is $f = 49.5$ Hz. THD varies in the range [1, 6] % with a step of 0.5%. The analyzed waveforms and the remaining simulation parameters are as in Fig. 1.

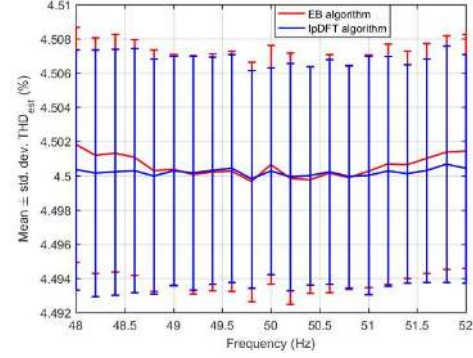


Fig. 1. Mean $\pm$ standard deviation of the THD measurements achieved by the EB and the two-point IpDFT algorithms based on the Hann window versus the waveform frequency. Analyzed waveforms affected by the first 15 harmonics and wideband noise of SNR = 50 dB. For each waveform frequency 1000 runs of $M = 4096$ samples each with random phases of all signal tones.

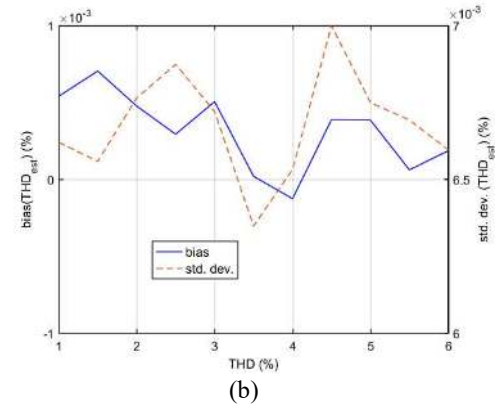
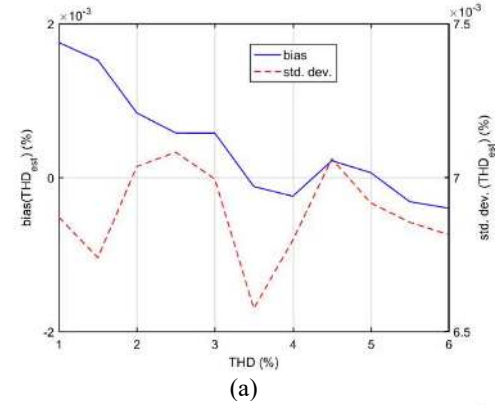


Fig. 2. Bias and standard deviation of the THD measurements achieved by the EB algorithm (a) and the two-point IpDFT algorithm (b) versus THD when the waveform frequency is $f = 49.5$ Hz. Analyzed waveforms affected by the first 15 harmonics and wideband noise of SNR = 50 dB. For each THD 1000 runs of $M = 4096$ samples each with random phases of all signal tones.

By analyzing the results shown in Fig. 2 it follows that the bias values and the standard deviations of the *THD* measurements achieved by both algorithms are very small. Thus, the bias achieved by the two-point IpDFT algorithm and the EB algorithm vary in the range $(-1.3 \cdot 10^{-4}, 7.1 \cdot 10^{-4})$ % and $(-4.0 \cdot 10^{-4}, 18 \cdot 10^{-4})$ %, respectively. Conversely, the standard deviation achieved by the above two algorithms vary in the range $(6.3 \cdot 10^{-3}, 7 \cdot 10^{-3})$ % and $(6.6 \cdot 10^{-3}, 7.1 \cdot 10^{-3})$ %, respectively.

It should be remarked that to achieve small *THD* standard deviations the acquisition length should be enough high, such as $M \geq 2048$ samples.

Since the bias values achieved by both algorithms are very small it follows that the mean values of the *THDs* achieved by both EB and two-point IpDFT algorithms are very close to the theoretical *THDs* and they can be used as reference value for the *THD* practical measurements. In the following we will consider as *THD* reference value the mean achieved by the two-point IpDFT algorithm based on the Hann window since it provides a smaller bias than the EB algorithm under off-nominal frequency condition.

III. THD MEASUREMENT ACCURACY ACHIEVED BY USING THE PICOSCOPE 2205A USB OSCILLOSCOPE

In this Section *THD* is measured in two cases, which are synthesized signals and power grid voltages, respectively. The nominal frequency is $f_n = 50$ Hz. The time/div coefficient of the oscilloscope is set to 20 ms/div. The sampling rate is equal to 24.4 kSa/s and the FFT length is set to $M = 4096$ samples. The PicoScope 7 software is used.

A. Synthesized signals

The synthesized signals are harmonically distorted sinewaves provided by a Siglent SDG2042X waveform generator. It has a 16-bit vertical resolution. The sinewaves provided by that generator have a maximum *THD* of 0.075%, a maximum non-harmonic spurious of -70 dBc, a signal output accuracy equal to $\pm(1\% + 1 \text{ mVpp})$ at 10 kHz, and a maximum frequency error of 1 ppm of setting [11]. The SDG2042X waveform generator can provide odd, even or both odd and even harmonics up to the 16th harmonic. The test signals have the frequencies equal to 49.5 Hz, 50.0 Hz, and 50.5 Hz, and the amplitude equal to 4 V. Each signal contains the first 15 harmonics. That number of harmonics has been set in the *Highest harmonic* box of the *THD* measurement menu of PicoScope 7 software [12]. The Hann window has been selected. The voltage range for the vertical axis of the oscilloscope is set to ± 5 V. The values considered for the *THD* are 1.125%, 2.25%, and 4.50%, respectively, which are much higher than the maximum *THD* of the test sinewaves. The harmonic amplitudes are chosen as in Section II. Thus, the amplitude A_2 is chosen equal to 5 mV, 10 mV, and 20 mV to obtain *THD* = 1.125%, 2.25%, and 4.50%, respectively. For each *THD* and frequency values 100 *THD* measurements

provided by the PicoScope oscilloscope are performed. Also, for each above *THD* and frequency values the signals are acquired by a NI USB-6259 acquisition board, which has a 16-bit resolution [13]. The used signal input range is ± 5 V. At this range the specified total error due to the Analog-to-Digital Converter (ADC) gain and offset errors, noise, and temperature at full scale is 1.01 mV [13]. 100 sets of $M = 4096$ samples each are considered. The sampling rate is equal to 20.48 kSa/s. The above number of samples corresponds to 10 signal periods. For each data set the *THD* is computed by means of the IpDFT algorithm based on the Hann window. For each frequency the mean value of the achieved *THDs* is computed, and used as reference. It is worth noticing that since the total error at 5 V is 1.01 mV, the total errors corresponding to the harmonic amplitudes, which are very small as compared with that of the fundamental component, are very small, and so their contributions on the *THD* reference values are negligible.

In Table 1 the achieved reference values for the *THD* are given, while in Table 2 the maximum of the absolute value of the relative errors of the *THD* measurements, $|\delta_{THD}|_{\max}$, in percentage, provided by the PicoScope 2205A oscilloscope are given for the considered *THD* and signal frequency values. Also, Figs. 3 and 4 show the achieved *THD* measurements and reference value, and the signal spectrum when the signal frequency is $f = 49.5$ Hz and *THD* = 1.125% (Fig. 3) and *THD* = 4.50% (Fig. 4).

Table 1. The *THD* reference values computed as the mean values of the *THD* measurements achieved using the IpDFT algorithm based on the Hann window. The analyzed signals are acquired by a 16-bit acquisition board, that is the NI USB-6259 acquisition board.

Signal frequency f (Hz)	<i>THD</i> reference values		
	<i>THD</i> = 1.125%	<i>THD</i> = 2.25%	<i>THD</i> = 4.50%
49.5	1.1145	2.2400	4.4910
50.0	1.1146	2.2399	4.4907
50.5	1.1145	2.2400	4.4909

Table 2. The relative errors $|\delta_{THD}|_{\max}$ for the *THD* measurements provided by the PicoScope 2205A oscilloscope when the signal frequency is equal to 49.5 Hz, 50.0 Hz, and 50.5 Hz, and *THD* is 1.125%, 2.25%, and 4.50%.

Signal frequency f (Hz)	Relative errors $ \delta_{THD} _{\max}$ for the <i>THD</i> measurements		
	<i>THD</i> = 1.125%	<i>THD</i> = 2.25%	<i>THD</i> = 4.50%
49.5	5.27	1.19	0.78
50.0	5.23	1.22	0.87
50.5	5.23	1.32	0.81

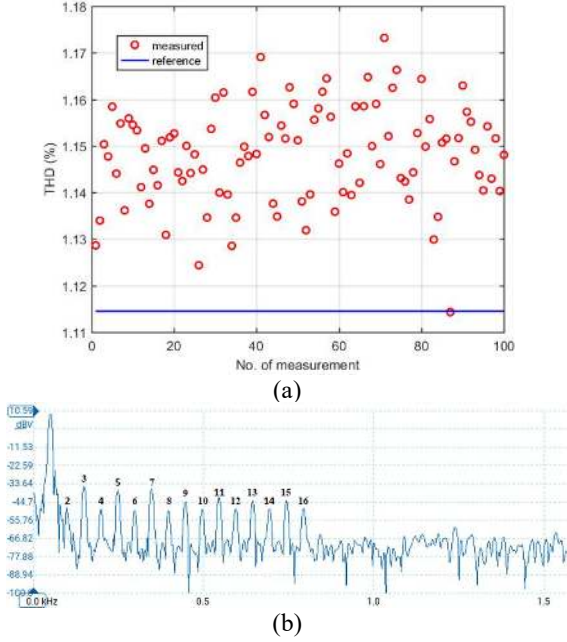


Fig. 3. Synthesized signals with $f = 49.5$ Hz and $THD = 1.125\%$: THD measurements provided by the PicoScope 2205A oscilloscope and the THD reference value (a) and signal spectrum achieved using the PicoScope oscilloscope with the indicated signal harmonics (b).

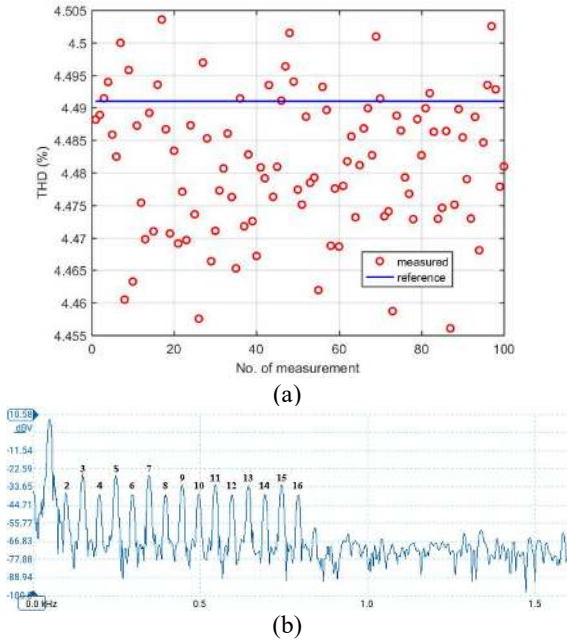


Fig. 4. Synthesized signals with $f = 49.5$ Hz and $THD = 4.50\%$: THD measurements provided by the PicoScope 2205A oscilloscope and the THD reference value (a) and signal spectrum shown using the PicoScope oscilloscope with the indicated signal harmonics (b).

For each considered THD and signal frequency values the THD reference values given in Table 1 are almost the same, as in the simulations performed in Section II. They are most close to the set THD as THD increases. From the results given in Table 2 it follows that the accuracy of the THD measurements provided by the oscilloscope increases as THD increases. That behavior occurs due to the contribution of the intrinsic harmonic distortion of the PicoScope oscilloscope, which for smaller $THDs$ has a more important contribution. Also, from the results presented in Table 2 it can be observed that for all considered frequencies almost the same accuracy is achieved. When $THD = 1.125\%$ the accuracy of the THD measurements is relatively low. Conversely, when $THD = 2.25\%$ and $THD = 4.50\%$ accurate THD measurements are achieved.

In Fig. 3(a) it can be observed that in the case of $THD = 1.125\%$ almost all THD measurements are higher than the reference THD value. Conversely, in Fig. 4(a) it can be seen that in the case of $THD = 4.50\%$ there are both higher and smaller THD measurements as compared with the reference THD value. Figs 3(b) and 4(b) clearly shown that all considered harmonics are higher than the noise floor of the PicoScope oscilloscope when $THD = 1.125\%$ and $THD = 4.50\%$, respectively. However, as expected when $THD = 1.125\%$ the harmonics are much closer to the noise floor as compared to the harmonics corresponding to $THD = 4.50\%$.

B. Power grid voltages

The voltages are achieved from the power grid by means of a step-down voltage transformer (220 V : 6 V). In order to avoid the contribution of the harmonics due to the voltage transformer the $THDs$ of the input and output voltages of the transformer have been measured by means of a very accurate power quality logger, that is the Fluke 1738 three-phase power quality logger [14], over 10 minutes. Fluke 1738 measures the THD on the first 50 signal harmonics and the related permissible error is equal to $\pm 0.5\%$ [14]. The difference between the average values of the measured $THDs$ of the output and the input voltages of the transformer, that is 2.21%, has been used in the following as a correction factor.

A hair dryer is connected to the same power source. The number of harmonics used to compute THD is $K_H = 51$ as it is suggested in IEEE 519:2022 Standard [1]. Fig. 5 shows 500 THD measurements provided by the PicoScope 2205A oscilloscope in both cases – when the hair dryer is off and on, respectively. Fig. 6 shows the spectra of both signals, while Fig. 7 shows the histograms of the THD measurements achieved in both cases.

In Fig. 5 it can be observed that when the hair dryer is on the THD increased as compared to the case when the hair dryer is not connected. In this case also the even harmonics increase as it can be observed in Fig. 6. Moreover, in Fig. 6 it can be seen that the most important harmonics of the

analyzed signals are the odd harmonics until the 23rd order in the case of the signal with hair dryer off, and the odd harmonics until the 21th order plus the 2nd and 4th harmonics in the case of the signal with the hair dryer on.

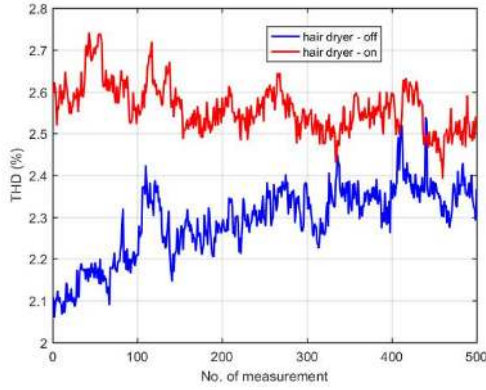


Fig. 5. Power grid voltages: *THD* measurements provided by the PicoScope 2205A oscilloscope when the hair dryer is off and on, respectively.

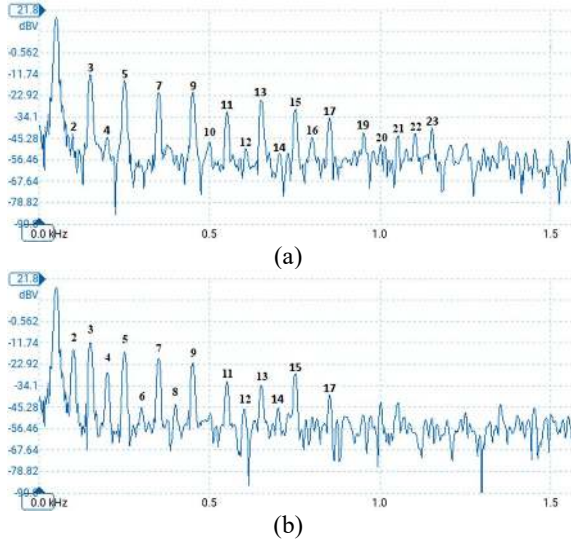
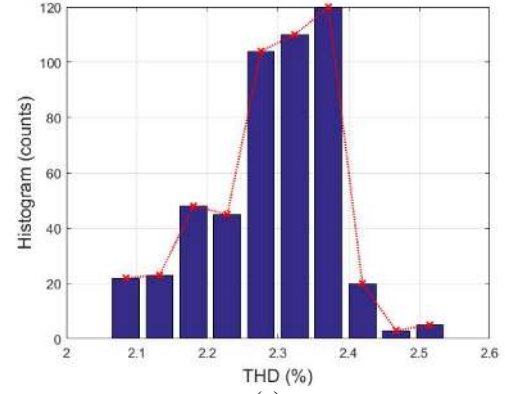


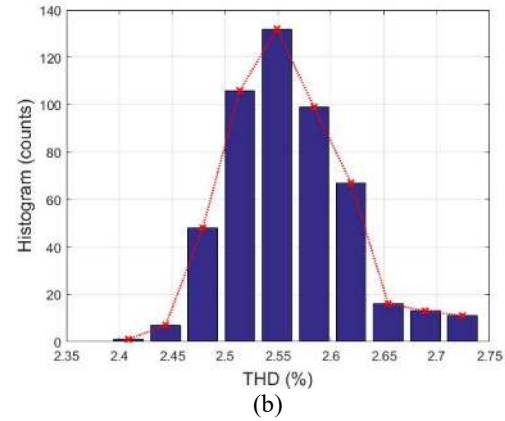
Fig. 6. Power grid voltages: signal spectra achieved by the PicoScope oscilloscope with the indicated signal harmonics when hair dryer is off (a) and on (b).

From Fig. 7 it follows that in most situations the *THD* values are between 2.3% – 2.4% when the hair dryer is off, and between 2.5% – 2.6% when the hair dryer is on.

It is worth noticing that the achieved *THD* is between 2.3% and 2.6%, and so, as it follows from the investigations performed in the previous subsection, it is relative accurately measured by the PicoScope 2205A oscilloscope.



(a)



(b)

Fig. 7. Power grid voltages: histograms of the *THD* measurements provided by the PicoScope 2205A oscilloscope when the hair dryer is off (a) and on (b), respectively.

Moreover, the *THD* measurements of the power grid voltages achieved by both PicoScope 2205A oscilloscope and Fluke 1738 are compared with each other. The measurements have been performed into a laboratory of the Faculty of Electrical and Power Engineering, University Politehnica Timișoara. The experimental setup is shown in Fig. 8 and the minimum, maximum, and average values of the *THD* measurements achieved by both instruments over 10 minutes with a step of 1 minute are shown in Fig. 9.

From Fig. 9 it follows that the minimum, maximum, and average *THD* values achieved by the PicoScope 2205A oscilloscope are about 0.1% smaller than those achieved by the Fluke 1738.

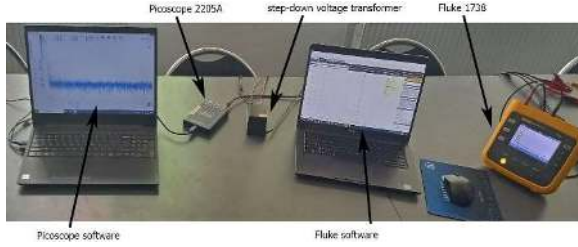


Fig. 8. Experimental setup used to measure the THDs of the power grid voltages by both PicoScope 2205A oscilloscope and Fluke 1738.

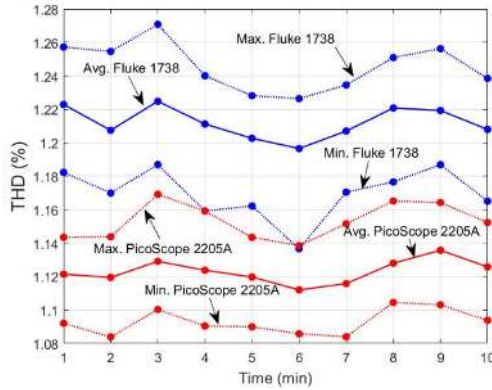


Fig. 9. Power grid voltages: minimum, maximum, and average THD measurements of the power grid voltages achieved by the PicoScope 2205A oscilloscope and the Fluke 1738 versus time.

IV. CONCLUSIONS

In this paper the accuracy of the *THD* measurement provided by the low-cost PicoScope 2205A oscilloscope has been investigated using harmonically distorted sinewaves. It has been shown through simulation results that the mean of the *THD* measurements achieved by the two-point IpDFT algorithm based on the Hann window can be used as reference value for the *THD* practical measurements. Moreover, it has been shown through experimental results that when *THD* is higher than 2% it can be accurately estimated by the PicoScope 2205A oscilloscope. Also, the *THD* of the power grid voltages have been measured by the PicoScope 2205A oscilloscope in two scenarios. In addition, it has been shown that the *THD* measurements achieved by the PicoScope 2205A oscilloscope are relative close to those achieved by the Fluke 1738 three-phase power quality logger, which has a much higher price than the PicoScope 2205A oscilloscope. Thus, this oscilloscope is well suited to be used for *THD* measurement in the case of low-voltage systems when *THDs* are higher than 2%.

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Model-based magnetization measurement in Fluxgate sensors

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Abstract – The magnetization curve of the fluxgate sensor core is estimated using measurements and calculations. The finite element method is used to estimate the magnetization curve from measurement data and a nonlinear magnetic analytical model. The fluxgate measurement sensitivity, measured voltage, and current are utilized in the inverse method to estimate the magnetization curve of the fluxgate sensor.

Keywords: magnetization curve; FEM; measurement; analytical model.

I. INTRODUCTION

Permeability and magnetization of magnetic materials can usually be measured on ring-shaped samples, which have two advantages [1]:

1. There is no demagnetization, so the magnetic field strength can be calculated from the current.
2. If the ring is slim, the magnetic field strength is homogeneous

To some extent, this idealization also applies to long samples in long solenoid coils and, in general, to ellipsoidal samples, for which the field is homogeneous, and the demagnetization factor can be calculated. However, the industry continues to require the measurement of the magnetic properties of parts with complex shapes. The only magnetic property that can be directly measured in samples of arbitrary shape is coercivity.

The magnetic amorphous tapes are sensitive to mechanical strain and bending, which cause changes in magnetic properties. The magnetic properties include the B - H curve, hysteresis loop, and coercivity. Using amorphous tape for a micro fluxgate is a challenging mechanical task when cutting the tape into a racetrack shape, and winding can alter the magnetic properties of the tape used in the core of the micro fluxgate. A measured magnetization curve obtained using a standard approach with a ring-shaped sample cannot be used to design and simulate a micro fluxgate, because the magnetic properties change during manufacturing. The manufacturing process can locally alter magnetic properties in the magnetic core,

where the direct B - H measurement method is not applicable. Estimating an equivalent B - H curve after the manufacturing process using an indirect method is recommended. Using finite element method (FEM) techniques and a nonlinear material model, the effective permeability of the sample and its magnetization characteristics can be reconstructed from a set of measurements via inverse methods [2]-[5]. Besides the external cause of uncertainty for the magnetic properties, caused by the mechanical process in the micro fluxgate, internal cause of uncertainty can be caused by heating. The heating is caused by winding copper losses and core losses, including eddy-current and hysteresis losses. The hysteresis loss in the core can be higher in the manufactured micro fluxgate because cutting the amorphous tape influences the magnetic domain, increasing hysteresis loss.

Using an accurate magnetization curve when designing a fluxgate magnetometer with a micro-dimensional magnetic core [5]-[6] is critical, especially when operating at excitation frequencies in the hundreds of kHz to MHz range. Skin effect can be accurately estimated at very high frequencies when an accurate magnetization curve is used in the modeling and simulations of a micro fluxgate. It depends on the relative magnetic permeability of the micro fluxgate core. The skin effect in the magnetic core affects the performance of the micro fluxgate and should be taken into account during the design process.

Material VITROVAC 6025 I50 [7] is used for the fluxgate sensor, which is a Co-based amorphous alloy with the highest magnetic permeability, the lowest coercivity, and very low saturation magnetostriction. It has a flat 'F' hysteresis loop in magnetic sensor applications. This paper presents an estimate of the appropriate magnetization curve for VITROVAC 6025 I50 used in the core of the microfluxgate [6], obtained using a time-stepping 3D FEM. In the proposed inverse method, an analytical model for the B - H curve is used, with its parameters estimated from measured microfluxgate sensitivity versus excitation current. The accurate B - H curve for the VITROVAC 6025 I50 can be estimated using the inverse method and the microfluxgate measurement results.

II. ANALYTICAL MODEL FOR MAGNETIZATION

Combined rational and power functions [2] are used to model the B - H curve of the magnetic core. Equation (1) shows the proposed analytical function:

$$(1) \quad B = J + \mu_0 H, \quad J = \frac{a \cdot H^b}{c \cdot H^b + 1}$$

where B , J , and H are the magnetic flux density, magnetic polarization, and magnetic field strength, respectively. a , b , and c are constants, which are calculated by curve fitting. The saturation value of magnetic polarization, J_s is calculated as follows:

$$J_s = \frac{a}{c}, \quad H \rightarrow \infty$$

(2)

The conditions for a , b , and c in (3) need to be satisfied in (1) to obtain a monotonic and continuous function for the B - H curve and relative permeability curve, μ_r - H , in the total range of magnetic strength, $0 \leq H \leq \infty$. Equation (4) presents the derivation of the differential permeability, μ_d versus H .

$$\mu_r = \frac{B}{\mu_0 H} = \frac{J}{\mu_0 H} + 1, \quad \mu_r = \frac{1}{\mu_0} \cdot \frac{a \cdot H^{b-1}}{c \cdot H^b + 1} + 1$$

$a, c \geq 0, \quad b \geq 1$

(3)

$$\mu_d = \frac{dB}{\mu_0 dH} = \frac{dJ}{\mu_0 dH} + 1, \quad \mu_d = \frac{1}{\mu_0} \cdot \frac{a \cdot b \cdot H^{b-1}}{(c \cdot H^b + 1)^2} + 1$$

(4)

The maximum μ_r and its corresponding H , depending on the parameters of a , b , and c :

$$\frac{d\mu_r}{dH} = \frac{-a \cdot c \cdot H^{2b-2} + a \cdot (b-1) \cdot H^{b-2}}{(c \cdot H^b + 1)^2} = 0$$

$$H = \sqrt[b]{\frac{b-1}{c}}$$

(5)

Fig. 1 presents the B - H and μ_r - H curves as functions of the parameters a , b , and c in (1). Increasing the a/c ratio increases saturation flux density according to (2) and also enhances maximum μ_r . The B - H curve becomes sharper for an increase in b , thereby increasing μ_r considerably.

Fig. 2 compares the measured B - H curve for VITROVAC 6025 I50 with the curve-fitting function. The curve-fitting function in Fig. 2, with the specified values of a , b , and c , will be used as an initial B - H curve for 3D FEM analysis of the fluxgate sensor. The maximum μ_r is about 4500.

The conductivity of VITROVAC 6025 I50 is approximately $\sigma=0.6$ MS/m. For example, the eddy current skin depth, $\delta = 1/(\pi f \sigma \mu_r \mu_0)^{0.5}$, is 31 μm and 10 μm for $\mu_r=4500$ at 100 kHz and 1 MHz, respectively. Higher and lower μ_r change skin depth, which affects micro fluxgate performance and sensitivity.

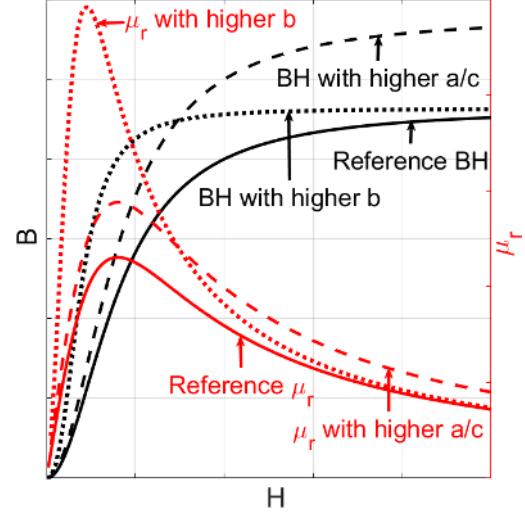


Fig. 1. Variation of B - H curve and μ_r versus parameters a , b , and c

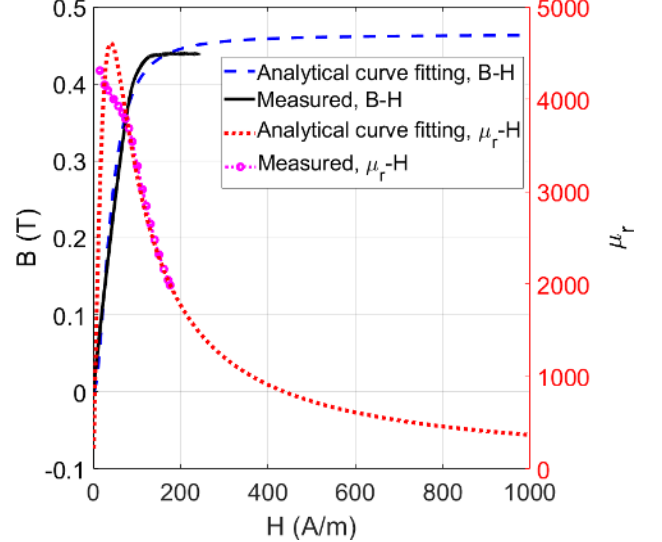


Fig. 2. Comparison of measurement and analytical model for the B - H and μ_r - H curves ($a=2.78\text{e-}4$, $b=2.012$, and $c=6.0\text{e-}4$).

III. MEASUREMENT OF THE B - H CURVE

The B - H loop of the material was measured on a sample that is a 4-times upscaled version of the micro fluxgate's core. The sample is therefore 32 mm long and 7 mm wide. Primary and secondary coils were wound around the core. The current in the primary coil and the induced voltage in the secondary coil were measured using a 16-bit PicoScope 4262.

The B field was obtained by digital integration. The "version 0" referential B - H curve was obtained by averaging and smoothing the four quadrants of the loop, while the other versions were fitted using the analytical model described earlier.

IV. SIMULATION MODEL OF THE FLUXGATE

The fluxgate consists of a racetrack-shaped core, one sensing (pick-up) coil, which is placed across the middle part, and four segments of the excitation coil, placed on the outer parts, as shown in Fig. 3. The figure also shows the simulated B field. There is a homogenous field of $50 \mu\text{T}$ around the fluxgate (which is the field to be measured by the fluxgate sensor) and the excitation field, which alternates through clockwise and counterclockwise directions through the core.

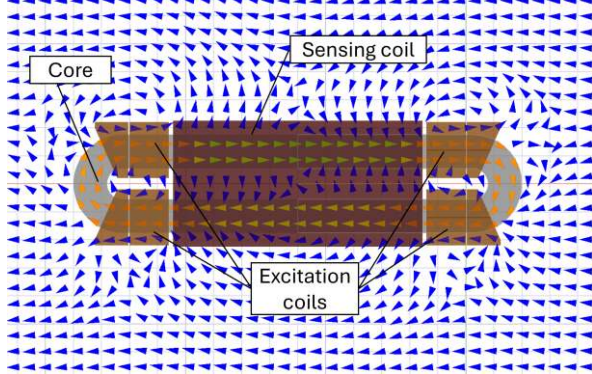


Fig. 3. Model of the fluxgate with geometry and B field displayed

Symmetry is used to simplify the model; there are two planes of symmetry (left-right and front-back in Fig. 3), so only one quarter is required to be simulated. Note that even though the geometry is also symmetric in the third plane (top-bottom in the figure), the fields are not, as the external field is tangential to that plane while the excitation field is normal.

V. SIMULATION OF FLUXGATE SENSITIVITY

We are using a transient (time-stepping) simulation in Ansys Maxwell 3D to determine the relation between sensitivity and excitation current at a given excitation frequency. Such a characteristic is important for determining the maximum open-loop sensitivity and the minimum current required to achieve sufficient saturation. This current is given by a plateau with a low slope behind the peak sensitivity. The signal at the second harmonics is most used to process the fluxgate's output. The second harmonic is extracted from the simulated induced voltage by emulating synchronous detection in post-processing. While it may be interesting to know the sensitivity of higher harmonics as well, these higher frequencies will be more affected by factors not considered in the simulation (e.g., parasitic capacitance of the sensing coil) and therefore more complicated.

The simulation results match the measurements well at 1 MHz, as shown in Fig. 4. It is now possible to obtain more accurate results by tuning the fitting curve coefficients.

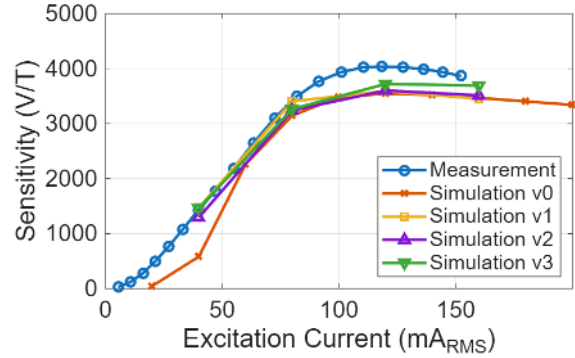


Fig. 4. Comparison of measurement and simulation of the micro fluxgate at 1 MHz excitation frequency

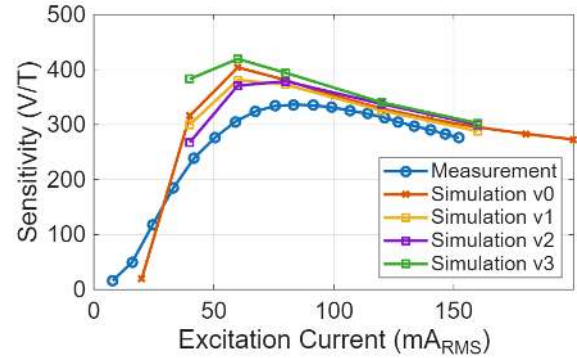


Fig. 5. Comparison of measurement and simulation of the micro fluxgate at 100 kHz excitation frequency

However, given that many other factors influence the output signal, some of which can't be realistically included in the simulation, we cannot expect a model that perfectly fits at one frequency to also fit at other frequencies. For example, when repeating the simulation at 100 kHz (Fig. 5), we obtain results that are similarly close, but in this case, the sensitivity tends to be slightly overestimated, unlike at 1 MHz, where it is underestimated. Furthermore, by testing multiple variations of the B - H curve (Table 1 and Fig. 6), we can see that the results vary, indicating that the differences are partially due to inaccuracies in the B - H curve.

Version 3 seems to be the closest to the measurement (especially at 160 mA), but the trend is better matched with v1 and v2, as v3 predicts an increase in range where the measured sensitivity already decreases with current.

Table 1. Coefficients of the B - H curves used in simulations (Versions 1 and 2 used in 100 kHz simulation, all three used in 1 MHz simulation)

	a	b	c
Version 1	2.78e-04	2.0119	6.00e-04
Version 2	3.05e-04	1.9628	6.00e-04
Version 3	8.32e-04	1.9628	1.50e-03

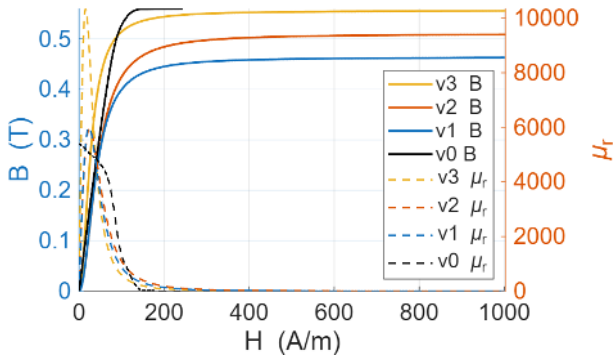


Fig. 6: B - H curves used in simulations (Versions 1 and 2 used in 100 kHz simulation, all three used in 1 MHz simulation)

VI. SIMULATION OF EXCITATION COIL VOLTAGE

Another interesting characteristic to simulate is the relation between voltage and current on the excitation coil to evaluate the estimated B - H curve. In the following measurement, the excitation coil is powered by a sine-wave signal from a signal generator. The generator has $50\ \Omega$ output impedance, plus there is a $10\ \Omega$ shunt to measure the current. Because of this high series resistance, the current is close to a sinusoid, while the voltage is not.

We start by recording the voltage and current waveforms (Fig. 7). In the simulation, we apply the recorded current waveform to the excitation coil and run a transient simulation to calculate the voltage waveform, which we compare with the measurement.

This simulation is much faster for multiple reasons: Model size can be reduced using symmetry from 1/4 to 1/8; the sensing coil does not need to be modeled; and the time step can be larger, as the voltage does not change as rapidly as in the case of the sensing coil signal in the previous simulation.

For these reasons, this method may be preferable for quickly verifying the accuracy of the B - H curve. There are, however, also downsides: The microfabricated coil has relatively high resistance (around $23\ \Omega$ at operating temperature) and therefore behaves as a mostly resistive load at low frequencies. We need a high frequency to see a significant effect of the inductance. Also, the waveform is affected by the material's coercivity, which cannot be accounted for in the simulation.

In Fig. 7 (upper), we see the recorded waveforms; the voltage is mostly determined by the coil's resistance, which needs to be removed. The DC resistance was measured to be $23\ \Omega$. In the bottom plot of Fig. 7, we subtracted the voltage from that resistance:

$$V_L = V - I \times 23\ \Omega$$

The simulation assumes that the coil is made of a material with negligible resistivity, so the real component of the simulated impedance arises solely from eddy currents in the core.

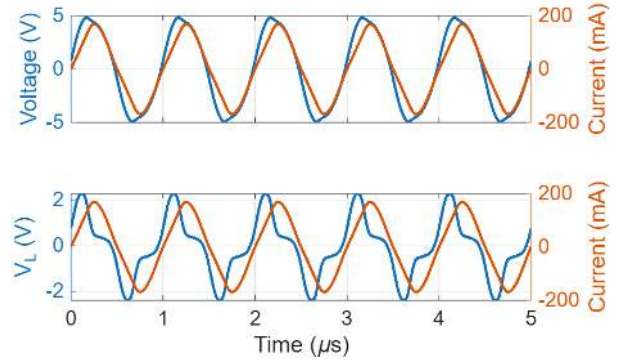


Fig. 7: Voltage and current waveforms on the excitation coil at 1 MHz. The upper plot is the actual voltage on the coil, and the lower plot is the voltage after subtracting the voltage at the DC resistance of the coil.

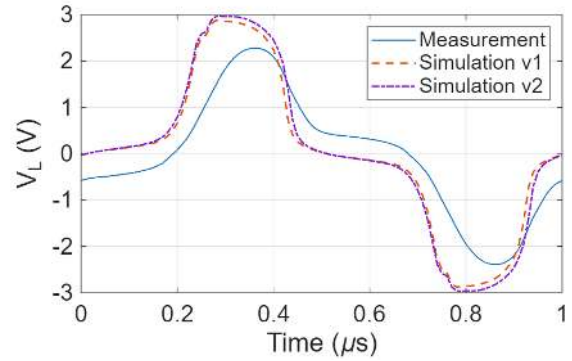


Fig. 8: Measured and simulated voltages on the excitation coil (without considering the DC resistance of the coil)

By the principle of fluxgate, the excitation coil operates in a nonlinear range (core being deeply saturated), so the inductance cannot be expressed as a single number. However, the results in Fig. 8 show that the simulation reproduced the shape and amplitude of the voltage waveform quite well. There appears to be a phase shift between the measured and simulated results, likely due to the material's coercivity, which wasn't accounted for in the simulation as mentioned earlier. Fig. 9 presents the harmonic contents for the measurements and simulation results. Further adjustment of the B - H curves can help to achieve adequate magnetization curve.

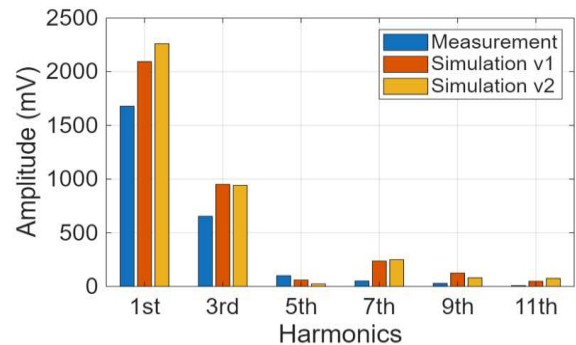


Fig. 9: Harmonic content of the V_L waveform

VII. CONCLUSION

Two experiments have been conducted to evaluate the accuracy of the estimated BH curve. One is the sensitivity analysis, which is useful for the design optimization of the fluxgate; the second is to apply a sinusoidal current to the excitation coil and measure the induced voltage, which is highly dependent on the BH curve and can be computed more quickly due to the lower complexity of the model.

For low excitation current, the BH curve estimated using the analytical model matches the measured sensitivity very well; however, at higher excitation current, further modification of the BH curve is required to improve the accuracy of the calculated sensor sensitivity.

Usage of the estimated BH curve will be very helpful for further design optimization of the miniaturized fluxgate, which is the main goal of our research.

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Estimation of torquer rod magnetic moment with fluxmetric method and compact sensing coils

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Abstract – We present a robust, noise-resistant method for calibrating magnetorquer rods for commercial spacecraft, utilizing a homogeneous, Merritt or Lee-Whiting coil and an analog integrator. After initial calibration with a moment coil, the magnetorquer rod calibration error due to its non-zero length was as low as 1% if the Lee-Whiting coil had approximately the same length as the torquer rod. This procedure is not only space saving compared to magnetometric methods, but could be utilized in environments with high magnetic noise (in order of 100 nT pp) and still exhibit a low standard deviation of the results (<0.5%).

Keywords: torquer rod; magnetic flux; fluxmeter.

I. INTRODUCTION

A magnetorquer rod consists of a thin cylinder-shaped, magnetically soft ferromagnetic core with one or more solenoid windings around it. The magnetic dipole moment is linearly proportional to the energizing current up to the point of saturation of the magnetic core, and thus creates rotational torque due to interaction with homogeneous field (such as the Earth's magnetic field). Such rotational torque control is used to change the attitude of satellites [1]. During the space qualification process the nominal torquer rod moment (active and remanent) needs to be determined for accurate application and also taken into consideration during calibrations of onboard magnetometers [2].

II. MOTIVATION AND PROCEDURE

Due to the substantial torquer rod length, the usual magnetic dipole equation used in magnetometric methods has to take its length L into consideration even at moderate distances [3]. The two-pole model representing two monopoles at L -distance [4] is perfectly valid theoretically; however, the more corresponding physical reality of a torquer rod is that of a homogeneously magnetized cylinder as presented later.

Apart from low signal-to-noise ratio at large distances, an issue in magnetometric methods utilized in [1] and [3]

is the effect of mechanical alignment of the rod relative to the magnetometer: the resulting uncertainty is dominated by distance uncertainty and uncompensated misalignment angles, affecting repeatability and calibration time.

In [5] we proposed two alternative approaches of moment estimation: a two-point magnetometric measurement and the fluxmetric method. The two-point method sounded promising, since the absolute distance to the torquer rod is not important and only the relative change in distance between the two measurements needed to be known and can be established with high precision. However, we did not see any significant improvement in repeatability or precision of results with this method [5].

The second alternative, which is a derivation of the legacy fluxmetric method for determining the magnetic dipole moment of permanent magnets [6] - [8], showed more promise. Thus the fluxmetric method is studied deeply in this contribution, mainly in terms of achievable error with compact, homogeneous sensing coils. Theoretically, this method should not only be the least sensitive to misplacements, but also robust in terms of suppressing environmental noise, allowing for its use in industrial practice.

III. MAGNETOMETRIC METHODS OF DIPOLE MOMENT MEASUREMENT

The simplest method of dipole moment measurement is based on an approximation of the torquer rod as a point-like, dipole source, with only one magnetic moment component along the rod axis z . Then the moment can be calculated from B_z at a distance Z from the torquer rod:

$$m = m_z = \frac{2\pi}{\mu_0} B_z Z^3 \quad (1)$$

Eq. 1 yields negligible error only at $Z/L > 10$, so for moderate distances ($\sim 2L-4L$) – where the measured field B_z can be obtained with reasonable signal-to-noise-ratio and the setup is not too bulky – the torquer rod length must be taken into account. The commonly used magnetometric methods [3], [4] use the following equation:

$$m = \frac{4\pi}{\mu_0} \frac{B_z}{\frac{Z/L - 1/2}{(Z^2 - ZL + L^2/4)^{3/2}} - \frac{Z/L + 1/2}{(Z^2 + ZL + L^2/4)^{3/2}}} \quad (2)$$

The rod is now considered to be a pair of magnetic monopoles with spacing L . Magnetic moment is then calculated using Eq. 2, where Z is the distance from the centre of the rod to the centre of the magnetic sensor, L is the length of the rod and B_z is the z component of the magnetic flux density.

IV. THE FLUXMETRIC METHOD - REAL COILS AND RODS

The legacy fluxmetric method with homogeneous (Helmholtz) coils [6] is widely utilized to establish the magnetic moment of permanent magnets [7], [8]. Ref [6] derives symmetric equations for a dipole source in Helmholtz coils, which is valid for small and strong sources like permanent magnets. The equation is conveniently simple and consists only of the flux change $\Delta\Phi$ seen by the fluxmeter (or an integrator, respectively) and field constant of the fluxmeter (Helmholtz) coil K_c :

$$m = \frac{\Delta\Phi}{K_c} = \frac{K_I \Delta rdg}{K_c} \quad (3)$$

where K_I is the integrator constant at the range used and Δrdg is the change in its reading.

The magnetic flux Φ threading the coil can be defined

$$\Phi = \int_{coil} B_z(z) dA \approx B_z A \quad (4)$$

for a constant area A and a short torquer rod, with B_z being the field due to the torquer rod placed axially in a homogeneous coil with constant K_c - see Fig. 1. The flux change $\Delta\Phi$ can be generated by either inserting the torquer rod into the coil, or by switching the rod current on/off, depending on whether the remanent moment should be established too.

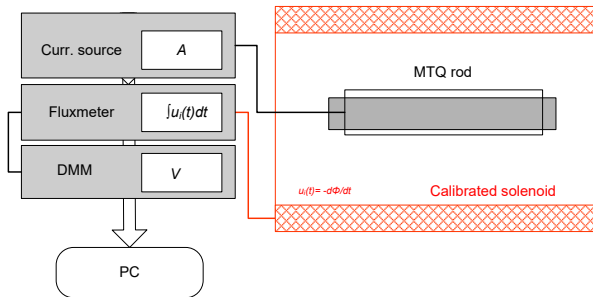


Fig. 1. The fluxmetric method with a solenoid coil and magnetorquer rod (MTQ) inserted in the centre

This simplified approach is valid for permanent

magnets or very short rods relative to the size of the coil. Furthermore, the torquer rod of length L is neither a point-like dipole source nor two monopoles at distance L , but instead – for a large enough core permeability – behaves like a magnetized cylinder of length L and radius R , producing a field $B_{rod}(z)$ along the z axis. Also, the sensing coil has its own field sensitivity profile $W_{coil}(z)$ due to its construction (simple solenoid or a coil system with larger homogeneity).

In this manner, the B_z of Eq. 2 is now just approximated by B_{zmeas} , which is smaller due to the inhomogeneous response of the coil $W_{coil}(z)$ and field profile of the rod $B_{rod}(z)$:

$$B_{zmeas} = \int W_{coil}(z) B_{rod}(z) dz \quad (5)$$

$B_{rod}(z)$ represents the field of a magnetized cylinder with length L , radius R and magnetization M (A/m):

$$B_{rod}(z) = \frac{\mu_0 M}{2} \left(\frac{z+L/2}{\sqrt{R^2+(z+L/2)^2}} - \frac{z-L/2}{\sqrt{R^2+(z-L/2)^2}} \right) \quad (6)$$

We can then define the “true” dipole moment m_{true} of the rod by knowing its volume V :

$$m_{true} = MV = M\pi R^2 L = \frac{1}{\delta} m_{meas} = \frac{1}{\delta} \frac{\Delta\Phi}{K_c} \quad (7)$$

where M is the (homogeneous) magnetization from Eq. 6, $\Delta\Phi$ is the change of flux as seen by the fluxmeter, K_c is the field constant in the homogeneous area of the coil and δ is the correction/error due to real coil field profile and real torquer rod dimensions.

So the L -dependent moment error $\delta(L)$ is actually:

$$\delta(L) = \frac{\int W_{coil}(z) B_{rod}(z, L) dz}{\int W_{coil}(z) B_{rod}(z, L \rightarrow 0) dz} \quad (8)$$

which could be solved numerically for a given coil and rod geometry in iterations of decreasing L (L cannot be 0 in any of previous equations). Alternatively, $\delta(L)$ could be estimated for a given coil system by measurements.

V. TORQUER ROD CALIBRATIONS IN A MAGNETICALLY QUIET AREA

In our initial evaluation, we have compared magnetometric and fluxmetric methods in terms of repeatability, and have found the fluxmetric method to be superior also due to its immunity to misalignment errors [5]. However a further question evolved: what would be the requirements for a 1% calibration target in terms of sensor to torquer rod distance (magnetometric) or coil size (fluxmetric).

For this test, we had 5 magnetorquer rods of different

sizes available at SANSA – see Fig. 2, and a 60 cm long, 35 cm diameter solenoid with a nominal constant $K_c=543.2 \mu\text{T/A}$. Since SANSA exhibits very low noise ($<1 \text{ nT p-p}$), we could establish the “true reading” with the magnetometric method and Eq. 2. The magnetometer (LEMI011) was at 2 m Z-distance, where the signal to noise ratio was still sufficient even for the 43 cm long rod. The alignment was done with the help of a laser pointer coaxial with the torquer rod and a mirror on the enclosure of a freshly calibrated magnetometer – see Fig. 3. The current was provided by a Yokogawa GS200 source-meter, with a trapezoidal waveform of the current (max 100 mA).



Fig. 2. The torquer rods for method comparison – the top one is 43 cm long



Fig. 3. Alignment for reference magnetic moment measurements with magnetometric method

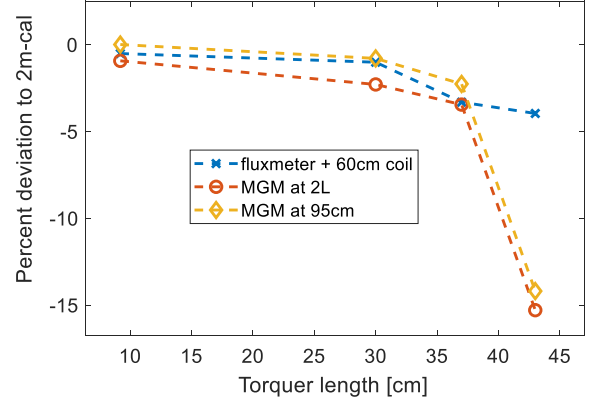


Fig. 4. Difference to reference magnetometric calibration with magnetometric and fluxmetric method (blue) – four torquer rods up to 43 cm rod length

From the results in Fig. 4, we can see that whereas at $2\times L$ and 95 cm distance all deviate significantly with increasing rod size down to -18%, the fluxmetric method in a 60 cm long solenoid yields less than 5% error for the 43 cm rod, and is below 1% for rods shorter than 30 cm. The integrator (Hirst IFM03) had to be calibrated as we found it to be 3% out of specifications - we used a known current profile generated by the source-meter, which roughly matched the rectangular voltage induced in the solenoid.

VI. FLUXMETRIC METHOD IN A NOISY LAB: ERRORS VS COIL/TORQUER ROD SIZE

In an area with high field noise (tens to hundreds of nT peak, such as our laboratory at the CTU), the fluxmetric method with as small coil as possible will be better suited than magnetometric measurements. Previously, we have shown our first results with a 60 cm long solenoid [5]. However, such a coil is unnecessarily bulky. If e.g. we would use the circular Lee-Whiting 4-coil system [9], it would give better homogeneity for the same dimensions. The same would also apply for Merritt coils [10].

In order to establish the behavior of the error δ vs. coil length L (Eq. 8), we used two 16 cm and 10 cm torquer rods (designated as 1L and 1S respectively) and 3 coils of similar designs and sizes, as shown in Table 1 and Fig. 5 below. The rods were energized with a trapezoidal 200 mA p-p bipolar current with a duration of 2000 ms, preprogrammed in a Keithley 2601B source-meter. The current shape and duration were chosen so that the derivative in sensing coil is well defined (square waves with low amplitude), the eddy current effect is negligible but the pulse is fast enough to be able to capture and compensate for drift. The EF-2D fluxmeter analog output was sampled at 2kSa/s by a 24-bit ADC module NI 9239.

Table 1. Parameters of the three coil systems

Coil type, dia/length [mm]	Nom. Kc [$\mu\text{T/A}$]	Area with <1% inhomogeneity [$\pm \text{ mm z}$]
Merritt, 500	122	175
Lee-Whiting, 155	685	75
Lee-Whiting, 125	1850	40

In order not to rely on individual coil constants and integrator calibration, we first calibrated the whole measurement chain (coil, integrator, current supply and DAQ) using a pre-calibrated moment coil supplied by the Czech Metrology Institute [11] with a nominal constant of $0.092682 \pm 0.000074 \text{ m}^2$. The response of the integrator in the large Merritt coils for 2 A_{pp} current can be seen in Fig. 6 – clearly the large size and low sensitivity of the Merritt coil introduce a challenge in data processing. Despite this, we managed to calibrate the relative responses of the three coil systems within 0.3% standard deviation. After that, the response of the torquer rods 1S and 1L was established in each of the three coils.

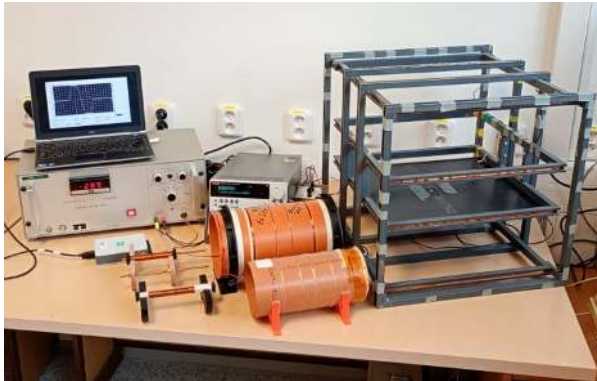


Fig. 5. Measurement setup with all three coils (Merritt and 2× Lee-Whiting) and the two torquer rods

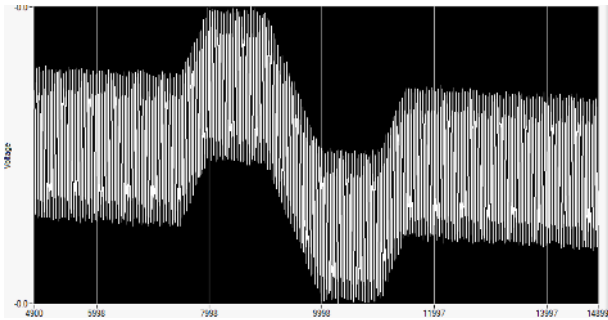


Fig. 6. The raw acquired signal during moment calibration in Merritt coils – 50 Hz and drift visible

Calibration deviation: 2 rod sizes, 2 L-W coils

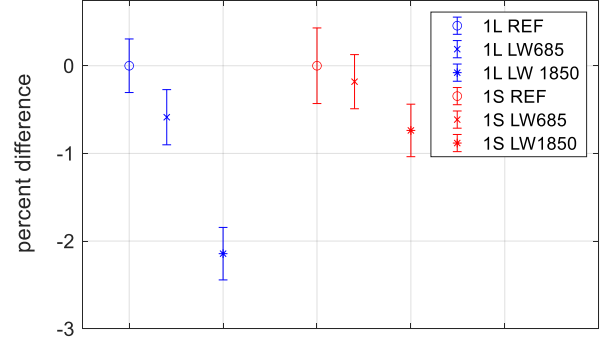


Fig. 7. The resulting calibration differences as coil size is reduced. We can see that for the 1S torquer rod, even the smallest Lee-Whiting coil was enough for the 1% target.

As the 1% homogeneous area of the Merritt coils is much larger than the torquer rod 1L length, we can take this calibration as a reference. The results are shown in Fig. 7, together with the standard deviation obtained from ten measurements at a single measurement point, and also from the standard deviation of the fluxmeter chain calibration done in the first step. We can derive that the torquer rod's length should be at most the overall length of the Merritt or Lee-Whiting coil in order to stay within 1% deviation.

VII. CONCLUSION

The torquer rod calibration error due to its non-zero length was as low as 1% if the Lee-Whiting coil had approximately the same length as the torquer rod (155 mm), while we regarded the initial measurement in the square 50 cm Merritt coils as reference. The same could be obtained from the 10 cm rod and 12.5 cm Lee-Whiting coil. An improvement of this method would be to design a spherical coil [12] where the increased homogeneity could further reduce the deviation for larger cores on smaller coil dimensions, with a target torquer rod length of about 40 cm.

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Compact modular cryogenic dielectric resonator for surface impedance measurements in high magnetic fields

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Abstract – We present a new dielectric-loaded resonator (DR) for cryogenic microwave measurements intended primarily for measurements of the complex surface impedance of conducting samples, including bulk, film, and additively manufactured specimens. The presented resonator is designed to work in a temperature range from 4.2 K to 300 K, to be compact, modular, non-magnetic, and to be orientable with respect to the applied external magnetic field. The prototype can operate in the TE₀₁₁ and TE₀₂₁ modes at 14.27 GHz and 25.15 GHz, respectively. We present the first tests and validation of the resonant cell with a superconducting YBa₂Cu₃O_{7-δ} sample and compare results with those from a reference resonator. We observe a greater than 30% increase in loaded quality factor (Q_l) with respect to the previous resonator and good repeatability, thus validating the concept for cryogenic operation.

Keywords: dielectric-loaded resonator; surface impedance; cryogenic measurements; magnetic field.

I. INTRODUCTION

Microwave resonant methods remain among the most versatile approaches for measurements of electromagnetic properties of conductive and dielectric samples. These methods are well established for nondestructive characterization of metals and superconductors, providing one of the most sensitive techniques for measurements of the surface impedance ($Z_s = R_s + jX_s$) [1–4]. Within this framework, dielectric-loaded resonators (DR) retain a distinct role because they combine field concentration, reduced volume, and a geometry that can be tailored either for surface perturbation on conducting samples or for volume perturbation on dielectric samples [5, 6]. In superconducting radio-frequency cavities for particle accelerators, for example, achieving ultra-low surface resistance (R_s) is critical to obtain high quality factors $Q \sim 10^{10}$ and reducing operational losses [7, 8]. Another fundamental area of development is the high-field magnets (for fusion reactors, high-resolution nuclear magnetic resonance [9, 10])

using high- T_c superconductors like ReBCO coated conductors. Superconductors' performance in the superconductor mixed state is significantly influenced by the dissipation associated with vortex motion, so understanding their surface impedance in strong fields is crucial for predicting future magnetic system performance and stability [1, 11].

However, conventional microwave measurement techniques face practical challenges: the majority of high-field magnets do not allow rotation of the field direction, making the study of the anisotropic properties of superconductors costly and challenging. Dielectric-resonator fixtures usually have a complex structure and often need to be disassembled to change samples or to replace the resonant measurement cell to be used in different experimental conditions or different cryogenic systems, leading to poor measurement repeatability [12].

The present work is motivated by a gap between the favorable features of dielectric resonators and the constraints imposed by cryogenic operation in a magnetic field, which limit their versatility across different cryogenic platforms. Surface-impedance measurements below the superconducting transition and, in particular, in the normal state, require a resonant structure that maintains stability after cool-down and during the variations of the external magnetic field and fits in the limited cryostat space available for a sample. A small resonant cell volume should fit various dry and wet cryostats. A further key requirement is the variable orientation of the resonant cell structure, to achieve angle-dependent measurements also within fixed orientation magnets. These requirements become even stricter when the resonator is expected to operate on more than one mode and to support comparative measurements on different samples under nominally identical conditions.

The resonant cell proposed here is a new DR prototype for cryogenic applications inspired by the idea of variable-orientation microwave fixtures for anisotropic surface-impedance measurements, as addressed in [13], based on a gapped cylindrical DR configuration as developed for room-temperature applications [3, 5, 6]. We do not simply reproduce or adapt the room-temperature resonator in [3] for cryogenic use; we redesign the structure and parts,

maintaining the idea of a sapphire (single crystal Al_2O_3) cylindrical DR with PTFE gaps between puck and bases, with a sample surface partially exposed to the DR resonant electromagnetic field using a thin mask with a circular hole. The new resonant cell consists of a single-crystal sapphire puck placed in the center of the cylindrical OFH copper cavity. The dielectric puck is placed in a fixed position at the center of the metallic cavity using two PTFE puck holders, which also create a symmetrical gap between puck and cavity bases or masks. The sample under study is exposed to the electromagnetic field of the resonant mode by placing it over the circular hole in the thin mask. To ensure that the sample remains in place, the mask is integrated into the sample holder, creating a stable, secure sample mounting. This configuration has already shown excellent performance with a good repeatability [3]. In the redesigned cryogenic resonator, we change the size of the cavity, sample fixing mechanism, and procedure, and redesign the coaxial-cable fixing scheme to allow more reliable coupling tuning.

The paper is organized as follows. Section II recalls the small-perturbation DR measurement principle. Section III describes the cryogenic resonator and the design choices. Section IV presents the first experimental validation of the cell based on the temperature and magnetic-field response of a superconducting $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ (YBCO) sample, with a direct comparison to the previous-generation resonator. Section V is the conclusion.

II. SURFACE IMPEDANCE MEASUREMENT PRINCIPLE

For conducting samples, the resonator is operated in an end-wall replacement configuration. Under the standard surface-perturbation approximation, the variation with respect to the reference value of the reference state surface impedance in the sample state or conditions $\Delta Z_s = Z_s - Z_{s,ref}$ is related to the resonator observables through the simple relation:

$$\Delta Z_s = \Delta R_s + i\Delta X_s = G_s \left[\Delta \left(\frac{1}{Q} \right) - 2i \frac{\Delta f_0}{f_0} \right] + \text{bg}, \quad (1)$$

where $Q = Q_l(1 + \beta)$, Q_l is the loaded quality factor, β is a coupling constant and f_0 is the resonant frequency, both obtained from the measurements of the scattering S -parameters as a function of frequency; G_s is a constant geometrical factor associated with the exposed sample part and resonant mode field distribution, and bg is the residual background contribution of the resonator cavity and feeding lines [3, 12].

In practical cryogenic measurements, it is of interest to investigate the variations on Z_s induced by varying the applied magnetic field, at fixed temperature, or vice versa. In some applications, comparing a sample's Z_s to another sample reference Z_s is useful, for example, when inves-

tigating homogeneity across multiple samples. In those cases, a round-robin measurement procedure, in which two samples can be placed in the same resonant body under the same exposed area and nominally the same field distribution, can also be exploited. Thus, the present cell was conceived to be capable of accommodating two samples. For these applications, the reliability of the measurements depends directly on the mechanical stability of the resonator and on the repeatability of the sample mounting.

To investigate samples with different sizes and geometries, a thin mask with a circular axial aperture is used to limit the interaction to the central part of the sample surface, thereby reducing the influence of border degradation and allowing to compare samples on a common reference area. In the prototype used for the present measurements, the comparison was performed with a 6.00 mm aperture in the new resonator and with a 12.00 mm aperture in the previous one, reflecting the different cavity diameters and fixing geometries. The resonator cell dimensions are mainly constrained by available cryogenic space and can be adapted to measurement requirements.

The same resonant body is intended to support dielectric materials characterization in a later development stage. In that case, the samples will perturb both the stored energy and the loss distribution of the dielectric-loaded cavity, and the pair (f_0, Q_l) can be interpreted in terms of complex permittivity through the usual cavity-perturbation framework [3, 12]. In this paper, we do not present results and validation related to permittivity data; the point relevant here is that the presented resonant cell was designed to keep validation of permittivity measurements in the same cell as a future extension.

III. CRYOGENIC RESONATOR DESIGN

A view of the resonant measurement cell is shown in Fig. 1, which summarizes the design idea. The assembly includes the metallic cavity manufactured from oxygen free copper, the dielectric puck with a diameter 7.30 mm and height 4.50 mm made of single-crystal sapphire (Al_2O_3) [14], the sample support, the masking element that defines the exposed circular area (with a 6.00 mm diameter) of the sample surface, the spring-loaded sample fixing parts and the external support used to install the cell in the cryogenic setup.

The structure is designed to be spring-loaded along the axial direction using a minimal number of parts to reduce uncertainty contributions due to thermal contraction and to be modular. The DR cell was designed to have the center of the dielectric puck placed at the axis of the cryostat sample space. Two samples can be placed on the two sides of the dielectric puck for round-robin measurements; when single-sample measurements are of interest, one side of the resonant cavity should be closed by the blank OFC base. The PTFE supports for the dielectric puck ensure stable po-

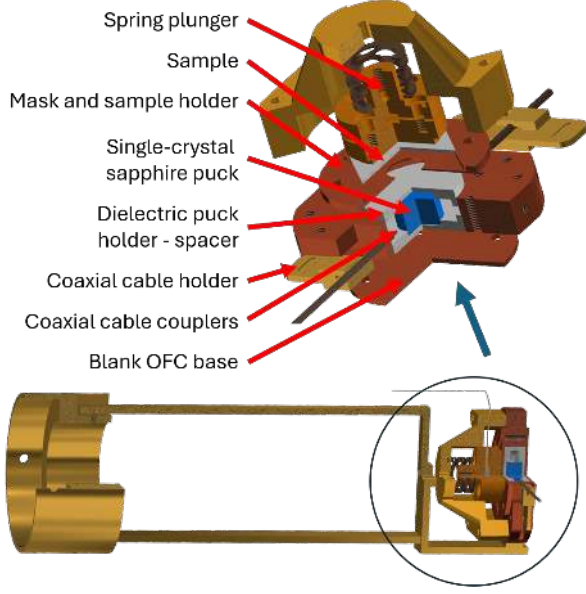


Fig. 1. Exploded view of the cryogenic dielectric-loaded resonator cell and its representation on bi-positional support.

sitioning of the dielectric puck, minimize its radial movement, and guarantee fixed symmetrical gaps between puck and bases/samples. In the presented configuration, PTFE gaps were designed to be 0.70 mm. Due to restrictions in the sample space of the cryostat, the minimum radius of curvature of the cryogenic coaxial cables, and the minimal acceptable size of available mass-produced non-magnetic screws, the resonant cell is optimized to fit a 50 mm diameter space when it is fully assembled and with coaxial cables installed.

The resonator support, for simplicity, was designed for two preset orientations 0° and 90° with respect to the cryostat sample space axis, allowing the cell axis to be oriented either parallel or perpendicular to the cryostat bore axis. Future development will be directed toward an *in situ* rotatable version.

Compared with the room-temperature prototype, We optimized the coupling method and coaxial cable bending to operate in a very strict space. Coupling to the external microwave line is achieved by a coaxial cable loop just barely inserted in the resonator cavity. The two coaxial cables are inserted in a movable part, allowing the fine tuning of the resonator coupling by varying the position of the coupling loop up to 2 mm radially. The outer part of the coaxial cables was formed into a spring to save space, accommodate thermal contraction without breakage, and permit mounting the DR cell at both 0° and 90° orientations while keeping the coaxial loop position, and therefore the coupling, unchanged

The support is designed to allow the resonant cell to

be adapted to different cryostats and different directions of the external magnetic fields. For example, in classical cryomagnets, a magnetic field is generated along the cryomagnet axis and, therefore, along the cryogenic insert, usually requiring the sample to be placed perpendicular to the field direction (0° position). In the magnetic configuration used for the experimental validation, where the used non-superconducting low-field magnet provides a field perpendicular to the cryostat axis, the sample should be placed parallel to the latter 0° position. Thus, the angle between the sample surface and the external magnetic field is chosen before cool-down through the mounting configuration.

For what concerns the resonant operation, the design was optimized for TE_{011} at ~ 14.27 GHz and TE_{021} at ~ 25.15 GHz operation.

IV. EXPERIMENTAL VALIDATION AND DISCUSSION

In the validation reported here, a conventional magnet mounted on a rotating base was used in order to vary the field direction with respect to the sample surface while keeping the resonator fixed. This choice allowed the electromagnetic concept to be validated in parallel and perpendicular configurations without introducing additional mechanical complexity.

The validation was performed on a YBCO de-peeled coated conductor sample ($12\text{ mm} \times 12\text{ mm}$) to explore a large Q_l dynamics, achieved thanks to the superconducting transition, and compare the results from the new resonator (denoted DR2) with those from the cryogenic DR already in operation (denoted DR1). The resonator DR1 shares with the new prototype discussed here the sapphire dielectric puck from the same bench and the mask (with hole diameter 6 mm). The metal cavity in DR1 has a diameter 20 mm, which is larger with respect to the 16 mm cavity in DR2, thus determining a different G_s . The DR2, having the PTFE components, is symmetrically gapped with a gap 0.7 mm while DR1 has no gaps.

Here we focus our comparison on the fundamental mode TE_{011} and defer TE_{021} -related analysis to future investigations. For the verification of the main signatures in the superconducting sample with $T_c \sim 91$ K, we compare two type of experimental results: first, Q_l around the superconducting transition and zero magnetic field, and second, magnetic field response for perpendicular magnetic field up to 1.2 T at fixed 78 K for both - DR1 and DR2 - resonators.

A. Transition at zero applied field

Fig. 2 compares the loaded quality factor measured during the superconducting transition in the DR2 and DR1 cells. The DR2 resonator yields a higher Q_l for TE_{011} mode mainly due to a symmetrically gapped structure and corresponding increased sample geometrical factor G_s .

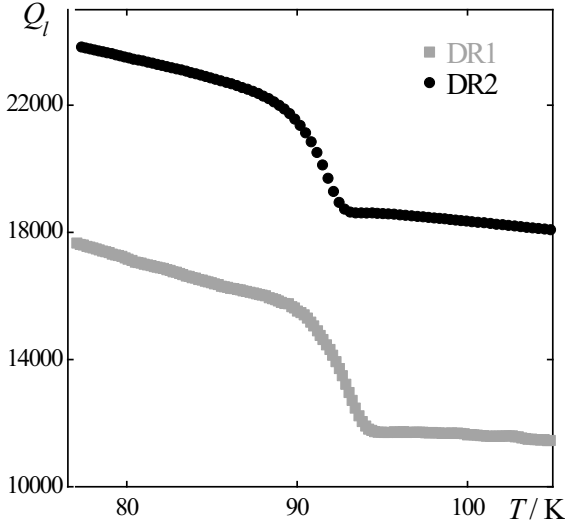


Fig. 2. Comparison of Q_l at the superconducting transition measured on the TE_{011} mode with the new DR2 resonator (frequency 14.27 GHz) and with the non-gapped DR1 resonator (frequency 16.23 GHz). The redesigned cell maintains higher Q_l values across the whole investigated temperature interval.

We obtain more than 30 % increase of Q_l with the DR2 cell throughout the whole superconducting transition. Q_l could be directly extracted from the measured transmission coefficient S_{21} and includes the losses introduced by coupling to the external microwave line (Eq. 1) and by each conductive or dielectric element of the measurement cell configuration. Performing several stress tests on the DR2 cell, we observed excellent superconducting transition repeatability for $Q_l(T)$ between cool-down and warm-up cycles.

The repeatability obtained performing several controlled heating-ups and cooling-downs confirms the usability of the presented DR2 design for Z_s measurements of the superconducting samples and validates the design choices. The repeated transition curves also confirm that DR2 provides a more reliable response than DR1 across different temperatures, magnetic fields, and sample-space environment conditions. This point is central to future optimizations of DR2, as the new cell is intended to serve as the baseline configuration for a resonator with a robust structure and improved measurement repeatability.

B. Magnetic-field sweep

Measurements obtained by performing magnetic field sweeps allows the extraction of the vortex motion parameters of superconducting samples as a function of magnetic field intensity and orientation, and temperature.

An example of field-dependent response at selected tem-

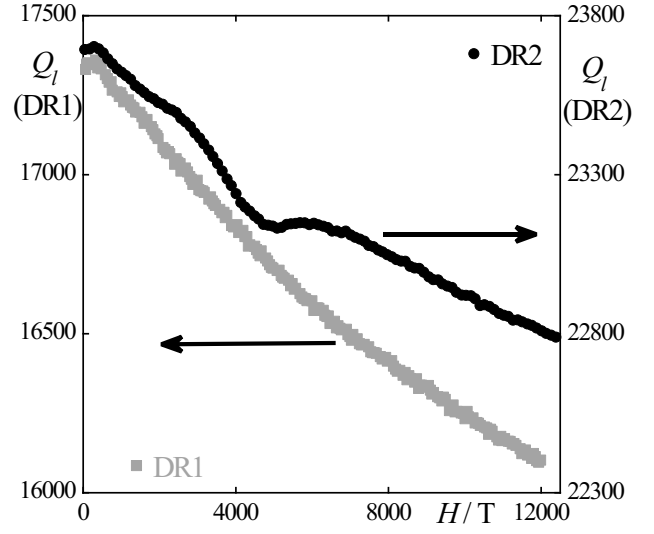


Fig. 3. Q_l as a function of magnetic field for the TE_{011} mode at 78 K with the new DR2 resonator and with the non-gapped DR1 resonator.

perature 78 K, with the external magnetic field applied perpendicularly to the sample surface, is reported in Fig. 3 for both resonators. The decrease of Q_l with the applied magnetic field is a signature of the increased dissipation to the vortex motion in the superconductor. The maximum Q_l corresponds to the zero field.

These values show that the DR2 preserves the expected field dependence while maintaining a higher resonator Q factor across the whole sweep. At the same time, two structures have different cavity diameters, different exposed sample conditions, including different geometrical factors. Thus, the absolute Q_l values cannot be translated directly into surface impedance comparison without a dedicated calibration of each cell. Even these preliminary results show that the new resonator provides a similar measurement observable under otherwise comparable cryogenic conditions.

Another point emerges during this test. Even with the $Q_l(H)$ variation at the same level for both resonators and increased DR2 performance, the paramagnetic-resonance contribution associated with Fe-ion inclusions [15] in the used single-crystal sapphire puck near 0.4 T becomes more pronounced. First of all, this confirms that the redesigned cell makes secondary field-dependent features well visible. Second, this indicates that a further optimization of resonant cell geometry and field distribution is required to minimize the paramagnetic contribution in the measurement.

The result of the validation can therefore be stated with appropriate caution and should be viewed as a first validated step of a measurement platform rather than as its final configuration. The prototype demonstrates that the

cryogenic concept is functional, that the TE_{011} mode is usable for quantitative measurements, and that the new mechanical architecture yields higher Q_l and better repeatability than the earlier configuration. The investigations of the multimodal TE_{011} and TE_{021} operation of DR2 configurations are the subject of future studies.

V. CONCLUSIONS

We presented a prototype of a new compact dielectric-loaded resonator for cryogenic microwave measurements and its experimental validation. The cell was developed primarily for surface-impedance measurements on conducting and superconducting samples using the surface-perturbation technique. The generalized configuration of DR cell allows the same resonant cell to be used for measurements of complex dielectric permittivity at a later stage. Applying best practices developed from previous resonator designs for a wide temperature range operation, the resonant cell for a limited volume space and cryomagnetic environment was designed and experimentally validated. The cryogenic tests demonstrated stable operation of the TE_{011} mode at 14.27 GHz. In the liquid nitrogen temperature range above 73 K, the new resonator provided consistently higher Q_l . At thermalization temperature 78 K, the field sweep preserved the expected dependence and kept an absolute Q_l advantage, thus validating the proposed resonator as a functional starting point for cryogenic surface-impedance measurements in a magnetic field. The future development of this DR is directed toward an improved selectivity of the higher-order modes and extension of the same resonant cell for measurements of the dielectric sample.

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A systematic aliasing suppression architecture for real-time FI-DAC based on linear-phase complementary filter and oversampling

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Abstract – Frequency interleaving digital-to-analog converter (FI-DAC) is a promising solution for synthesizing low-distortion, wideband signals in modern electronic measurement. While traditional offline FI-DAC is effective, advanced testing scenarios increasingly demand Real-Time FI-DAC (RTFI-DAC) architectures. However, these systems inevitably suffer from spectral aliasing caused by the non-ideal transition bands of digital and analog filters during subband decimation and reconstruction. To address this, we propose a systematic architecture combining Linear-Phase Complementary Filters (LPCF) and an oversampling-based guard band method to structurally avoid aliasing. We establish quantitative relationships among the oversampling ratio (OSR), filter characteristics, and local oscillator (LO) frequency, providing a clear design methodology. An equivalent resource-efficient structure for the LPCF is also presented. Simulation results demonstrated a 1.6 times enhancement in bandwidth, with aliasing suppressed by 70 dB at an OSR of 1.25. The trade-off between resources and performance is also discussed. This research holds positive engineering significance for the realization of real-time FI-DAC and the advancement of wideband signal source performance.

Keywords: DAC; Frequency Interleaving; Aliasing suppression; Wideband signal generation.

I. INTRODUCTION

The synthesis of wideband signals is a fundamental challenge in modern electronic measurement, where the bandwidth requirements of emerging testing scenarios, such as advanced communications, have outpaced the capabilities of single digital-to-analog converters (DACs) [1]. Overcoming the bandwidth limitations imposed by semiconductor technology is critical for developing next-generation measurement instruments [2, 3]. Multi-channel parallel architecture is a key strategy for achieving this, with Frequency Interleaving DAC (FI-DAC) emerging as a particularly promising approach. Through partitioning

a wideband signal in the digital domain and recombining the subbands in the analog domain, FI-DAC effectively circumvents the bottleneck, such as zero-order-hold (ZOH) that limits other interleaving concepts. This offers a viable path to wideband signal generation [4, 5, 6].

Nowadays, demands for signal generation capabilities go beyond merely higher output bandwidth, increasingly trending toward real-time and dynamic features. Current FI-DAC research has predominantly focused on offline systems, addressing impairments like spectral aliasing and channel mismatches [7, 8, 9] with computationally intensive, compensation-based techniques such as digital pre-distortion and various complex equalization methods [10, 11, 12, 13]. System-level optimization frameworks have automated the design process to balance complex trade-offs [14, 3]. While these methods have enabled remarkable demonstrations [5, 15, 16], they fundamentally rely on computationally intensive offline processing.

This leaves a research gap for Real-time FI-DAC (RTFI-DAC) systems, which process continuous data streams. The main challenge this paper addresses is the spectral aliasing introduced during subband decimation and analog subband reconstruction due to the non-ideal characteristics of filters. To solve this, we propose a systematic aliasing suppression architecture for a two-channel RTFI-DAC. Our approach structurally avoids the aliasing by combining an oversampling-based guard band with a linear-phase complementary filter (LPCF) bank for perfect subband decomposition. We derive the necessary constraints and analyze trade-offs among the oversampling ratio (OSR), filter transition bandwidth, and system parameters, including an equivalent architecture to reduce resource overhead. Simulation results verify the architecture, confirming its potential for high-performance, real-time wideband signal generation.

II. PRINCIPLE OF SYSTEM

The FI-DAC is a hybrid multirate signal generation system that utilizes multiple DACs in parallel to synthesize an analog signal from a digital data stream. Theoretically, the output bandwidth of an M-channel FI-DAC system is M

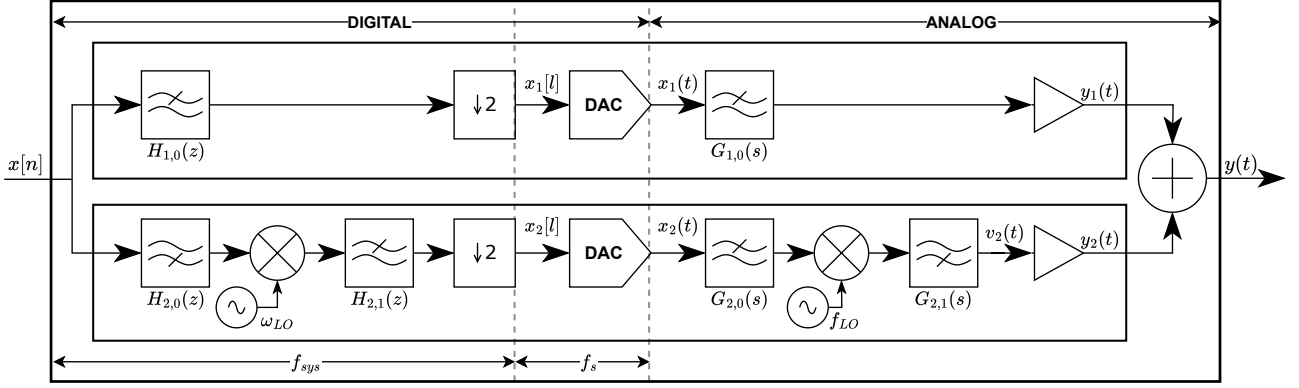


Fig. 1. Block diagram of the proposed two-channel RTFI-DAC architecture

times that of a single DAC.

The most critical digital signal processing stage in an FI-DAC system is subband decomposition. It can be implemented using two main methods. One is based on FFT for subband decomposition in the frequency domain, which is suited for offline, non-real-time FI-DAC system. For the RTFI-DAC system that needs to process continuous, non-predefined data streams, the FFT-based decomposition method is not applicable due to problems such as its inherent latency and the spectral leakage introduced by non-integer period truncation. Therefore, RTFI-DAC system is well-suited for using digital filter banks for real-time subband decomposition.

Fig. 1 shows a dual-channel RTFI-DAC system where the digital signal, at a sampling rate of f_{sys} , is decomposed into low-frequency and high-frequency subband signals by the subband decomposition filters $H_{1,0}(z)$ and $H_{2,0}(z)$. The high-frequency subband is shifted to baseband via digital down-conversion. Next, both subband signals are decimated by a factor of 2, reducing the sampling rate to the DAC's sampling rate f_s , and are then converted into analog signals by their respective DACs. In the digital domain, the output of the m -th channel after subband filtering and decimation, $X_m(e^{j\omega_s})$, can be expressed as:

$$X_m(e^{j\omega_s}) = \frac{1}{M} \sum_{k=0}^{M-1} X \left(e^{j \left(\frac{\omega_s - 2\pi k}{M} + \omega_{LO,m} \right)} \right) H_{m,0} \left(e^{j \left(\frac{\omega_s - 2\pi k}{M} + \omega_{LO,m} \right)} \right) H_{m,1} \left(e^{j \left(\frac{\omega_s - 2\pi k}{M} \right)} \right). \quad (1)$$

This equation holds for all $m = 1, \dots, M$ where $\omega_{LO,1} = 0$ and $H_{1,1} = 1$.

In the analog domain, the subband, having been digitally down-converted, is returned to its original frequency band by analog up-conversion. Finally, the two signals are combined using a combiner, achieving a signal output with

twice the bandwidth of a single DAC.

$$Y_m(j\Omega) = \frac{A_m}{2} \left[X_m \left(e^{j(\Omega - \Omega_{LO,m})T_s} \right) H_{zoh}(j(\Omega - \Omega_{LO,m})) G_{m,0}(j(\Omega - \Omega_{LO,m})) + X_m \left(e^{j(\Omega + \Omega_{LO,m})T_s} \right) H_{zoh}(j(\Omega + \Omega_{LO,m})) G_{m,0}(j(\Omega + \Omega_{LO,m})) \right] G_{m,1}(j\Omega). \quad (2)$$

This equation holds for all $m = 1, \dots, M$ under ideal conditions ($\Omega_{LO,1} = 0$, $G_{1,1} = 1$, $A_1 = 1$, $A_{m \geq 2} = 2$).

$$Y(j\Omega) = \sum_{m=1}^M Y_m(j\Omega) \quad (3)$$

III. PROPOSED METHOD

The non-ideal transition bands of the filters introduce spectral aliasing from two primary sources: Besides the inter-band aliasing resulting from the image components of the analog-domain DAC and mixer [12, 14], for example, as shown in Fig. 2(c), the output images of the DAC cannot be fully suppressed; RTFI-DAC that utilize subband decomposition filters encounter intra-band aliasing within the digital domain. Fig. 2(a) shows the stopband edge frequency of the filter is greater than $\pi/2$. The components in the transition band will cause digital spectral aliasing in the subband signal during critical decimation.

Prior to handling aliasing distortion, it is necessary to clarify the subband decomposition approach.

Assuming distortion-free subband synthesis in the analog domain, as shown in Fig. 3, there can be three methods—A, B, and C—depending on the overlap of the filter transition bands. In method A, the transition bands of the two adjacent subband decomposition filters do not overlap, which results in a loss of signal information after decomposition. In method B, the passband edge frequen-

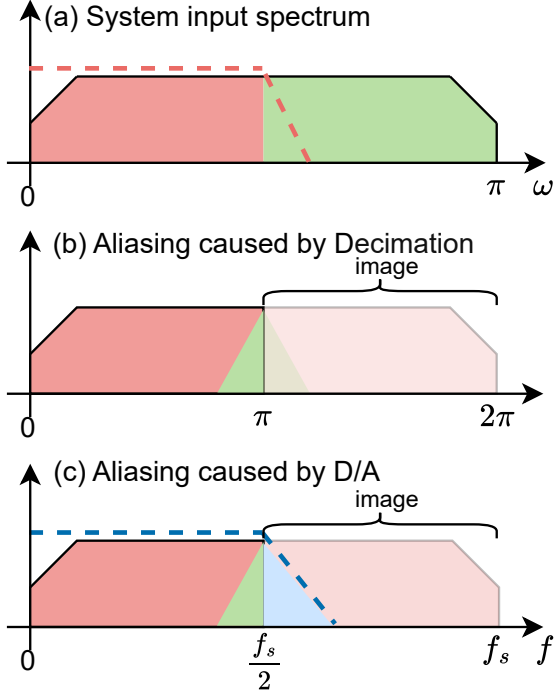


Fig. 2. Sources of aliasing in an FI-DAC system. (a) System input spectrum. (b) Digital spectrum aliasing caused by the filter's transition band during decimation. (c) Aliasing resulting from imperfect suppression of DAC images by a non-ideal analog filter.

cies of the two filters are equal, leading to distortion in the amplitude-frequency characteristics of the signal during subband synthesis. In contrast, method C, in which the transition bands perfectly overlap and are complementary, can avoid the problems.

A. LPCF for Subband Decomposition

A linear phase (constant group delay) is a necessary condition for distortion-free signal transmission. Therefore, Linear-Phase Complementary Filters (LPCF) offer notable advantages for the requirements of distortion-free subband decomposition. In a dual-channel RTFI-DAC, we define the subband decomposition filters $H_{1,0}$ and $H_{2,0}$ to satisfy:

$$H_{1,0}(e^{j\omega}) + H_{2,0}(e^{j\omega}) = e^{-j\omega D}. \quad (4)$$

Equation 4 shows that the sum of the frequency responses of the two filters is a pure delay, where $D = N/2$, and N is the filter order. More specifically, a Type I linear-phase FIR filter [17] that satisfies Equation 4 can meet the requirements. Let the filter length be L ; L must be an odd number to ensure an integer-sample group delay.

We use the Parks-McClellan algorithm to design the low-pass filter $h_{1,0}[n]$. High-pass $h_{2,0}[n]$ can then be determined using the constraint relationship between their

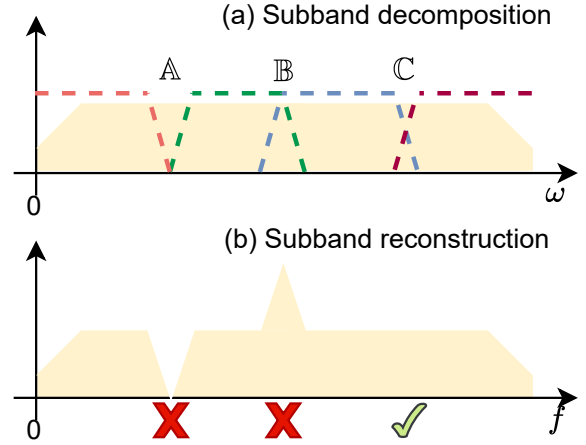


Fig. 3. Comparison of subband decomposition methods. (a) Non-overlapping filters resulting in signal loss. (b) Overlapping passbands causing magnitude distortion. (c) Ideal decomposition using complementary filters.

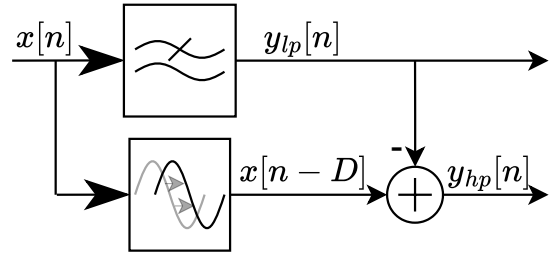


Fig. 4. Proposed resource-efficient equivalent architecture

impulse responses:

$$h_{2,0}[n] = \delta[n - D] - h_{1,0}[n]. \quad (5)$$

A.1 Equivalent architecture

While the HPF impulse response $h_{2,0}[n]$ can be explicitly calculated and implemented as a separate filter, a more efficient realization arises from (5). The output $y_{2,0}[n]$, when filtering an input $x[n]$, can be expressed as:

$$\begin{aligned} y_{2,0}[n] &= x[n] * h_{2,0}[n] \\ &= x[n] * (\delta[n - D] - h_{1,0}[n]) \end{aligned}$$

Recognizing that convolution with a delayed impulse $\delta[n - D]$ simply delays the signal, and that $x[n] * h_{1,0}[n]$ is the output of the LPF, $y_{1,0}[n]$, we arrive at:

$$y_{2,0}[n] = x[n - D] - y_{1,0}[n]. \quad (6)$$

The delay D for the input signal $x[n]$ ensures proper phase alignment with the LPF output $y_{1,0}[n]$ before subtraction, which is critical for accurately realizing the complementary characteristic. This structure is illustrated in

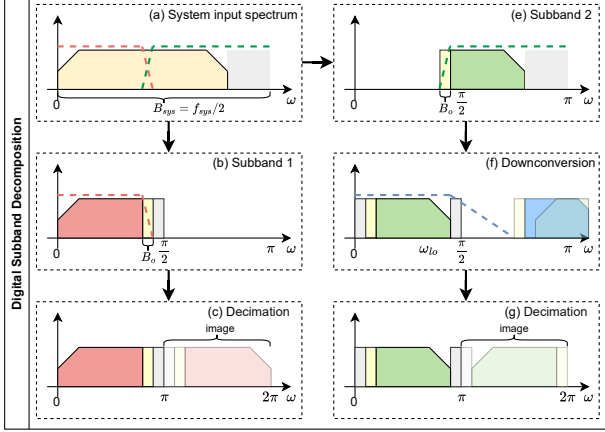


Fig. 5. Digital domain signal processing for the proposed method.

Fig. 4. This method can be further generalized to systems with $M = 2^J$ subbands, where J is a positive integer.

B. Oversampling-based guard band method

To suppress aliasing in both digital and analog domains, we introduce a systematic guard band method enabled by oversampling. This is achieved by creating three distinct spectral guard bands.

Let the equivalent sampling rate of the system be f_{sys} , so the system bandwidth is $B_{sys} = f_{sys}/2$. To mitigate aliasing caused by the non-ideal characteristics of the subband decomposition filter, a guard band G_o is set, with its width B_o equal to the transition bandwidth of filter. To mitigate aliasing caused by the transition bands of the DAC's anti-imaging filter and the mixer's output sideband selection filter, guard bands G^U and G^L are set, respectively. Reference [12] discusses the parameter selection for the analog domain transition bands, where their widths B_g^U and B_g^L are half that of the respective transition bands.

The oversampling method we propose is illustrated in Fig. 5 and 6. Fig. 5 shows the digital domain part of the oversampling method. The guard band G_o is established by confining the transition band of the subband 1 decomposition filter within the first Nyquist zone. As seen in Fig. 5(c), due to the presence of this guard band, the decimated subband signal no longer experiences aliasing. The space between $\pi/2$ and the filter edge provides room for the guard band G^U . The guard band G^L is implemented by controlling the local oscillator (LO) frequency, ω_{LO} , of the digital down-conversion. Compared to a system without a guard band, ω_{LO} is reduced so that subband 2 is no longer shifted down to the DC frequency. This reserved bandwidth constitutes G^L .

Fig. 6 illustrates the analog domain of the system. During subband synthesis, aliasing is well suppressed because

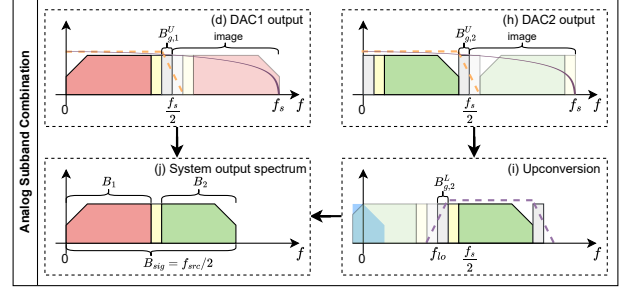


Fig. 6. Analog domain signal synthesis for the proposed architecture.

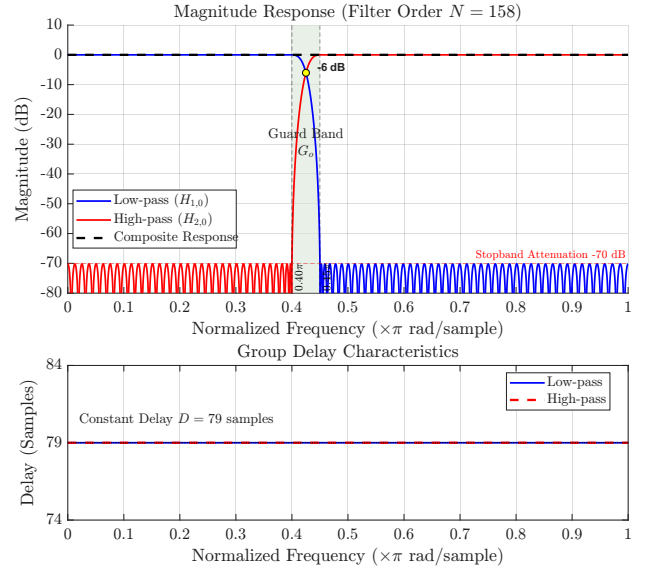


Fig. 7. Magnitude response and group delay of the designed FIR filters used for subband decomposition.

of the established guard bands. Neither the DAC images nor the unwanted sidebands from mixing affect the adjacent channel.

We can calculate the total guard band bandwidth used for alias suppression as:

$$B_{guard} = 2B_g^U + B_o + B_g^L. \quad (7)$$

The effective bandwidth of the system is:

$$B_{sig} = B_1 + B_o + B_2. \quad (8)$$

Finally, we can obtain the OSR of the system:

$$OSR = \frac{B_{guard} + B_{sig}}{B_{sig}}. \quad (9)$$

IV. RESULTS AND DISCUSSION

To validate the efficacy of the proposed methodology, a dual-channel FI-DAC system operating at MHz frequencies was constructed through simulation.

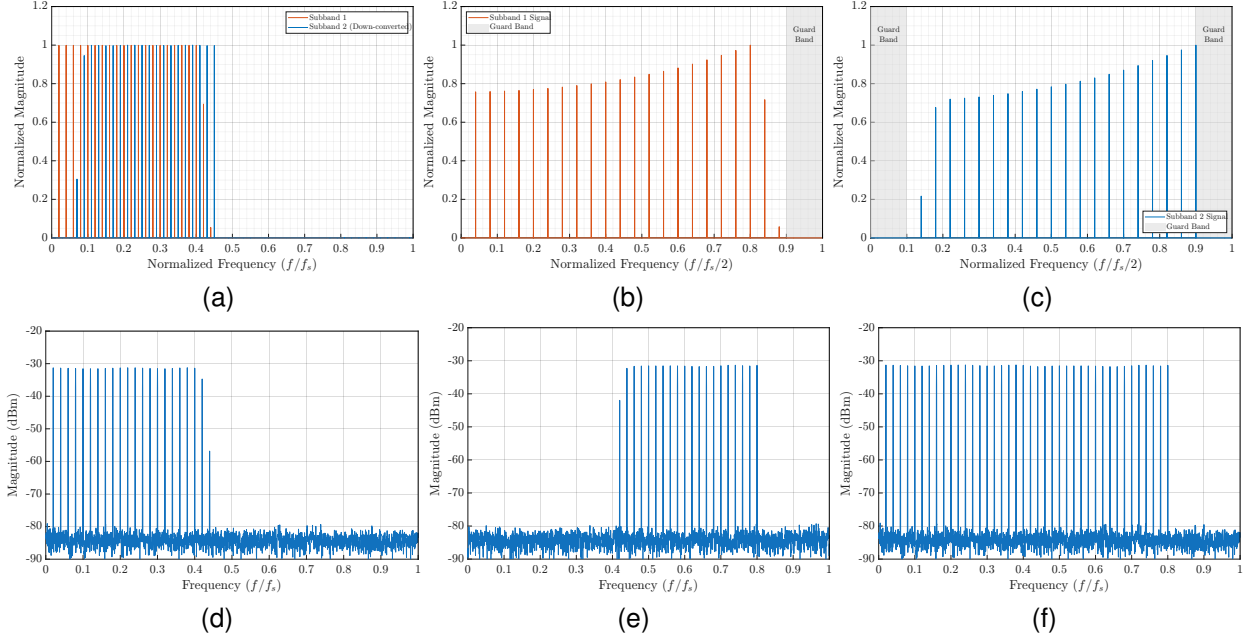


Fig. 8. System simulation results. (a) Subband signal after subband decomposition and down-conversion. (b) Spectrum of the signal input to DAC1. (c) Spectrum of the signal input to DAC2. (d) Subband 1 output. (e) Subband 2 output. (f) Combined spectrum at the system output.

The DAC sampling rate, f_s , was configured to 200 MHz with a vertical resolution of 16 bits. A 9th-order low-pass elliptic filter, with a passband cutoff frequency of 90 MHz, was employed as the analog anti-imaging filter. The side-band selection filter was implemented as a 9th-order band-pass elliptic filter, with a passband from 80 MHz to 160 MHz. As depicted in Fig. 7, the designed linear-phase complementary filter has an order of 158. Its passband and stopband cutoff frequencies are 0.4π and 0.45π , respectively, with a stopband attenuation of 70 dB.

Based on the analysis in Section III, the system parameters B_g^L , B_g^U , and B_o were determined to be 10 MHz, 10 MHz, and 0.05π (10 MHz at $f_{sys} = 400$ MHz), respectively. The LO frequency was set to 70 MHz. Consequently, the effective bandwidth of the system is 160 MHz.

A multi-tone signal was generated according to this system configuration, with frequency intervals of $0.02f_s$, comprising 40 single tones within the range $[DC, 0.8f_s]$. The simulation results are presented in Fig. 8. Fig. 8(a) illustrates the outcome of subband decomposition and down-conversion. Fig. 8(b) and 8(c) display the frequency spectra of the decimated subband signals input to DAC1 and DAC2, respectively. The spectra indicate that guard bands are reserved in the high-frequency regions of both Subband 1 and Subband 2, as well as in the low-frequency region of Subband 2. Both subbands have been processed by an inverse-sinc filter. Fig. 8(d) and 8(e) show the outputs of Subband 1 and Subband 2 from the FI-DAC system,

respectively, demonstrating complementarity in their overlapping region and no mutual crosstalk. Fig. 8(f) presents the final combined signal output spectrum.

The results demonstrate that an aliasing suppression of 70 dB is achieved with an OSR of 1.25. If the filters within the system could realize steeper transition bands, the OSR could approach unity more closely, thereby reducing the sacrificed effective bandwidth. Under this methodology, a trade-off exists between the filter transition bandwidth and the system's effective bandwidth, necessitating careful parameter balancing in practical system design.

The core of this method is this trade-off: moderately sacrificing effective bandwidth in exchange for optimized aliasing suppression. By establishing a quantitative relationship among the guard band width, filter transition bandwidth, LO frequency, analog filter parameters, and the OSR, the method ensures that for a given aliasing suppression target, the OSR is minimized while the effective bandwidth is maximized. Compared to compensation-based methods that rely on precise system models, this approach offers lower design complexity and greater robustness.

V. CONCLUSION

This paper proposes a systematic aliasing suppression method for the RTFI-DAC system by combining LPCF-based subband decomposition with an oversampling guard-band strategy. The method provides clear relationships among OSR, transition bandwidth, and LO fre-

quency. Simulation results demonstrated the effectiveness of the proposed architecture, achieving a 1.6-fold increase in bandwidth with 70 dB of aliasing suppression at an OSR of 1.25. This research contributes to the practical realization of real-time FI-DAC and the advancement of wide-band signal generation capabilities, offering substantial engineering value.

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Parameter estimation of complex-valued sinewaves by a frequency-domain linearized sine-fit algorithm

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Abstract – In this paper a frequency-domain linearized three-parameter sine-fit (FL3PSF) algorithm is proposed for the estimation of the frequency, amplitude, and phase of a complex-valued noisy sinewave. The algorithm linearly interpolates a small number of Discrete Fourier Transform (DFT) samples exploiting the Gauss-Newton approach. The analytical expressions for the proposed parameter estimators are provided and their accuracies are compared with those of some Discrete-Time Fourier Transform (DTFT)-based algorithms through computer simulations.

Keywords: discrete Fourier transform; parameter estimation; sine-fit algorithm; statistical analysis.

I. INTRODUCTION

Real-time accurate sinewave parameter estimates are often required in many application domains such as radar, communications, power systems, and instrumentation. To that aim Discrete Fourier Transform (DFT)-based algorithms are widely employed. When complex-valued sinewaves are of concern and rectangular windowing is adopted, DFT-based parameter estimates are affected by multiple uncertainty sources: the leakage effect due to the finite acquisition length [1], [2], the picket-fence effect due to the finite number of processed samples [2], the possible algorithm intrinsic approximation [3], and wideband noise [3]. The leakage effect can be limited by windowing [1] while the picket-fence effect can be strongly reduced by applying interpolated Discrete-Time Fourier Transform (DTFT) algorithms [4]-[9]. The algorithm intrinsic approximation can be compensated through a proper derivation of the parameter estimators [10]. Noise contribution can be minimized by properly selecting the DTFT interpolation points through an iterative procedure [8].

A time-domain linearized sine-fit algorithm for parameter estimation of real-valued noisy sinewave affected by harmonics has been proposed in [11]. Based on a similar idea, in this paper a simple frequency-domain linearized three-parameter sine-fit (FL3PSF) algorithm is proposed for the estimation of complex-valued noisy sinewave parameters. Rectangular data windowing is considered, and linearization is performed by applying the Newton-Gauss approach [12]. The proposed algorithm exploits samples of the signal DFT symmetrically located around the spectrum peak location. The algorithm accuracy is analysed as a function of both the number of the interpolated DFT samples and the number of the analysed signal samples. Its performance is compared with those of some state-of-the-art Interpolated DTFT (IpDTFT) algorithms through computer simulations.

The remaining of the paper is organized as follows. In Section II the FL3PSF algorithm is presented. Its accuracy is analysed and compared with those of other IpDTFT algorithms in Section III. The main conclusions of this work are mentioned in Section IV.

II. THE FL3PSF ALGORITHM

Let's consider a discrete-time complex-valued noisy sinewave, modelled as:

$$y(m) = x(m) + \zeta(m) = Ae^{j(2\pi \frac{\nu}{M}(m - \frac{M-1}{2}) + \phi)} + \psi(m), \quad m = 0, 1, 2, \dots, M - 1 \quad (1)$$

where $x(\cdot)$ represents a pure complex-valued sinewave of amplitude A , normalized frequency ν (expressed in bins), and phase ϕ , while $\psi(\cdot)$ is a Gaussian noise with zero mean and variance σ^2 . M is the number of the analysed samples and the Signal-to-Noise Ratio of (1) is $\text{SNR} \triangleq \frac{A^2}{\sigma^2}$.

The normalized frequency ν represents the number of acquired signal cycles, composed by an integer part l and a fractional part δ ($-0.5 \leq \delta < 0.5$):

$$\nu = l + \delta. \quad (2)$$

If SNR is higher than the breakdown threshold [13], the integer part l of the number of analysed cycles is exactly determined with very high probability as the maximum of the signal discrete spectrum. Thus, the frequency estimation error coincides with the error of the fractional frequency δ .

For convenience, the following notation is adopted in the following: $Y_k \triangleq Y(l+k)$, $X_k \triangleq X(l+k)$, and $\Psi_k \triangleq \Psi(l+k)$, where $Y(\cdot)$, $X(\cdot)$, and $\Psi(\cdot)$ are the DFTs of the signals $y(\cdot)$, $x(\cdot)$, and $\zeta(\cdot)$, respectively, and $|k| \leq K < M/2 - l$. We have:

$$Y_k \triangleq \sum_{m=0}^{M-1} y(m) e^{-j2\pi \frac{l+k}{M} (m - \frac{M-1}{2})} = X_k + Z_k \quad (3)$$

$$= AW_k(\delta) e^{j\phi} + \Psi_k, |k| \leq K$$

where $X_k(\cdot)$ is given by:

$$X_k \triangleq \sum_{m=0}^{M-1} x(m) e^{-j2\pi \frac{l+k}{M} (m - \frac{M-1}{2})} = AW_k(\delta) e^{j\phi}, \quad (4)$$

in which $W(\cdot)$ is the DTFT of the rectangular window:

$$W_k(\delta) \triangleq W(k - \delta) \quad (5)$$

$$\triangleq \sum_{m=0}^{M-1} e^{-j2\pi \frac{k-\delta}{M} (m - \frac{M-1}{2})} = \frac{\sin(\pi(k - \delta))}{\sin(\frac{\pi(k - \delta)}{M})}$$

From (4) and (5) it follows that X_k is a function of the unknown parameters δ , A , and ϕ , that is $X_k = X_k(\delta, A, \phi)$.

Let's denote with $\hat{\delta}$, $\hat{A}$, and $\hat{\phi}$ the initial estimates of the related parameters. By applying the Taylor's series to $X_k(\hat{\delta}, \hat{A}, \hat{\phi})$ centred around $\hat{\delta}$ and truncating the expansion to the first order term, since $A \cong \hat{A}$ and $\phi \cong \hat{\phi}$, from (3) we obtain:

$$Y_k \cong \hat{A} W_k(\hat{\delta}) e^{j\hat{\phi}} + \hat{A} W'_k(\hat{\delta}) e^{j\hat{\phi}} \Delta\delta + \Psi_k, \quad (6)$$

$$|k| \leq K$$

where $\Delta\delta \triangleq \hat{\delta} - \delta$ and

$$W'_k(\hat{\delta}) \triangleq \left. \frac{dW_k(\delta)}{d\delta} \right|_{\delta=\hat{\delta}} = \quad (7)$$

$$\frac{-\pi \cos(\pi(k-\hat{\delta})) \sin(\frac{\pi(k-\hat{\delta})}{M}) + \frac{\pi}{M} \sin(\pi(k-\hat{\delta})) \cos(\frac{\pi(k-\hat{\delta})}{M})}{\sin^2(\frac{\pi(k-\hat{\delta})}{M})}.$$

By defining:

$$\varepsilon_k \triangleq Y_k - \hat{A} W_k(\hat{\delta}) e^{j\hat{\phi}}, \quad |k| \leq K \quad (8)$$

and applying the least-squares approach to (6), the following expression for $\Delta\delta$ results:

$$\Delta\delta \cong \text{Re} \left\{ \frac{\sum_{k=-K}^K W'_k(\hat{\delta}) \cdot \varepsilon_k}{\hat{A} e^{j\hat{\phi}} \sum_{k=-K}^K [W'_k(\hat{\delta})]^2} \right\}, \quad (9)$$

where $\text{Re}\{\cdot\}$ is the real part of the complex-valued argument.

By expanding $X_k(\hat{\delta}, \hat{A}, \hat{\phi})$ through the Taylor's series centred around $\hat{A}$ and truncating to the first order term, since $\delta \cong \hat{\delta}$ and $\phi \cong \hat{\phi}$ we obtain:

$$Y_k \cong \hat{A} W_k(\hat{\delta}) e^{j\hat{\phi}} + W_k(\hat{\delta}) e^{j\hat{\phi}} \Delta A + \Psi_k, \quad (10)$$

$$|k| \leq K$$

from which it follows:

$$\varepsilon_k \cong W_k(\hat{\delta}) e^{j\hat{\phi}} \Delta A + \Psi_k, \quad |k| \leq K \quad (11)$$

where $\Delta A \triangleq \hat{A} - A$.

By applying the least-squares approach to (10) the following expression for ΔA results:

$$\Delta A \cong \text{Re} \left\{ \frac{\sum_{k=-K}^K W_k(\hat{\delta}) \cdot \varepsilon_k}{e^{j\hat{\phi}} \sum_{k=-K}^K [W_k(\hat{\delta})]^2} \right\}. \quad (12)$$

By expanding $X_k(\hat{\delta}, \hat{A}, \hat{\phi})$ through the Taylor's series centred around $\hat{\phi}$ and truncating to the first order term, since $A \cong \hat{A}$ and $\delta \cong \hat{\delta}$ we have:

$$Y_k \cong \hat{A} W_k(\hat{\delta}) e^{j\hat{\phi}} + j \hat{A} W_k(\hat{\delta}) e^{j\hat{\phi}} \Delta\phi + \Psi_k, \quad (13)$$

$$|k| \leq K$$

from which it follows:

$$\varepsilon_k \cong j \hat{A} W_k(\hat{\delta}) e^{j\hat{\phi}} \Delta\phi + \Psi_k, \quad |k| \leq K \quad (14)$$

where $\Delta\phi \triangleq \hat{\phi} - \phi$.

From (13), the least-squares approach provides:

$$\Delta\phi \cong \text{Re} \left\{ -j \frac{\sum_{k=-K}^K W_k(\hat{\delta}) \cdot \varepsilon_k}{\hat{A} e^{j\hat{\phi}} \sum_{k=-K}^K [W_k(\hat{\delta})]^2} \right\}. \quad (15)$$

Expressions (9), (12), and (15) enable the proposal of the following FL3PSF algorithm:

step 1: acquire M samples of complex-valued signal $y(\cdot)$;
step 2: determine the integer part l of the number of analysed signal cycles through a peak search procedure applied to the DFT signal spectrum;
step 3: determine the initial parameter estimates by applying the IpDFT algorithm based on the rectangular window [5], [7]:

$$\hat{\delta} = \begin{cases} -\frac{|X_{-1}|}{|X_{-1}| + |X_0|}, & \text{if } |X_{-1}| \geq |X_1| \\ \frac{|X_1|}{|X_1| + |X_0|}, & \text{if } |X_{-1}| < |X_1| \end{cases}$$

$$\hat{A} = \frac{|X_0|}{|W_0(\hat{\delta})|}, \quad \hat{\phi} = \arg\{X_0\}.$$

step 4: compute the initial estimation errors $\Delta\delta$, ΔA , and $\Delta\phi$ by (9), (12), and (15);
step 5: estimate the unknown parameters as $\tilde{\nu} = l + \hat{\delta} + \Delta\delta$, $\tilde{A} = \hat{A} + \Delta A$, $\tilde{\phi} = \hat{\phi} + \Delta\phi$.

The proposed FL3PSF algorithm requires two phases. In the first one the initial parameter estimates are obtained by applying the IpDFT algorithm. The second phase updates the initial estimates exploiting the expressions derived for the parameter estimation errors (8), (9), (12), and (15). Observe that the DFT samples used in (8) are computed in step 2. Moreover, the required values of the window transforms and its derivative can be easily computed using (5) and (7). Thus, the processing time required by the FL3PSF algorithm is dominated by the calculation of the DFT samples through the FFT algorithm, which is of the order $O(M\log_2 M)$.

III. ACCURACY EVALUATION OF THE FL3PSF ALGORITHM

The FL3PSF algorithm requires the computation of $2K + 1$ DFT samples symmetrically located around the location l of the spectrum peak. In the following the algorithm accuracy is analysed as a function of both K and the number of signal samples M . Moreover, it is compared through simulations with those of the classical two-point IpDFT algorithm [5], [7], the AM algorithm [8], and the Candan's algorithm [9]. The signal amplitude is $A = 1$ and $l = 7$ cycles are observed. Only positive fractional frequency δ values are considered due to the symmetry of the estimates around l . For each variable parameter the signal phase ϕ takes random values in the range $[0, 2\pi)$ rad, and 10,000 runs of M samples each are performed. Accuracy is quantified by the estimator Root Mean Square Error (RMSE) or the Normalized RMSE (NRMSE), defined as the ratio between the RMSE and the square root of the related Cramér-Rao Lower Bound (CRLB) for unbiased estimators, which are given by the following expressions:

$$(\sigma_{\delta}^2)_{CR} = \frac{3}{2\pi^2 M \cdot \text{SNR}}, \quad (\sigma_A^2)_{CR} = \frac{\sigma^2}{2M}, \quad \text{and} \quad (\sigma_{\phi}^2)_{CR} = \frac{1}{2M \cdot \text{SNR}} \quad [12].$$

A. Accuracy analysis

Fig. 1 shows the NRMSE of the frequency estimator $\tilde{\nu}$ provided by the FL3PSF algorithm as a function of the fractional frequency δ which varies in the range $[0.02, 0.5)$ with a step of 0.02 cycles. $K = 3, 4, 5$, and 6 DFT samples, $M = 128$ signal samples, and $\text{SNR} = 40$ dB.

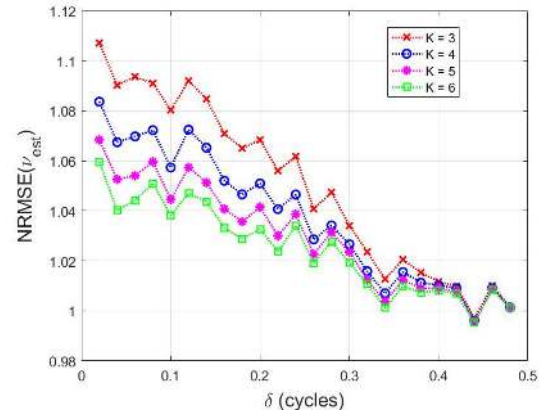


Fig. 1. NRMSEs of the frequency estimator provided by the FL3PSF algorithm versus δ . $K = 3, 4, 5, 6$; $l = 7$ cycles; $\text{SNR} = 40$ dB. 10,000 runs of $M = 128$ signal samples for each value of δ . Phase ϕ selected at random.

As shown in Fig. 1, the NRMSE of the estimator $\tilde{\nu}$ improves as K increases. Moreover, the effect of K is more significant for small values of δ . No significant improvement is obtained when K is greater than 5 (i.e., when more than 11 DFT samples are interpolated). Behaviour very similar to those shown in Fig. 1 has been observed for the amplitude and the phase estimators.

Fig. 2 shows the NRMSEs of the frequency estimator $\tilde{\nu}$ as a function of δ when $M = 8, 64$, and 512 samples, $K = 5$ and $\text{SNR} = 40$ dB.

As it can be seen, the normalized accuracy of the FL3PSF algorithm is almost independent of the number of analysed samples M . Further simulations showed that the same conclusion holds also for the FL3PSF amplitude and phase estimators, for which RMSE values very close to the related $\sqrt{\text{CRLB}}$ are obtained for all values of δ .

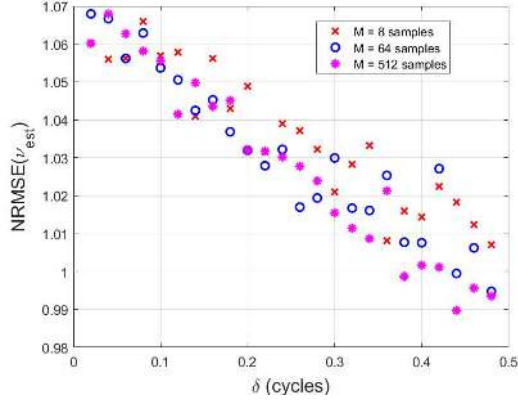


Fig. 2. NRMSEs of the frequency estimators provided by the FL3PSF algorithm versus δ $M = 8, 64, 512$ signal samples; $K = 5$; $l = 7$ cycles; $\text{SNR} = 40$ dB. 10,000 runs of M samples for each value of δ . Phase ϕ selected at random.

B. Accuracy comparison

Fig. 3 shows the behaviour of different estimator NRMSEs as a function of δ . Fig. 3(a) deals with the frequency estimators returned by the two-point IpDFT, the AM, the Candan's, and the FL3PSF algorithms, while Fig. 3(b), (c) relates to the amplitude and the phase estimates provided by the two-point IpDFT and the FL3PSF algorithms. The simulation conditions are the same as in Fig. 2, except that $M = 128$ samples. Complex-valued DFT samples are considered in the AM algorithm and two iterations are performed [8]. Fig. 3(a) shows that the frequency FL3PSF estimates outperforms the two-point IpDFT and the Candan's estimates, ensuring almost statistical efficiency as the AM algorithm. Figs. 3(b) and (c) show that the FL3PSF amplitude and phase estimators are almost efficient and largely outperforms the two-point IpDFT algorithm.

Fig. 4 shows the RMSEs of the considered estimators and the corresponding $\sqrt{\text{CRLB}}$ as a function of SNR, which varies in the range $[-20, 90]$ dB with a step of 5 dB. Fig. 4(a) shows the results related to frequency estimates returned by the two-point IpDFT, the AM, the Candan's, and the FL3PSF algorithms. Fig. 4 (b), (c) report the RMSEs for the amplitude and the phase estimates returned by the two-point IpDFT and the FL3PSF algorithms. The simulation conditions are the same as in Fig. 3, except that $\nu = 7.25$ cycles. In addition, in the insets the NRMSEs are shown as a function of the SNR. Observe that, when the SNR is above the breakdown threshold (equal to about -10 dB [13]), only the AM and the FL3PSF algorithms ensure frequency estimate RMSEs close to the related $\sqrt{\text{CRLB}}$. When the amplitude and the phase estimates are of concern, the FL3PSF algorithm outperforms the IpDFT algorithm and it provides almost efficient estimates for SNRs above the breakdown thresholds.

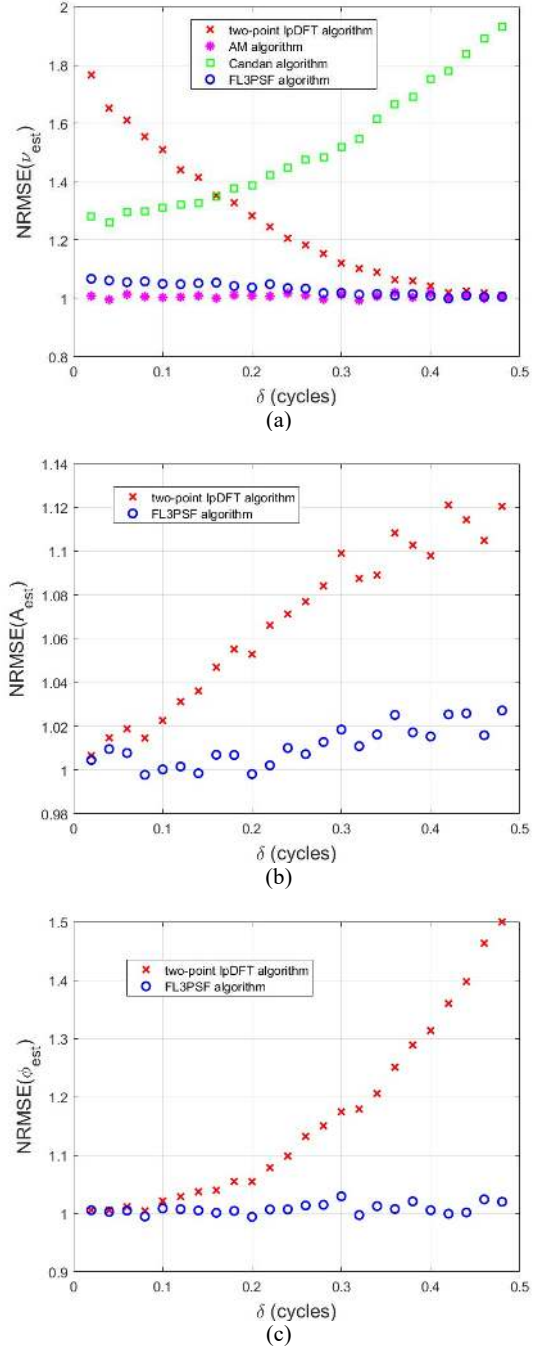


Fig. 3. NRMSEs of considered estimators versus δ . (a) frequency estimators provided by the two-point IpDFT, the AM, the Candan, and the FL3PSF algorithms; (b), (c) amplitude and phase estimators provided by the two-point IpDFT and the FL3PSF algorithms. $M = 128$ signal samples, $l = 7$ cycles, and $\text{SNR} = 40$ dB. 10,000 runs of M samples for each δ . Phase ϕ selected at random.

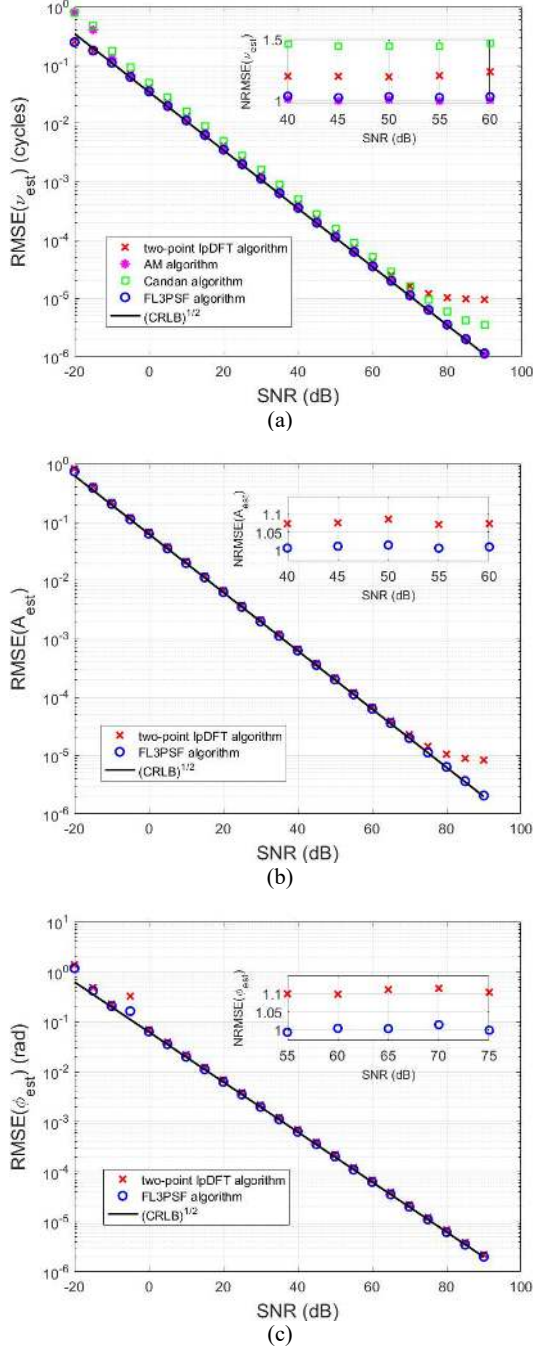


Fig. 4. RMSEs of the considered estimators and related CRLBs versus SNR. (a) frequency estimators provided by the two-point IpDFT, the AM, the Candan, and the FL3PSF algorithms; (b), (c) amplitude and phase estimators provided by the two-point IpDFT and the FL3PSF algorithms. $M = 128$ signal samples and $\nu = 7.25$ cycles. The insets show the related NRMSEs.

IV. CONCLUSIONS

In this paper the FL3PSF algorithm for the estimation of complex-valued sinewave parameters is proposed. Simulation results show that it outperforms the two-point IpDFT and the Candan algorithms, while its accuracy is very close to that ensured by the AM algorithm. Moreover, all the FL3PSF estimate MSEs are very close to the related CRLBs. In addition, the proposed algorithm is very simple to be implemented and its computational burden is very low.

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Benchmarks for IIR Model Identification Based on Binary Measurements

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Abstract – This paper addresses the identification of IIR systems from binary measurements, a challenging problem due to the need to jointly reconstruct the continuous output and the model parameters, leading to a nonlinear optimization problem. An incremental identification algorithm is considered, where estimates obtained from the first k samples are used to initialize the update at $k + 1$, enabling progressive refinement of the estimates. Within this framework, the role of the thresholding policy is investigated. Three strategies are analyzed: fixed, continuous, and quantized thresholds. The performance is evaluated on six benchmark IIR systems. Results show that a 3-bit quantized threshold provides the most accurate and reliable estimates, achieving reconstruction errors of about 1% and parameter convergence close to the true values, outperforming a state-of-the-art method.

Keywords: Binary measurement, System identification.

I. INTRODUCTION

Measurement systems are undergoing a progressive transformation driven by the increasing demand for scalability, robustness, and efficiency. In many modern applications, ranging from embedded platforms to large-scale distributed systems, reducing power consumption has become a critical requirement. This trend is particularly evident in data acquisition architectures, where the use of high-resolution sensing is increasingly challenged by constraints on cost, energy, and reliability.

Within this context, *low-resolution measurement strategies*, and in the extreme case *binary sensing*, are emerging as a viable alternative to traditional approaches [1]. Binary measurements, typically obtained through simple thresholding devices such as comparators, drastically simplify the acquisition chain while enabling high sampling rates and robust operation. Despite the severe information loss induced by quantization, recent studies have shown that meaningful estimation and reconstruction tasks can still be performed when appropriate algorithms are employed [2, 3]. These considerations have stimulated significant interest in the problem of system identification from quantized and binary data. While this problem is relatively well understood for systems with a finite impulse response (FIR) [4], where the absence of feedback

preserves a linear regression structure, the identification of systems with internal dynamics remains considerably more challenging. In particular, for infinite impulse response (IIR) models such as ARX or ARMA structures, the presence of feedback introduces unobserved variables and nonlinear dependencies between measured data and model parameters [5]. As a result, the identification task becomes a nonlinear estimation problem involving latent variables. Several methodologies have been proposed to address this challenge, including maximum-likelihood approaches, Bayesian techniques, and set-membership formulations [6]. Although effective, these methods often rely on probabilistic assumptions on noise or require iterative procedures with non-negligible computational cost, which may limit applicability in real-time or resource-constrained environments.

This paper addresses these challenges by adopting a combined methodological and comparative perspective. The work focuses on the identification of linear IIR systems from quantized measurements and investigates the critical role played by the thresholding strategy in determining the achievable estimation performance, thus extending prior work by the authors in [7, 8]. In particular, an incremental identification algorithm for ARX models with quantized outputs is considered, analyzing different threshold selection policies namely constant, continuous time-varying, and quantized time-varying thresholds. The performance of the model identification procedure is evaluated over a set of benchmark systems. Performance is assessed in terms of reconstruction accuracy of the continuous output signal, as well as consistency with the observed binary measurements. The results highlight the strong impact of the threshold design on the identification process. Finally, the proposed approach is compared with a state-of-the-art identification method for binary-output IIR systems [2]. The comparison demonstrates that the proposed framework achieves improved accuracy and robustness.

II. THE LINEAR IIR MODEL

In this study, a discrete-time linear IIR ARX structure is considered. At time step k , the system is described by the difference equation

$$y(k) + \sum_{i=1}^n a_i y(k-i) = \sum_{j=0}^m b_j u(k-j) + \eta(k), \quad (1)$$

where $y(k)$ is the system output, $u(k)$ is the input, $\eta(k)$ is the modeling error, the Auto Regressive part is of order n and the Moving Average part of order m (where $n \geq m$). The model coefficients a_i and b_j are unknown and have to be estimated from input-output data. Introducing the backward shift operator z^{-1} the system's discrete-time transfer function associated to (1) can be readily achieved and is given by (neglecting the noise term):

$$G(z) = \frac{Y(z)}{U(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_m z^{-m}}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_n z^{-n}} \quad (2)$$

It is assumed that only a binary measurement of $y(k)$ is available. This implies that the actual output signal is:

$$y_q(k) = \begin{cases} 1 & \text{if } y(k) < T(k) \\ 0 & \text{if } y(k) \geq T(k) \end{cases} \quad (3)$$

where $T(k)$ is a known threshold signal that may be constant or time dependent according a logic. This logic can be defined using different strategies, as we will show in Section III.C..

III. SYSTEM IDENTIFICATION WITH BINARY MEASUREMENTS

A. ARX regression form

For system identification purposes, the ARX model introduced in (1) is rewritten in linear regression form as

$$y(k) = \phi^T(k)\theta + \eta(k) \quad (4)$$

where the regression vector $\phi(k) \in \mathcal{R}^{n+m}$ is defined as

$$\phi(k) = [-y(k-1) \dots -y(k-n) \ u(k) \dots u(k-m)]^T \quad (5)$$

and the unknown parameter vector $\theta \in \mathcal{R}^{n+m}$ is given by

$$\theta = [a_1 \dots a_n \ b_0 \dots b_m]^T. \quad (6)$$

If the identification must be performed from binary measurements, this implies that the continuous time output signal $y(k)$ is not directly available. Consequently, the true output and its lagged values cannot be used to construct the regression vector $\phi(k)$. This limitation has a significant impact on the identification procedure. Indeed in this setting the problem boils down in a nonlinear parameter estimation problem involving latent variables. Specifically, the unknown output $y(k)$ must be replaced by an estimate $\hat{y}(k)$; the same holds for the lagged output values $\hat{y}(k-i)$. Consequently, the regression vector used for estimation becomes

$$\hat{\phi}(k) = [-\hat{y}(k-1) \dots -\hat{y}(k-n) \ u(k) \dots u(k-m)]^T \quad (7)$$

where $\hat{\phi}(k) \in \mathcal{R}^{n+m}$. The noise-free predicted output is therefore

$$\hat{y}(k) = \hat{\phi}^T(k)\hat{\theta}(k). \quad (8)$$

It is observed that both vector terms in the right hand side of (8) are unknown. Stacking the regressors in (8) from $k = 1$ to time k , this leads to the matrix form

$$\hat{Y}_k = \hat{H}_k \hat{\theta}(k) \quad (9)$$

where

$$\hat{H}_k = [\hat{\phi}(1) \ \hat{\phi}(2) \ \dots \ \hat{\phi}(k)]^T \quad (10)$$

and $\hat{Y}_k \in \mathcal{R}^k$, $\hat{H}_k \in \mathcal{R}^{k \times (n+m)}$ and $\hat{\theta} \in \mathcal{R}^{n+m}$.

Unlike the classical ARX identification problem, here the elements of output vector $\hat{Y}_k$ and a part of the regression matrix $\hat{H}_k$ are unknown, as well as the parameter vector $\hat{\theta}(k)$. This defines a *latent variable* or *hidden state estimation* problem from *binary measurements*, where the true outputs are hidden and must be inferred jointly with the model parameters. The only available information for this estimation consists of the binarized measurements $y_q(k)$, the known input $u(k)$ and threshold $T(k)$. Our study addresses this problem by proposing a methodology to jointly estimate the latent outputs $\hat{y}(k)$ and the parameters $\hat{\theta}(k)$.

B. Incremental System Identification Procedure

The proposed algorithm incrementally estimates both the latent outputs $y(k)$ and the ARX parameters θ from binary measurements $y_q(k)$. As many identification algorithms in this context, the identification problem is divided into two alternate phases that are repeated until convergence, typically operated in batch mode. The algorithm proposed in this study introduces the following significant variations:

- At time instant k it uses only the previous samples from $k = 1$ until k and therefore it is also suitable for *on-line* operation.
- The two-phase optimization problem is solved incrementally, that is, the dimension of the overall optimization problem initially has a small dimensionality that increases with time index k .
- The solution computed at time k is used as *hot-starter* for the solution at time $k + 1$.
- To improve the convergence property, for a given k , the two phase optimization is repeated not only a single time but for a specified number of times N_{iter} .

The procedure, summarized in Algorithm 1, starts by initializing the estimate of the system parameter vector to an initial guess $\hat{\theta}(0) = [a_{1,0} \dots a_{n,0} \ b_{0,0} \dots b_{m,0}]^T$ and setting $\hat{y}(0) = \hat{y}(-1) = \dots = \hat{y}(-n) = 0$ and $u(0) = u(-1) = \dots = u(-m) = u(1)$. Then, for each sampling time k , the input $u(k)$ and the corresponding binary output $y_q(k)$ are considered. Using such acquired samples, the following two step procedure is repeated N_{iter} times, where N_{iter} is a parameter chosen by the user:

Step-1 The matrix $\hat{H}_k = [\hat{\phi}(1) \dots \hat{\phi}(k)]^T$ is constructed and the estimate $\hat{Y}_k = \hat{H}_k \hat{\theta}(k-1)$ is obtained. Such estimate of the hidden full-resolution output sequence $\hat{Y}_k$ is then binarized as in (3), thus producing the binary output estimated sequence $\hat{Y}_{q,k} = [\hat{y}_q(1) \dots \hat{y}_q(k)]^T$.

The estimated binary sequence is compared to the actual measured binary sequence $Y_{q,k} = [y_q(1) \dots y_q(k)]^T$ and, whenever a mismatch between the elements of $\hat{Y}_{q,k}$ and $Y_{q,k}$ is detected, the corresponding predicted sample is corrected by substituting it with the value of the threshold using the following simple logic

$$\hat{Y}_k(i) = \begin{cases} \hat{Y}_k(i) & \text{if } \hat{Y}_{q,k}(i) = Y_{q,k}(i) \\ T(i) & \text{otherwise} \end{cases}, i = 1 \dots k. \quad (11)$$

Step-2 The corrected sequence $\hat{Y}_k$ in (11) is used to update the estimate of the system parameters by solving the following ordinary least squares optimization problem:

$$\hat{\theta}(k) = \underset{\theta}{\operatorname{argmin}} \|\hat{Y}_k - \hat{H}_k \theta\|^2 \quad (12)$$

It is well-known that the closed form solution of problem (12) is given by:

$$\hat{\theta}(k) = (\hat{H}_k^T \hat{H}_k)^{-1} \hat{H}_k^T \hat{Y}_k. \quad (13)$$

The overall algorithm is reported in 1.

C. Threshold computation strategy

Different strategies can be adopted to compute the threshold $T(k)$. In this study, three approaches are considered and evaluated.

Constant threshold: The simplest approach consists of fixing the threshold to a known constant value. This value should be selected in order to induce a sufficiently large number of 0–1 transitions, thereby maximizing the information content of the binarized measurements. The selection of an appropriate threshold requires some prior knowledge of the output signal range and its statistical characteristics. In this case, the threshold is defined as:

$$T(k) = T_0 \quad (14)$$

Continuous Time-Varying Threshold: A time-varying threshold is introduced to better investigate the variation domain of the output signal. To enforce a high rate of 0–1 transitions in the quantized signal, the measured output $y(k)$ is compared with its estimate $\hat{y}(k)$. The rationale is that when $\hat{y}(k)$ closely tracks the true output, even small discrepancies between $y(k)$ and $\hat{y}(k)$ induce frequent sign changes in the resulting quantized signal, thereby increasing the transition activity. Accordingly, the time-varying

Algorithm 1 Iterative Parameter Estimation Algorithm

```

1: Initialization:
2:  $\hat{\theta}(0) = [a_{1,0} \dots a_{n,0} b_{0,0} \dots b_{m,0}]^T$ 
3:  $\hat{y}(0) = \hat{y}(-1) = \dots = \hat{y}(-n) = 0$ 
4:  $u(0) = u(-1) = \dots = u(-m) = u(1)$ 
5: for  $k = 1$  to  $N$  do
6:   Read  $u(k)$  and  $y_q(k)$ 
7:   Compute threshold  $T(k)$  as in Section C.
8:   for  $\ell = 1$  to  $N_{\text{iter}}$  do
9:     Step 1: Output reconstruction
10:     $\hat{H}_k = [\hat{\phi}(1) \dots \hat{\phi}(k)]^T$ 
11:     $\hat{Y}_k = \hat{H}_k \hat{\theta}(k-1)$ 
12:    Binarize  $\hat{Y}_k$  to obtain  $\hat{Y}_{q,k}$ 
13:    for  $i = 1$  to  $k$  do
14:      if  $\hat{Y}_{q,k}(i) \neq Y_{q,k}(i)$  then
15:         $\hat{Y}_k(i) \leftarrow T(i)$ 
16:      end if
17:    end for
18:    Step 2: Parameter update
19:    Update  $\hat{H}_k$  using the new  $\hat{Y}_k$ 
20:     $\hat{\theta}(k) \leftarrow (\hat{H}_k^T \hat{H}_k)^{-1} \hat{H}_k^T \hat{Y}_k$ 
21:  end for
22: end for

```

threshold is defined as

$$T(k) = \hat{y}(k) = \hat{\phi}^T(k) \hat{\theta}(k) \quad (15)$$

Quantized Time-Varying Threshold: From a hardware implementation perspective, it is often feasible to realize only a finite set of threshold values. In such cases, the continuous time-varying threshold is quantized using a fixed step size Δ , which is selected based on the range of the output signal and on the desired number of quantization levels. This leads to a discretized approximation of the threshold, defined as:

$$T(k) = \Delta \cdot \text{round}\left(\frac{\hat{y}(k)}{\Delta}\right) \quad (16)$$

IV. THE SET OF TESTING SYSTEMS

The proposed algorithm was tested on a set six discrete time IIR systems compatible with the TF in (2). The six systems share the same parametric model structure. In particular, the input–output dynamics is described by the following discrete-time transfer function:

$$G(z) = \frac{b z^{-1}}{1 - \theta_1 z^{-1} - \theta_2 z^{-2} - \theta_3 z^{-3} - \theta_4 z^{-4}}. \quad (17)$$

This representation corresponds to an AR model of order four with a single delayed input term. The parameter vector $\theta = [\theta_1, \theta_2, \theta_3, \theta_4, b]^T$ fully characterizes the system dynamics, where the coefficients θ_i determine the pole

Table 1. Parameter values and dynamical characteristics (Type) of the 6 models.

N_{iter}	θ_1	θ_2	θ_3	θ_4	b	Type
G_1	0.99	-0.97	0.30	-0.25	0.10	LD1
G_2	1.60	-0.96	0.28	-0.03	0.10	DR
G_3	1.75	-1.52	0.66	-0.12	0.10	HO
G_4	0.98	-0.94	0.29	-0.23	0.10	LD2
G_5	1.45	-0.85	0.32	-0.08	0.10	MX
G_6	1.82	-1.63	0.74	-0.15	0.10	MD

LD1: lightly damped 1; DR: dominant real pole; HO: highly oscillatory;
LD2: lightly damped 2; MX: mixed dynamics; MD: moderately damped.

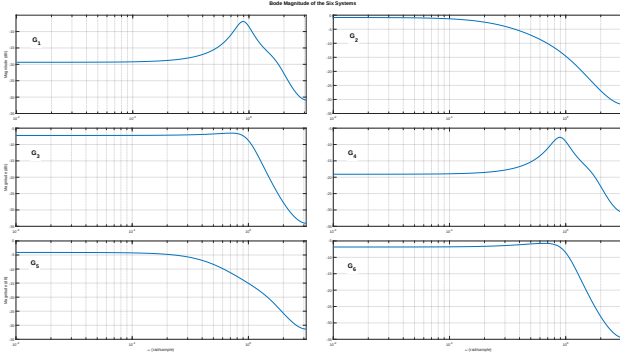


Fig. 1. Frequency response of the 6 models.

locations, while b defines the input gain. Systems characterized by different dynamics responses have been considered whose model parameters are reported in table 1; the corresponding frequency response are reported in figure 1 (assuming a sampling period $\Delta t = 1s$).

V. PERFORMANCE EVALUATION

A. The role of the threshold policy

The first experiment investigates the influence of the thresholding policy as a function of the iteration parameter N_{iter} . To this end, all benchmark systems G_i have been excited using the same random input sequence $u(k)$ of length $N = 1000$, drawn from a zero-mean unit-variance Gaussian distribution. For the constant threshold policy, this is selected as a fixed positive offset proportional to the output operating range. Specifically, the range is defined as

$$\text{range}(y_i) = \max_{1 \leq k \leq N} y_i(k) - \min_{1 \leq k \leq N} y_i(k) \quad i = 1 \dots 6$$

and the corresponding threshold is set to $T_{0i} = \text{range}(y_i)/6$ for all considered systems. For the continuous threshold policy, the threshold is updated online as $T_i(k) = \hat{y}_i(k)$, thus requiring no additional tuning parameters. Finally, for the quantized threshold policy, a 3-bit quantization scheme is assumed. For each system the quantization step is defined as $\Delta_i = \text{range}(y_i)/2^3$. Performance was assessed in terms of the percent relative recon-

Table 2. Fixed Thr.: E_{rec} Performance vs N_{iter} .

N_{iter}	1	5	10	15
G_1	2.9	2.1	2.1	1.3
G_2	20.1	19.8	19.8	2.3
G_3	3.9	6.4	1.9	5.7
G_4	8.9	8.9	9.0	8.1
G_5	7.1	3.8	3.2	5.1
G_6	5.3	2.5	2.5	2.3

struction RMS error associated with the continuous output $y_i(k)$ defined as:

$$E_{rec_i} = 100 \cdot \frac{\sqrt{\frac{1}{N} \sum_{k=1}^N (y_i(k) - \hat{y}_i(k))^2}}{\sqrt{\frac{1}{N} \sum_{k=1}^N y_i^2(k)}} \quad (18)$$

Table 2 reports E_{rec} error for $N_{iter} \in \{1, 5, 10, 15\}$, evaluated over the six models using $N = 1000$ data samples. It can be observed that increasing N_{iter} generally leads to a reduction in the percentage reconstruction error, although the trend is not strictly monotonic across all cases. In particular, for model G_2 the error is significantly large for small values of N_{iter} , while it decreases to approximately 2.3% when $N_{iter} = 15$. This non-uniform behavior is attributed to the complex interaction between the coarse binary quantized output and the iterative reconstruction algorithm, which may lead to different convergence rates and local minima.

Table 3 reports the E_{rec} for the continuous threshold policy. It can be observed that this strategy generally improves performance with respect to the constant threshold case. This behavior is expected, since the time-varying threshold allows the reconstruction algorithm to better explore the dynamic range of the output signal. Similarly to the previous case, increasing N_{iter} generally has a positive effect on E_{rec} , although the improvement is not strictly monotonic. It is also observed that, for $N_{iter} = 15$, a degradation in performance may occur. Also in this case this behavior is not monotone and is likely due to the complex interaction between the relative position of the true output $y(k)$ and its estimate $\hat{y}(k)$ during the iterative reconstruction process.

Table 7 reports the E_{rec} for the quantized threshold policy. It can be observed that this case yields a significant performance improvement, with the E_{rec} remaining below 1% in most configurations. In addition, with few exceptions, increasing N_{iter} leads to a general reduction in E_{rec} . This notable improvement is attributed to the role of the quantization mechanism, which effectively acts as a dithering-like process centered around the estimated continuous threshold $\hat{y}(k)$. This behavior has a beneficial effect, as it induces a larger number of threshold crossings, which are essential to ensure accurate system identification from binarized measurements.

Table 3. Continuous Thr.: E_{rec} Performance vs N_{iter} .

N_{iter}	1	5	10	15
G_1	3.3	0.7	0.5	0.4
G_2	0.9	1.8	1.3	1.4
G_3	1.2	1.3	9.4	18.8
G_4	1.7	1.6	1.6	1.7
G_5	14.0	7.0	7.4	14.7
G_6	2.4	18.1	1.5	18.1

Table 4. Quantized Thr.: E_{rec} Performance vs N_{iter} .

N_{iter}	1	5	10	15
G_1	0.8	0.5	0.4	0.3
G_2	0.7	0.4	0.4	0.3
G_3	0.8	0.6	1.2	1.5
G_4	0.8	0.3	0.2	0.1
G_5	1.1	0.6	0.5	0.5
G_6	2.2	2.7	0.5	1.0

B. Analysis of the algorithm for quantized thresholds

In this section, a deeper analysis of the proposed quantized-threshold policy is presented. Figure 2 shows the incremental evolution of the identified parameters for system G_1 as a function of the iteration index k . It can be observed that the estimates converge smoothly toward nearly steady-state values that are close to the true parameters, despite the quantization effects. It is worth emphasizing that, at each sample k , the proposed algorithm exploits only the information available from time instants 1 to k . Consequently, for small values of k , the estimation relies on a limited number of samples and is therefore less accurate. As the number of samples increases, the estimates become progressively more consistent, reaching an approximately steady-state condition around $k = 300$. This behavior was observed for all six considered models.

Table 5 reports, for all considered models, the final values of the identified parameters evaluated at $k = 1000$ for $N_{\text{iter}} = 10$. It can be observed that, in all cases, the estimated parameters closely match the corresponding true values reported in Table 1. These results confirm the capability of the proposed algorithm to accurately recover the actual system parameters despite the quantized measures.

It is also of interest to evaluate the capability of the algorithm to reconstruct the true quantized output signal $y_q(k)$. Indeed, the identification procedure iteratively updates the estimate $\hat{y}_q(k)$ in order to progressively reduce the mismatch between $\hat{y}_q(k)$ and $y_q(k)$. If the two signals coincide over all the available training samples N , the identified model generates exactly the same binary sequence as the true system. From a system identification perspective, such a model is indistinguishable from the actual one given the available training data set. In other words, the optimization procedure has reached a global minimum, and no further improvement can be achieved on the considered data. In general, however, the set of parameter vectors producing identical quantized sequences is not necessarily

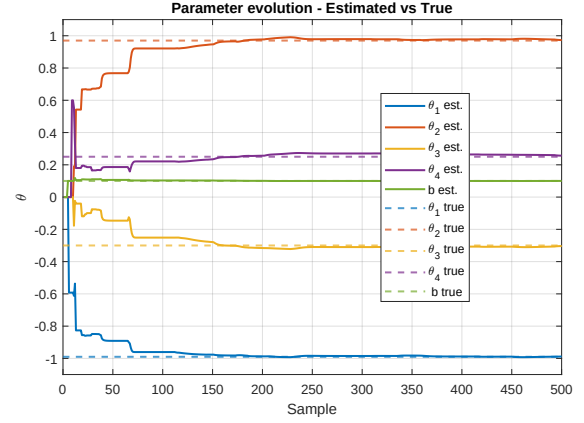


Fig. 2. Evolution of the model parameter estimates for system G_1 as a function of the time step k for $N_{\text{iter}} = 10$.

Table 5. Estimated parameter values obtained for the six considered models for $N_{\text{iter}} = 10$ (quantized thr.).

Model	θ_1	θ_2	θ_3	θ_4	b
G_1	0.9873	-0.9670	0.2974	-0.2504	0.1003
G_2	1.5917	-0.9494	0.2781	-0.0316	0.1008
G_3	1.7622	-1.5524	0.6938	-0.1334	0.1000
G_4	0.9816	-0.9424	0.2920	-0.2304	0.1000
G_5	1.4526	-0.8489	0.3130	-0.0766	0.0997
G_6	1.8188	-1.6309	0.7438	-0.1522	0.1003

unique. Table 6 reports the number of samples at which a discrepancy between $\hat{y}_q(k)$ and $y_q(k)$ is detected at the end of the identification process, namely for $k = 1000$. It can be observed that increasing N_{iter} generally has a beneficial effect, leading to a reduction in the number of mismatch instants. It is worth noting that, for the G_2 model, the algorithm was able to reach the global optimum, yielding zero mismatches in two cases.

C. Validation

All the identified models were validated using as input $u(k)$ a different white noise sequence of length $N = 1000$. This was used as input for the identified models $G_i(z)$. The simulated output is therefore $y_{\text{sim},i}(z) = G_i(z)u(z)$. The capability of the identified models to replicate the true output $y(k)$ in simulation is evaluated using the metric defined in 18, with $y_{\text{sim},i}(k)$ replacing $\hat{y}_i(k)$. It is observed that for $N_{\text{iter}} \geq 10$, the reconstruction error E_{rec} falls below 2% for all systems, confirming the accuracy of the identified models.

D. Comparison with a state of the art method

The proposed algorithm was evaluated against the state-of-the-art method for binarized output IIR system identification proposed in [2], where the identification problem is formulated as an iterative procedure based on the solution of a constrained quadratic programming problem

Table 6. Number of mismatches between the binarized signals $\hat{y}_q(k)$ and $y_q(k)$ evaluated over the $N = 1000$.

N_{iter}	1	5	10	15
G_1	4	3	3	3
G_2	2	1	0	0
G_3	4	2	4	6
G_4	3	3	2	1
G_5	2	2	1	1
G_6	17	12	1	3

Table 7. Quantized Threshold: Simulation E_{rec} vs N_{iter}

N_{iter}	1	5	10	15
G_1	2.1	2.0	2.0	2.0
G_2	1.1	1.0	1.0	1.0
G_3	1.4	1.2	1.6	1.8
G_4	2.1	2.0	2.0	2.0
G_5	1.6	1.2	1.2	1.2
G_6	2.3	2.9	1.1	1.4

with linear inequality constraints. For comparison purposes, all methods were tested under identical conditions, namely the same random Gaussian input sequence used for testing our algorithm in case of uniform quantization step $\Delta = \text{range}(y)/2^3$ for the threshold signal. Table 8 reports the index E_{rec} as a function of the number of iterations n_{iter} . It is important to emphasize that n_{iter} has a fundamentally different interpretation compared to our approach. In our method, it denotes the number of internal optimization iterations performed within each optimization cycle at sample k , whereas in [2] it represents the number of global iterations performed over the entire dataset of length N . It is observed that, in some cases, the algorithm in [2] fails to converge; such cases are indicated by missing entries in the table. In general, increasing n_{iter} improves performance, leading to a reduction of E_{rec} . However, for systems G_3 and G_6 , the method in [2] exhibits severely degraded performance. Finally, a comparison with Table 7, obtained using our method under the same 8-level threshold quantization setting, shows that our approach outperforms approach [2] in most cases. Moreover our algorithm does not exhibit instability or divergence for any tested value of N_{iter} .

VI. CONCLUSIONS

This paper investigated the identification of linear IIR systems from binary measurements, with emphasis on the impact of threshold selection. The proposed incremental ARX identification framework was evaluated on six benchmark systems under three thresholding strategies: constant, continuous, and quantized. Results show that the threshold policy is a key factor for performance. The quantized time-varying threshold consistently provides the best results, achieving reconstruction errors below 1% in most cases and below 2% in all validation scenarios for $N_{iter} \geq 10$. In contrast, fixed and continuous thresholds

Table 8. Reconstruction error E_{rec} for the quantized system versus n_{iter} for the approach in [2].

n_{iter}	5	10	20	30	40	50
G_1	9.6	1.3	0.4	0.4	0.2	0.1
G_2	158.4	68.6	6.3	4.7	4.6	4.7
G_3	70.1	—	—	159.3	—	180.6
G_4	7.6	1.3	0.4	0.2	0.2	0.2
G_5	105	10.9	1.9	1.0	0.7	0.8
G_6	192.7	76.7	192.7	—	76.7	76.7

exhibit higher variability and, in some cases, significantly larger errors. The proposed method demonstrates accurate parameter convergence across all tested systems and outperforms the considered state-of-the-art approach, which shows poor performance in some cases. The incremental nature of the model identification algorithm also makes it suitable for online applications.

VII. ACKNOWLEDGMENT

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Comparative Analysis of Search Algorithms for Automatic L-type Matching Network

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Abstract –

Impedance matching is paramount to increase the power transferred to a load. However, whenever the load impedance is unknown or non-static, impedance matching can become a laborious process that requires constant adjustment within varying environmental factors. Therefore, automation of a L-network impedance matching is proposed herein. The automated impedance matching system is composed of banks of reactive elements activated by relays and was applied to electromagnetic acoustic transducers as loads. Two algorithms were applied to obtain the network's best configuration, namely, an adapted bidimensional Golden Section search and Nelder-Mead Simplex-based search. These algorithms were compared to an exhaustive search using simulations and experimentally, with distinct transducers and frequencies. The proposed solution was able to consistently obtain optimal network configuration and experimentally achieved at least 65% of the voltage gain of an exhaustive search while taking less than 6% of its search time.
Keywords: Impedance Matching; Automation; EMAT

I. INTRODUCTION

Impedance matching techniques are vital to ensure maximum power transfer between source and load; consequently, increasing Signal to Noise Ratio [1, 2]. The L-network technique is one of the simplest impedance matching networks; it uses two reactive components, such as capacitors and inductors [2], whose values are calculated based on the impedance of the load and source. Here, we address the problem of impedance matching for Electromagnetic Acoustic Transducers (EMATs) acting as loads.

EMATs are strictly non-contact ultrasonic transducers which are able to generate and receive less conventional

ultrasonic wave modes, such as shear waves, in conducting material [3, 4]. However, compared to the most common type of transducer, which employs piezoelectric elements [5], EMATs exhibit a lower transduction efficiency, making impedance matching paramount for EMAT systems. Their impedance is strongly influenced by external variables, such as temperature, the lift off, or distance between the transducer and ferromagnetic objects [6, 1, 7]. Consequently, in practical applications, the EMAT's impedance is often unknown a priori. Therefore, one often needs to tune the network components empirically.

To streamline this process, automated impedance matching systems have been developed [8, 9, 10] to eliminate the need for repetitive tuning and reduce the susceptibility to human error. However, to match a wide gamut of loads, the automated network must search for its optimal configuration in a bidimensional discrete space with tens of thousands of points. Therefore, the tuning process of L-networks under unknown load value can be modeled as a mathematical optimization problem. This study evaluates two derivative-free optimization algorithms, namely, a Nelder-Mead Simplex based method and an adapted Golden Section Search for the two-dimensional space, which are implemented in an automated L-network impedance matching applied to EMAT systems.

II. BACKGROUND THEORY

A. L-network Impedance Matching

According to the Maximum Power Transfer Theorem, to transfer the maximum amount of energy from a source to a load, the load impedance must be the complex conjugate of the source's output impedance [11]. Usually, the source impedance is designed to be purely real with a standard value of 50Ω ($Z_S = 50$).

The L-network impedance matching network works by

associating two reactive elements between the source and the load, which changes the impedance seen by the source [2], aiming to maximize transmitted power [2]. Figure 1 shows the L network topology configuration for load's resistance less than the source's ($R_0 < R_s$), adopted herein.

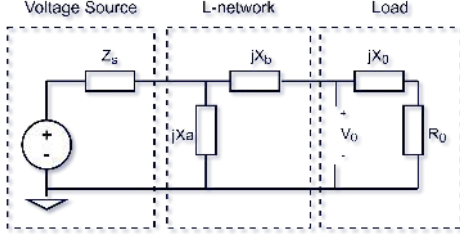


Fig. 1. Electrical equivalent circuit diagram illustrating the connection of a voltage source to a load through an L-network impedance matching circuit. The source's output impedance is Z_s , and the load (Z_0) is composed of a resistance, R_0 , and a reactance, X_0 . The matching network has parallel X_a and a series X_b element, both reactive.

For a purely resistive source, the ideal L-network's reactance values are given by [2]

$$X_a = \pm \frac{50\Omega}{\sqrt{\frac{50\Omega}{R_0} - 1}}, \quad (1)$$

$$X_b = \pm R_0 \sqrt{\frac{50\Omega}{R_0} - 1} - X_0, \quad (2)$$

where X_a and X_b are the parallel and series reactances of the L-network, respectively, and R_0 and X_0 are the resistance and reactance components of the load, that is $Z_0 = R_0 + jX_0$.

Resorting to theoretical values for X_a and X_b , as per Eqs. 1 and 2 presents a major challenge whenever the load impedance is unknown or is subject to variation, which often occurs in practical applications, such as with EMAT systems. That is, in such scenarios, one cannot simply calculate the ideal X_a and X_b since the load impedance is unknown a priori. Instead, the process becomes a combinatorial optimization problem, searching for the network's configuration that maximizes voltage over the transducer. Moreover, in practical applications, the search space is noisy due to environmental factors and presents uncertainty due to component tolerances.

B. Electromagnetic Acoustic Transducers

EMATs typically consist of a coil positioned beneath a permanent magnet [3]. When an alternating current is injected into the coil, it induces a current density in the skin-depth of a conductive medium. Its interaction with the static magnetic field provided by the magnet generates Lorentz forces inside the medium [3], which can

generate ultrasonic waves directly within the conductive medium [3] without physical contact, unlike piezoelectric transducers, which require mechanical coupling [5]. However, it also provides EMATs with lower efficiency, making impedance matching paramount [8, 9].

EMATs can be approximately modeled as an inductive load, Z_0 .

However, Z_0 may vary with operational frequency, temperature, lift-off, and significantly differs among distinct transducers [8, 12]. This dynamic behavior renders fixed matching networks less effective, hence an automated reconfigurable impedance matching network is proposed. Here, since the main electric element of an EMAT is a coil, the load has a lower resistance than the source ($R_0 < R_s$), which informs the use of the topology of the L-network where the parallel element is connected directly to the source [2], shown in Fig 1.

III. MATCHING NETWORK HARDWARE

An automated L-matching network was designed with two banks of components, one acting as X_a and the other as X_b , as shown in Fig 1. Each component is connected or disconnected by an electromechanical relay. The X_a bank is composed of eight different capacitors, [0.24, 0.5, 1, 1.9, 3.3, 6, 11, 23 nF], that can be associated into 256 (2^8) distinct values. The X_b bank has four capacitors, [0.47, 1, 2, 3.9 nF] and four inductors, [0.5, 1, 2, 6 μ H], totaling 32 ($2^4 + 2^4$) possible reactances, without mixing capacitors and inductors. Each possible reactance for X_a and X_b is associated with a respective index, which ultimately corresponds to an on/off state of each relay. The relays are actuated by a microcontroller, which is also able to read the voltage over the EMATs (V_0 , Fig 1) via a peak detector circuit and 16-bit ADC [9]. The design and construction of this network is non-trivial, due to the necessity of withstanding the high peak power of an EMAT system and introducing low noise. Figure 2 shows a schematic and a photograph of the network.

The EMAT systems considered herein are excited by tone-bursts lasting from 0.6 to 50 μ s. Each excitation occurs within a repetition period of 100 ms. Consequently, the relays must be switched in a time interval between the end of the excitation tone-burst and its rise in the next excitation cycle. Therefore, performing a brute-force search, meaning an exhaustive method that systematically tests every single one of the 8192 (256×32) possible relay configurations one by one, would require several minutes, encouraging the use of optimization algorithms capable of obtaining the global maximum load peak voltage with a minimal number of switching operations or burst cycles.

IV. OPTIMIZATION STRATEGIES

The impedance matching task is formulated as a maximization problem namely to find the optimal configuration

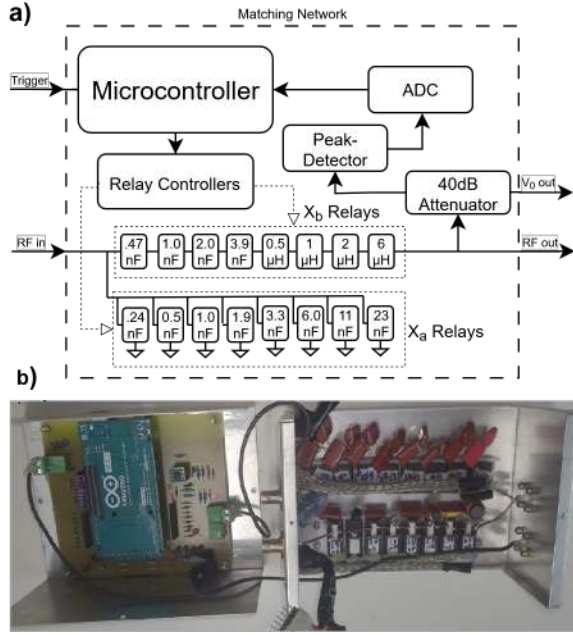


Fig. 2. Automatic matching network built. (a) Schematic diagram: the microcontroller commands the relay controllers, activating each bank of 8 relays. Each relay is attached to a corresponding reactive element, altering the impedance in the path from RF in through RF out. The latter is connected externally to the load and is measured via a 40dB attenuator, peak detector circuit, and ADC. (b) Photograph of the fabricated automated impedance matching network.

vector $x = [X_a, X_b]$ within the discrete search space that maximizes the peak voltage at the load (V_0). Since the expression for the analytical gradient of V_0 is not available in real-time, standard gradient-based methods cannot be applied [13]. This requires derivative-free optimization methods that rely solely on objective function evaluations. Two algorithms were implemented.

A. Golden Section Search

The Golden Section Search finds the maximum or minimum of a strictly unimodal function by iteratively narrowing the search interval [13]. It relies on the golden ratio $\varphi = (1 + \sqrt{5})/2 \approx 1.618$, selecting two interior probe points such that the ratio of intervals remains constant, guaranteeing linear convergence toward the peak.

To address the need for a multidimensional search while accommodating the specific hardware constraints of the automated relay banks, a hybrid optimization strategy was proposed. While the standard Golden Section Search is inherently one-dimensional, its application was extended to the two-dimensional (X_a, X_b) surface of the L-network by integrating it with a coordinate descent framework [14]. The proposed 2D hybrid Golden Search specifically ex-

ploits the asymmetric resolution of the hardware design. Since the X_a bank possesses a vast number of potential states (256 combinations), it requires a highly efficient convergence algorithm. Conversely, the X_b bank is limited to only 32 indices. Therefore, the custom algorithm systematically alternates between two distinct search methods along the axes of the search space.

First, the algorithm fixes the X_b value or index and treats V_0 as a one-dimensional function with respect to X_a . Then, the standard Golden Section Search is deployed to rapidly find the optimal X_a configuration. Once the best X_a is identified, this index is temporarily fixed. The algorithm then performs a complete exhaustive sweep across all 32 discrete indices of the X_b bank. This step guarantees the maximum along the secondary axis without the risk of the algorithm converging prematurely on a false local peak due to the coarse discretization of the components.

These two steps are applied alternately, and the resultant iterative coordinate-descent loop continues until a complete cycle yields no change in either the X_a or X_b value possible on the hardware, or until a previously evaluated state configuration is revisited. This secondary condition acts as a fail-safe against infinite oscillatory limit cycles, confirming that the system has successfully navigated the 2D surface and settled on the best hardware approximation of the global maximum.

B. The Nelder-Mead Simplex Method

The Nelder-Mead algorithm optimizes multidimensional functions without derivatives [13]. For a 2D problem, it maintains a polygon of 3 vertices (named simplex) over the search grid, here each representing a relay configuration with an associated V_0 . At each iteration, the algorithm ranks these vertices based on the resultant V_0 value to identify the best (greatest V_0) and worst (lowest V_0) configurations. The algorithm improves the worst vertex through four operations, namely (i)-reflection, which projects the worst vertex through the centroid of the remaining points; (ii)-expansion, which extends further in a promising direction if reflection yields a significant gain; (iii)-contraction, which shrinks the simplex locally if reflection fails to improve the voltage; and (iv)-shrink, which reduces the entire simplex towards the best vertex if the contraction step also fails to find an improvement.

Since standard geometric operations work in the continuous domain, each newly calculated coordinate passes through a discretization function (`discretizeXa` and `discretizeXb`), approximating it to the nearest physical hardware component before triggering the relays. The algorithm's parameterization utilizes canonical coefficients, namely, reflection, α , equals 1.0, expansion, γ , equals 2.0, contraction, ρ , equals 0.5, and shrink, θ , equals 0.5. To prevent premature convergence on local maxima, the simplex is initialized using the extreme physical

bounds for X_a and X_b , covering the largest possible area of the search surface. Finally, the search terminates based on explicitly defined convergence and plateau criteria. The convergence criteria are met either geometrically, when all vertices collapse to the same hardware combination, or numerically, when the absolute difference of V_0 between the best and worst vertices is lower than 10^{-6} . Conversely, the plateau criteria are triggered if five consecutive transformations produce no index change, indicating search stagnation. A hard limit of 100 iterations serves as a final fail-safe.

V. METHOD

A. Computational Modeling and Simulation

To assess the validity of the proposed algorithms, a simulation environment was developed utilizing measured EMAT impedance data obtained with a Precision Impedance Analyzer (Agilent 4294A). This simulation also replicates the physical constraints of the automated matching unit regarding the available states of the relay banks. To evaluate the matching efficacy, the voltage transfer to the load is calculated for every combination of the network elements, X_a and X_b . Based on the implemented L-network, the equivalent input impedance of the network in association with the load is:

$$Z_{eq} = jX_a \parallel (jX_b + Z_0), \quad (3)$$

where the symbol $\parallel$ denotes the parallel association of the reactance X_a with the series branch containing X_b and Z_0 .

The normalized voltage across the transducer, which is the quantity to be maximized, is given by:

$$V_0 = \frac{Z_{eq}}{Z_{eq} + Z_{source}} \frac{Z_0}{jX_b + Z_0}. \quad (4)$$

By iterating through all 8192 network combinations in a brute-force fashion, the simulation identifies the specific combination that results in the peak magnitude of $|V_0|$, providing a theoretical benchmark. Alongside this exhaustive search, both the Nelder-Mead Simplex and the 2D Golden Section Search algorithms were implemented within the simulator to quantitatively evaluate their convergence speed and accuracy against this theoretical baseline. Furthermore, for each load impedance scenario, distinct signal-to-noise ratios (SNR) were considered in the simulation, including the ideal case (no noise), followed by 40, and 20 dB SNR.

B. Experimental Setup

To assess the performance of the adopted optimization algorithms, an experimental setup was used, as illustrated in Figure 3. Two nominally identical EMATs operate in through-transmission on a 25.4mm thick steel block. The excitation signal is generated by a Tektronix AFG3251 function generator and amplified by a power

amplifier (E&I 2100L), which is connected to the matching impedance system. An oscilloscope is used to monitor the waveforms at the transmitter and receiver transducers, even though the values measured at the matching network are used for the search algorithms. Two different models of transducer were evaluated, namely, RPDR1000 and HWS2205, each driven at 1 and 2 MHz, generating four distinct loads, namely, $Z_{0\#1}$ and $Z_{0\#2}$ indicate the RPDR driven at 1MHz, and 2MHz, respectively, $Z_{0\#3}$ and $Z_{0\#4}$ indicate the HWS at 1 and 2MHz, respectively.

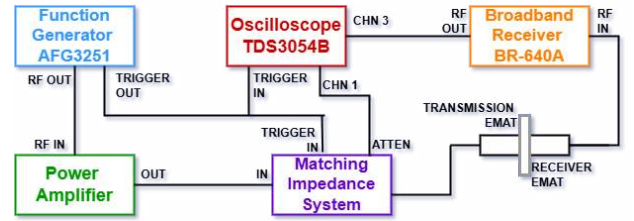


Fig. 3. Block diagram of the experimental setup. The waveform from the function generator is amplified, then passes through the impedance matching system before the transmission EMAT. The Oscilloscope reads this attenuated signal and the transmission through a broadband receiver.

C. Performance Analysis

To quantitatively evaluate and compare the efficiency of the implemented optimization algorithms against standard exhaustive search methods, two key performance parameters were defined. The first one is the total number of points evaluated, defined as the cumulative count of points measured on the $X_a - X_b$ plane. It is the main parameter for optimization, given that each point takes one cycle, meaning it is directly proportional to the search time. Minimizing it from the brute-force starting point of 8192 makes the system more adaptable to environmental changes and also minimizes the wear on the electromechanical relays. The second metric is the voltage optimization error, given by:

$$\Delta V_0 = \left(\frac{V_{global} - V_{alg}}{V_{global}} \right) \cdot 100\%, \quad (5)$$

where V_{global} is the global maximum voltage determined via an exhaustive search, and V_{alg} is the peak voltage achieved by the optimization algorithm being evaluated.

Since the primary control objective is to maximize V_0 , the accuracy of the algorithm is quantified by this parameter, which calculates the relative deviation between the voltage level achieved by the proposed algorithm (V_{alg}) and the true global maximum voltage (V_{global}) achievable by exhaustive search.

VI. SEARCH ALGORITHMS PERFORMANCE

Figure 4 shows an example of the experimental waveforms at the receiver and transmitter EMATs, comparing the exhaustively matched and unmatched cases. The excitation signal is visible from 0 to 2.5 μs in the emitter, and

its interference in the receiver, which also shows the signal transmitted through the steel form around 8,5 to 11 μ s. It is possible to observe that the transmitter matching increases the voltage over the receiver.

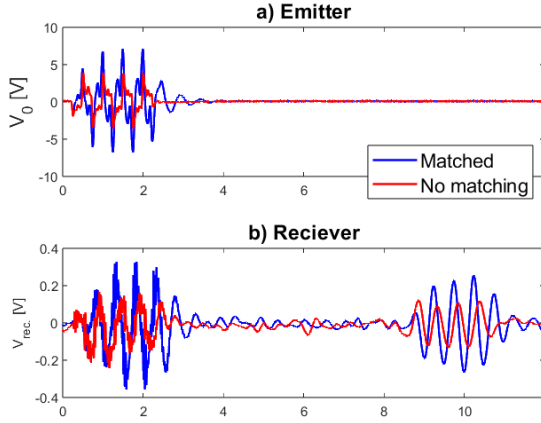


Fig. 4. Waveforms for the HWS2205 transducer, excited with 2MHz with 4-cycle tone-burst, at (a) transmitter, V_0 and (b) over the receiver. Red lines stand for unmatched signals and blue lines for optimally matched (brute force).

Table 1 summarizes results for each load, including ΔV_0 and the number of iterations for both search methods employed. It is worth mentioning that the optimum configuration obtained with the exhaustive search is used as a reference for comparison with the search algorithms, as opposed to a baseline obtained with the unmatched circuit, which would provide a much lower voltage. The equivalent ΔV_0 for the unmatched circuits are reported in each headline of the table.

Under ideal, noise-free simulated conditions (∞ dB SNR), both algorithms successfully navigate the search space, converging to the global maximum ($\Delta V_0 = 0.00\%$). However, a stark contrast in computational cost is immediately apparent. The Simplex method consistently requires significantly fewer iterations (ranging from 18 to 34) compared to the 2D hybrid Golden Search, which has a minimum of 86 iterations.

As moderate noise (40 dB SNR) is introduced, both algorithms maintain excellent accuracy, keeping the voltage optimization error well below 1%. However, the computational demand of the 2D hybrid Golden Search spikes as it exhaustively evaluates the noisy surface, peaking at 516 iterations for $Z_{0\#1}$. The Simplex method, conversely, maintains a stable and low iteration count of about 36.

Under higher noise conditions (20 dB SNR), the structural limitations of the algorithms become evident. The Simplex method demonstrates vulnerability to local optima, experiencing severe convergence failures, yielding a 16.05% error for $Z_{0\#1}$ and a high error of about 58.35% for $Z_{0\#3}$. The 2D hybrid Golden Search proves much

Table 1. Performance evaluation of the implemented search algorithms. Results indicate the voltage optimization error (ΔV_0) and total iterations required to reach convergence for different load configurations ($Z_{0\#1}$ to $Z_{0\#4}$), comparing simulated scenarios against experimental data. Unmatched ΔV_0 are shown in each respective load header.

Test	Golden Search		Simplex	
SNR [dB]	ΔV_0 [%]	#Iter.	ΔV_0 [%]	#Iter.
$Z_{0\#1}$ - RPDR, 1 MHz				
(unmatched $\Delta V_0 = 58.8\%$)				
Sim. ∞	0	86	0	31
Sim. 40	0	516	0.01	37
Sim. 20	0.35	301	16.05	24
Experim.	20.63	86	18.66	15
$Z_{0\#2}$ - RPDR, 2 MHz				
(unmatched $\Delta V_0 = 61.1\%$)				
Sim. ∞	0	86	0	18
Sim. 40	0.58	215	0.41	41
Sim. 20	0	129	0.08	20
Experim.	13.3	190	2.25	15
$Z_{0\#3}$ - HWS, 1 MHz				
(unmatched $\Delta V_0 = 54.3\%$)				
Sim. ∞	0	86	0	34
Sim. 40	0.11	86	0	33
Sim. 20	5.83	215	58.35	20
Experim.	0	104	6.10	17
$Z_{0\#4}$ - HWS, 2 MHz				
(unmatched $\Delta V_0 = 68.1\%$)				
Sim. ∞	0	86	0	27
Sim. 40	0.20	129	0.16	32
Sim. 20	1.47	86	0.20	31
Experim.	0.16	104	4.79	18

more resilient to this noise level, containing the maximum error to 5.83% ($Z_{0\#3}$), though it still exacts a heavy computational toll.

For ideal, noise-free conditions, they were consistently able to find the optimal values, and experimentally, they provided ΔV_0 values of at most 21%, whereas the unmatched ΔV_0 is approximately 60%. Therefore, the search algorithms provide around 65% ($100\% - 21\%/60\%$) of the possible performance gain of impedance matching.

In the experimental environment, the Simplex algorithm demonstrates exceptional convergence speed, requiring only 15 to 18 switching cycles across all test cases. Its accuracy on the physical hardware shows moderate baseline deviations ranging from 2.25% to 6.88% for $Z_{0\#2}$, $Z_{0\#3}$, and $Z_{0\#4}$, and a more significant error of 18.66% related to $Z_{0\#1}$. The Golden Search, while significantly slower, demanding from 86 to 190 physical switching iterations, showed highly load-dependent accuracy. For the

$Z_{0\#3}$ and $Z_{0\#4}$ configurations, it achieved near-perfect physical matches, with errors of only 0.00% and 0.16%, respectively. Conversely, it performed similarly to the Simplex method on the $Z_{0\#1}$ setup, resulting in a higher error, about 20%.

Figure 5 shows V_0 in the $X_a - X_b$ plane for the noiseless simulation, and experimental results for the $Z_{0\#4}$ load, showing good agreement. The resulting points on the plane from both search methods are also indicated by circles and are situated at the higher points of V_0 .

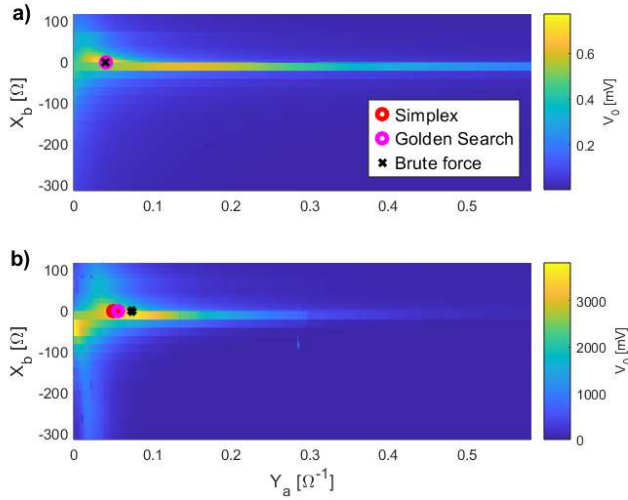


Fig. 5. Color plot for V_0 over the $X_a - X_b$ plane a) Simulation (∞ SNR), and b) Experimental result of $Z_{0\#4}$

VII. CONCLUSION

Here, an impedance matching L-network was automated, and two algorithms were developed to find its optimal configuration, namely, an adapted bidimensional Golden Section search and Nelder-Mead Simplex based search. They were applied to EMAT systems and evaluated in simulations and experimentally. Both search algorithms were able to find near-optimal configurations of impedance matching applied to electromagnetic acoustic transducers. For ideal, noise-free conditions, they were consistently able to find the optimal values. Experimentally, they achieved less than 21% voltage optimization error, implying more than 65% of the possible performance gain provided by impedance matching. This was achieved with less than 6% of the required iterations to perform a brute force or exhaustive search. These search algorithms can also enable the expansion of the matching network, including more elements and values without significant drawbacks, increasing performance whenever impedance matching is required, such as in ultrasound systems. While the Simplex method is significantly faster, it was shown to be less resilient to a noisy environment, both in simulation and in the experimentally.

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Airborne Navigation During GNSS Outages Using Tightly Coupled IMU/DVB-T2 Fusion

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Abstract – This paper presents a tightly coupled navigation system integrating a low-cost Inertial Measurement Unit (IMU), GNSS, and terrestrial Digital Video Broadcast (DVB-T2) signals to provide robust resilience during GNSS-denied scenarios. To mitigate the unbounded drift of a stand-alone IMU, the proposed Error-State Kalman Filter utilizes DVB-T2 Signals of Opportunity (SoP) as an independent source of navigation corrections. Real-world flight test results demonstrate that during simulated GNSS outages, the IMU/DVB-T2 fusion bounds the average position error to tens of meters over a 5-minute outage, and maintains it at approximately 100 meters after a 10-minute outage. In stark contrast, a stand-alone IMU yields kilometer-level errors under identical conditions. Ultimately, this research confirms that DVB-T2 signals offer a practically viable, independent fallback system for manned and unmanned aviation, ensuring safe mid-flight navigation.

Keywords: GNSS-denied navigation; Kalman Filter; DVB-T2; Signals of Opportunity;

I. INTRODUCTION

While GNSS-aided Inertial Navigation Systems (INS) are the industry standard, the increasing global proliferation of GNSS jamming and spoofing necessitates resilient alternatives. Radio positioning based on Digital Video Broadcasting-Terrestrial (DVB-T2) Signals of Opportunity (SoP) offers a robust secondary navigation source utilizing Time Difference of Arrival (TDoA) techniques with high resilience to jamming. Since these signals were designed primarily for broadcasting and communication, their application in navigation represents a typical secondary use case for SoP methods.

Existing literature, e.g., [1], demonstrates DVB-T2 positioning with 100-150 m accuracy within a Single Frequency Network (SFN). However, these studies primarily address static terrestrial scenarios, which suffer from severe non-line-of-sight (NLOS) signal degradation (diffraction caused by the Earth's surface and man-made struc-

tures), rendering stand-alone implementation impractical.

To overcome these terrestrial limitations, this paper extends DVB-T2 positioning described in [1] to the aviation domain, taking advantage of favorable line-of-sight (LOS) propagation. A tightly coupled Error-State Kalman Filter (ESKF) architecture is proposed that fuses low-cost IMU, GNSS, and DVB-T2 measurements. This approach delivers a resilient dynamic navigation solution, specifically designed to limit position, velocity, attitude, and inertial drifts to maintain operational safety for manned or unmanned vehicles during prolonged GNSS outages.

II. INERTIAL NAVIGATION, DATA FUSION

At the core of every Inertial Navigation System lies the Inertial Measurement Unit (IMU). [2] However, especially when using low-cost sensors, a stand-alone INS suffers from rapid, unbounded position drift. This divergence is caused by the inherent double integration of uncompensated stochastic and deterministic errors – mainly sensor biases, scale factor errors, and random walk. [3]

To estimate and correct these inertial errors, this study utilizes an Error-State Kalman Filter [4, 5, 6, 7] within a tightly coupled strapdown architecture. Unlike loosely coupled configurations, our system directly fuses raw observations – namely GNSS pseudoranges (ρ), pseudorange rates ($\dot{\rho}$), and DVB-T2 TDoA measurements ($\Delta\tau$). The critical advantage of this approach is the ability to perform filter updates even when the number of available GNSS satellites or DVB-T2 transmitters falls below the minimum required for an independent position fix. Although processing raw measurements increases computational complexity and necessitates a larger state vector, the resulting robustness in degraded environments is essential for reliable airborne navigation.

A. Kalman filter – proposed state vector

Depending on the available measurement update, the ESKF state vector takes one of two forms. For GNSS measurements, the 18-state vector $\delta\mathbf{x}_{GNSS}$ takes the form

$$\delta\mathbf{x}_{GNSS} = [\delta\mathbf{p}^n, \delta\mathbf{v}^n, \delta\phi, \delta\mathbf{f}^b, \delta\boldsymbol{\omega}^b, \delta t_r, \delta\dot{t}_r, \delta t_{con}]^T, \quad (1)$$

where $\delta \mathbf{p}^n = [\delta p_N^n, \delta p_E^n, \delta p_D^n]^T$ and $\delta \mathbf{v}^n = [\delta v_N^n, \delta v_E^n, \delta v_D^n]^T$ are the position and velocity errors, respectively, in the navigation (North, East, Down – NED) frame; $\delta \phi = [\delta \phi, \delta \theta, \delta \psi]^T$ is the attitude error; $\delta \mathbf{f}^b = [\delta f_x^b, \delta f_y^b, \delta f_z^b]^T$, $\delta \boldsymbol{\omega}^b = [\delta \omega_x^b, \delta \omega_y^b, \delta \omega_z^b]^T$ are the accelerometer bias error and the angular rate sensor bias error in the body frame, respectively; δt_r is the receiver clock error and $\delta \dot{t}_r$ is its time derivative (clock drift), and δt_{con} is the GNSS inter-system time bias (in this case, between GPS and Galileo time).

For DVB-T2 updates, the state vector $\delta \mathbf{x}_{DVB-T2}$ consists of $15 + p$ states, where p is the number of available DVB-T2 transmitter pairs at a given epoch

$$\delta \mathbf{x}_{DVB-T2} = [\delta \mathbf{p}^n, \delta \mathbf{v}^n, \delta \phi, \delta \mathbf{f}^b, \delta \boldsymbol{\omega}^b, \delta \mathbf{l}]^T, \quad (2)$$

where $\delta \mathbf{l} = [\delta l_1, \delta l_2, \dots, \delta l_p]^T$, δl_p is the TDoA distance bias error for p -pair of DVB-T2 transmitters. This is defined as $l_p = c\delta(\Delta\tau_p)$, where c is the speed of light and $\delta(\Delta\tau_p)$ is the TDoA time bias error. This bias accounts for errors in transmitter clock synchronization, environmental propagation effects, and other unmodeled delays.

B. Kalman filter – proposed observation model

For the GNSS component, the standard tightly coupled observation model for pseudorange and pseudorange rate measurements is utilized, as detailed in [8, 9].

For the DVB-T2 updates, the TDoA observation model is defined as follows. The measured distance difference for the p -th pair of transmitters, $\tilde{r}_{TDoA_p}$, is modelled as

$$\tilde{r}_{TDoA_p} = c\Delta\tau_p = \tilde{d}_p + \tilde{l}_p + \epsilon_r, \quad (3)$$

where $\tilde{d}$ is the true geometric distance difference between the receiver and two DVB-T2 transmitters, $\tilde{l}$ is the true distance bias, and ϵ_r is random measurement noise.

The corresponding predicted TDoA distance, r_{TDoA_p} , based on the INS-derived position for the p -th transmitter pair is

$$r_{TDoA_p} = \left| \|\mathbf{p}_r^n - \mathbf{p}_{tx_{p1}}^n\| - \|\mathbf{p}_r^n - \mathbf{p}_{tx_{p2}}^n\| \right| + l_p, \quad (4)$$

where $\mathbf{p}_r^n$ is the INS-derived receiver position in the NED frame, and $\mathbf{p}_{tx_{p1}}^n$ and $\mathbf{p}_{tx_{p2}}^n$ are the known NED positions of the respective transmitters.

The second component of the DVB-T2 observation model utilizes non-holonomic constraints (NHC) to aid in velocity correction. This approach leverages the fact that the forward kinematic motion of an aircraft is predominantly aligned with the longitudinal axis (v_x^b) of its body frame. Although the lateral and vertical velocity components (v_y^b and v_z^b) are assumed to be theoretically zero under ideal conditions, practical aerodynamic factors—such as the angle of attack, crosswinds, and turbulence—introduce

non-zero disturbances. Therefore, the NHC must be formulated as a soft constraint to accurately reflect these dynamic flight realities.

The measurement innovation (residual) vector $\mathbf{z}_k$ in the proposed Kalman filter is a column vector of dimension $(p+2) \times 1$:

$$\mathbf{z}_k = \begin{bmatrix} (\tilde{r}_{TDoA_1} - r_{TDoA_1}, \dots, \tilde{r}_{TDoA_p} - r_{TDoA_p})^T \\ (-v_{y_{INS}}^b, -v_{z_{INS}}^b)^T \end{bmatrix}, \quad (5)$$

where $v_{y_{INS}}$ and $v_{z_{INS}}$ are the lateral and vertical velocities of the receiver derived from the INS in the body frame.

The Equation (5) can be rewritten in the standard form

$$\mathbf{z}_k = \mathbf{H}_k \delta \mathbf{x}_k + \epsilon_k, \quad (6)$$

where $\mathbf{H}_k$ is the observation matrix with dimensions $(p+2) \times (15+p)$

$$\mathbf{H}_k = \begin{bmatrix} -\mathbf{u}_1 & \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times 6} & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ -\mathbf{u}_p & \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times 6} & 0 & \cdots & 1 \\ \mathbf{0}_{1 \times 3} & \mathbf{h}_{v,y} & \mathbf{h}_{\theta,y} & \mathbf{0}_{1 \times 6} & 0 & \cdots & 0 \\ \mathbf{0}_{1 \times 3} & \mathbf{h}_{v,z} & \mathbf{h}_{\theta,z} & \mathbf{0}_{1 \times 6} & 0 & \cdots & 0 \end{bmatrix}, \quad (7)$$

where $\mathbf{u}_p = (\mathbf{p}_r^n - \mathbf{p}_{tx_{p1}}^n) / \|\mathbf{p}_r^n - \mathbf{p}_{tx_{p1}}^n\| - (\mathbf{p}_r^n - \mathbf{p}_{tx_{p2}}^n) / \|\mathbf{p}_r^n - \mathbf{p}_{tx_{p2}}^n\|$ is the line-of-sight gradient vector of the p -th TDoA measurement with respect to the receiver's position. The terms $\mathbf{h}_{v,y}$ and $\mathbf{h}_{v,z}$ correspond to the second and third rows of the direction cosine matrix (DCM) from the NED to the body frame (C_n^b), respectively. The terms $\mathbf{h}_{\theta,y}$ and $\mathbf{h}_{\theta,z}$ are defined as the respective rows of the matrix product $C_n^b[\mathbf{v}^n \times]$, where $[\mathbf{v}^n \times]$ denotes the skew-symmetric matrix of the velocity vector in the NED frame. The diagram of the system architecture is shown in Fig. 1.

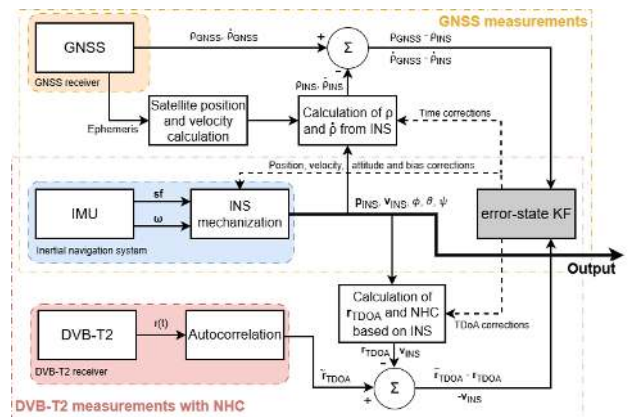


Fig. 1. Schematic diagram of the proposed system with IMU and KF with GNSS (up) or DVB-T2 (down) updates.

As established, the residual vector $\mathbf{z}_k$ and the observation matrix $\mathbf{H}_k$ must be dynamically constructed at each update epoch k , scaling with the number p of available DVB-T2 measurements.

III. DVB-T2

Digital Video Broadcast (DVB) systems as a potential SoP for navigation resource emerged around 2005 – 2010 [10]. Various regional broadcasting standards have been taken into account, e.g., the Advanced Television Systems Committee (ATSC) in North America, Digital Terrestrial Multimedia Broadcast (DTMB) in Asia.

Then Digital Video Broadcasting – Terrestrial (DVB-T/T2) was under focus in Europe [1, 11, 12, 13]. An important feature for its utilization for navigation is the existence of a Single-Frequency Network option [14].

The SFN architecture is a prerequisite for a practical implementation of Time-of-Arrival (ToA) or Time-Difference-of-Arrival positioning [1, 12, 13], as illustrated in Fig. 2. In the context of this paper, the primary motivation for exploiting these signals is to continuously aid the IMU during GNSS outages.

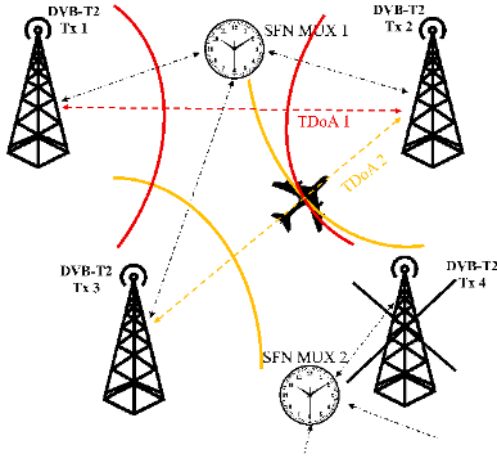


Fig. 2. DVB-T2 positioning using TDoA ranging in Single Frequency Network.

A. DVB-T2 – processing method

The DVB-T2 measurement relies, according to [1], on a computationally efficient signal processing algorithm. The modified autocorrelation function $R[\Delta\tau]$ (Eq. 8) of the received signal $r(t)$ (Eq. 9) itself is used.

This method exploits the fundamental principle of an SFN, also described by the model (Eq. 9): the received signal $r(t)$ is simply a superposition of time-delayed copies of the identical signal $s(t)$ broadcast by multiple transmitters within the SFN, combined with additive noise $w(t)$.

$$R[\Delta\tau] = \sum_{n=0}^{\frac{N}{T_{\Delta}}} \left(\sum_{t=nT_{\Delta}+1}^{(n+1)T_{\Delta}} r[t] \cdot r^*[t - \tau] \right)^2, \quad (8)$$

$$r(t) = \sum_i s(t - \tau_i) + w(t), \quad (9)$$

where N is the total number of samples, and the outer

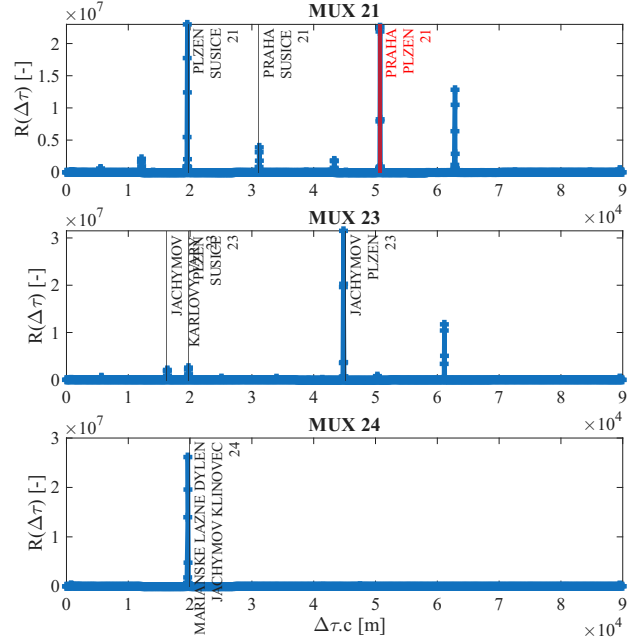


Fig. 3. Resulting cross-correlation according Eq. 8 for signals from three DVB-T2 multiplexes (SFNs) available.

sum represents the non-coherent integration over frames of length T_{Δ} .

Therefore, the resulting autocorrelation function (Fig. 3) contains many peaks on the delays corresponding to the difference between the propagation of individual SFN transmitter's (Tx_i) signals to receiver (Rx). This time difference is therefore denoted as $\Delta\tau$, and its physical meaning represents the time difference of arrival (TDoA).

The $R[\Delta\tau]$ in Fig. 3 presents actual signals captured during the flight test described below. The DVB-T2 transmitters corresponding to individual peaks are marked and listed in Table 1 with their Equivalent Isotropically Radiated Power (EIRP). These transmitters belong to distinct DVB-T2 multiplexes (e.g., MUX 21–24), which operate as independent SFNs. Their geographical distribution, along with the trajectory of the aircraft, is shown in Fig. 4.

Table 1. DVB-T2 transmitters/SFN-MUX used

Transmitter name	MUX	EIRP (kW)	Altitude
Praha	21	100	399
Plzen	21	100	710
Jachymov	23	50	1243
Plzen	23	100	710
Susice	23	100	823
Mar.Lazne-Dylen	24	50	936
Jachymov-Klinovec	24	40	1236

All DVB-T2 SoP navigation methods have difficulties in the reliable identification of individual transmitters and association of measurements [12]. This problem worsens

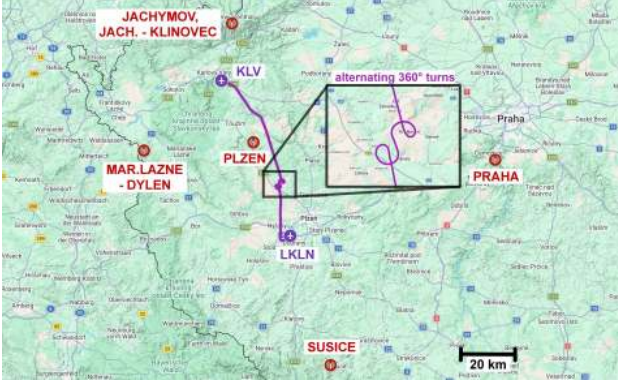


Fig. 4. Map with the flight trajectory (purple) and with the DVB-T2 transmitters (red).

with an increasing number of transmitters in the same SFN.

B. DVB-T2 – signal tracking

The $\Delta\tau$ measurement, corresponding to the distance $\Delta\tau \cdot c$ (in meters) and used as an input for the ESKF, is extracted by tracking the $R[\Delta\tau]$ peaks.

To mitigate the data association ambiguity discussed previously, the tracking algorithm restricts its search window to a proximity of ± 2 km around the expected peak position. For robust peak detection, the Peak-to-Noise Ratio (PNR), defined in (Eq. 10), is employed. This PNR value is used in the noise covariance matrix $\mathbf{R}$ of the ESKF.

$$PNR = 10 \log_{10} \left(\frac{\max R[\Delta\tau]}{\text{mean } R[\Delta\tau]} \right) \quad (10)$$

The tracking result for the pair "PRAHA-PLZEN 21" transmitter, highlighted in red in Fig. 3, is presented in Fig. 5. During the in-flight phase, the resulting error is less than 100 m, sometimes below 10 meters – with the exception of the alternating 360° turns caused by high flight dynamics. The most significant error is observed while the aircraft is at the airport, primarily due to the NLOS channel.

IV. HARDWARE SETUP AND DATA ACQUISITION

Experimental validation was conducted during a 30-minute flight aboard a Tecnam P2006T aircraft (Fig. 6) between the Karlovy Vary (KLV) and Plzen-Line (LKLN) airports (Fig. 4). The core hardware comprised a low-cost Analog Devices ADIS16465 IMU (operating at 200 Hz) and a u-blox NEO-F10T GNSS receiver (GPS/Galileo at 2 Hz) installed in the baggage compartment. DVB-T2 signals were acquired using an ADALM-PLUTO software-defined radio (SDR) equipped with a bandpass filter and an amplifier (Fig. 7). Both the GNSS and DVB-T2 antennas were mounted inside the cockpit (Fig. 8).

The SDR sequentially sweeps three distinct DVB-T2 frequency channels (multiplexes). For each multiplex, it

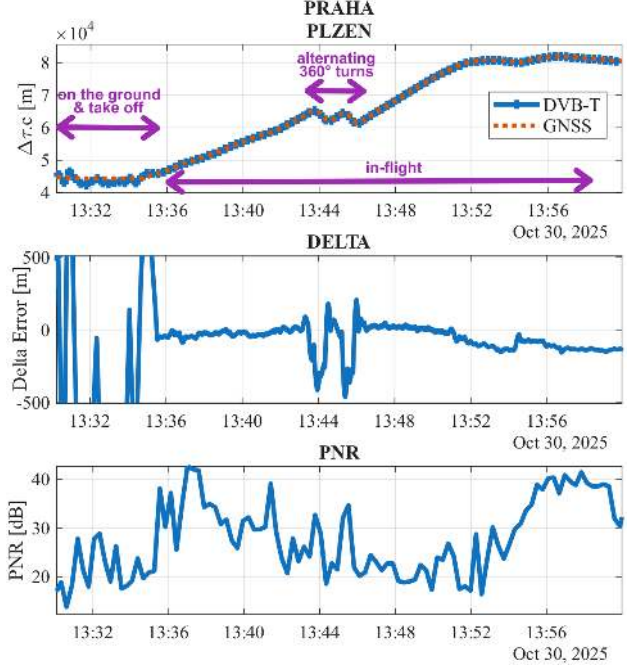


Fig. 5. Tracking of DVB-T2 TDoA PRAHA-PLZEN peak (highlighted in red in Fig. 3) during experiment.



Fig. 6. The Tecnam P2006T aircraft used for experiments.

captures a 50 ms data snapshot, followed by a 5.5 s interval required for data storage and hardware retuning. Due to this sequential sweeping process, correction updates from any specific DVB-T2 multiplex are available at an interval of approximately 17 seconds.

The proposed ESKF processes these multi-rate updates asynchronously, dynamically constructing the design matrix $\mathbf{H}$ and the noise covariance matrix $\mathbf{R}$ based on real-time signal availability and outlier rejection using a fixed residual threshold to discard incorrect associations. Navigation accuracy is evaluated using the Mean Absolute Error (MAE) and Standard Deviation (STD) in the local North-East-Down (NED) frame, alongside the maximum magnitude of the 3D position error ($\max \|\Delta \mathbf{p}\|$) to quantify the worst-case system divergence. These error metrics were calculated against position solutions derived from a Septentrio AsteRx-m2a multi-frequency, multi-constellation GNSS receiver, which served as the reference

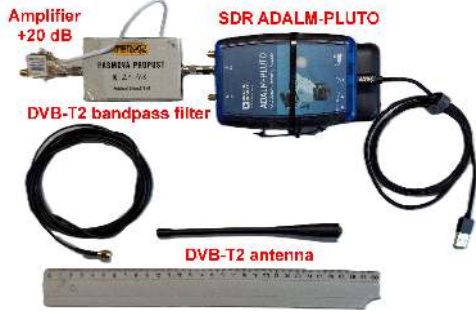


Fig. 7. ADALM-PLUTO with amplifier and bandpass filter.



Fig. 8. DVB-T2 and GNSS antennas inside the cockpit.

ground truth during the experiments.

V. RESULTS

The proposed multi-sensor navigation system, integrating a low-cost IMU with GNSS and DVB-T2, is primarily intended to provide resilience during GNSS outages. Under nominal conditions with uninterrupted GNSS coverage, the performance gains achieved by incorporating DVB-T2 data are marginal compared to a standard IMU/GNSS configuration (see Table 2). This limitation stems from the inherently lower precision of DVB-T2 measurements relative to GNSS positioning.

Table 2. Comparison of system accuracy without GNSS outages: IMU/GNSS vs. IMU/GNSS/DVB-T2 fusion. STD is approximately 1 m in both configurations.

IMU corrections	MAE [m]			$\max\ \Delta\mathbf{p}\ $ [m]
	Δ_N	Δ_E	Δ_D	
GNSS	2.1	1.3	1.1	5.5
GNSS/DVB-T2	2.1	1.2	1.1	5.7

Nevertheless, the core objective of the proposed integration is to validate system performance during in-flight GNSS outages, relying exclusively on DVB-T2 updates. To simulate these degraded environments, GNSS observations were intentionally masked for durations of 60, 300, and 600 seconds. The navigation results from these simulated outage scenarios, benchmarked against a stand-alone low-cost IMU solution, are summarized in Table 3.

As observed in Table 3, the IMU/DVB-T2 fusion significantly reduces the horizontal position drift compared to the stand-alone IMU across all outage scenarios. However, for short outages (e.g., 60 seconds), the stand-alone IMU exhibits a lower MAE in the vertical channel (Δ_D) than the fused solution. This behavior highlights a practical limitation of the method: terrestrial transmitters form a nearly co-planar network below the aircraft, resulting in poor vertical observability (high VDOP). This geometry introduces noise into the vertical state estimation before the inherent IMU drift fully dominates. For longer outages (300 s and 600 s), the unbounded integration error of the stand-alone IMU far exceeds this initial noise, making the DVB-T2 updates crucial for constraining the 3D position.

Quantitatively, the IMU/DVB-T2 integration bounds the MAE to roughly 10 m (with a maximum 3D error of 99 m) during a 60-second outage, and to tens of meters (with a maximum 3D error of 302 m) during a 5-minute outage. Furthermore, even after a 10-minute period without GNSS, the horizontal MAE is constrained to approximately 100 meters, with the maximum 3D position error reaching 441 meters. In contrast, relying solely on the stand-alone low-cost IMU leads to position errors on the order of tens to hundreds of kilometers during the 5- and 10-minute outages. Figure 9 illustrates the estimated trajectories during the 10-minute outage, demonstrating that the fused system remains stable even during dynamic aircraft maneuvers.

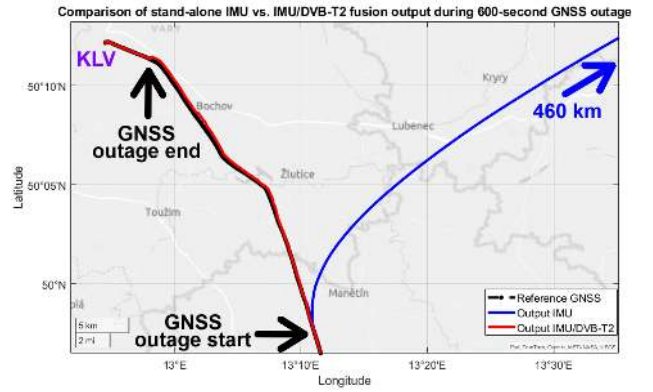


Fig. 9. Trajectory comparison of stand-alone IMU (blue) vs. IMU/DVB-T2 fusion (red) against the reference ground truth (black) during a 600-second GNSS outage.

Ultimately, these results fully align with the expected accuracy capabilities and physical limitations of positioning based on DVB-T2 signals of opportunity. More importantly, this achieved precision is practically viable for aviation applications, serving as a reliable fallback system to maintain situational awareness and ensure safe mid-flight navigation during prolonged GNSS-denied scenarios.

VI. CONCLUSION

This paper evaluated a tightly coupled navigation system integrating a low-cost IMU, GNSS, and terrestrial DVB-T2

Table 3. Comparison of proposed system accuracy during GNSS outages: stand-alone IMU vs. IMU/DVB-T2 fusion.

Data source	GNSS outage [s]	Mean Absolute Error [m]			Standard Deviation [m]			$\max\ \Delta\mathbf{p}\ $ [m]
		Δ_N	Δ_E	Δ_D	σ_N	σ_E	σ_D	
IMU	60	57.9	32.7	7.3	58.4	46.0	4.0	264
IMU/DVB-T2		14.9	7.0	10.7	23.8	8.2	13.6	99.3
IMU	300	501	1007	80.0	554	1208	228	47.6×10^3
IMU/DVB-T2		34.5	67.0	47.1	42.5	72.2	22.5	302
IMU	600	33.8×10^3	101×10^3	19.1×10^3	35.3×10^3	121×10^3	28.4×10^3	465×10^3
IMU/DVB-T2		78.4	119	75.8	94.3	89.1	54.3	441

signals to provide robust resilience during GNSS-denied scenarios. Through real-world airborne experiments, the proposed Error-State Kalman Filter demonstrated its ability to effectively mitigate the unbounded drift of the stand-alone IMU, which otherwise yielded kilometer-level errors within minutes.

Quantitatively, the IMU/DVB-T2 fusion bounded the horizontal position error to tens of meters during a 5-minute GNSS outage, and contained the maximum 3D error to roughly 440 meters after 10 minutes. Ultimately, this research confirms that DVB-T2 signals serve as a practically viable, independent fallback system for manned and unmanned aviation, ensuring essential situational awareness and safe mid-flight navigation when GNSS is compromised (jammed or spoofed).

Future work will focus on utilizing a multi-channel DVB-T2 receiver to overcome the 17-second sequential sweeping interval, enabling simultaneous signal acquisition across all available multiplexes and deploying the system for real-time operation. To enable a more precise performance assessment, additional flight experiments will be conducted using a professional-grade reference INS. Furthermore, to enhance overall accuracy and robustness during prolonged GNSS outages, the multi-sensor fusion framework will be expanded to incorporate additional aiding sensors, e.g., barometric altimeters or visual odometry.

VII. ACKNOWLEDGMENT

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Microcontroller based software defined instruments as an economical substitute of virtual measurement instruments

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Abstract – This paper presents a concept of low-cost, multifunctional Software Defined Instruments (SDIs) designed for education and rapid prototyping. Originating from the need to provide students with affordable access to measuring instruments, the SDI platform utilizes modern microcontrollers (MCUs) with suitable peripherals to implement devices such as oscilloscopes, voltmeters, counters, and function generators, entirely in software. The solution combines MCU (e.g., STM32 Nucleo) with custom firmware and a PC application, enabling hands-on experimentation in laboratories and at home. Various generations of SDIs have been developed, ranging from terminal-based tools to more advanced GUI-driven applications, and hundreds of students have successfully used multiple courses in them. The SDI concept supports flexibility, scalability, and cost-effectiveness, making it ideal for teaching basic measurement techniques, circuit behavior, and embedded system development.

Keywords: *Virtual; Measurement; Instrument; Microcontroller; Software*

I. INTRODUCTION

In technical experiments, whether in university lectures or development laboratories, standard measuring instruments such as voltmeters, oscilloscopes, signal generators, and counters are commonly used. However, their variants, in the form of Virtual Instruments (VI), are increasingly becoming an integral part of modern experimental setups. A notable example includes the Analog Discovery 2 and 3 modules from Digilent, whose technical parameters often surpass the basic requirements for laboratory experiments [1]-[3]. Another example is National Instruments NI Elvis, which integrates VI technology directly into an experimental workstation with a breadboard, making it a valuable tool for teaching electronics and fundamental circuit analysis [4] or [5]. These virtual instruments offer excellent technical capabilities, making them well-suited for educational purposes as well as simple development laboratories.

Despite their advantages, a major drawback of VIs is their cost, which makes it impractical to provide, for

example, each student with their own system for both in-class and home experiments. One possible alternative is remote access to laboratory equipment, allowing students to interact with real devices from a distance [6]. However, this approach lacks the hands-on experience of physically assembling circuits, troubleshooting errors, and refining their designs, skills that are essential for a development engineer. Therefore, finding a cost-effective yet practical solution that combines the benefits of VI with hands-on experimentation remains a significant challenge in engineering education. The proposed Software Defined Instruments (SDI) based on microcontrollers is one of the possible solutions.

II. MOTIVATION

More than a decade ago, the challenge of providing students with sufficient measurement instruments was addressed at the Department of Measurement, Czech Technical University (CTU) in Prague. In a microprocessor instrumentation course, students designed a 4½-digit digital voltmeter based on dual-slope integration as their final project, requiring work in labs and at home. However, the limited access to oscilloscopes outside the lab hindered their ability to debug hardware and software. Although devices like the Analog Discovery 2 would be ideal, their cost made them impractical. This led to the development of a virtual oscilloscope using the STM32F303RC Discovery kit, later replaced by the Nucleo STM32F303RE. With support from STMicroelectronics, each student receives two Nucleo boards, one for control and one running the LEO (Little Embedded Oscilloscope) software, functioning as an oscilloscope, voltmeter, signal generator, and power source [7]. Combined with a quality breadboard, this accessible solution eliminates the need for PCBs. All instrument functions are implemented in software, forming the basis for the Software Defined Instrument (SDI) concept.

III. DEVELOPMENT AND VARIANTS OF SDI SOLUTIONS

Using SDI to meet the need for affordable measuring instruments proved successful. Initially, 50 SDI-based oscilloscopes were used in the "Practical Electronics Course." In subsequent years, the number grew to 130,

allowing students to demonstrate the fundamental principles of signal behavior in electronic circuits. Additionally, it was important to ensure that each student could keep the SDI for future independent projects throughout their studies. A solution was sought to integrate the SDI directly on the breadboard to achieve this. The result was a minimized SDI variant implemented using a compact STM32F042 MCU with a USB interface [8], see Fig. 1.

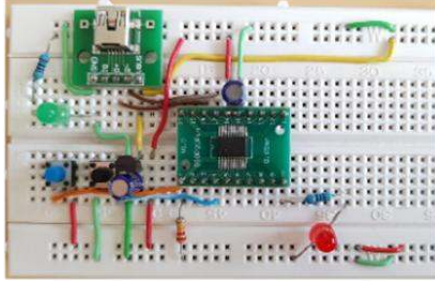


Fig. 1 MCU STM32F042 with PCB adaptor on the breadboard

Aside from the MCU mounted on a universal adapter, the setup includes a 3.3V regulator, blocking capacitors, and an LED. Power and control are provided via a USB interface to the PC. The Zero eLab Viewer PC application (Fig. 2) was developed for SDI control, offering functions like a basic oscilloscope, including sampling speed adjustment, trigger level settings, and averaging [8]. It also features a simple FFT spectrum analyzer, and recorded signal data can be exported as a .csv file for further processing.

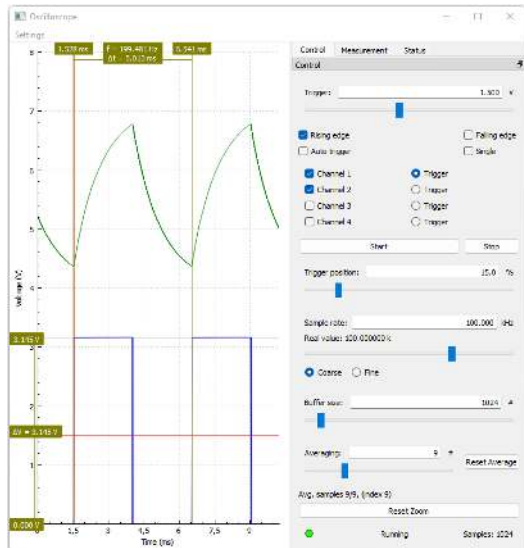


Fig. 2 PC application Zero eLab Viewer

The MCU connects to the PC via an internal USB interface using the CDC, appearing as a Virtual COM Port (VCP). This enables easy adaptation to an external, low-cost UART/USB converter for MCU communication over UART. This approach led to the development of the

smallest SDI implementation yet, featuring the STM32G030J6M6 in an ultra-compact 8-pin package that communicates via UART. This SDI significantly outperforms the original design, offering a sampling frequency of up to 2 MS/s, 12-bit ADC resolution, and a recording length of 2 kS. The STM32F042-based SDI has been successfully used in undergraduate preparatory courses for over nine years, benefiting more than 1,000 students [9]. While its parameters are somewhat limited, they have proven sufficient for basic experiments. The SDI solution and other implementations discussed in this article, including firmware and PC applications, are available for download at www.leo.fel.cvut.cz.

IV. SOFTWARE SOLUTIONS

4.1. Dedicated PC Application

Each SDI requires a dedicated PC application alongside its firmware. A critical aspect of development is the selection of a suitable communication protocol, including control commands and message structure. In practice a single PC application was successfully used across multiple SDI implementations based on different MCUs (STM32F042, STM32F103, STM32G030, STM32L072, STM32C071), with only the firmware adapted to ensure compatibility [9].

However, protocol limitations can restrict further functionality. For example, early SDIs used a two-byte format to transmit frequency parameters for a pulse generator, which was sufficient for basic applications but inadequate for modes like Equivalent Time Sampling (ETS), when SDI oscilloscopes began to use this mode, that require finer resolution. Similarly, new MCUs with enhanced capabilities often cannot be fully utilized due to constraints of the original protocol. Thus, selecting a well-designed, scalable communication protocol is essential. While standard protocols such as SCPI (Standard Commands for Programmable Instruments) could address many limitations, their complexity and longer message formats may place an undue burden on SDIs based on simpler microcontrollers.

A. The Usage of the Terminal Program

For SDIs that output numerical values rather than waveforms, such as voltmeters, counters, or RC meters, or for devices that require only parameter input (e.g., function or pulse generators), the PC application can be replaced by a terminal program supporting ANSI escape sequences (e.g., PuTTY). These sequences allow control of cursor position, text overwriting, font and background colors, enabling a basic yet interactive user interface [10], see Fig. 3.

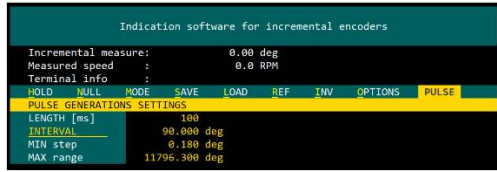


Fig. 3 SDI using PuTTY and ANSI Escape Sequences

Terminal programs are available across virtually all operating systems, eliminating the need to resolve cross-platform compatibility issues. Another benefit is simplified communication: control is handled through character input directly from the keyboard, based on a displayed menu, without the need for a custom communication protocol between firmware and PC. A minor drawback is that, due to SDI simplicity, most control commands are limited to single-character inputs (except for numerical values). However, using a terminal significantly accelerates SDI development, as only firmware changes are required—no adjustments to a PC application are necessary. The main limitation of this approach is the inability to display signal waveforms.

B. Combined PC application

The challenge of creating a universal PC application compatible with SDIs based on various MCUs was addressed by designing a semi-universal solution. As part of a student project, the Data Plotter application was developed, combining features of a dedicated PC application and a terminal interface. Unlike typical SDI applications, which act as a communication master (initiating and managing interaction), Data Plotter functions more like a slave; it passively receives messages from the MCU, processes them, and displays the results [11].

The format of these messages, sent from the MCU to the PC, is predefined and structured to support various types of measured data. This includes formats suitable for oscilloscope or logic analyzer outputs, where resolution, time parameters, and record length are embedded directly in the message. As a result, the waveform can be rendered on the PC without requiring further interaction, enabling a more streamlined and flexible development process. The Data Plotter PC application includes a terminal window with ANSI escape sequence support for SDI control. This window allows users to input single characters or multi-character text commands sent from the PC to the MCU. Responses from the MCU are displayed directly in the terminal window. Mouse-based interaction is implemented using a unique method: characters in the terminal are color-coded and rendered as invisible text (e.g., white font on a white background). Clicking on a specific position returns the corresponding character command to the MCU. An example is in Fig. 4, where

various characters are in illegible white font on a white background.

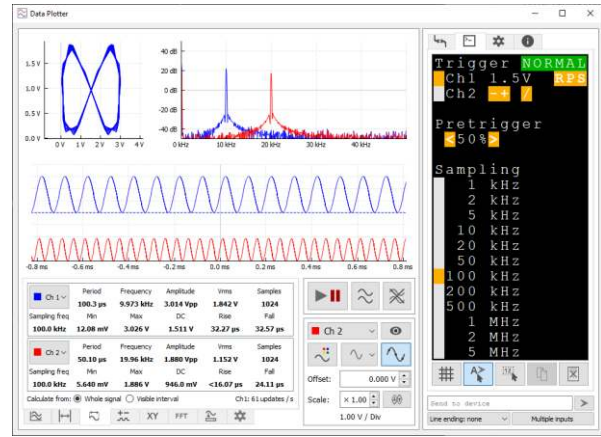


Fig. 4 PC application Data Plotter

The MCU fully controls the layout and behavior of this interactive terminal field via messages sent to the PC application. While mouse interaction is limited to sending single-character commands, this limitation is addressed by switching between multiple interactive panels within the terminal. Each panel interprets the same character differently, effectively encoding commands by both panel index and character. The Data Plotter application has successfully implemented a range of SDIs, including signal generators (with waveform previews), oscilloscopes, logic analyzers, and counters. The application provides powerful tools such as horizontal and vertical zoom, data export, and FFT spectrum analysis for more advanced instruments like oscilloscopes. Available at www.leo.fel.cvut.cz, Data Plotter enables the development of both simple and advanced SDI without requiring dedicated PC software. It has proven valuable in education and the rapid prototyping of specialized virtual measuring instruments.

V. MCU-BASED SDI IMPLEMENTATION AND ITS LIMITATIONS

SDI implementations primarily utilized MCUs with ARM Cortex-M cores (M0, M0+, M3, M4F), specifically from the STM32 family, due to their suitable peripherals such as high-speed ADCs, analog watchdogs, multi-channel DMA, and advanced timers. An exception was using the Arduino module based on the ATmega328 MCU, which, despite its limited performance and peripheral set, demonstrated that even basic microcontrollers can support simple SDI implementations. This included functionality such as an oscilloscope, logic analyzer, voltmeter, pulse generator, and reciprocal counter, sufficient for illustrating fundamental measurement principles in educational settings.

Some SDI realizations were based on ready-made development boards such as the Nucleo F303RE, Raspberry Pi Pico, and BluePill STM32F103. However, the majority used individual MCU chips (e.g., STM32F042, STM32G030, STM32G431, etc.) soldered onto PCB adapters and inserted into high-quality solderless breadboards, as can be seen in Fig. 1. This approach proved to be both fast and cost-effective. For PC connectivity, the USB interface was used natively through the MCU or via an external USB/UART converter in cases where the MCU lacked an integrated USB controller.

A. Limitations of SDI

Implementing devices based on SDI using an MCU inherently involves several limitations. Nevertheless, SDI remains applicable in many areas and offers valuable functionality. On the other hand, some limitations can be improved by special methods. For example, for certain measurement applications, such as assessing the circuit's response to a step input, the maximum sampling frequency (from hundreds of kHz to units of MHz) may become inadequate. The ETS method was validated for most of the implemented SDIs, where the input signal of the circuit is driven by a pulse generated by the PWM block in the MCU. The circuit's output signal is then digitized using an ADC, with an analogue bandwidth ranging from approximately 20 to 50 MHz for the MCUs (except the Arduino with the ATmega328).

B. Oscilloscope Limitations

Based on our experience, the most used SDI functionality is the oscilloscope. The key parameters are the sampling rate, record length, and number of channels. For newer MCUs with increasing SRAM capacity, the record length is typically not an issue. However, the sampling frequency poses a different challenge. MCUs that typically use CMOS SAR ADCs with 12-bit resolution usually face limitations in the range of 500 kS/s to 5 MS/s. Simpler MCUs not designed for measurement applications often feature a single ADC with an input multiplexer. This setup allows multiplexed signal recording across multiple channels but reduces overall sampling frequency. In contrast, MCUs designed for measurement tasks often include multiple independent ADCs, ranging from two to four, which addresses this limitation. For example, the STM32F303RE can achieve up to 4x 4 MS/s. For MCUs with multiple ADCs, the maximum sampling frequency can be increased by reducing the channels and utilizing the interleave mode, where two or more ADCs alternately sample the same input signal. With this approach, the STM32F303RE can achieve up to 12 MS/s with a single channel [13]. However, as the sampling rate increases, the available sampling time (the time for charging the internal sampling

capacitors in the SAR ADC) decreases, necessitating a low internal resistance in the signal source, especially if no analog buffer is used. Therefore, some MCUs designed for measurement applications include an integrated analog buffer to address this challenge.

C. Counters Limitations

MCUs typically include timers that can be utilized to implement universal counter functions. These functions include direct frequency measurement through pulse counting and reciprocal frequency measurement by measuring the duration of one or more pulse periods. Another function is measuring pulse delays across two channels. An input capture block can be employed for pulse length and duty cycle measurements linked to the DMA (Direct Memory Access) block, which enables fast storage of edge positions in memory without CPU involvement.

The frequency of externally counted pulses is typically limited to half the system clock frequency. This enables the implementation of counters with a maximum frequency of up to 70 MHz, even in simple MCUs. In MCUs equipped with high-speed timers, the maximum measurable input frequency can be even higher. This range is generally sufficient for basic experimental setups.

D. Function Generators

Function generator requires an internal Digital to Analog Converter (DAC) in the MCU, which is usually not the case with basic MCUs. DAC speed allows for generating signals with a sample rate of typically up to 2 MS/s. Generation is usually done using a DMA block and a timer based on samples stored in memory. Another method uses a program-defined DDS (Direct Digital Synthesis) with a fixed table of values. A special case is calculating the generated signal's sample values (e.g., using a polynomial function) in real time. This method requires relatively high computing power and is limited to lower signal frequencies.

The method of generating a signal from FLASH memory stored values allows for changing the frequency of the generated signal in steps determined by the timer division ratios. Signal generation methods based on samples stored in the internal SRAM memory were used to ensure the possibility of fine adjustment. Fine frequency adjustment is achieved by changing the length of the table of function values for a given desired signal frequency, which is calculated for a given frequency either by a PC application or these functional values are determined directly by the MCU [7].

VI. OVERCOMING THE LIMITATIONS OF SDI

As already mentioned, the fundamental limitation of an SDI oscilloscope that relies on an internal SAR ADC is its relatively low sampling rate, typically ranging from a few hundred kS/s to a few MS/s.

If an SDI oscilloscope has to be used for circuit behaviour analysis, e.g. to measure the step response, see Fig. 5, a specific technique can be applied. It leverages the fact that both the PWM signal used to generate the step input signal, and the ADC sampling control signal share the same timing reference derived from the MCU's internal clock generator.

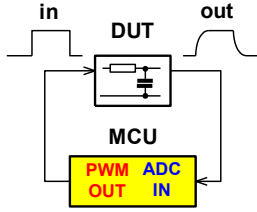


Fig. 5 A step response measurement

Assuming a sufficiently wide analog bandwidth (ABW) at the ADC input, equivalent-time sampling (ETS) can be employed. Using ETS, the maximum equivalent sampling frequency can theoretically approach the frequency of the MCU's clock generator. As a result, even very simple and low-cost MCUs based for example, on the ARM Cortex-M0+ core can achieve an equivalent sampling rate f_{SEQ} on the order of tens of MHz [12].

The principle of ETS for step-response analysis is based on generating an excitation signal using a PWM generator at frequency f_{PWM} , while the signal is sampled at a rate f_{SAMP} that is only slightly lower than f_{PWM} (1).

$$f_{SEQ} = \frac{f_{PWM}}{f_{PWM} - f_{SAMP}} \cdot f_{SAMP} = \frac{f_{PWM}}{\Delta f} \cdot f_{SAMP} \quad (1)$$

The control signals for the PWM generator are derived from a clock source with frequency f_{DAQ} , which is also used to generate the ADC sampling control signal. The frequency f_{DAQ} may, but does not necessarily have to, match the maximum system clock frequency of the MCU. This is the case, for example, with the Raspberry PI Pico MCU.

For instance, as (1), with an MCU operating at $f_{SYS} = 64$ MHz, using $f_{PWM} = 10$ kHz and $f_{SAMP} = 9998.437744$ Hz (which corresponds to 9998.437744 S/s) yields a maximum achievable ETS 64 MS/s. With $f_{SAMP} = 9996.875976$ Hz, the resulting equivalent sampling rate is 32 MS/s.

The key idea is that the PWM signal is generated by dividing the f_{DAQ} frequency by a constant N , while the ADC sampling control signal is generated by dividing the same clock source by a constant M . Then, the equivalent sampling frequency is given by

$$f_{SEQ} = \frac{f_{DAQ}}{M-N} \quad (2)$$

where M and N are integers very close to each other. This implies that an SDI oscilloscope can operate with an equivalent sampling rate of

$$f_{SEQ} = \frac{f_{DAQ}}{n} \quad (3)$$

where $n = M - N$, or equivalently $M = n + N$.

A prerequisite for the practical use of ETS in SDI devices is a sufficient ABW. Assuming the input is driven by a low-impedance source, the typical ABW ranges from 20 MHz to 40 MHz. It can be approximately estimated based on the behavior of the equivalent RC circuit, with the time constant $\tau = R \cdot C$, where R is the resistance of the sampling switch (typically hundreds of ohms, or the multiplexer resistance if present) and C corresponds to the sampling capacitor of the ADC input circuit (order of a few pF).

This approach allows for certain experiments to partially overcome the limitations imposed by the maximum achievable sampling rate of the internal SAR ADC in the MCU, see Tab. 1.

TABLE 1. HIGHEST SAMPLING RATE USING ETS FOR SELECTED MCU

MCU type	RTS (MS/s)	ETS (MS/s)
STM32F303RE	4	72
STM32G431	4	52
STM32F042	1	48
STM32C071	2	48
STM32G031	2	64
RP2040	0.5	48

Tab. 1 presents a comparison of MCU sampling performance under real-time sampling (RTS) and equivalent-time sampling (ETS) based on the configurations used in performed experiments. An interesting example is the Raspberry Pico 2040, where the use of ETS increases the equivalent sampling rate approx. hundred times.

VII. DISCUSSION

SDI instruments have various limitations, but their key advantage is that, apart from the MCU itself, they require very few additional components. This allows, for example, students, to assemble SDI on a breadboard with minimal guidance. This approach is based on more than ten years of teaching experience at Czech Technical University in Prague, Faculty of Electrical Engineering, Department of Measurement. Thanks to simplicity, practical tasks could continue in selected courses even during the special situations like during COVID-19 pandemic. Each student assembled an SDI device at home (e.g., voltmeter,

oscilloscope, pulse generator) and performed a series of experiments according to the assignment.

An interesting observation was that students demonstrated greater independence in problem-solving than they typically would at a university's laboratories with teachers and standard measurement instruments.

Furthermore, since the SDI devices use a PC for control and visualization, remote instruction was feasible. Through MS Teams, instructors could observe the signals from the students' SDI oscilloscopes, and when combined with a camera monitoring the student's workspace, it was possible to provide guidance and direct troubleshooting in real time.

SDI proved to be a very low-cost alternative to virtual instruments, suitable not only for use in university laboratories but also for student home projects.

VIII. CONCLUSION

As modern MCUs continue to evolve in performance and features, their potential applications have greatly expanded. Laboratory experiments have shown that MCU-based SDIs are a practical, multifunctional alternative to conventional measuring instruments—combining functions like generator, voltmeter, counter, and oscilloscope in a single module. This approach provides a cost-effective solution for equipping large groups of students in hands-on electronics and instrumentation courses. SDIs can also be integrated into larger systems, offering in-situ diagnostics and fault analysis without requiring additional equipment.

Furthermore, by employing the ETS method, even a relatively simple MCU like the STM32G030 can achieve an equivalent sampling frequency of 64 MS/s, which has proven sufficient for various experiments.

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Exploring a new indirect approach to latency measurement

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Abstract – This paper explores a novel indirect approach to measure latency for digital signal processing systems. Current approach of measuring real-time DSP systems is by calculating the correlation of the input and output signal in time domain which requires high computational power and without interpolation delivers only 10.41 μs latency precision with 48 kHz sampling frequency. Our methodology suggests an indirect latency measurement which requires minimal computational power, uses basic lab equipment and delivers uncertainty $u_{95\%} = 0.135 \mu\text{s}$; two orders higher than non-interpolated correlation.

Keywords: Latency measurement, digital signal processing, Daisy Seed, frequency analysis.

I. INTRODUCTION

Measuring the latency of signal processing systems can be complicated, and requires high computational power with an industry-standard approach convolves the input and output signals in the time domain. In this article, we present a novel indirect approach to measuring the signal path's latency that delivers high-precision results with minimal required hardware (an oscilloscope with frequency analysis).

II. STATE OF THE ART

Latency on the signal path is a crucial parameter that affects the user experience in many scenarios. Latency affects the timing and phase of the audio signal arriving at the user and needs to be set with precision, for example in multi-channel audio systems (theatres), or limited below user's acceptable threshold (for example for performing music). [1], [2], [3]

The current methodology for measuring signal latency between two sampled signals is mostly based on either non-interpolated correlation, where the latency resolution corresponds to the sampling frequency (with 48 kHz, minimal delay resolution 20.83 μs), or interpolated correlation, which can deliver sub-sample precision. Both of these methods require calculating the correlation of two signals, which requires significant computational

power. [4] - [9]

III. PROPOSED METHOD

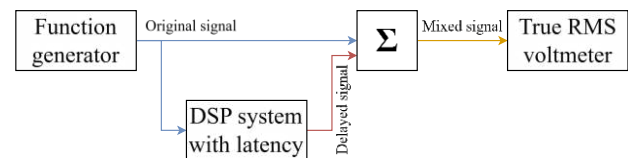


Figure 1: Block diagram of the measurement principle.

The methodology principle exploits the delay time propagating through the system as a phase shift. When the original and delayed signals are summed (see the measurement block diagram in Figure 1), the output's RMS is proportional to the signal's phase shift, e.g., its latency. In Figure 2, we can see an example of 1ms latency affecting the output added signal of 750 Hz. When the input and delayed signals are summed, the resulting signal's RMS value corresponds to their phase shift (e.g., latency). A special case occurs when the signals sum with opposite phase shifts, and the output RMS drops to zero. These special cases can be used in a simplified measurement methodology; see the Discussion chapter.

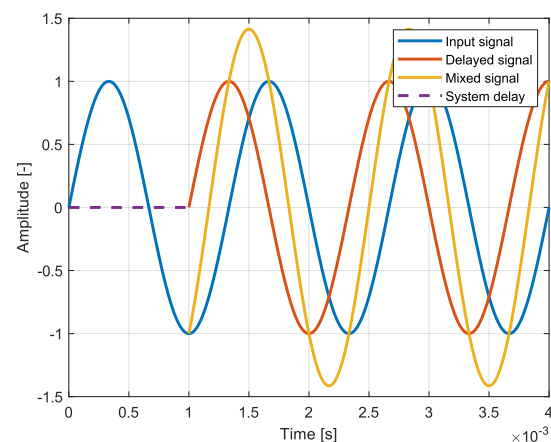


Figure 2: Example of latency affecting the phase and the summed output signal.

The output (additively mixed) signal is equivalent to Equation (1).

$$y_{\Sigma,rms}(f, t, t_D) = \text{RMS}(\sin(2\pi ft) + \sin(2\pi f(t + t_D))) \quad (1)$$

Where f is the signal's frequency, t is time and t_D is the delay. When we measure the RMS over complete cycles, the Equation can be adjusted for a direct RMS value of the additively mixed signals dependent on frequency into Equation (2).

$$y_{\Sigma,rms}(f, t_D) = \sqrt{2}|\cos(\pi f t_D)| \quad (2)$$

One can see the frequency spectrum of the complete measuring system in Figure 3 with 1ms and 0.3ms delay in the audible spectrum (20 Hz to 20 kHz). We can see that the delay time defines the period in the frequency spectrum.

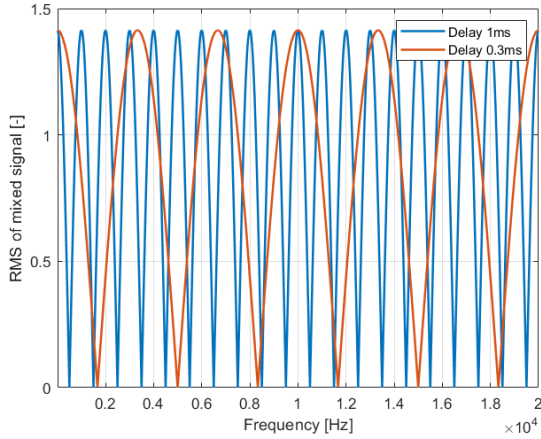


Figure 3: Frequency spectrum of the measurement system based on Equation (2).

Since we know the relation between the frequency and latency (from Equation (2)), we can analyze the frequency spectrum of the resulting system with a frequency analyzer (oscilloscope with a generator) and fit the resulting data with the mathematical model.

IV. EXPERIMENT

To confirm the hypothesis, we set up a measurement experiment (see the wiring block diagram in Figure 4). The measurement uses the frequency analysis of the Keysight EDUX1052G oscilloscope in a dual-channel ratiometric configuration and a custom DSP program running on a Daisy Seed microcontroller.

The Daisy Seed microcontroller has a built-in asynchronous stereo audio codec, PCM3060, with 24-bit Delta-Sigma ADC and DAC set to $f_s = 48$ kHz sampling frequency. The program copies the input signal to the output with an adjustable delay using the delay buffer with

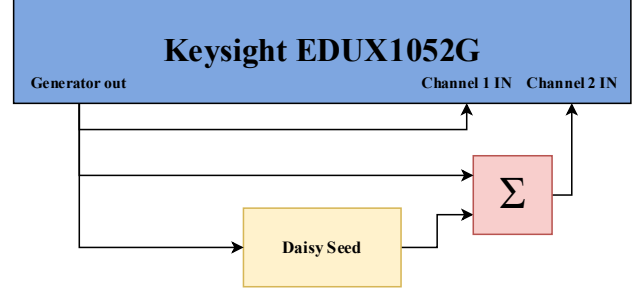


Figure 4: Block diagram of the measurement experiment using dual-channel ratiometric configuration.

a resolution of the sampling period $T_s = 1/48000 \text{ s} = 20.83 \mu\text{s}$. There is a basic delay $T_{D,b} \sim 880 \mu\text{s}$ happening due to the program's window buffer size and then additional adjustable delay T_D is added. Therefore, the overall delay is equal to:

$$T_{D,total} = T_{D,b} + T_D \doteq (880 + T_D) \mu\text{s} \quad (3)$$

The Σ summing block is constructed by two $R_\Sigma = 10\text{k}\Omega$ resistors in a passive additive mix configuration.

The signal from the oscilloscope's generator output is split into three branches. The channel 1 input of the oscilloscope for remeasuring the frequency and amplitude of the generated signal, the Daisy Seed microcontroller input, and the passive additive mix. The Daisy Seed delays the signal, then mixes it in the passive mix with the original signal, and sends it to the channel 2 input of the Keysight oscilloscope. The oscilloscope measures the amplitude and corresponding frequency and creates Bode plot data for further processing.

We measured the transfer function of the Daisy Seed microcontroller on its own first, to select the frequency bandwidth for further processing. The results and the selected frequency bandwidth (from 50 Hz up to 20 kHz) are in Figure 5.

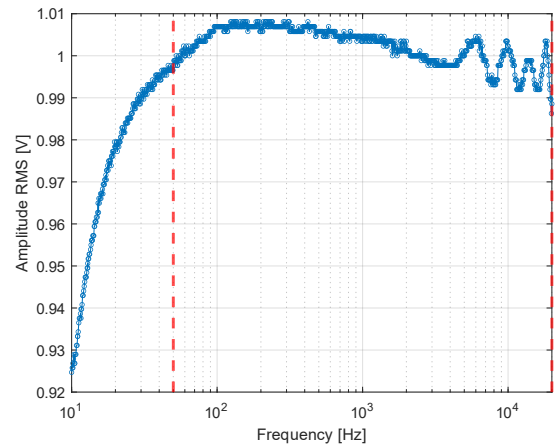


Figure 5: Daisy seed transfer function.

The frequency analysis for the latency measurement was set to 1 V_{pp}, frequency bandwidth from 50 Hz up to 20 kHz and to 1000 frequency points of measurement. We selected an additional delay as $T_D = 0, 1, 3, 5, 10$ ms, measured each frequency analysis and fitted the resulting data with the model (from Equation (2)) using the MATLAB software and curve fitting toolbox. An example of the fitted transfer function for the total delay of $T_{D,total} = 880 \mu\text{s}$ is in Figure 6.

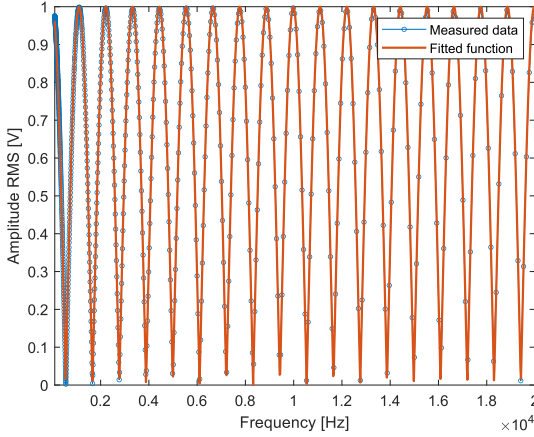


Figure 3: Measured data with fitted model for total delay $T_{D,total} = 880 \mu\text{s}$.

From the fitted model, we exported the measured delay $T_{D,meas}$ and the 95% uncertainties $u_{95\%}$ (half of the 95% confidence interval).

The measured latency for the $T_{D,total} = 880 \mu\text{s}$ was measured as $T_{D,meas,880\mu\text{s}} = 901.397 \mu\text{s}$. This is due to the SW structure of the measured DSP with adjustable latency. It introduces a minimum added latency of one step of the resolution frequency $T_s = 1/48000 \text{ s} = 20.83 \mu\text{s}$ to the overall delay $T_{D,total}$, therefore the $T_{D,meas,880\mu\text{s}} = (901.397 - 20.83) \mu\text{s} = 880.567 \mu\text{s}$. The results are in Table 1.

Table 1: Results

$T_{D,total} [\mu\text{s}]$	$T_{D,meas} [\mu\text{s}]$	$u_{95\%} [\mu\text{s}]$
880	880.567	0.045
1880	1880.288	0.077
3880	3879.661	0.118
5880	5879.037	0.135
10880	10877.500	0.123

V. DISCUSSION

The methodology exhibits high-precision latency measurements for digital signal processing systems with minimal hardware requirements (two resistors and a frequency analyzer). Processing the results can be achieved with less computational power than the

convolution standard by fitting the data with a goniometric model depending on frequency (which corresponds to the desired latency).

The methodology offers a simple way to find initial conditions by finding a first minimum in the measured frequency spectrum f_{min} and calculating the corresponding delay estimation by adjusting the equation (2) to $T_{D,est} = 1/(2f_{min})$. Due to the precise initial conditions, the MATLAB algorithm solves the problem as a direct, closed-form Linear Least Squares (LLS) problem, which drastically lowers the computational complexity. Therefore, the computational complexity reduces to $O(M)$, where M is the number of measured frequency points ($M = 1000$). To acquire same precision, the correlation would have to be interpolated by a factor of $L = \frac{T_s}{u_{95\%}} = \frac{20.83}{0.135} \approx 154$. The computational complexity would be $O(LN \log(LN))$, where the typical window size is $N = 2048$.

A simpler algorithm for fitting the measured frequency analysis can be achieved and will be further researched. Fitting algorithms are very sensitive to the initial frequency estimate; therefore, estimating the frequency from the local minima in the spectrum would improve and robustify the measurement.

Searching for the period in the resulting frequency analysis (e.g., the delay) is potentially resistant to the signal being affected in the measured DSP system, for example, non-unity gain, clipping, frequency filtering, or different signal shapes from the generator. This behavior will be further researched.

We see an opportunity to use this phenomenon in another measurement type: by tuning the generator's frequency, searching for the minimum of the combined mixed output signal, and computing the delay based on the minimum's frequency. This algorithm would require minimal computational power and would be ideal for on-site latency measurements.

VI. CONCLUSION

The measurement approach was pitched with its basic mathematical fundamentals. The experiment was set up, and multiple delays were measured using this approach, with the expanded ($k = 2$) uncertainty $u = 0.135 \mu\text{s}$ corresponding to a sampling rate of 7.4 Mhz used for the convolution standard. To compare, the basic resolution of noninterpolated correlation at 48 kHz sampling frequency is $20.83 \mu\text{s}$. Therefore, this methodology delivers results two magnitudes better than the basic resolution; moreover, without the need to calculate correlation.

ACKNOWLEDGEMENT

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Detection of Mercury Ion in Water Samples Using a Low-Tech EPS-Coated Interdigitated Capacitive Sensor through S-Parameter Analysis

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Abstract – This work presents the experimental characterization of a low-tech interdigitated capacitive (IDC) sensor designed for mercury (Hg^{2+}) detection in liquid media. The sensor response was evaluated using scattering parameters at 10 MHz, with and without a sensitive extracellular polymeric substance (EPS) coating. As results, statistical analysis based on Student's t-test showed that the uncoated sensor detects Hg^{2+} concentrations as low as 10^{-6} M, whereas the EPS-coated sensor reached a detection limit of 10^{-5} M. Although the EPS layer reduced the detection limit, it improved sensitivity by at least four times. Also, standard deviation values remained below 0.06 dB and 0.7° for S_{11} , and below 1.3 dB and 2.2° for S_{21} , indicating good repeatability. Overall, the proposed IDC sensor offers a competitive detection limit at an exceptionally low fabrication cost of USD 0.05, without requiring advanced technological facilities, making it suitable for decentralized and low-resource environments.

Keywords: extracellular polymeric substance; interdigitated capacitive sensor; mercury detection; low-tech.

I. INTRODUCTION

The Amazon Basin, covering approximately 7 million square kilometers, is one of the most important hydrographic regions in South America and a vital water source for local populations, especially riverside communities that often rely directly on untreated water from natural sources [1]. Among the main pollution sources in this basin is illegal mining, which uses mercury

for amalgamation and releases it into the aquatic environment, contaminating rivers, sediments, biota and exposing human populations through water and fish consumption [2-6]. A systematic review was carried out in [3] highlighting that hair-mercury concentrations exceeded the safe threshold of 6×10^{-6} $\mu\text{g/g}$, as recommended by the World Health Organization (WHO), leading to both cognitive impairments in children and neuro-behavioral effects in adults.

Given this context, continuous monitoring of the waters in Amazon Basin is essential to detect and quantify mercury concentrations to prevent environmental and human health damage. To be effective in remote or resource-limited regions, such monitoring must rely on low-cost, easy-to-operate devices that can be fabricated in basic laboratories, with detection limits compatible with regulatory standards.

Accordingly, some studies have developed devices such as gold-coated microcantilevers for Hg^{2+} detection, achieving thresholds of approximately 10^{-11} M, but requiring gold film deposition, masks, and high-cost instrumentation [7]. Other studies have employed interdigitated capacitive (IDC) sensors for mercury detection based on the allowable limit by WHO of 10^{-6} M in liquids [8]. However, the materials used for fabrication, such as gold/platinum electrodes [9] or complex nanostructures, like graphene [8] or graphite oxide [10] or carbon nanotubes [11], and sophisticated deposition processes, such as magnetron discharge sputtering [12], increased the cost and demanded specialized personnel or microfabrication infrastructure.

As a lower-cost alternative, previous studies have demonstrated IDC sensors using copper electrodes and

chitosan films, but with analysis limited to the S_{21} parameter [13], which may limit the interpretation of sensor behavior, particularly regarding reflection phenomena and impedance matching at the sensing interface.

Another recent approach used an extracellular polymeric substances (EPS) functional layer on a Love-wave acoustic sensor, achieving a wide detection range ($\sim 10^{-2}$ M to 10^{-10} M) and good correlation between S_{11} and S_{21} , even in turbid water [14]. However, this sensor still presented a relatively high cost, around 70 USD, and needed to be fabricated in a microelectronics laboratory with advanced technical facilities. To overcome the high-cost and high-tech devices, we propose the development and characterization of a low-cost (<1 USD) IDC sensor functionalized with an EPS layer, using S-parameters responses of the device. The objective is to achieve sensitivity and detection limits compatible with potability standards of WHO [8], enabling preventive and continuous monitoring of mercury in vulnerable basin waters, thus facilitating application in remote areas where laboratory infrastructure is limited.

II. METHODOLOGY

The methodology adopted in this work was organized into four main stages using tools and facilities available in our standard academic laboratory: manufacture of the IDC sensor, preparation of mercury samples, preparation of EPS sensitive coating, and experimental procedure.

A. Manufacture of the Interdigitated Capacitive Sensor

A two-port IDC sensor with five pairs of interdigitated fingers was designed and fabricated on a 0.8 mm thick FR4 Printed Circuit Board (PCB) with copper conductors using a Technodrive 3 CIF three-axis CNC milling machine, aiming at low-cost production.

The low-cost nature of the proposed solution comes from the use of standard FR4 material, conventional copper PCB conductors, and a milling-based fabrication process, which eliminates the need for cleanroom facilities, photolithography, or specialized substrates [19].

The sensor architecture enables the measurement of S-parameters (S_{11} , S_{22} , S_{12} , and S_{21}), allowing both transmission and reflection analysis. The sensor manufactured can be seen on Fig. 1.

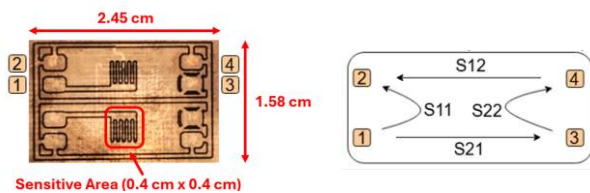


Fig. 1 – Manufactured two ports IDC sensor and its S parameters. Adapted from [18].

The sensitive area of the IDC sensor is formed by the interdigitated electrodes, where the electric field is concentrated between adjacent fingers. This area is directly exposed to the analyte, allowing detection through variations in the local dielectric properties.

The IDC sensor dimensions were defined to fit a dedicated low cost and low-tech test cell ($8.7 \text{ cm} \times 5.4 \text{ cm} \times 1.5 \text{ cm}$) developed by the IMS Laboratory in Bordeaux (Fig. 2), enabling precise deposition of small liquid volumes (on the order of a few tens of microliters) onto the sensitive area via a polydimethylsiloxane (PDMS) microfluidic chamber [15]. The sensor is integrated into a microstrip line as a discontinuity element, where the interdigitated electrodes are connected between the signal line and ground through pogo pins, acting as a variable capacitive load [15].

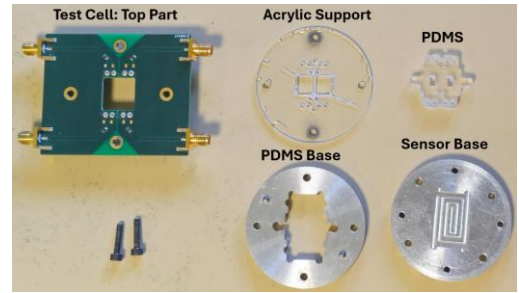


Fig. 2 – Sensor test cell developed in IMS in Bordeaux.

B. Mercury's Samples Preparation

For mercury testing, Hg^{2+} solutions were prepared by dissolving HgCl_2 (Sigma-Aldrich, St. Louis, MO, USA) in deionized water (DIW). An initial solution was obtained by dissolving 6.6 mg of HgCl_2 in 10 mL of DIW, resulting in a concentration of 2.43×10^{-3} M, followed by mixing in a vacuum mixer for 13 minutes. Serial dilutions were then carried out to achieve concentrations ranging from 2.43×10^{-10} M to 2.43×10^{-3} M, with each dilution step also mixed for 13 minutes. Finally, all samples were covered with aluminum film to prevent light exposure.

C. EPS' Preparation

EPS were obtained from the cultivation of *Graesiella sp.* by the Laboratory of Blue Biotechnology & Aquatic Bioproducts (National Institute of Marine Sciences and Technologies, Tunisia), following previously reported methods [16-17].

For the sensitive layer preparation, 4 mg of EPS were mixed with 4 mL of ethanol (99% purity, Sigma-Aldrich) using sequential agitation: 5 minutes with a spatula, 5 minutes with a micropipette, and 30 minutes in a vacuum mixer. The resulting solution was then filtered using a $22 \mu\text{m}$ filter (Sartorius, Bohemia, New York, USA) and transferred into a syringe. Subsequently, 50 μL of the filtered solution was deposited onto the IDC sensor positioned inside the test cell. The system was covered with aluminum foil to prevent light exposure and left to

stabilize for 24 hours.

Light shielding was applied during preparation to preserve the chemical integrity of the EPS layer. Although no significant variations were observed under standard laboratory lighting, the potential effects of prolonged exposure cannot be completely excluded. Two PCB (four sensors) were prepared with the sensitive layer which can be observed in Fig. 3.

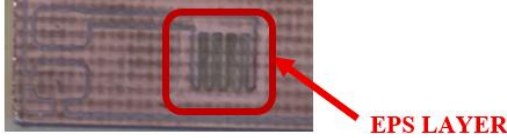


Fig. 3 – IDC sensor with EPS layer.

For the functionalized sensor showed in Fig. 3, the sensing mechanism is based on changes in the effective dielectric properties of the sensitive area due to the interaction between Hg^{2+} ions and the EPS-functionalized surface. These interactions modify local permittivity and conductivity, affecting the capacitance of the interdigitated structure. Consequently, the characteristic impedance of the microstrip line changes, leading to measurable variations in both the magnitude and phase of the S-parameters [13], [18-19].

D. Experimental Procedure for S-parameters Measurement

The experimental procedure for the measurements of S parameters can be seen in Fig. 4.

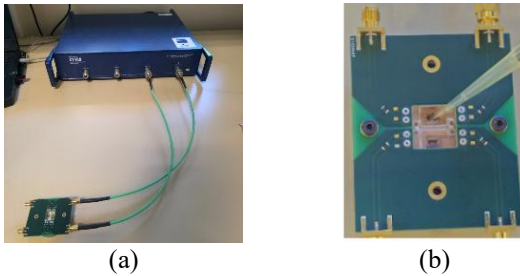


Fig. 4 – Setup for S-parameters measurements with the IDC sensor: (a) connection with the test cell and (b) liquid sample deposition on the IDC sensor.

Measurements were performed using a COBALT C1409 VNA connected to a computer running S4VNA software via USB, with the test cell linked through SMA cables, as shown Fig. 4(a). Initially, a calibration procedure was carried out using a custom calibration kit developed by the IMS Laboratory (Bordeaux), removing systematic errors from the cables and test cell. The measurements were set over a frequency range from 1 MHz to 1 GHz with 1,000 points.

After calibration, the test cell was opened to insert the IDC sensor, and baseline S-parameter measurements were first recorded in air. Then, 30 μL of DIW was deposited

onto the interdigitated electrodes with a micropipette, as shown Fig. 4(b). After removing the DIW, mercury solutions were sequentially applied, ranging from 2.43×10^{-10} M to 2.43×10^{-3} M. A stabilization time of 2 minutes was allowed before each measurement. All measurements were conducted on three sensors and stored as .s2p files. To evaluate the effect of the EPS layer, the same procedure was repeated on sensors without the coating layer.

Data were processed using Python and the scikit-rf library, focusing on S_{11} and S_{21} specifically at 10 MHz to emphasize the capacitive behavior of the IDC sensor, as explained in [18].

Although the test cell had four terminals, measurements were performed in a single-ended two-port configuration with unused ports left open. The performance of sensors with and without the EPS layer was compared using student's t-test [20], considering S_{11} and S_{21} responses, reproducibility, sensitivity, and detection limits. The magnitude and phase sensitivities (S) were calculated between each pair of consecutive concentrations according to (1). Then, average sensitivity was computed by taking the mean of all pairwise sensitivity values obtained across the full concentration range.

$$S = \frac{\Delta S}{\Delta[x]}, \quad (1)$$

where ΔS represents the magnitude and phase variation of the S-parameter, and $\Delta[x]$ denotes the variation in mercury concentration. For this parameter, the mercury concentration in DIW was 0, while air was not included in the calculation.

III. RESULTS

To highlight the capacitive behavior of the IDC sensor, the analysis was done at 10 MHz as explained in [18]. Fig. 5, 6, and 7, 8 show respectively the magnitude and phase of S_{11} , and of S_{21} , according to the concentration of Hg^{2+} for EPS-coated and uncoated sensor. Also, Table 1 shows the average sensitivity of the system.

As shown in Fig. 5, Hg^{2+} concentrations of 10^{-3} M, 10^{-4} M, and 10^{-5} M can be clearly distinguished for both EPS-coated and uncoated IDC sensors, with student's t-test p-values below 0.01, indicating statistically significant differences and very strong evidence against the null hypothesis. For the pair 10^{-5} M and 10^{-6} M, a marginally significant difference was observed for the uncoated sensor ($p \approx 0.09$), while no significant difference was found for the EPS-coated sensor ($p \approx 0.26$). For all other comparisons, p-values were higher, indicating no significant differences. These results suggest that the sensor responses become statistically indistinguishable below 10^{-6} M for the uncoated sensor and 10^{-5} M for the coated sensor. About the distinction between EPS-coated and non-coated sensors, marginally significant differences were observed across all concentrations ($p < 0.10$).

In terms of standard deviation, the higher values

occurred at 10^{-3} M (0.24 dB with EPS and 0.21 dB without EPS layer) and in DIW with EPS (0.18 dB); for all other conditions, it remained below 0.06 dB.

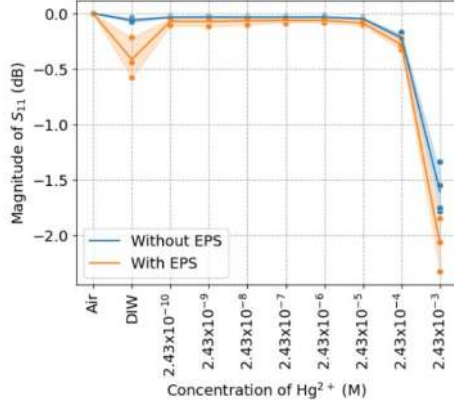


Fig. 5 – S_{11} magnitude of the manufactured IDC sensor in terms of Hg^{2+} concentration at 10 MHz.

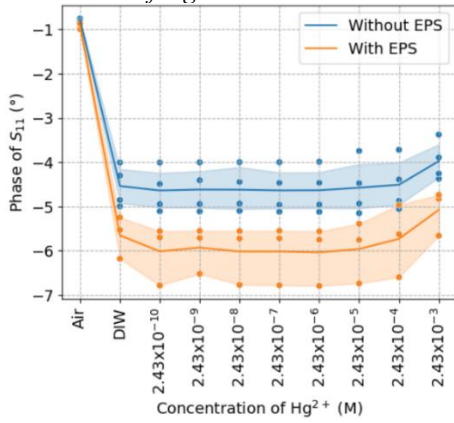


Fig. 6 – S_{11} phase of the manufactured IDC sensor in terms of Hg^{2+} concentration at 10 MHz.

In Fig. 6, the phase of S_{11} shows a maximum standard deviation of 0.82° at 10^{-4} M in the EPS-coated sensor. For all other cases, excluding air, the standard deviation ranges from 0.4° to 0.7° for both EPS-coated and uncoated sensors. Due to this variability, the student's t-test shows a statistically significant difference only between air and DIW for both sensors ($p < 0.01$). For the remaining cases, the p-values were above 0.20, indicating no significant difference. Regarding EPS identification, marginally significant differences were observed ($p < 0.07$).

Fig. 7 shows that the magnitude of S_{21} discriminates air from DIW, as well as air from the Hg^{2+} sample at 10^{-3} M, with p-values below 0.01 in both comparisons. For the remaining sample pairs, the p-values were above 0.76, indicating no significant difference. When comparing sensor configurations, the EPS-coated and uncoated devices showed marginally significant differences in all tested conditions ($p < 0.07$), suggesting a measurable effect of the EPS layer on the S_{21} response. The standard deviation was highest at 10^{-4} M for the EPS-coated sensor (1.33 dB) and at 10^{-5} M for the uncoated sensor (1.27 dB),

while for the remaining conditions it ranged from 0.8 dB to 1.30 dB.

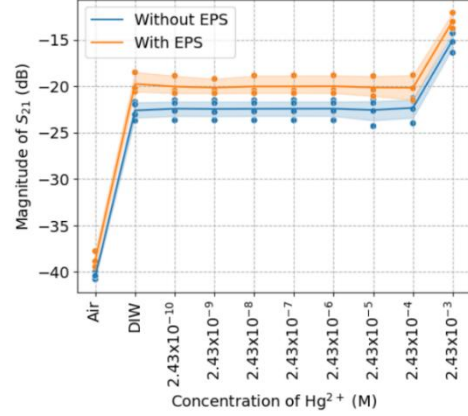


Fig. 7 – S_{21} magnitude of the manufactured IDC sensor in terms of Hg^{2+} concentration at 10 MHz.

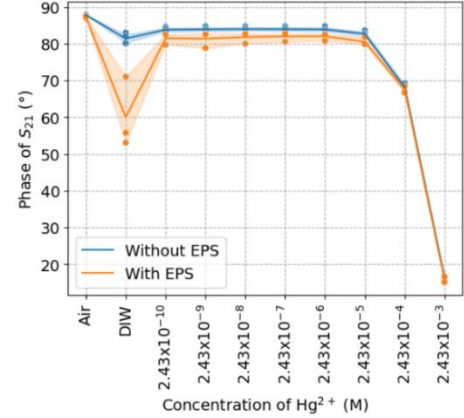


Fig. 8 – S_{21} phase of the manufactured IDC sensor in terms of Hg^{2+} concentration at 10 MHz.

Fig. 8 shows that the S_{21} phase discriminates Hg^{2+} concentrations of 10^{-3} M, 10^{-4} M, and 10^{-5} M, with p-values below 0.01, indicating very strong evidence against the null hypothesis. For the pair 10^{-5} M and 10^{-6} M, a statistically significant difference was observed for the uncoated sensor ($p \approx 0.05$), but not for the EPS-coated sensor ($p \approx 0.17$). For all other comparisons, p-values were above 0.79, indicating no significant differences. These results suggest detection limits of approximately 10^{-6} M for the uncoated sensor and 10^{-5} M for the coated sensor.

Regarding EPS identification, marginally significant differences were observed in most conditions ($p \approx 0.07$), with weaker evidence at 10^{-3} M and 10^{-4} M ($p \approx 0.36$). The highest variability occurred in DIW, with standard deviations of 1.36° (uncoated) and 9.68° (EPS-coated). For other concentrations, the standard deviation ranged from 0.07° to 0.81° (uncoated) and from 0.29° to 2.17° (EPS-coated).

According to Table 1, the average sensitivity increased by 10.5 times for the S_{11} magnitude, 4.4 times for the S_{11}

phase, and 7.46 times for the S_{21} phase for the EPS-coated sensor compared with uncoated sensor. In contrast, no significant change was observed for the S_{21} magnitude. These results indicate that the inclusion of the EPS layer enhances the sensor response, mainly by improving the interaction between the electromagnetic field and the sensing region. Consequently, higher sensitivity also leads to a lower limit of detection, since smaller variations in the analyte concentration produce larger measurable changes in S-parameters, increasing the standard deviation.

Table 1 – Average sensitivity of IDC sensor with and without EPS layer.

S	Without EPS layer	With EPS layer
Magnitude of S_{11}	3.72×10^7 dB/M	4.29×10^8 dB/M
Phase of S_{11}	8.58×10^7 °/M	4.63×10^8 °/M
Magnitude of S_{21}	3.60×10^9 dB/M	3.45×10^9 dB/M
Phase of S_{21}	3.18×10^9 °/M	2.69×10^{10} °/M

After the data analyses, we can compare this work with other works in literature according to limit detection and cost, as shown in Table 2.

Table 2 – Limit detection of Hg^{2+} in water.

Device	Limit Detection (M)	Fabrication Cost (USD)	Work
Microcantilever sensor	10^{-11}	~30.00	[7]
IDC sensor	10^{-6}	~1.00	[8]
IDC sensor	10^{-20}	~1.00	[9]
IDC sensor	10^{-11}	~2.50	[10]
IDC sensor	10^{-6}	~4.50	[11]
IDC sensor	10^{-11}	~10.00	[12]
IDC sensor	10^{-6}	~0.30	[13]
Love Wave sensor	10^{-10}	~70.00	[14]
IDC sensor	10^{-6}	~0.05	This work
High technological level Low technological level			

According to Table 2, the proposed sensor does not achieve the lowest detection limit among the reviewed approaches; however, it presents a favorable trade-off between cost, fabrication simplicity, and sufficient sensitivity for mercury detection in water above the WHO threshold [8]. Compared with microcantilever [7] and Love wave devices [14], the proposed IDC sensor is much simpler to fabricate, as it does not require cleanroom-based microfabrication, photolithography, or sputtering processes. Compared with other IDC-based solutions [8–13], the present work reaches a similar detection limit to [13] while offering a substantially lower fabrication cost.

One important point is that the experimental validation was carried out using a VNA, a high-cost laboratory instrument. Therefore, the main contribution of this work is the development of a low-cost sensing element, not a complete low-cost measurement system. For this reason, Table 1 reports only the fabrication cost of the sensor itself.

The S-parameters (S_{11} and S_{21}) of IDC sensors are affected by the dielectric properties of the medium under

test. Thus, ionic species can modify the effective permittivity and conductivity, causing variations in magnitude and phase. As a result, the proposed sensor is not intrinsically selective at the electrical level, since different ions may produce similar responses. In addition, the EPS layer exhibits affinity toward heavy metal ions due to its functional groups, meaning that although the sensor is sensitive to Hg^{2+} , similar interactions may occur with other metals [14].

However, all measurements were performed under identical controlled conditions, using the same preparation protocol, with Hg^{2+} concentration as the only variable. This minimizes uncontrolled.

IV. CONCLUSIONS

This work presented the experimental analysis of an interdigitated capacitive (IDC) sensor for Hg^{2+} detection in liquid media. The performance was evaluated through the magnitude and phase of the S_{11} and S_{21} parameters, for both EPS-coated and uncoated configurations.

The statistical analysis using the student's t-test showed that the sensor can discriminate Hg^{2+} concentrations down to 10^{-5} M. At this concentration, both the S_{11} magnitude and S_{21} phase yielded p-values below 0.01, indicating very strong evidence against the null hypothesis. At 10^{-6} M for the uncoated sensor, the S_{21} phase showed $p = 0.05$, indicating statistical significance at the 5% level, while the S_{11} magnitude showed $p < 0.1$, indicating marginal evidence against the null hypothesis. Also, the EPS coating improved the sensitivity of the sensor but reduced the detection limit to 10^{-5} M. Nevertheless, the statistical analysis revealed marginal evidence of differences between coated and uncoated configurations in most cases ($p < 0.1$), suggesting that the EPS layer has a measurable but weak effect on the sensor response. The standard deviation remained below 0.06 dB and 0.7° for S_{11} , and below 1.3 dB and 2.2° for S_{21} in most conditions, indicating good repeatability.

Despite not achieving the lowest detection limit among other studies, the proposed device reached a detection threshold above the minimum mercury levels established by the World Health Organization (WHO), with a fabrication cost of only USD 0.05, significantly lower than comparable sensors. Future work should focus on improving EPS-coated sensor sensitivity, investigating selectivity in the presence of competing ions, and extending the approach to other heavy metals such as Pb^{2+} , Cd^{2+} , and Cu^{2+} .

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Development and Experimental Evaluation of a Three-Lead ECG Holter System

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Abstract – This paper presents the design and experimental evaluation of a compact three-lead ECG Holter system for portable low-power cardiac monitoring. The system acquires standard bipolar leads I, II, and III using an analog front-end with amplification, filtering, and common-mode interference suppression. The signals are digitized by a microcontroller-based unit and further processed by digital filtering. Experimental verification was performed on a healthy volunteer under resting conditions, focusing on signal quality, inter-lead consistency, and temporal stability. Channel agreement was assessed using Einthoven's law, while additional analysis included R-peak detection, RR interval evaluation, and signal-to-noise ratio estimation. The results confirmed stable three-lead acquisition and sufficient signal quality for technical ECG monitoring. The prototype provides a suitable platform for future implementation of compressive sensing methods.

Keywords: ECG, Cardiac monitoring, ARM Cortex-M0

I. INTRODUCTION

Cardiovascular diseases remain a major global health problem and continue to contribute substantially to premature mortality and long-term disease burden. This creates a persistent need for accessible methods capable of supporting earlier detection, follow-up, and long-term observation of cardiac activity in both clinical and out-of-hospital settings [1]. Electrocardiography remains one of the most important non-invasive tools for assessment of cardiac electrical activity. However, conventional short-duration recordings may not always capture transient or intermittent rhythm abnormalities. For this reason, ambulatory and wearable ECG systems have become increasingly important, as they enable prolonged monitoring under more realistic everyday conditions and may improve the likelihood of capturing clinically relevant events. At the same time, ongoing advances in miniaturized electronics, low-power embedded systems, and digital signal processing continue to expand the practical usefulness of portable ECG devices [2]. Although

ECG acquisition circuits and Holter recorders are well-established, the development of compact multi-lead ECG acquisition systems remains relevant not only for monitoring applications but also for research into new sensing and processing approaches. In particular, transparent prototype platforms are valuable because they allow verification of the complete acquisition chain, assessment of signal quality, direct access to raw multi-lead data, and preparation for later implementation of more advanced methods. Compared with established diagnostic systems, such prototypes are typically simpler and are not intended to replace certified medical equipment; however, they offer higher flexibility for hardware modification, embedded processing, and experimental evaluation. One such direction is compressive sensing, which has been widely investigated in ECG processing as a promising approach for reducing data acquisition, transmission, and storage requirements in resource-constrained wearable systems [3]. This paper presents the design and experimental evaluation of a compact three-lead ECG Holter prototype intended for portable low-power monitoring and for future use as an experimental platform for compressive sensing (CS) research [4]. The main contribution of this work lies in the development and experimental verification of a functional, modifiable prototype that enables evaluation of the complete acquisition chain and provides a suitable basis for further investigation of compressed acquisition and embedded ECG processing methods.

II. HARDWARE DESIGN OF A THREE-LEAD ECG HOLTER SYSTEM

This section describes the hardware architecture of the proposed three-lead ECG Holter system for long-term monitoring of cardiac activity. The system consists of an analog front-end for signal acquisition and conditioning, a digital subsystem for sampling and storage, and a power management unit designed for low-power operation. The design reflects typical requirements of wearable biomedical devices, including high common-mode rejection, low noise, and efficient energy utilization.

A. Amplifier Stages Design and Implementation

The proposed ECG Holter employs three identical acquisition channels corresponding to leads I, II, and III. Each channel implements a two-stage amplification structure with a total gain of approximately 2240, which is suitable for amplifying low-amplitude ECG signals to levels compatible with the ADC input range under the intended operating conditions. The circuit diagram of lead I is shown in Fig. 1. The same circuit topology is used for all three leads.

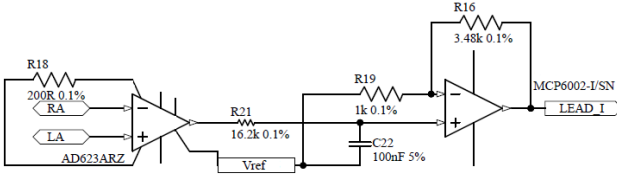


Fig. 1. Implementation of the two-stage ECG amplification channel comprising an AD623 instrumentation amplifier and an MCP6002 operational amplifier.

The first stage is realized using the instrumentation amplifier AD623, which offers high input impedance (2 G Ω), low input bias current (17 nA), and high CMRR (approx. 100 dB) [5]. These properties are beneficial for ECG acquisition, particularly for suppression of common-mode power-line interference [6], while motion artifacts remain strongly dependent on the electrode-skin interface [7]. The gain is set by a resistor of 200 Ω , resulting in a gain of approximately 500. This value is consistent with typical biopotential front-end designs, where moderate initial amplification is used to preserve dynamic range and reduce the risk of saturation in the presence of electrode offset potentials and other low-frequency disturbances, while their residual effect is attenuated by the high-pass coupling network. The amplifier operates with a single-supply voltage (0–3.3 V), with its reference pin biased at 1.12 V. This reference ensures that the bipolar ECG signal is shifted into the ADC input range without clipping. This approach is standard in low-voltage biomedical instrumentation. The second amplification stage is implemented using a rail-to-rail operational amplifier, MCP6002, in a non-inverting configuration. The gain is determined by resistors of 3.48 k Ω and 1 k Ω , resulting in a gain of approximately 4.48. The amplifier maintains the signal centered around the 1.12 V reference. The combined gain of both stages is therefore approximately 2240, which aligns with the expected amplification requirements for ECG signals. The selected MCP6002 amplifier, with a gain-bandwidth product of 1 MHz, is sufficient for the second amplification stage, since ECG signals are typically band-limited to approximately 150 Hz in wearable monitoring systems [8]. In addition, the rail-to-rail output capability improves utilization of the available voltage swing and helps maximize the effective dynamic range at the ADC input [9], [10]. A passive low-pass RC filter ($R = 16.2$ k Ω , $C = 100$ nF) is placed between the two

stages to suppress high-frequency noise before further amplification. This placement is appropriate, as it prevents amplification of unwanted spectral components. Overall, the chosen architecture (instrumentation amplifier followed by a low-power operational amplifier) is consistent with established designs in wearable ECG systems and provides a suitable compromise between performance, power consumption, and implementation complexity.

B. Analog Filters and Driven Right Leg Circuit

Effective filtering is essential for maintaining signal quality in ECG systems. The analog front-end implements a band-pass characteristic using first-order passive RC filters. A high-pass filter ($C = 4.7$ μ F, $R = 100$ k Ω) is placed at the input of the instrumentation amplifier.

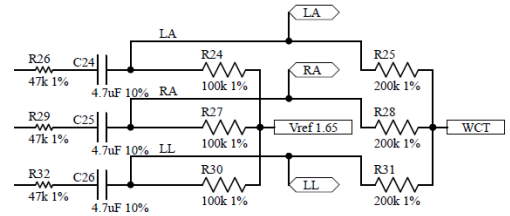


Fig. 2. Configuration of the analog input stage featuring high-pass RC filters and the Wilson Central Terminal (WCT) network.

This configuration yields a cutoff frequency of approximately 0.34 Hz, which is appropriate for removing baseline wander caused by respiration and electrode polarization while preserving clinically relevant low-frequency components of the ECG signal (typically above 0.05 Hz). A low-pass filter ($R = 16.2$ k Ω , $C = 100$ nF) introduces a cutoff frequency of approximately 98 Hz. This effectively attenuates high-frequency noise, including electromyographic interference and external electromagnetic disturbances. The resulting analog passband of approximately 0.3–100 Hz provides a reasonable compromise for prototype ECG acquisition and rhythm-oriented monitoring [11], [12]. Although this range is narrower than the bandwidth typically recommended for standard diagnostic ECG recording, additional digital filtering is applied in the proposed system to improve suppression of residual interference and high-frequency noise. Although first-order analog filters provide only a gentle roll-off of 20 dB/decade, they are adequate for preliminary analog conditioning because additional filtering is performed digitally.

To suppress common-mode interference, particularly at 50 Hz, a Driven Right Leg (DRL) circuit is implemented (Fig. 3). The DRL uses a Wilson Central Terminal formed by averaging the potentials of electrodes RA, LA, and LL via three 200 k Ω resistors. This averaged signal is inverted and fed back to the patient through the right leg electrode. The DRL amplifier, based on MCP6002, provides a gain of approximately 20 at low frequencies, improving the

effective CMRR of the system by approximately 26 dB. This significantly enhances immunity to power-line interference, which is critical in ambulatory monitoring environments. The inclusion of stabilization components in the feedback loop ensures loop stability and prevents oscillations, which is a known design challenge in DRL implementations.

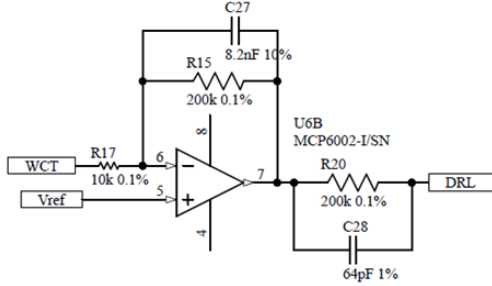


Fig. 3. Implementation of the Driven Right Leg (DRL) circuit for active common-mode interference suppression.

C. Power Supply System

As Fig. 4 shows, the system is powered by a 3.7 V Li-Ion battery (18650 type) with a capacity of 3000 mAh. Two LDO regulators are used: LD2980 for the analog section and LD2981 for the digital section. This separation reduces coupling of digital noise into sensitive analog circuits. The analog front-end consumption is approximately 1.78 mA, while the microcontroller consumption is conservatively estimated at 80 mA. The total system current is therefore below 100 mA. This results in a theoretical battery life of approximately 30 hours, which is sufficient for standard 24-hour Holter monitoring with a margin for inefficiencies. The use of low-noise LDO regulators and proper decoupling (1 μ F and 10 μ F capacitors) ensures stable supply voltages and minimizes noise propagation. The VCC output is used as the power supply of amplifiers, and VCC_uC is used for the microcontroller.

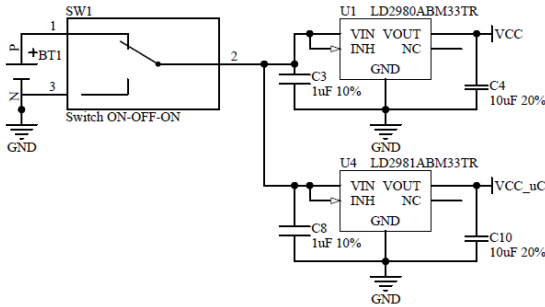


Fig. 4. Power management subsystem utilizing independent LDO regulators for analog (VCC) and digital (VCC_uC) domains.

A reference voltage of 1.12 V is generated using a precision resistor divider (10 k Ω , 0.1% and 5.1 k Ω , 0.1%) followed by a buffer amplifier, as shown in Fig. 5. This provides a stable virtual ground for analog signal processing.

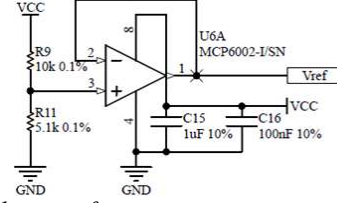


Fig. 5. Voltage reference circuit generating a buffered 1.12 V reference voltage for single-supply signal conditioning.

D. Microcontroller and Analog-to-Digital Conversion

The digital core of the system is based on the XMC1100 microcontroller, utilizing an ARM Cortex-M0 architecture operating at 32 MHz. For the total analog gain used in the proposed system, the 12-bit ADC quantization step of approximately 0.8 mV corresponds to about 0.36 μ V referred to the original ECG signal; however, this value expresses only the theoretical input-referred quantization step, not the effective resolution of the acquisition chain. Signal sampling is implemented using a timer-driven approach, where a periodic interrupt defines a fixed sampling frequency of 500 Hz (2 ms period). This frequency satisfies the Nyquist criterion for the selected 100 Hz bandwidth and ensures sufficient temporal resolution for accurate waveform reconstruction. All three channels (leads I, II, and III) are sampled sequentially within a single acquisition cycle. The ADC conversion time of approximately 1.4 μ s is negligible compared to the sampling period, ensuring temporal alignment between channels and preventing timing conflicts. To decouple the time-critical acquisition from slower storage and communication tasks, acquired samples are temporarily stored in a 32-record circular buffer. Each data frame has a size of approximately 21 bytes. At a sampling rate of 500 Hz, the resulting data throughput is approximately 10.5 kB/s (approx. 84 kbps). Measured data are stored on an SD card via SPI (400 kHz) and simultaneously transmitted via UART configured at 256 kbit/s. This configuration provides a significant throughput margin (over 60%), ensuring continuous and lossless data acquisition suitable for long-term Holter monitoring.

III. DIGITAL SIGNAL PROCESSING

The previous sections focused on the hardware design and its essential configuration, including sampling frequency, data acquisition, and storage mechanisms. This section addresses the digital signal processing (DSP) applied to the acquired ECG signals. Prior to digitization, the analog ECG signals from all three leads (I, II, III) are conditioned using a first-order band-pass filter with a passband approximately from 0.3 Hz to 100 Hz. This filtering stage suppresses baseline wander caused by respiration, reduces low-frequency components of baseline wander and motion-related disturbances, and eliminates DC offsets originating from electrode polarization. Additionally, a DRL circuit is employed to attenuate common-mode

interference, particularly power-line noise at 50 Hz, by actively feeding back an inverted common-mode signal to the patient. However, due to non-ideal electrode impedance matching and residual common-to-differential mode conversion, the DRL circuit alone cannot fully eliminate power-line interference. Therefore, an additional notch filter centered at 50 Hz is required to further suppress this disturbance. To simplify the analog front-end and enhance design flexibility, the notch filtering is implemented in the digital domain using a second-order IIR filter with a center frequency $f_0 = 50$ Hz at a sampling frequency $f_s = 500$ Hz with a quality factor $Q = 15$. This approach yields a narrow and precisely defined attenuation band centered at 50 Hz, with -3 dB cutoff frequencies at approximately 48.40 Hz and 51.69 Hz, ensuring selective suppression of power-line interference while preserving adjacent spectral components. Furthermore, a first-order IIR low-pass filter is applied to the digitized signal to reduce high-frequency noise and improve waveform smoothness. The filter is implemented using the difference equation $y[n] = 0.67x[n] + 0.33y[n-1]$, which, for a sampling frequency of 500 Hz, corresponds to an approximate -3 dB cutoff frequency of 100 Hz. Compared with higher-order filtering approaches, this implementation offers low computational complexity, low memory requirements, and smooth attenuation characteristics, while introducing only limited phase distortion [13]. The selected cutoff frequency reflects a trade-off between noise suppression and preservation of ECG morphology, favoring robust long-term signal acquisition and reliable beat detection over preservation of higher-frequency components. The application of IIR filters, despite introducing nonlinear phase distortion, represents a reasonable engineering trade-off to accommodate the strict computational and memory constraints of the ARM Cortex-M0 core, while sufficiently preserving R-peak localization for subsequent processing stages.

IV. EXPERIMENTAL RESULTS

The experimental validation of the proposed system included measurements performed on a human subject. The volunteer was a healthy individual with no known history of cardiovascular disease. The experimental procedure was conducted in accordance with the Declaration of Helsinki [14], and informed consent was obtained from the participant before data acquisition. During the measurement, the subject remained resting in a seated position to minimize motion artifacts and ensure stable recording conditions. All three standard bipolar leads were simultaneously recorded using disposable Ag/AgCl electrodes. Before electrode placement, the skin was cleaned to reduce contact impedance and improve signal quality. The measurement setup was designed in accordance with safety practices for biopotential recording systems. The device was powered from an isolated battery supply (3.7 V Li-Ion), ensuring galvanic isolation from the

mains and eliminating the risk of leakage currents. The analog front-end utilizes high-impedance instrumentation amplifiers, as well as current-limiting resistors (47 k Ω), which were implemented at the analog front-end to ensure signal integrity and patient safety. The overall design follows general principles outlined in standards such as IEC 60601-1 for medical electrical equipment, particularly regarding electrical isolation in battery-operated devices. The recorded signals were used for the technical evaluation of signal quality, stability, and noise performance, without any clinical interpretation. These conditions provided representative data for assessing the performance of the proposed ECG Holter system under controlled laboratory conditions. The performance of the proposed system was evaluated through a continuous 5-minute recording, with the initial 3-second transient period excluded to account for the settling time of the 0.34 Hz analog high-pass stage and digital IIR filters. For offline evaluation, the recorded ECG data were analyzed in MATLAB. The mean value of each lead over the evaluated interval was subtracted in order to remove the global DC offset prior to further analysis. A detailed analysis of Lead I was performed to assess the effect of the digital signal processing. The average QRS complex shown in Fig. 6 indicates a mild asymmetry around the R-peak, consistent with phase distortion introduced by the applied IIR filters. Specifically, the mean signal level in the local pre-peak region was -0.132 V, whereas the corresponding mean signal level in the local post-peak region was -0.183 V. The minimum amplitude in the pre-peak region reached approximately -0.502 V, compared to -0.587 V in the post-peak region. The average R-peak amplitude was 1.514 V, and the relative difference between the pre- and post-peak mean signal levels corresponded to approximately 3.37 % of the R-peak amplitude. The local pre- and post-peak regions were defined as 80 ms intervals before and after the detected R-peak, excluding the R-peak sample itself. These observations confirm a measurable but limited phase-related distortion, while the temporal localization of the R-peaks remains preserved.

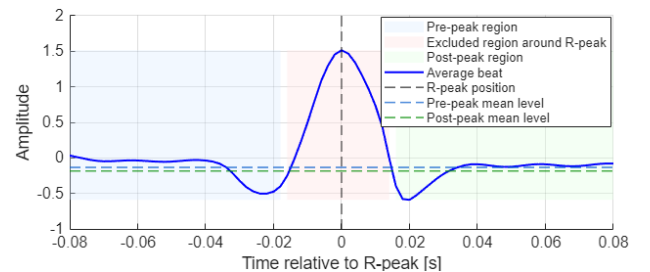


Fig. 6. Average QRS complex of Lead I aligned to detected R-peaks, illustrating the asymmetry introduced by filtering.

A representative segment of the acquired signals is shown in Fig. 7, demonstrating good morphology and consistent temporal alignment across the recorded channels.

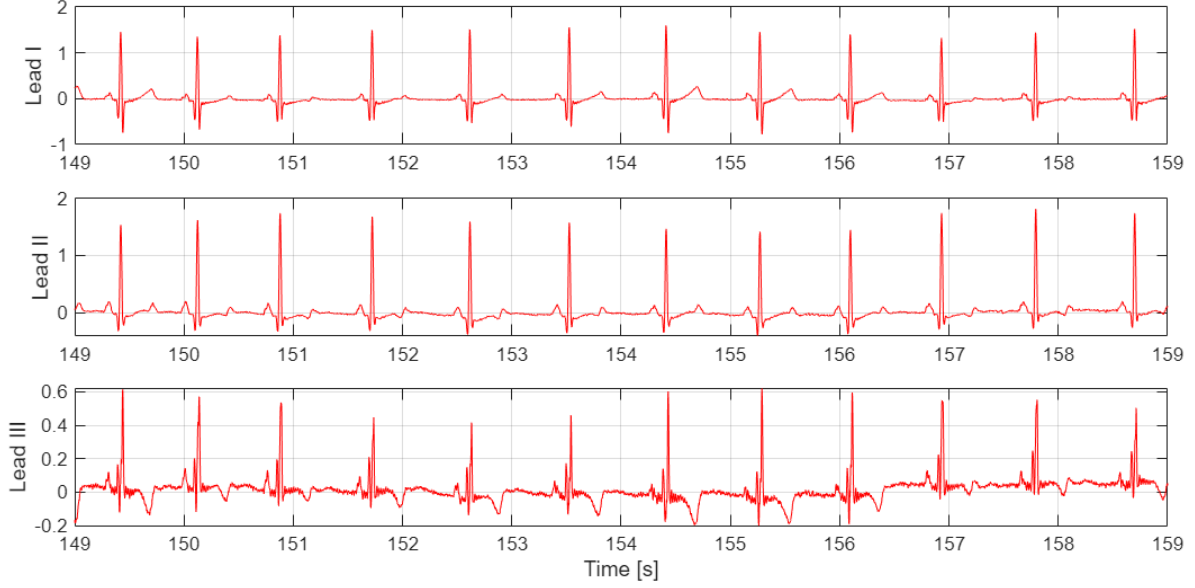


Fig. 7. Representative segment of three-lead ECG signals (I, II, III) acquired by the proposed Holter system, demonstrating signal quality, baseline stability, and absence of power-line interference.

To quantitatively validate the precision of the analog front-end and gain matching between channels, Einthoven's law [15], [16] was verified across the entire dataset, yielding a mean absolute error of 0.0026 V with a standard deviation of 0.0021 V. Given the 1.5 V R-peak amplitude, the 0.17% residual error is negligible, confirming good gain matching and stable differential signal acquisition. No visible clipping was observed in the processed signals during the evaluated recording. The overall signal morphology remained stable over most of the recording, although occasional local variations and minor artefactual disturbances were present in some intervals, which is consistent with practical ECG acquisition conditions. To further quantify signal quality and temporal stability, additional metrics were evaluated for all three leads. The signal-to-noise ratio (SNR) was estimated as the ratio between the mean R-peak amplitude A_R and the RMS noise level σ_n measured in baseline segments as eq. (1) shows.

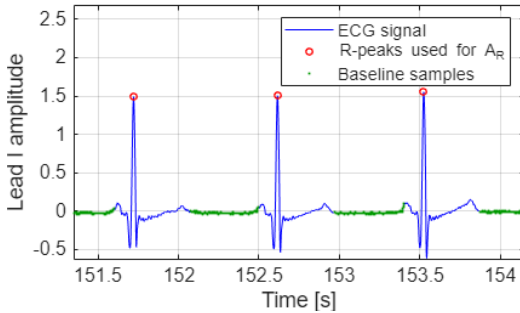


Fig. 8. Lead I ECG signal with detected R-peaks and baseline segments used for SNR estimation.

Baseline segments were defined as signal intervals outside the intervals from 120 ms before to 350 ms after each

detected R-peak, to minimize the influence of QRS complexes. Fig. 8 shows the ECG signal of Lead I with three QRS complexes, together with the detected R-peaks and the baseline segments used for SNR estimation.

$$SNR = 20 \log_{10} \left(\frac{A_R}{\sigma_n} \right) \quad (1)$$

The resulting SNR values were 34.01 dB for Lead I, 31.06 dB for Lead II, and 22.64 dB for Lead III. These values confirm effective noise suppression in all channels, with the best signal quality achieved in Lead I and lower SNR in Lead III due to its smaller signal amplitude and higher relative noise contribution. Temporal stability of the recording was evaluated using RR interval analysis based on R-peak detection in Lead I. A total of 346 beats were detected in all three Leads. The mean RR interval in Lead I was 858 ms with a standard deviation of 74.12 ms, corresponding to a mean heart rate of 69.93 bpm with a standard deviation of 6.376 bpm. These values indicate stable rhythm and reliable temporal acquisition under resting measurement conditions. Amplitude stability was evaluated from the detected R-peak amplitudes. The mean R-peak amplitudes were 1.51 V, 1.55 V, and 0.52 V for Leads I, II, and III, respectively, with corresponding standard deviations of 0.16 V, 0.15 V, and 0.11 V. The relatively low amplitude variability confirms stable acquisition and good signal quality. Overall, the results demonstrate stable channel behavior, good inter-lead consistency, and sufficient signal quality for technical ECG monitoring and further signal analysis.

V. CONCLUSION

This paper presented the design and experimental validation of a compact three-lead ECG Holter system.

The proposed solution integrates a low-power analog front-end, digital signal processing, and microcontroller-based acquisition within a battery-operated architecture suitable for wearable applications. The analog front-end provides sufficient gain and noise performance for ECG acquisition, while first-order analog filtering and the DRL circuit suppress baseline wander and common-mode interference. Additional digital conditioning, including a second-order IIR notch filter for 50 Hz interference and a first-order IIR low-pass filter, further improves signal quality. Experimental results obtained from a 5-minute recording on a healthy subject confirmed stable operation and good signal quality across all three leads under resting conditions. Quantitative validation using Einthoven's law demonstrated good gain consistency between channels, while SNR estimation, RR-interval analysis, and R-peak amplitude evaluation confirmed reliable temporal acquisition and stable signal morphology. The applied digital filtering introduced only mild phase-related distortion without compromising R-peak localization. Compared with established Holter and diagnostic ECG systems, the proposed device is not intended to replace certified medical equipment. Its bandwidth and validation scope are limited to technical prototype evaluation. Its main advantage lies in the transparent hardware structure, direct access to synchronized raw data from three bipolar leads, low-power operation, and simple modification of the acquisition and processing stages. These properties make the system suitable as an experimental platform for future CS research. The selected ARM Cortex-M0 architecture represents a realistic resource-constrained platform for wearable devices and can support the acquisition side of CS experiments, including controlled sampling, buffering, lightweight preprocessing, and evaluation of data-rate reduction strategies.

VI. ACKNOWLEDGEMENT

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Multiparametric metrological characterization of an inductive sensing element for mechanical wear estimation

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Abstract – Online wear debris monitoring is essential for preventive maintenance of mechanical systems, allowing early fault detection and the prevention of catastrophic failures. Inductive debris sensors are widely used, however their sensing elements are often not characterized from a metrological perspective. This paper presents a multiparametric experimental analysis of an inductive sensing element to measure the mass of accumulated ferromagnetic material. The influence of the main geometrical parameters, coil diameter, number of turns, and excitation frequency on sensor response is investigated through frequency sweep measurements. A calibration function relating the impedance variation to the captured debris mass is then derived, including a detailed uncertainty analysis that accounts for instrumentation effects and particle distribution. The obtained results allows the optimal design to be identified and, within the investigated mass range, show that the main influence quantity is the particle distribution while the particle size has a negligible effect.

Keywords: Wear debris monitoring; Inductive sensing; Multiparametric analysis; Uncertainty.

I. INTRODUCTION

Wear debris monitoring is becoming important in many fields where it is needed to predict the life of mechanical equipment, to prevent catastrophic failures that may lead to costly downtime and eventual injuries.

The employed methods can be subdivided into offline and online. The former requires to periodically send oil samples to specialized laboratories, where a ferrography analysis [1] is performed: it consists in separating the ferromagnetic particles from the oil and examine them under a microscope to determine size, concentration and morphology. However, this is a costly, time consuming and not real time operation. On the contrary, the latter, despite being less accurate, can detect the wear process in real time, which is an important requirement especially in functional safety applications.

In an industrial context, such devices have been already

employed [2], and their main working principle is based on a magnetic effect. Specifically, these objects are made of a Permanent Magnet (PM) that attracts the ferromagnetic particles on the top of a copper coil that generates an alternating magnetic field thanks to an alternating current injected by an external circuit. From theory [3, 4, 5] it is known that a metal passing in the proximity of an AC powered coil causes a change of the coil impedance. It can be either a resistive or an inductive change or both, depending on the metal properties, specifically its magnetic permeability and electrical conductivity, and the excitation frequency of the coil circuit. This change can be related to the mass of debris captured and hence to the evolution of the wear.

Although there have been several studies [6, 7, 8, 9, 10, 11] on this subject, they mainly focus on the design of the conditioning circuit and the definition of a calibration function. However, they lack a proper study of the sensing element itself as they do not show how its geometrical parameter and the excitation frequency affect the sensor response. Furthermore, a complete metrological characterization of the sensing element, that accounts for all influence quantities such as temperature and particle size, is absent.

In this article, an experimental analysis is proposed to study the aforementioned aspects. Specifically, first a multi-parametric experiment is performed to assess the sensor response as its geometric configuration varies. Subsequently, a calibration function is derived with uncertainty calculations, and the effect of the influence quantities is studied. Through this study, it is aimed to provide the necessary knowledge to choose the inductive sensing element configuration and to see its application limits, which is a required step for the successive design of the acquisition interface.

II. METHOD

A. Test setup

The sensing element is based on the inductive principle, i.e. a coil placed over a PM. The PM captures the ferromagnetic particles, and since the coil is AC powered they change the coil impedance, which is related to the accumulated debris mass.

Fig. 1 shows the experimental setup. It must be un-

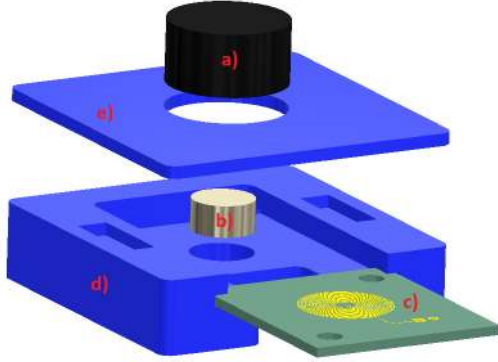


Fig. 1. 3D model of the inductive sensing element. a) particle's tablets b) PM c) PCB coil d) and e) case

derlined that it is not the definitive system suitable for the industrial application, but it was designed to verify the working principle, optimize the parameters and perform a controlled metrological characterization. The system under study is made of a NdFeB N35 PM, i.e b), which attracts the ferromagnetic particles accumulating on the top of the coil c), which is printed on a PCB. A PCB coil is less dense, in terms of the number of turns, and less compact with respect to a coil wrapped around, which would be used in the real application. However, since its geometrical features, such as the number of turns, can be controlled in a fine way, it represents an alternative and easier solution to study the effects of the main coil parameters on the sensor response. The center of the coil lies on the same axis as the PM. The PCB is tightly inserted inside a 3D printed case d) made of ABS-GF (Glass Fiber reinforced); this would allow to perform high temperature tests without the case melting. In the middle of hole d) is the PM. Object e) is a top cover that is used to center the particles with respect to PM and coil. Objects a) are the tablets containing a known particles quantity. Because the sensor response depends on the particle mass as well as particles distribution, to improve repeatability, some tablets have been embedded in resin. This allows to compare various configurations on the same particle mass and distribution, and hence on the same setup. These tablets are useful for multi-parametric frequency response analysis. However, when it comes to identify the calibration function, the particle distribution must be considered as one of the influence quantities. Thus, for this latter analysis the particles are only placed inside the 3D printed black case without the blocking resin. The tablets containing resin were prepared according to the following procedure:

- 3D print a case containing the resin.
- Place the case on a scale, pour an initial controlled amount of resin, and wait until it becomes dense.
- Place the case again on the scale and add the particles



Fig. 2. Real test setup

until the desired weight is reached.

- Add another layer of resin on top and wait until everything solidifies.

The manufactured tablets are directly inserted on top of the coil through the hole of e), as depicted in Fig. 1. In this work the particle's material is 99.7% Fe powder since it is commercially available in known size. The scale used to weigh the resin and the particles has range 200 g and the following metrological characteristic: resolution 0.1 mg, repeatability 0.05 mg and linearity 0.2 mg. The coil impedance is measured through an LCR meter with an excitation frequency of up to 10 MHz.

Fig. 2 shows the real test setup. The blue case is the calibration case, while the black cylinders on the left are the tablets with resin.

B. Experimental plan

A two step analysis has been performed.

- Identification of the main geometrical parameters affecting the inductive sensing element response and frequency analysis by changing them.
- Characterization of the sensor response for the best configuration, i.e. captured mass vs impedance change, with uncertainty calculation and effect of the influence quantities.

In the first phase, two geometrical parameters were varied: coil's external diameter (Φ_c), coil's number of turns per layer (N). For each configuration a frequency sweep with and without the ferromagnetic particles was performed. During acquisitions, the mass of debris and the influence quantities were kept constant. Specifically, the debris tablet embedded in resin, as explained in section ii.A., was chosen as $m = 500.3$ mg made of particles of $d = 45 \mu\text{m}$ diameter at room temperature. Three coil's external diameters $\Phi_c = [10 \ 15 \ 20]$ mm and three coil's

number of turns per layer $N = [10 \ 15 \ 20]$ were varied one at the time. For $\Phi_c = 10$ mm only the configuration with $N = 10$ was tested due to the manufacturing limits of the PCB coil, for a total of 7 different combinations. Each PCB was made of two layers of coil, i.e. top and bottom. During the frequency sweeps the inductance and resistance of the equivalent series inductor model were recorded. The frequency was changed from 10 kHz to 10 MHz, for a total of 3000 acquisitions.

In the second phase, the best geometrical configuration was chosen, and the sensor response was measured as the debris mass on top of it increased. The response was studied only at room temperature with debris meshes $45 \mu\text{m}$ and $150 \mu\text{m}$. The sensor characteristics were derived at a constant excitation frequency.

III. RESULTS

A. Multi parametric experiment

Figs 3, 4 and 5 show the results of the multi parametric experiment.

It can be seen that as the frequency increases both the unloaded impedance, i.e. the impedance when no particle deposition is present, and the impedance with the material deposited have their resistive part that increases and the inductive part that initially decreases. This is expected since from theory it is known that an alternating magnetic field in a conductive medium, that in this case are both the magnet and the particles, induces in it eddy currents that are proportional to the square of the excitation frequency. These currents manifest as an increase of equivalent series resistance and reduction of equivalent series inductance of the coil originating the alternating magnetic field. As showing in Figs. 3, 4 and 5, considering the relative inductance and absolute resistance changes caused by particle addition it can be seen that the former tends to initially reach a minimum and then to increase while the latter increases. This behavior is the same for every combination of investigated geometrical parameters.

If the external coil diameter is fixed, the greater the number of turns per layer the greater are both the relative inductance change and resistance change. Instead, by fixing the number of turns, as the external coil's diameter increases the relative change of inductance and absolute change in resistance decreases. Both behaviors are related to the surface density of the turns. A higher surface density, that considers both the number of turns and the surface on which are distributed, provides a higher impedance change. As a results, during the design phase this parameter should be maximized considering also the other eventual geometrical constraints.

The behavior that has been seen so far is due to a shift of the Self Resonant Frequency (SFR) of the coil. With the aim of studying this effect, the sensing element has been modified to move the SFR in the frequency range of the

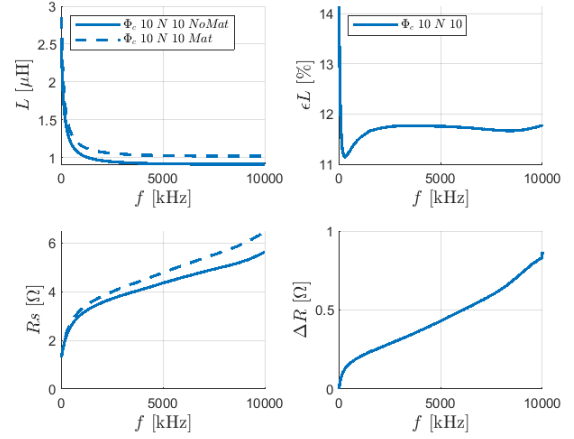


Fig. 3. Frequency response coil with $\Phi_c = 10$ mm

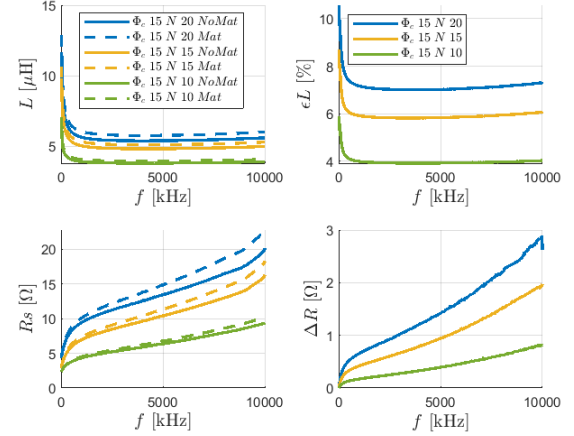


Fig. 4. Frequency response coil with $\Phi_c = 15$ mm

available LCR meter. As can be seen in Fig. 6, the addition of the debris table causes a shift of the SFR. As a result, it might be expected that a greater shift of the SFR causes at low frequency a greater impedance change.

B. Sensor calibration

In this section, the sensor response is evaluated for the configuration that exhibited the highest relative inductance change. Specifically, the coil with $\Phi_c = 15$ mm and $N = 20$ is considered. As discussed in section ii.A., the tablets used for characterizing the sensing element are the ones without the resin, to evaluate the effect of the particle position as an influence quantity. The calibration was performed at room temperature. The quantity of interest is the impedance change, with respect to the no particle condition, caused by the addition of debris on the top of the coil. It can be divided into its real and imaginary part, i.e. $\Delta Z = \Delta R + j2\pi f \Delta L$.

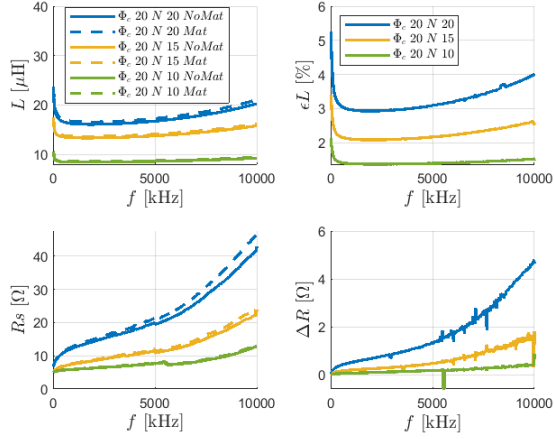


Fig. 5. Frequency response coil with $\Phi_c = 20$ mm

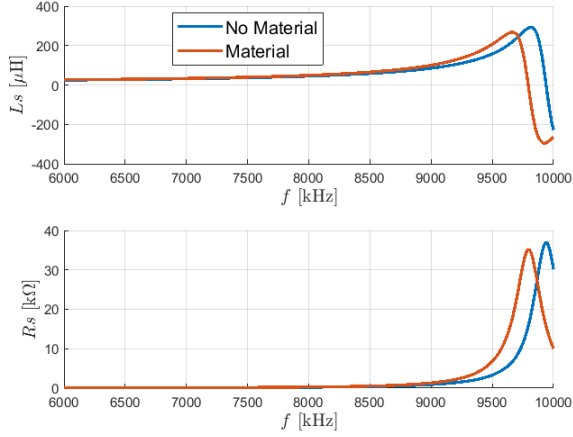


Fig. 6. Frequency response with modified SFR

Fig. 7 shows the results at excitation frequency $f = 1$ MHz for both fine and coarse debris sizes. As expected, as the quantity of debris increases, the change of impedance increases. The curves have a linear behavior, in the range analyzed, that is expected to become quadratic since from a certain mass value the sensor reaches saturation. The blue and red bars are the uncertainty intervals at one sigma. Furthermore, in the mass range analyzed, it is not possible a fine comparison of the two particle meshes because the uncertainty intervals are superimposed.

C. Uncertainty calculation

In this section the uncertainty calculations shown in Fig. 7 is explained. The measurement model is the following.

$$\Delta L = L - L_0 ; \Delta R = R - R_0 \quad (1)$$

where L_0 and R_0 are the coil's unloaded inductance and re-

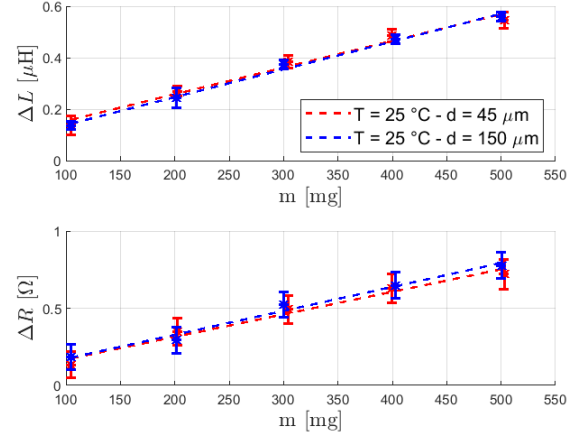


Fig. 7. Sensor response @ $f = 1$ MHz for debris size $45 \mu\text{m}$ and $150 \mu\text{m}$

sistance at a certain excitation frequency. The uncertainty contributions taken into account are:

- Instrumental uncertainty of the LCR meter: $u_I(R)$ and $u_I(L)$.
- System repeatability, due to noise superimposed to the signal: $u_r(R)$ and $u_r(L)$.
- System reproducibility, related to the particle distribution: $u_{rr}(R)$ and $u_{rr}(L)$.

The system repeatability was evaluated by taking subsequent measurements with the tablet always inserted in its position. For the system reproducibility the series of measurements were taken by removing each time the tablet from the case and shaking it. The system repeatability and reproducibility were calculated by means of the experimental standard deviation of ten repeated measurements. The LCR meter used also processes the impedance magnitude and phase to get the values of the resistive and inductive parts, these were used for the calculations of the repeatability and reproducibility uncertainty contributions. As a result the total uncertainty of R_0 , R , L_0 and L are the following:

$$u(R_0) = \sqrt{u_I^2(R_0) + u_r^2(R_0)} \quad (2)$$

$$u(R) = \sqrt{u_I^2(R) + u_r^2(R) + u_{rr}^2(R)} \quad (3)$$

$$u(L_0) = \sqrt{u_I^2(L_0) + u_r^2(L_0)} \quad (4)$$

$$u(L) = \sqrt{u_I^2(L) + u_r^2(L) + u_{rr}^2(L)} \quad (5)$$

Regarding the instrumental uncertainty, the manufacturer provides only the uncertainties on the impedance

magnitude and impedance phase, i.e. $\delta|Z|$ and $\delta\theta$, according to the deterministic approach. Thus, the standard uncertainties of $|Z|$ and θ are $u(|Z|) = \delta|Z|/\sqrt{3}$ and $u(\theta) = \delta\theta/\sqrt{3}$ respectively. However, the ones on the resistive and inductive parts can be derived applying the formula of propagation of uncertainty to the following measurement model:

$$R = |Z| \cos(\theta) ; L = \frac{|Z| \sin(\theta)}{2\pi f} \quad (6)$$

thus,

$$u_I(R) = \sqrt{\cos^2(\theta)u^2(|Z|) + |Z|^2 \sin^2(\theta)u^2(\theta)} \quad (7)$$

$$u_I(L) = \frac{\sqrt{\sin^2(\theta)u^2(|Z|) + |Z|^2 \cos^2(\theta)u^2(\theta)}}{2\pi f} \quad (8)$$

At the frequency studied, $\delta|Z| = 0.00264|Z| \Omega$ and $\delta\theta = 0.159^\circ$. By applying the rule of propagation of uncertainty to Eq. 1 the uncertainty of ΔL and ΔR is found:

$$u(\Delta R) = \sqrt{u^2(R) + u^2(R_0) - 2 \cdot \rho \cdot u(R) \cdot u(R_0)} \quad (9)$$

$$u(\Delta L) = \sqrt{u^2(L) + u^2(L_0) - 2 \cdot \rho \cdot u(L) \cdot u(L_0)} \quad (10)$$

Where ρ is the correlation factor between R_0 and R and between L_0 and L . For an initial analysis the case with $\rho = 0$, that is the worst one, is considered. Tabs. 1 and 2 show the main uncertainty contributions for particles of size $45 \mu\text{m}$. The uncertainties for particles of size $150 \mu\text{m}$ are similar, hence they are not reported. The uncertainty associated to the system repeatability is lower than $1 \text{ m}\Omega$ and 0.1 nH , for the reactive and inductive parts respectively, hence they can be neglected. The instrumental uncertainty associated to the unloaded values, i.e. $u_I(R_0)$ and $u_I(L_0)$, are similar to the ones of the tables, where the particles are added, hence they are not indicated as well. In Tabs. 1 and 2, $u(\Delta R)$ and $u(\Delta L)$ are calculated using Eqs. 9 and 10, respectively.

Table 1. Uncertainty contributions of ΔR - particles size $45 \mu\text{m}$

m [mg]	$u_{rr}(R)$ [mΩ]	$u_I(R)$ [mΩ]	$u(\Delta R)$ [mΩ]
104.8	15	59	83
202.4	25	60	86
304	34	61	90
399.8	32	64	91
503.1	46	63	97

Table 2. Uncertainty contributions of ΔL - particles size $45 \mu\text{m}$

m [mg]	$u_{rr}(L)$ [nH]	$u_I(L)$ [nH]	$u(\Delta L)$ [nH]
104.8	35	8.9	37
202.4	13	9.1	18
304	22	9.3	25
399.8	19	9.7	23
503.1	30	9.5	33

It can be seen that both the instrument and particles contribution should be accounted for, as they are comparable in magnitude. Specifically, the inductance uncertainty of the instrument is less than the effects of the influence quantity. Furthermore, the total uncertainty on the inductive part is smaller than the one on the resistive part.

IV. FUTURE DEVELOPMENTS

Once the metrological characterization of the sensing element is completed, a conditioning circuit will be designed that is able to measure the impedance changes due to the accumulated debris mass. About the sensor characterization, the last step consists in the evaluation of the temperature effects, which is now in progress.

For the design of the conditioning circuit, the results summarized in Fig. 7 suggest that an LC oscillator could be a suitable solution, since the changes in the imaginary part of the impedance are dominant with respect to the changes in the corresponding real part. In this way, the oscillation frequency measurement can be related to the inductance changes ΔL of the sensing element, which in turns can be converted into the accumulated debris mass. To this aim, the input (m)/output(ΔL) relationship of the sensing element is identified by means of a linear model:

$$\Delta L = a \cdot m + b \quad (11)$$

The model parameters a and b are identified through a Least Square algorithm:

$$\beta = \begin{bmatrix} a \\ b \end{bmatrix} = (X^T \cdot X)^{-1} \cdot X^T \cdot Y = M \cdot Y \quad (12)$$

where X is the input (m) matrix and Y is the output (ΔL) array. The following values of a and b have been found: $a = 0.0011 \mu\text{H}/\text{mg}$, $b = 0.04 \mu\text{H}$. The uncertainty on the mass values, i.e. on X , can be neglected while the uncertainty on the impedance change i.e. Y , as in Tab. 2, must be considered. The uncertainty in the fitting parameters a and b can be found from the covariance matrix of β , i.e. $u(\beta)$:

$$u(\beta) = M \cdot u(Y) \cdot M^T = \begin{bmatrix} u^2(a) & u(a,b) \\ u(a,b) & u^2(b) \end{bmatrix} \quad (13)$$

where $u(Y)$ is the covariance matrix of Y , which is a diagonal matrix, since successive measurements of ΔL are supposed to be independent, with diagonal elements $u^2(\Delta L)$ from Tab. 2 for particle size $45 \mu\text{m}$. From inversion of Eq. 11 the calibration function is eventually obtained:

$$m = \frac{\Delta L}{a} - \frac{b}{a} \quad (14)$$

The rule of propagation of uncertainty can be applied to Eq. 14 to find the uncertainty on the mass, i.e. $u(m)$. The uncertainty contributions to consider are $u(a)$, $u(b)$, $u(a,b)$ and $u_{LC}(\Delta L)$, where the latter is the uncertainty related to the inductance change measured through the LC oscillator. Its value can reasonably be estimated as $u(\Delta L) = 40 \text{ nH}$ if the effects of the particle distribution is considered as the main uncertainty contribution. If this assumption was true, the expected mass standard uncertainty would be on the order of tens of milligram.

V. CONCLUSIONS

This article presents a formal methodology to design and characterize an inductive sensing element to measure ferromagnetic material that is captured during the wear process of mechanical equipment.

At first, a multi-parametric experimental analysis is carried out to find the best sensor geometry compatible with the existing mechanical limits.

The sensing element found was then characterized. The effect of particle size as influence quantity was also studied. It was seen that its uncertainty, in the mass range analyzed, can be neglected. However, the particle distribution has a non negligible contribution to the uncertainty. Furthermore, it was seen that in the studied range the response is almost linear

Future works will include the evaluation of temperature effect, and also debris masses in lower ranges will be considered. Furthermore, a more detailed analysis of the conditioning circuit will be provided.

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Detection of Physical Layer Anomalies and Eavesdropping in Real World Data Cables

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Abstract – The reliability and security of communication cables are critical in Cyber-Physical Systems (CPS), where physical-layer anomalies such as damage and unauthorized tapping can compromise system integrity. This work presents a passive, non-invasive approach for anomaly detection in Ethernet cables, targeting low-cost edge deployment. Unlike active diagnostic methods, the proposed approach operates on naturally transmitted signals without interrupting communication. A system is developed, integrating a custom differential amplifier, and a Red Pitaya acquisition platform. Real-world datasets have been collected from 10BASE-T and 100BASE-TX Ethernet links under controlled normal and faulty conditions, including air exposure, water exposure, and tapping. Initial signal analysis using amplitude histograms reveals distinct statistical variations across cable conditions. The current results establish a validated data acquisition pipeline and demonstrate the feasibility of passive signal monitoring. This work lays the foundation for scalable, edge-based cable diagnostics in industrial and critical infrastructure systems.

Keywords: Passive Diagnosis, Anomaly Detection, Cable Security

I. INTRODUCTION

Cyber-Physical systems (CPSs) are new types of intelligent systems at the core of modern technologies such as autonomous vehicles, smart grids, industrial automation, and advanced robotics [1]. They integrate computing, control, and communication technologies, bridging the cyberspace and physical world [2]. CPSs can hence be seen as the integration of computation, networking, and physical processes. While these systems greatly enhance the efficiency, reliability, and functionality of infrastructure, they also bring new security challenges that need to be addressed [3]. An important part of any CPS infrastructure is the physical layer of the network, which encompasses

the hardware elements responsible for data transmission and reception, including cables, connectors, and associated equipment. Monitoring the physical layer enables the timely detection of signal degradation, physical damage, and unauthorized access attempts such as cable tapping [3]. In cable communication security, physical layer monitoring and diagnosis typically target three key objectives: anomaly detection, localization, and classification [4]. The type of anomalies can be classified as follows: i) hard and ii) soft [5]. Hard anomalies are severe and typically cause complete failure of the cable. These anomalies are characterized by physical damage or significant degradation that interrupts the electrical or signal continuity (e.g., open and short circuits) [5]. Soft anomalies are less severe and usually cause degradation in performance rather than complete failure. These anomalies can result in intermittent issues or degraded signal quality (e.g., insulation damage, tapping, and connector issues) [5].

Monitoring methods adopted for physical layer include reflectometry [6], and signal quality assessment [7]. Reflectometry is a technique for diagnosing cable conditions, which exploits the signal characteristics of the reflections of a transmitted signal in the wireline channel [4]. It is particularly effective for detecting anomalies, discontinuities, and physical degradation in cables and connectors [8]. Traditional reflectometry methods are active in nature, meaning they involve injecting a test signal into the cable and analyzing the resulting reflections in various domains [4]. Based on the type of incident signal and the considered domain of analysis, reflectometry can be divided into (i) time-domain reflectometry (TDR) [9], (ii) frequency-domain reflectometry (FDR) [10], and (iii) time-frequency-domain reflectometry (TFDR) [8], [11]. However, active reflectometry methods typically require interruption of the communication system's operation. To address this limitation, passive techniques have been introduced, relying on link quality indicators such as bit error rate, state transitions, eye patterns, and signal-to-noise ratio (SNR) [8]. These methods can detect partial or full open circuits or short circuits but cannot provide good information about soft anomalies (e.g., insulator degradation) [12]. Kallel et al.

[12] proposed a cost-effective, ML-based distributed monitoring system using histogram-based techniques to detect and classify anomalies, though the types considered were limited to hard and soft open/short circuits. Balestrieri et al. [4] introduced a novel approach for detecting, diagnosing, and localizing three types of soft anomalies: (i) tapping, (ii) water exposure, and (iii) air exposure on Cat5e cables, using various frequency-domain features. Subsequent work by Dhakal et al. [13] relied on histogram-based feature extraction to classify soft anomalies in Cat5e Ethernet cables, demonstrating that such features remain effective even at significantly reduced sampling rates.

The present research builds upon their [4, 13] works, aiming to extend passive cable anomaly detection into a more realistic operational setting by utilizing an actual communication channel. Unlike previous studies that relied primarily on synthetically generated signals from Arbitrary Waveform Generators (AWGs) [4], the approach captures and analyzes real network traffic over Ethernet protocols, specifically 10BASE-T and 100BASE-TX, between a router and a Raspberry Pi. Furthermore, dependence on expensive laboratory-grade oscilloscopes is reduced through the use of a cost-effective data acquisition setup based on Red Pitaya.

The article is structured as follows. Section II provides an overview of the active and passive diagnostic methods and discusses the novelties of the research activity. Section III describes the designed and realized experimental setup and the dataset generation. The resultant dataset and its properties are discussed in Section IV. Section V draws conclusions and delineates future work activities.

II. RELATED WORKS

Active methods rely on signal injection for detecting, classifying, or localizing anomalies in wireline channels. TDR techniques have low sensitivity for soft anomalies due to weak reflections [14]. Time-reversal multiple signal classification (TR-MUSIC) improves this by using an AWG and oscilloscope to estimate scattering parameters, achieving 100% classification accuracy and a 0.33% location error on a 51 m RG58/U coaxial cable [14]. TFDR techniques, employing Gaussian chirp signals, offer high spatial resolution. A Wigner-Ville-based algorithm enhanced anomaly localization, with 100% classification accuracy and a maximum location error of 4.6% [15]. Chang and Kwon [16] introduced a TFDR method using an LSTM-VAE model. Reflected signals processed through this network achieved 100% classification accuracy and 1% localization error for 35 m cable configurations. Impedance Spectroscopy (IS) identifies anomalies via input impedance analysis. A Fourier transform spike indicates the anomaly's location, with a 40 cm error over a 400 m distance [5].

Passive methods rely on monitoring naturally transmit-

ted signals without injecting test signals, making them less invasive but incapable of localization. In [7], features like signal attenuation and SNR margins were used with a random forest classifier to identify bridge tap, open/short circuits, and partial faults with 97% accuracy. Kallel et al. [12] used a support vector classifier (SVC) on peak voltage features acquired via a low-cost microcontroller, achieving 100% classification accuracy for Ethernet cable faults. Wang et al. [3] proposed a passive intrusion detection method for Fieldbus networks based on impedance shifts. By comparing real-time voltage-amplitude features to a stored baseline, it detected unauthorized access with 99% accuracy across three network points.

The research [4] used the passive measurement for the diagnosis of cable anomalies using frequency-based features. They implemented many features and trained the Decision Tree classifier at sampling rates ranging from 62.5 MHz to 6.25 GHz. The classification accuracy at 312.5 MHz was found to be 99.05%. They also conducted a sensitivity analysis of features relative to anomaly length, demonstrating the potential of using passive methods to estimate the lengths of soft anomalies.

Unlike active methods like TDR, which requires the injection of test signals and may interrupt normal communication, the proposed system passively monitors the cable during regular operation, enabling continuous and non-intrusive monitoring. However, a key limitation of the proposed approach is that it does not provide fault localization, which is one of the main advantages of TDR.

This work advances passive detection by moving from controlled laboratory signal generation to real Ethernet communication environments. Unlike prior studies that relied on AWG-generated synthetic signals and high-end oscilloscopes, the proposed approach analyzes live 10BASE-T and 100BASE-TX traffic between networked devices using a low-cost Red Pitaya acquisition platform.

III. METHODOLOGY

A. Hardware Architecture Design

Two experimental datasets were acquired using: (i) high-resolution oscilloscope-based acquisition and (ii) a low-cost edge acquisition system based on Red Pitaya and a differential amplifier. The setup consists of a Raspberry Pi and a router connected via a Cat5E Ethernet cable (see Fig. 1). The oscilloscope is connected to the cable via a differential probe and operates at a 1.25 GSa/s sampling frequency. The data-acquisition system, based on the Red Pitaya stage, consists of a differential amplifier to extract the voltage difference between signal pairs while rejecting common-mode noise. The differential amplifier was implemented using the AD8009 operational amplifier as depicted in Fig. 2), selected for its high bandwidth (1 GHz) and fast slew rate, which makes it appropriate for capturing high-frequency Ethernet signals such as those in 10BASE-

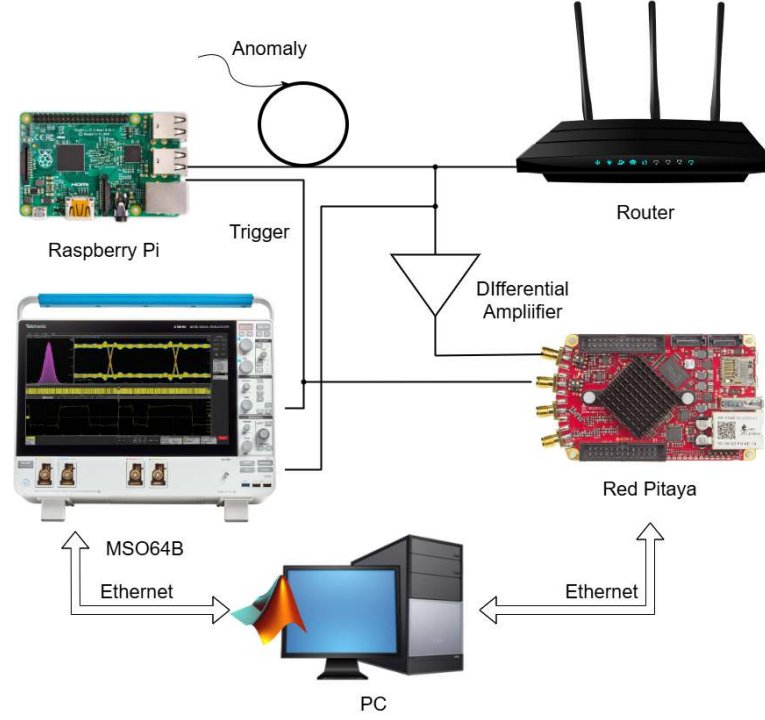


Fig. 1. Experimental Setup for dataset collection using Red Pitaya and oscilloscope

T and 100BASE-TX communication. The resistor network was carefully designed to balance input impedance and amplifier gain while ensuring compatibility with the Red Pitaya input range. The Red Pitaya features fast analog inputs with a full-scale voltage range of ± 1 V in the low-voltage mode. Hence, the input resistor values were selected such that a nominal 2.5 V signal from a 10BASE-T cable is attenuated to approximately ± 1 V at the Red Pitaya input. Specifically, $R_1 = 470 \Omega$ was used as the input resistor, while $R_3 = 180 \Omega$ served as the feedback resistor to establish the required amplifier gain. The amplifier output is given by:

$$V_{out} = \frac{R_3}{R_1}(V_2 - V_1) \quad (1)$$

The current prototype presents a differential input impedance of approximately 50Ω rather than the nominal 100Ω of the Ethernet line. However, this mismatch did not affect communication during the experiments, and the proposed method. Future work will focus on improving the front-end design to achieve a closer impedance match and further reduce loading effects.

The analog-to-digital conversion is performed using either the High-resolution oscilloscope Tektronix MSO64B (for reference dataset) or Red Pitaya ADC (125 MS/s) for the edge-compatible dataset. The STEMLab 125-14 Red Pitaya features a 50 MHz analog bandwidth and a dual-channel 14-bit ADC capable of sampling at 125 MS/s [17].

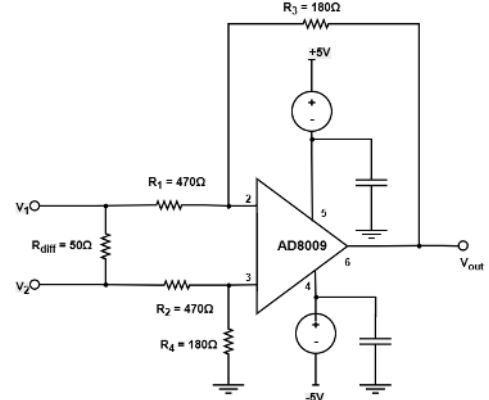


Fig. 2. Differential amplifier circuit using AD8009 for passive Ethernet signal acquisition.

The physical trigger shown in Fig. 1 is employed for synchronization during 100BASE-TX data collection, whereas for 10BASE-T, the trigger is not used. Unlike 10BASE-T, a 100BASE-TX link is never truly idle due to continuous signaling. Therefore, a specialized physical trigger was developed on a separate channel using GPIO pins in the Raspberry Pi to synchronize data acquisition with the exact moment a packet is transmitted. This approach ensures that the Red Pitaya or oscilloscope captures the relevant portion of the Ethernet signal without relying on idle periods. The oscilloscope and the Red Pitaya were

both connected to a PC through Ethernet for remote control and automated data collection using MATLAB.

B. Data Collection

Python scripts running on the Raspberry Pi generate controlled Ethernet traffic for 100BASE-TX and 10BASE-T. The payload depended on the protocol: (i) 94 random bytes for 10BASE-T and (ii) 250 random bytes for 100BASE-TX. Each captured waveform was transferred from the oscilloscope or the Red Pitaya to the PC, converted into voltage values, and stored in MAT files for later analysis.

The considered anomaly types include air-exposed conductors, water-exposed conductors, and tapping. To generate air-exposed conductor anomalies, the insulation of both wires in the cable under test was manually removed over lengths ranging from 5 cm to 50 cm, with increments of 5 cm. The exposed conductors were carefully kept separated to prevent short circuits. For water-exposed conductor anomalies, the same insulation removal procedure was applied. The exposed section of the cable was then fully submerged in a water container. The anomaly lengths were identical to those used in the air-exposed scenario. In the case of tapping anomalies, an additional two-wire cable was soldered at the midpoint of the main cable. The branch length of the tapping cable was varied between 0.5 m and 2 m, enabling the simulation of different in-

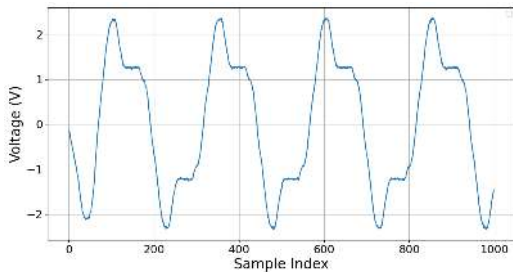


Fig. 3. Differential signal captured by the Oscilloscope from the Ethernet link under normal conditions for 10BASE-T.

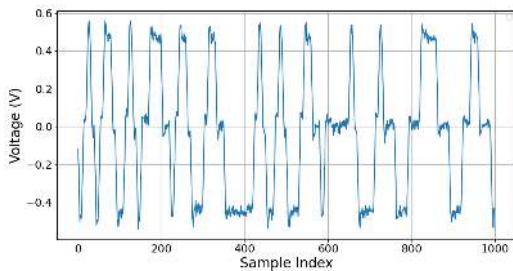


Fig. 4. Differential signal captured by the Oscilloscope from the Ethernet link under normal conditions for 100BASE-TX.

trusion conditions. The dataset is organized according to Ethernet standard and anomaly type. For 10BASE-T, each class contains approximately 12,500 samples, with around 1,250 anomaly samples for each air and water exposure anomaly length, and approximately 2,500 samples for each tapping length. For 100BASE-TX, each class contains approximately 10,000 samples, with around 1,000 anomaly samples for each air and water exposure anomaly length, and approximately 2,500 samples for each tapping length.

IV. RESULTS AND DISCUSSION

The collected data serves as the baseline for future anomaly detection. The differential signals captured from the Ethernet link under normal conditions, for 10BASE-T and 100BASE-TX, are shown in Figs. 3 and 4, respectively. 10BASE-T uses Manchester encoding, where each bit is represented by a transition at the center of the bit period to provide clock synchronization. A logical '1' is represented by a low-to-high transition, while a logical '0' is represented by a high-to-low transition. The 10BASE-T waveform exhibits comparatively lower frequency content with smoother transitions and larger amplitude variations due to Manchester encoding, with a differential amplitude of approximately $\pm 2.5V$. 100BASE-TX uses Multi-Level Transmit-3 (MLT-3) encoding, which operates using three voltage levels (-1 , 0 , and $+1$). The signal changes state only when transmitting a logical '1', helping to reduce bandwidth requirements for higher-speed Ethernet communication. The 100BASE-TX signal shows denser and more rapidly changing waveform patterns, reflecting its higher data rate and more complex signaling characteristics, with a differential amplitude of approximately $\pm 1V$.

Figures 5 and 6 show representative histograms of the differential signals acquired by the oscilloscope. The number of peaks observed in the captured waveforms (10BASE-T) is four, as observed from the time-domain signal in Fig. 3.

For 10BASE-T, air exposure has very little effect on the signal, with the original waveform shape remaining mostly unchanged except for small fluctuations. Tapping the cable

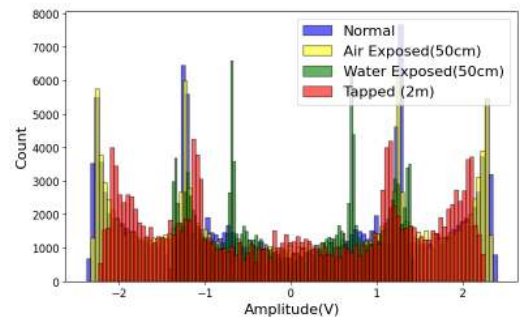


Fig. 5. Amplitude Histogram for 10BASE-T signals acquired by the oscilloscope.

causes signal dispersion and energy loss, making the peaks broader and less distinct while shifting them toward the center. Water exposure has the strongest impact, heavily attenuating the signal and compressing the voltage distribution into a narrower and lower-voltage range.

For 100BASE-TX, water exposure reduces the pulse amplitude, producing smaller secondary peaks at lower voltage levels and showing consistent signal attenuation. Air exposure introduces greater noise and variation, causing the histogram to spread out and the voltage levels to overlap across a wider range. Physical tapping has the strongest effect, spreading the signal distribution into a broad Gaussian-like shape and significantly reducing the distinct voltage levels, which indicates severe signal degradation.

The Figs. 7 and 8 show that the amplitude histograms for data collected through Red Pitaya exhibit similar statistical properties to those obtained using an oscilloscope; more importantly, the histograms corresponding to different cases are distinctly different, demonstrating the ability to differentiate various anomalies.

After preprocessing and feature extraction, a 30-dimensional feature vector consisting of 10 time-domain features and 20 frequency-domain features was extracted from the Fast Fourier Transform (FFT) spectrum, following the methodology described by Balestrieri et al [4]. A

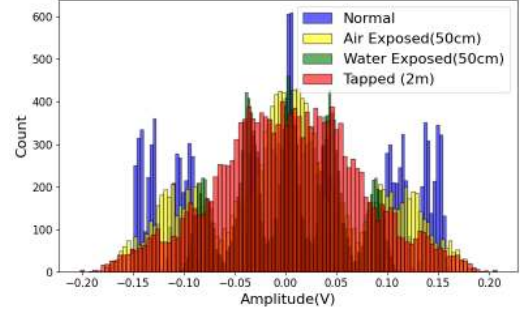


Fig. 8. Amplitude Histogram for 100BASE-TX Data Collected through Red Pitaya.

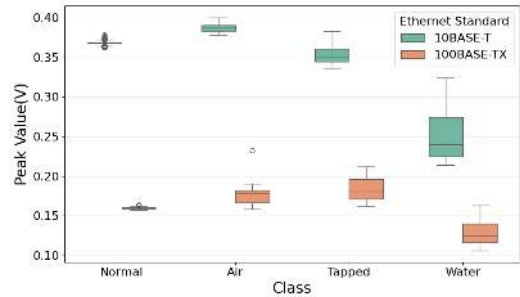


Fig. 9. Peak Voltage Across Cable Conditions for 10BASE-T and 100BASE-TX

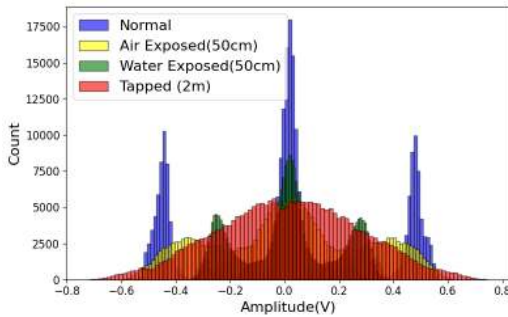


Fig. 6. Amplitude Histogram for 100BASE-TX signals acquired by the oscilloscope.

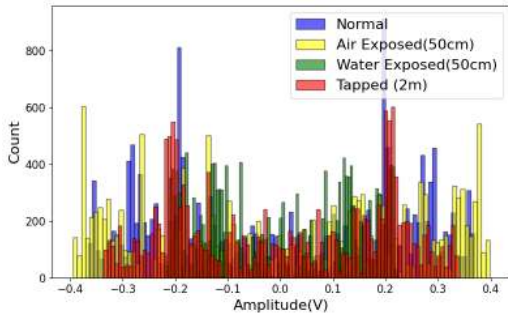


Fig. 7. Amplitude Histogram for 10BASE-T Data Collected through Red Pitaya.

One-Way Analysis of Variance (ANOVA) test was performed on both datasets to identify the sensitivity of these features to anomaly classes. For 100BASE-TX, the top-ranked feature was peak ($F = 44$), followed by the 1st harmonic peak amplitude, standard deviation (std), root mean square (rms), and 4th harmonic peak amplitude. For 10BASE-T, peak again ranked highest with a notably larger F-score ($F = 115$), followed by crest factor, std, rms, and impulse factor.

Figure 9 further examines the peak feature, which was identified as the most discriminative feature by ANOVA for both datasets. The boxplots show clear differences in peak voltage across the cable conditions, with water-exposed cables producing the lowest peak values and normal or air-exposed cables exhibiting the highest. The limited overlap between class distributions demonstrates that peak amplitude could be used for anomaly detection.

V. CONCLUSION

This research presented provides a critical advancement in the field of physical layer security by transitioning from laboratory-grade validation to practical, edge-deployable diagnostics. By demonstrating that passive signal acquisition and analysis can effectively identify soft anomalies like air exposure, water exposure and unauthorized tapping, this study establishes a low-cost framework for

continuous monitoring in Cyber-Physical Systems. This dataset offers a foundational dataset for future research into ML Classification, model compression and sub-Nyquist sampling. Ultimately, this work bridges the gap between signal processing theory and engineering practice, providing a scalable roadmap for enhancing the resilience and security of critical network infrastructure through real-time, non-invasive hardware diagnostics.

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Sustainability-oriented optimization of a sensing board for applications as a cyber-physical measurement system (CPMS)

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Abstract – Within the Industry 5.0 paradigm, Cyber-Physical Systems (CPSs) represent a key technological enabler through the integration of physical processes, sensing, communication, and computation. In this context, traditional Measurement and Monitoring Systems (MMSs), typically devoted to data acquisition and transmission, can evolve into autonomous and intelligent Cyber-Physical Measurement Systems (CPMSs). Starting from this perspective, this paper presents a sustainability-oriented methodology for CPMSs, aimed at providing a *self-adaptation* and *self-configuration* capabilities: these features are in fact crucial in 5.0 CPMSs. In particular, the proposed methodology addresses the self-adaption of sensing and communication configurations according to the operating condition in order to reduce energy demand and the related operational environmental impact. As a case study, the methodology is implemented on an *ST SensorTile.box PRO*, a compact multi-sensor platform for environmental and inertial monitoring: this device is widely adopted as a smart sensing device in cyber-physical applications. To this end, the implemented sustainability-oriented strategy relies on the autonomous identification of quiescent and moving states and dynamically re-configure sensing and transmission configurations accordingly. Environmental quantities are prioritized in quiescent conditions, whereas inertial data are emphasized during motion. Experimental results, expressed through mean values and standard uncertainties, show an energy saving of approximately 4.77 %, with a corresponding reduction in the operational Carbon Footprint. These results highlight the potential of sustainability-oriented CPMSs for more adaptive and resource-aware measurement infrastructures in line with the Industry 5.0 vision.

Keywords: Adaptive sensing; Carbon Footprint; Cyber-Physical Measurement Systems; Industry 5.0; Measurement and Monitoring Systems; Sustainability.

I. INTRODUCTION

The transition from Industry 4.0 to Industry 5.0 is fostering the evolution of measurement systems toward intelligent, connected, and context-aware cyber-physical nodes [1]. In particular, Cyber-Physical Systems (CPSs) have established themselves as the most representative technological paradigms, thanks to the tight integration of physical processes, sensing, communication, and computation. Within this framework, conventional Measurement and Monitoring Systems (MMSs) are typically used only as supporting subsystems, mainly devoted to data acquisition and transmission.

However, such MMSs can also incorporate the enabling technologies of Industry 4.0 and evolve into autonomous and intelligent entities. This evolution leads to Cyber-Physical Measurement Systems (CPMSs), in which sensing, local processing, communication, and adaptive decision-making coexist within the same measurement-oriented architecture [6, 2]. Compared with traditional MMSs, CPMSs extend the role of the sensing node from mere data source to active element capable of reacting to the observed operating condition. In this regard, the node can exhibit basic *Self-X* capabilities, such as self-awareness and self-configuration, coherently with the 5C architecture of CPMS, namely *Connection*, *Conversion*, *Cyber*, *Cognition*, and *Configuration* [7, 5]. This perspective is particularly relevant in Industry 5.0, where sustainability complements connectivity, intelligence, and autonomy as a key design requirement [8]. In this context, adaptive measurement nodes are of interest not only because they can react to the operating condition, but also because such adaptation may reduce unnecessary sensing and communication activities, thus improving resource efficiency.

Starting from this consideration, the present work investigates the evolution of an *ST SensorTile.box PRO* [10], typically used as a conventional MMS node for environmental and inertial monitoring, into a sustainability-oriented CPMS node. The proposed CPMS node adapts the operation according to its current condition: when the board is stationary, environmental information is priori-

tized, whereas when motion is detected, inertial information is given priority. This adaptive behaviour is implemented at the sensing edge through the autonomous reconfiguration of data transmission and sensor sampling frequency.

The contribution of the paper is twofold. First, it provides a practical implementation of self-adaptation on an embedded sensing node, showing how an MMS-oriented platform can be turned into a basic CPMS. Second, it experimentally evaluates the effect of this adaptive operation in terms of energy consumption and associated environmental impact, thus linking CPMS self-adaptation to sustainability-oriented measurement design in the perspective of Industry 5.0.

The paper is organized as follows. Section ii. describes the adopted methodology, Section iii. presents the case study, Section iv. discusses the experimental results. Finally, conclusions are drawn and future work is outlined.

II. METHOD

The adopted methodology is schematically summarized in Fig. 1. Starting from a sensing node capable of acquiring both environmental and inertial quantities, the proposed approach combines three main stages: i) state recognition through an on-board motion-detection algorithm; ii) state-dependent self-adaptive operation, implemented by reconfiguring sensing and transmission policies according to the detected condition; and iii) experimental assessment of the resulting energy demand and environmental impact.

A. State recognition via motion detection

The first stage of the proposed methodology concerns the recognition of the operating condition of the node. This task is performed by an on-board motion-detection algorithm based on accelerometer and gyroscope measurements. For each acquired inertial triplet (x_i, y_i, z_i) , the corresponding *Euclidean norm* is computed as

$$\|\mathbf{v}_i\| = \sqrt{x_i^2 + y_i^2 + z_i^2}. \quad (1)$$

This scalar quantity provides an orientation-independent representation of the instantaneous magnitude of the inertial signal. The resulting norm values are stored in circular buffers and processed through a moving-average filter over a sliding window:

$$\bar{v}_k = \frac{1}{N} \sum_{i=k-N+1}^k \|\mathbf{v}_i\|, \quad (2)$$

where N denotes the window length. This operation reduces short-term fluctuations of the inertial signal by averaging the most recent N values, thus providing a more stable indicator for motion-state detection. A *moving* condition is identified when the filtered norm exceeds experi-

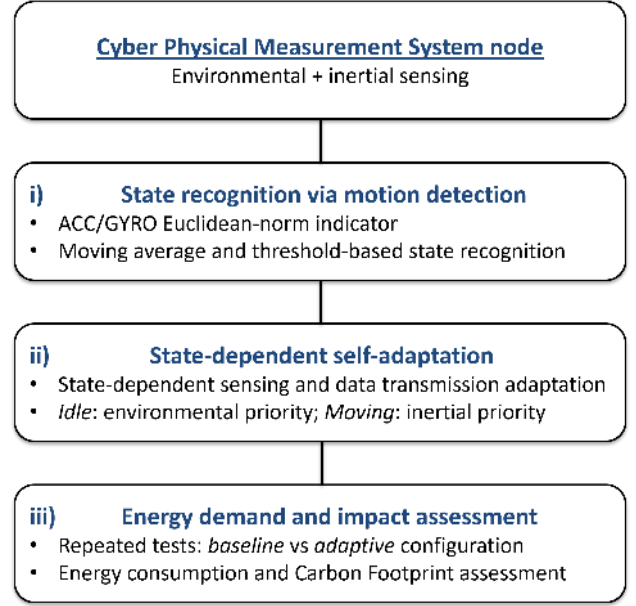


Fig. 1. Overview of the implemented methodology: a CPMS node performs environmental and inertial sensing, identifies its operating state through motion detection based on inertial-data processing, adaptively reconfigures sensing and data transmission configurations, and is finally assessed in terms of energy demand and environmental impact.

mentally calibrated threshold values. According to the outcome of this decision rule, the node is recognized as being either in a stationary or in a moving condition, which then drives the subsequent adaptive operation. The algorithm parameters were determined empirically as a trade-off among noise rejection, responsiveness, and motion detectability.

B. State-dependent self-adaptation

Once the operating condition has been recognized, it is associated with one of the two states used by the adaptive strategy, namely *idle* for stationary conditions and *moving* for motion conditions. For the application scenario considered in this work, these two states correspond to different priorities in the measured quantities: environmental variables, such as temperature and pressure, are considered more informative in the *idle* state, whereas inertial quantities become more relevant in the *moving* state. Accordingly, the node autonomously switches between these states and reconfigures its operation so that sensing and communication activities are adapted to the current condition. The adaptation is implemented mainly by modifying the transmission rate of the sensed quantities according to the detected state. The selected transmission rates were determined empirically as a trade-off between information availability, timely state-dependent reconfigu-

ration, and energy saving.

In this way, the proposed approach preserves the capability to detect state transitions while limiting redundant sensing and communication activities. From a CPMS perspective, this behaviour represents a basic form of *Configuration*-level adaptation.

C. Energy demand and impact assessment

To assess whether the proposed self-adaptive CPMS strategy provides an actual sustainability benefit, the energy demand of the node is compared under two operating conditions: i) a *baseline* condition, representative of a traditional non-self-adaptive MMS operation, in which sensing and transmission settings remain fixed throughout the test, and ii) an *adaptive* condition, corresponding to the proposed CPMS operation, in which sensing and transmission are adjusted according to the detected state (*idle* or *motion*). The comparison is carried out over the same observation interval, so that any difference in the measured energy can be attributed to the adopted operating approach. The monitored quantity is the energy absorbed during the test interval, expressed in kWh.

Let E_i denote the energy measured during the i -th test, with $i = 1, \dots, N$. The sample mean energy is computed as

$$\bar{E} = \frac{1}{N} \sum_{i=1}^N E_i, \quad (3)$$

where N is the number of repeated tests. The experimental dispersion of the measured energy values is described by the sample standard deviation

$$s_E = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (E_i - \bar{E})^2}. \quad (4)$$

Assuming repeated observations under nominally identical conditions, the standard uncertainty associated with the mean energy is evaluated through a Type A approach as

$$u(\bar{E}) = \frac{s_E}{\sqrt{N}}. \quad (5)$$

Once the average energy demand had been estimated, the environmental impact associated with the use phase was assessed. In particular, the operational *Carbon Footprint*, defined here as the amount of greenhouse gas emissions expressed in $\text{CO}_{2,\text{eq}}$ attributable to the considered operational phase [9], was evaluated by means of the following measurement model [3, 11]:

$$\overline{\text{CO}_{2,\text{eq}}} = \bar{E} \cdot \overline{EF} \quad (6)$$

where:

- $\overline{\text{CO}_{2,\text{eq}}}$ is the estimated operational Carbon Footprint over the considered test interval;
- $\bar{E}$ is the mean measured energy consumption, expressed in kWh;
- $\overline{EF}$ is the mean electricity conversion factor, expressed in $\text{g CO}_{2,\text{eq}}/\text{kWh}$.

In this work, the electricity conversion factor is derived from the monthly electricity-emission values reported by Electricity Maps [12] for Italy over the latest 12 months, and is represented by its mean value and associated standard uncertainty. These quantities are evaluated according to the statistical formulation given in (3)–(5).

Assuming independence between the estimated mean energy consumption and the estimated mean emission factor, the standard uncertainty of the operational Carbon Footprint is obtained by uncertainty propagation [?]:

$$u(\text{CO}_{2,\text{eq}}) = \sqrt{(\overline{EF} u(\bar{E}))^2 + (\bar{E} u(\overline{EF}))^2}. \quad (7)$$

This procedure makes it possible to assess, in a metrologically consistent way, whether the proposed transition from a fixed-operation MMS logic to a self-adaptive CPMS logic leads to a measurable reduction not only in operational energy demand, but also in the corresponding estimated environmental impact.

III. CASE STUDY

The experimental activity was carried out on the *ST SensorTile.box PRO* board [10], a compact multi-sensor node based on an STM32U585AI microcontroller and equipped with Bluetooth Low Energy (BLE) connectivity. In the present work, the board was used in *Pro Mode*, namely the advanced operating mode intended for custom firmware development.

The board was configured to acquire two classes of quantities:

- environmental quantities, namely temperature and pressure;
- inertial quantities, namely acceleration and angular velocity.

A custom firmware was developed in *STM32CubeIDE* [13] by adapting ST-provided software components to the proposed self-adaptive logic, and was then uploaded to the board through *STM32CubeProgrammer* [14]. On the host side, a Python script was used to receive BLE data and to support the timing of the experimental procedure.

A. Compared firmware configurations

Consistently with the comparison introduced in Section ii., the board was tested under two operating conditions: a *baseline* condition, representative of a traditional non-self-adaptive MMS with fixed sensing and transmission settings, and an *adaptive* condition, representative of the proposed CPMS implementation, in which sensing and transmission are adjusted according to the detected operating state. Specifically, two firmware configurations were experimentally compared:

- *Baseline firmware*: a fixed-operation configuration representative of a conventional MMS, in which both environmental and inertial data are transmitted at a constant rate of 50 sps;
- *Adaptive firmware*: a self-adaptive configuration representative of the proposed CPMS, in which the transmitted quantities depend on the detected operating state. In this implementation, the two operating states introduced in Section ii. were instantiated as follows:
 - *Idle state*: environmental data are transmitted at 50 sps while accelerometer and gyroscope data are transmitted at 10 sps;
 - *Moving state*: accelerometer and gyroscope data are transmitted at 50 sps, while environmental data are transmitted at 10 sps.

The values of 50 sps and 10 sps were selected as a practical compromise between information availability and energy reduction. A transmission rate of 50 sps was adopted as the high-rate condition because it provides a sufficiently dense data stream for dynamic observation while remaining compatible with the timing constraints of the BLE acquisition chain. Conversely, 10 sps was adopted as the low-rate condition because it still provides one transmitted sample every 0.1 s, which was considered sufficient to preserve awareness of the operating condition. In particular, during the *idle* state, this rate allows the inertial stream to retain sufficient sensitivity to detect macroscopic board motion and trigger the transition toward the *moving* state. Similarly, during the *moving* state, a 10 sps environmental stream remains sufficient to preserve contextual information while reducing the number of transmitted packets. Table 1 summarizes the tested configurations.

Overall, the adaptive configuration preserves the capability to detect motion and retain contextual awareness, while reducing the transmission rate of the less relevant sensing stream in each operating state.

B. Motion-detection configuration

The transition between the *idle* and *moving* states was implemented through the motion-detection approach described in Section ii., based on the *Euclidean norm* of ac-

Table 1. Tested firmware configurations.

Firmware	State	Sensor	Tx [sps]
Baseline	–	Env	50
Baseline	–	Acc	50
Baseline	–	Gyro	50
Adaptive	Idle	Env	50
Adaptive	Idle	Acc	10
Adaptive	Idle	Gyro	10
Adaptive	Moving	Env	10
Adaptive	Moving	Acc	50
Adaptive	Moving	Gyro	50

celerometer and gyroscope samples followed by moving-average filtering. In the adaptive firmware, the employed configuration was based on:

- Window size: $N = 25$ samples;
- Step size: 5 samples;
- Threshold comparison on the filtered accelerometer and gyroscope norm values.

The choice of $N = 25$ was made empirically as a compromise between noise rejection and responsiveness. Since the active inertial stream operates at 50 sps, a 25-sample window corresponds to a temporal span of 0.5 s. This duration was found to be sufficient to smooth impulsive fluctuations and reduce false triggers, while still preserving a sufficiently fast reaction to macroscopic board movements. With a step size of 5 samples, the window is updated every 0.1 s, thus yielding an effective update rate of about 10 sps and an 80 % overlap between consecutive windows. The thresholds adopted in the final implementation were determined through repeated acquisitions in quiescent conditions followed by empirical refinement, in order to obtain a satisfactory compromise among motion sensitivity, immunity to sensor noise, and timely state switching.

C. Experimental setup for energy measurement

Energy measurements were carried out with the board powered through USB-C and monitored by means of a *USB-C power tester*. The tester provided cumulative energy readings in *mWh*, which were used as the main experimental energy metric. Before each run, the battery level was initialized at 100 %. To verify this condition, the firmware was extended so as to expose the battery state both at the beginning and at the end of each test.

Each test lasted 30 minutes. In order to reproduce a repeatable mixed operating profile, the board was manually moved for approximately 1 minute around minutes 9, 19, and 29. The motion timing was coordinated through feedback from the Python acquisition script. For each firmware, the final energy result was expressed as the mean value over 5 repeated runs, together with the correspond-

ing standard uncertainty. Table 2 summarizes the adopted test conditions, while Fig. 2 shows the experimental setup.

Table 2. Features of the experimental setup.

Parameter	Value
Board	ST SensorTile.box PRO
Initial battery level	100 %
Measuring instrument	USB-C power tester
Main energy metric	mWh
Test duration	30 minutes
Number of runs / firmware	5
Forced motion events	3
Motion timing	9, 19, 29 min
Motion duration / event	~1 min



Fig. 2. Experimental setup adopted for the energy assessment: the host laptop runs the Python-based BLE acquisition script, the SensorTile.box PRO executes the tested firmware and transmits the acquired data via BLE, and the USB-C power tester measures the absorbed energy.

IV. RESULTS AND DISCUSSION

The obtained results are discussed through three performance indicators: mean energy consumption, relative energy saving, and operational Carbon Footprint.

A. Energy-consumption results

The energy measured over 30 minutes, computed over 5 repeated runs, was equal to 85.06 ± 0.17 mWh for the baseline firmware and to 81.00 ± 0.11 mWh for the adaptive firmware. The corresponding average energy saving is therefore $\Delta E = 4.06$ mWh, which corresponds to a relative reduction of ≈ 4.77 %. These results show that the proposed self-adaptive CPMS logic leads to a systematic and measurable reduction in the energy absorbed by the sensing node with respect to the fixed-operation MMS-like baseline, providing the main positive performance indicator supporting the sustainability-oriented goal of the work.

B. Environmental impact assessment

Following the methodology described in Section ii., the electricity emission factor EF was represented through the mean value and standard uncertainty of the monthly values reported by Electricity Maps for Italy over the last 12 months. This yielded $EF = 285 \pm 14$ g $\text{CO}_{2,\text{eq}}/\text{kWh}$. By applying the measurement model in (6), with the associated standard uncertainty evaluated through (7), the corresponding operational Carbon Footprint over 30 minutes was estimated as:

- *Baseline*: 0.0243 ± 0.0012 g $\text{CO}_{2,\text{eq}}$;
- *Adaptive*: 0.0231 ± 0.0011 g $\text{CO}_{2,\text{eq}}$.

The corresponding difference is: $\Delta\text{CO}_{2,\text{eq}} = 0.0012 \pm 0.0016$ g $\text{CO}_{2,\text{eq}}$. As expected, the reduction in operational Carbon Footprint is coherent with the reduction in energy consumption. Although the absolute difference is small for a single 30-minute experiment, this effect may become more relevant over longer operating times or across multiple deployed devices.

C. Discussion

The obtained results show that a sensing node (namely, a traditional MMS) can evolve toward a basic CPMS behaviour through state-dependent self-adaptation, and that this transition can produce a measurable sustainability-related benefit.

More specifically, the experimental comparison shows that the proposed adaptive logic reduces the energy demand of the node by about 4.77 % with respect to the fixed-operation baseline. This result is consistent with the implemented strategy, since the adaptation acts by reducing redundant sensing and communication activities associated with the less relevant measurement stream in each operating state. The observed 4.77 % energy reduction should not be interpreted as a theoretical upper limit, but as the result of the selected transmission rates. More aggressive frequency reduction could further decrease energy demand, at the cost of reduced temporal resolution and information availability. Overall, these findings demonstrate that a context-aware reconfiguration of node operation can translate into a measurable system-level energy benefit. Also, with regard to the environmental impact assessment, it can be noticed that the estimated Carbon Footprint values follow the same trend as the measured energy results, which is coherent with the adopted measurement model. Although the absolute difference between the two conditions remains small over the considered test interval, the result provides a consistent indication of the environmental effect associated with the reduced energy demand. Therefore, the Carbon Footprint result should be interpreted as a supporting sustainability indicator, useful to complement the energy analysis within the proposed methodological framework.

The experimental campaign also presents some additional aspects that should be considered when interpreting the results. In particular, the comparison was based on repeated 30-minute tests, which were sufficient to reveal a systematic difference between the two operation strategies under repeatable test conditions, while longer tests could further characterize the adaptive behaviour over extended operating periods. Second, the adopted USB-C power tester provides a practical and repeatable estimate of the overall energy absorbed by the board, although it does not isolate the BLE transmission subsystem from the rest of the node. Accordingly, the reported values should be interpreted as system-level measurements of the complete sensing node, which is consistent with the objective of assessing the overall effect of the proposed adaptive strategy.

V. CONCLUSIONS

This paper addresses the evolution of conventional MMSs toward sustainability-oriented CPMS nodes, with the aim of showing how *self-adaptive* and *self-configuring* operations can contribute to reducing resource consumption during the use phase. This objective, which is consistent with the Era 5.0 paradigm, was pursued through a case study based on the SensorTile.box PRO board, which was upgraded from a fixed-operation measurement node to a basic self-adaptive CPMS node. The proposed approach combines motion-based state recognition with adaptive sensing and transmission reconfiguration. The experimental results showed that, with respect to a fixed-operation baseline representative of a traditional MMS, the proposed solution reduces the average energy consumption from 85 mWh to 81 mWh over 30-minute tests, corresponding to an energy saving of about 4.77 %. The estimated operational Carbon Footprint showed a consistent decreasing trend, in agreement with the measured energy reduction. These results indicate that even a basic self-adaptive behaviour can support the evolution from conventional MMSs toward more sustainable CPMS nodes. Future work will address longer experimental campaigns, more advanced energy-management strategies, and the integration of Embedded AI approaches to enable more flexible and autonomous context recognition and self-adaptation at the edge.

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Vector Fitting-Based Feature Extraction from Electrochemical Impedance Spectroscopy for SOC and SOH Estimation

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Abstract – This paper analyses the use of vector fitting parameters obtained from electrochemical impedance spectroscopy (EIS) data to estimate battery state variables. The impedance frequency response is rationally approximated using a vector fitting algorithm, and the resulting parameters, together with temperature, are used as inputs to predict the battery states. Specifically, two models are developed to estimate State Of Charge (SOC) and State Of Health (SOH). The evaluation is based on repeated random train-validation-test splits to obtain robust performance estimates. The results show strong predictive performance, with SOC estimation achieving average R^2 values above 90%, while SOH estimation reaches an average R^2 of approximately 73 %. Overall, the results indicate that vector fitting parameters provide a useful feature representation for battery state estimation when combined with non-linear models.

Keywords: electrochemical impedance spectroscopy; vector fitting; state of charge; state of health

I. INTRODUCTION

The growing integration of renewable energy systems has increased the reliance on lithium-ion batteries as primary energy storage solutions. Their high energy density and efficiency make them suitable for applications ranging from portable electronics to large-scale energy storage. However, ensuring reliable operation requires accurate monitoring of battery condition over time. In particular, battery management systems require reliable estimation of internal states that cannot be measured directly.

Two key indicators are commonly used to describe battery condition: the State Of Charge (SOC), which reflects the battery charge level at a given time, and the State Of Health (SOH), which characterizes long-term

degradation. The SOH is defined as the ratio between the measured capacity at a given stage and the nominal capacity of the battery. While SOC provides short-term operational information, SOH captures aging effects and is essential for predicting Remaining Useful Life (RUL) and ensuring safe operation. Accurate estimation of both quantities remains a challenging task due to the nonlinear and time-varying nature of battery, including the effect of temperature.

A wide range of methods has been proposed for SOC and SOH estimation. Time-domain approaches rely on features extracted from charge and discharge curves, such as voltage profiles or incremental capacity analysis [1]. Electrochemical Impedance Spectroscopy (EIS) has emerged as a powerful alternative, providing rich information about internal electrochemical processes across a wide frequency range [2]. In this context, two main strategies can be identified. On one hand, feature-based approaches extract a limited number of indicators from the impedance spectrum [3]. On the other hand, recent data-driven methods use the full impedance spectrum directly as input to machine learning models [3, 4]. While the latter avoids manual feature selection and achieves high predictive accuracy, it introduces high-dimensional inputs and requires consistent frequency values across datasets.

Another common approach is based on Equivalent Circuit Models (ECM), which approximates the impedance response using a predefined circuit model. These models provide a compact and physically interpretable representation and have been successfully combined with machine learning techniques for state estimation [5]. However, ECMs rely on a priori assumptions regarding the circuit structure, which may not adequately capture complex battery behavior and can lead to model mismatch under varying operating conditions. Moreover, the estimation of ECM parameters typically involves nonlinear optimization and may require a large number of parameters to accurately describe the impedance response [6].

Despite these advances, several challenges remain. In particular, existing methods involve trade-offs between accuracy, computational complexity, and generalization, with no single approach providing a universally optimal solution [7]. Methods based on the full impedance spectrum suffer from high dimensionality and limited flexibility when applied to datasets with different frequency ranges. Feature-based and ECM approaches rely on predefined representations that may not generalize well across different batteries or operating regimes. These limitations highlight the need for representations that are both compact and adaptable, while preserving the essential information contained in the impedance response.

In this work, the use of vector fitting as a flexible and compact representation of EIS data is proposed. Unlike equivalent circuit models, vector fitting does not impose a predefined structure, while still providing a parametric description of the frequency response. This allows the transformation of high-dimensional impedance spectra into a reduced set of parameters that retain the underlying system dynamics and can be consistently applied across datasets with different measurement frequencies. Building on this representation, a machine learning framework is proposed for SOC and SOH estimation based on vector fitting parameters. The proposed approach combines the advantages of full-spectrum methods and compact feature-based models, enabling efficient and robust prediction across varying experimental conditions.

The remainder of this paper is organized as follows. Section 2 describes the data and preprocessing, followed by the vector fitting methodology in Section 3 and the machine learning framework in Section 4. The results and discussion are presented in Section 5.

II. DATA

In this study, a publicly available dataset of electrochemical impedance spectroscopy (EIS) measurements for lithium-ion batteries is used [8]. The dataset was designed for SOH estimation and includes impedance spectra collected under controlled experimental conditions across different states of health, states of charge and temperatures.

The dataset comprises measurements from 25 cylindrical lithium-ion cells with a nominal capacity of 5 Ah, five for each SOH level, which were subjected to accelerated ageing through repeated charge-discharge cycling. Specifically, the cells were charged at 0.5C and discharged at 1C until the predefined SOH breakpoints of 100 %, 95 %, 90 %, 85 %, and 80 % were reached.

EIS measurements were performed at five SOC levels (5 %, 20 %, 50 %, 70 %, and 95 %) and three temperature conditions (15 °C, 25 °C, and 35 °C). Each measurement consists of complex impedance values recorded over a range of excitation frequencies (0.01 Hz - 10 kHz), capturing the

electrochemical behavior of the cells.

The complete dataset has a total of 360 impedance spectra for analysis. Relevant metadata were extracted from the file naming convention, including battery identifier, temperature, SOC, nominal SOH, and measured SOH. These variables are combined with the parameters estimated from the impedance spectra via the vector fitting resulting in a structured dataset used for analysis and predictive modeling.

III. METHODS

A. Vector Fitting Algorithm

Vector fitting is a robust numerical method for approximating frequency-domain responses with rational functions. It was originally introduced by Gustavsen and Semlyen [9, 10] and is widely used for modeling impedance data.

Given a set of measured impedance values $Z(j\omega)$, the method approximates the response using a rational function of the form

$$Z(s) \approx \sum_{k=1}^N \frac{r_k}{s - p_k} + D + Es, \quad (1)$$

where the approximation is evaluated at $s = j\omega$, p_k are the N poles, r_k are the corresponding residues, and D and E are constant terms accounting for the system asymptotic behavior.

The vector fitting algorithm iteratively relocates the poles to improve the approximation. Starting from an initial set of poles, chosen as real values distributed over the frequency range of interest, the method solves a linear least-squares problem to estimate the residues and auxiliary coefficients. The poles are then updated based on these coefficients, and the procedure is repeated until convergence is achieved.

In this study, the impedance data are first transformed into the Laplace domain using $s = j2\pi f$, where f denotes the frequency from the EIS measurement files. The vector fitting procedure is then applied to obtain a rational approximation of each impedance spectrum. The resulting parameters provide a compact representation of the original high-dimensional data.

B. Implementation Details

The vector fitting procedure is applied to each impedance spectrum using a Python implementation adapted from the original MATLAB algorithm provided by SINTEF [11], specifically the open-source library [12].

A key parameter in vector fitting is the number of poles N used in the rational approximation. To determine an appropriate model order, configurations ranging from 4 to 10 poles are evaluated. For each configuration, the root mean square error (RMSE) is computed.

As shown in Fig. 1, the RMSE decreases significantly when increasing the number of poles from 4 to 7, after which the improvement becomes marginal. Based on this trade-off between model complexity and accuracy, a value of 7 poles is selected for all subsequent analyses.

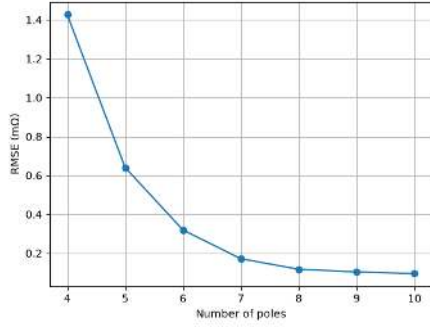


Fig. 1. Fitting RMSE as a function of the number of poles.

The initial poles are chosen as real-valued negative numbers logarithmically spaced over the frequency range of interest. This choice ensures stability of the resulting rational approximation and provides a suitable initial distribution for the iterative pole relocation process. The fitting is performed iteratively, with pole relocation at each step. Convergence is assessed based on the relative change in pole locations between successive iterations. Specifically, it is considered that the algorithm converged when the maximum relative change in poles falls below a predefined tolerance. In this work, a tolerance of 10^{-2} is used, with a maximum of 20 iterations.

The quality of the fit is illustrated in Fig. 2, where the measured impedance is compared with the fitted response in the complex plane for one of the cells. The fitted curve closely follows the measured data, indicating that the vector fitting model provides an accurate representation of the measured impedance.

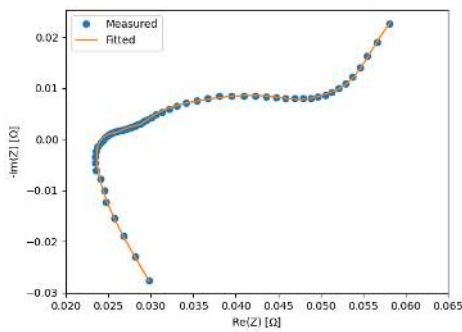


Fig. 2. Nyquist plot of measured and fitted impedance.

Further validation is provided in Fig. 3, which shows the measured and fitted impedances and the corresponding fitting error across the frequency range. The error remains consistently low, confirming the accuracy of the rational approximation.

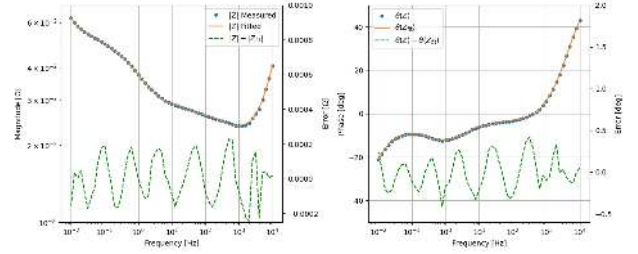


Fig. 3. Measured and fitted impedance magnitude, impedance phase and fitting errors.

C. Parameter Analysis

After convergence, the fitted model yields a set of parameters consisting of poles, residues, and constant terms. These parameters are combined into a feature vector for each impedance spectrum and used as inputs for the machine learning step. The extracted parameters are plotted against the SOH in Fig. 4 and SOC in Fig. 5 to assess their relevance.

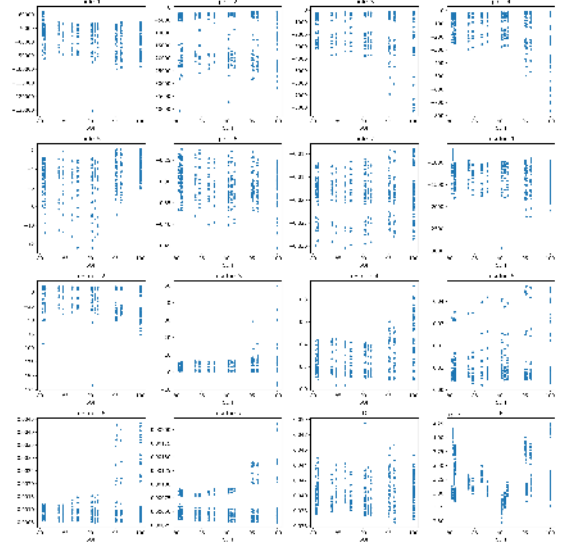


Fig. 4. Vector fitting parameters as a function of SOH.

The vector fitting parameters exhibit a nonlinear and highly scattered relationship with both SOC and SOH. This suggests that while individual parameters do not provide direct predictive insight, their combined use within a nonlinear model can capture the underlying relationships.

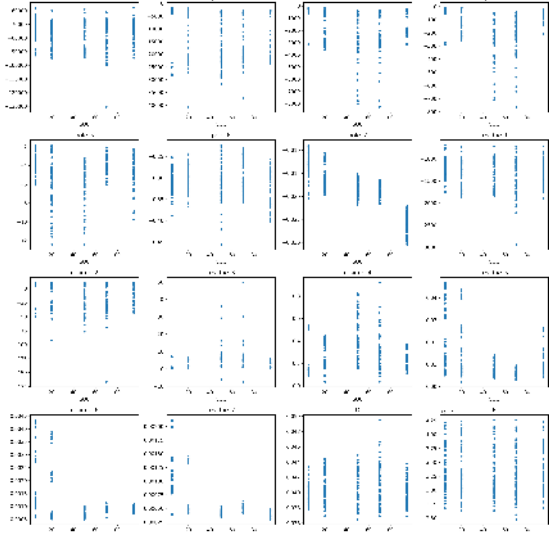


Fig. 5. Vector fitting parameters as a function of SOC.

IV. RESULTS

A. Machine Learning Framework

To estimate SOH and SOC, two separate Artificial Neural Networks (ANNs) models were developed using the vector fitting parameters and temperature as input features. Each input vector consisted of 17 parameters, which corresponds to the poles, residues, constant terms, and temperature, while the target variable was either the battery SOH or SOC. The same data preprocessing and training procedure was applied in both cases, ensuring a consistent evaluation framework across the two tasks. Both ANNs are fully connected feedforward models composed of three layers: an input layer that receives the 17 input features; a hidden layer with 64 neurons using the Rectified Linear Unit (ReLU) activation function; and an output layer that produces the predicted SOH or SOC values.

To ensure a balanced representation across different SOH and SOC levels, the dataset was first partitioned according to the nominal value of the target variable. This stratification ensures that all ranges of SOH or SOC are represented in each subset, preventing the model from being trained or evaluated on a limited subset of the target space. Within each category, a two-stage split was applied: 70 % of the data was assigned to the training set, while the remaining 30 % was divided equally into validation and test sets. The subsets obtained from each category were then pooled to form the final training, validation, and test datasets.

Given the limited number of observations, a single split may lead to overly optimistic or pessimistic performance estimates. To mitigate this effect, the procedure was repeated 50 times, using a different random split. This approach provides a robust assessment of model performance by capturing the variability induced by the data partitioning.

B. SOH Estimation Results

The performance of the ANN model for SOH estimation was evaluated over 50 independent runs using the procedure described in the previous subsection. The aggregated results are summarized in Table 1, including the RMSE, the Mean Absolute Error (MAE) and coefficient of determination R^2 .

Metric	Mean	Standard Deviation	Min	Max
RMSE	3.738	0.544	2.606	5.374
MAE	2.861	0.350	2.131	3.740
R^2	0.728	0.080	0.472	0.863

Table 1. ANN performance metrics for SOH estimation over 50 runs.

The results indicate that the model achieves a moderate level of predictive accuracy, with an average coefficient of determination $R^2 = 0.728$, suggesting that a substantial proportion of the variability in SOH is explained by the extracted vector fitting parameters. The relatively low standard deviations across the reported metrics indicate a reasonably consistent training process, although some sensitivity to the specific data split remains.

The range of results further highlights the impact of data partitioning. While the best-performing run achieves an R^2 of 0.863, the lowest value of 0.472 shows that performance can vary depending on the composition of the training and test sets. This variability justifies the use of repeated random splits instead of relying on a single partition and suggests that more measurements should be used.

To illustrate the predictive capability of the model, the results corresponding to the best-performing model (in terms of R^2) are presented in Figures 6 and 7.

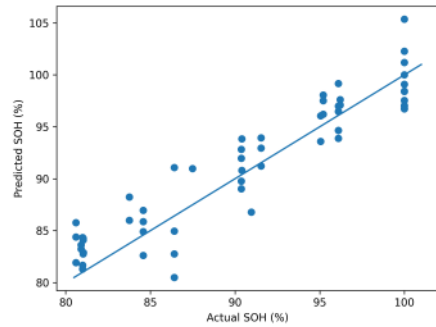


Fig. 6. Predicted versus actual SOH values for the best-performing model.

The predicted versus actual plot (Fig. 6) shows that the model captures the overall increasing relationship between predicted and true SOH values, with most points distributed around the identity line. A certain level of dispersion is observed, indicating that prediction errors are present but remain within a reasonable range given the limited characteristics of the dataset.

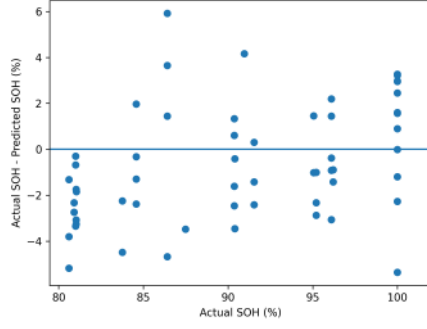


Fig. 7. Residual plot for SOH predictions.

The residual plot (Fig. 7) provides additional insight into the model behavior. Residuals are generally centered around zero, suggesting the absence of significant global bias. A mild structure can be observed, with a tendency for the model to slightly underestimate SOH values.

These effects can be partially attributed to the relatively narrow range of SOH values in the dataset (80 % to 100 %), which limits the model’s learning capabilities. In contrast to SOC, which spans a much wider range, the reduced spread in SOH makes the estimation task inherently more challenging.

Overall, this ANN captures the main relationship between the vector fitting parameters and SOH, as reflected by the average R^2 . While some dispersion and structured residual behavior remain, the results indicate that the model provides a reasonable approximation of SOH within the constraints of the available data.

C. SOC Estimation Results

The same modeling framework was applied for SOC estimation, following the procedure described previously. The dataset was partitioned within each SOC category, and the resulting subsets were combined to form the training, validation, and test sets. The aggregated results of the model performance over the 50 runs are summarized in Table 2.

Metric	Mean	Standard Deviation	Min	Max
RMSE	9.304	2.251	6.387	15.438
MAE	6.505	1.479	5.034	11.640
R^2	0.914	0.044	0.776	0.962

Table 2. ANN performance metrics for SOC estimation over 50 runs.

Compared to the SOH estimation, the results obtained for SOC indicate improved predictive performance. This is reflected in both the higher average R^2 and the reduced variability across runs, suggesting that the relationship between the input features and SOC is more consistently captured by the model. The results corresponding to the best-performing model are illustrated in Figures 8 and 9.

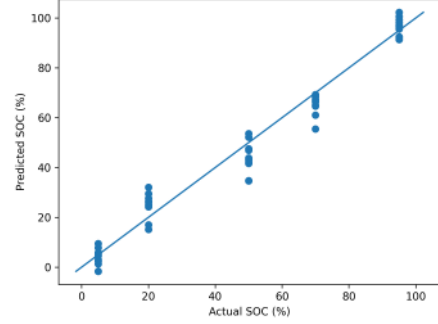


Fig. 8. Predicted versus actual SOC values for the best-performing model.

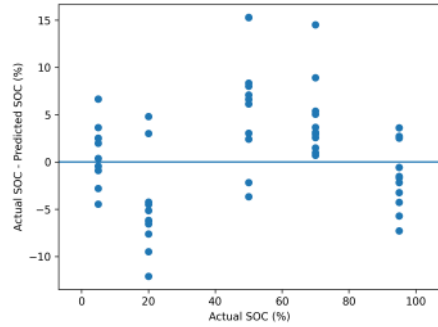


Fig. 9. Residual plot for SOC predictions.

The predicted versus actual values show a stronger concentration of points around the identity line, indicating a closer agreement between predicted and true SOC values. This alignment is particularly evident at lower and higher SOC levels, where the spread of predictions is relatively limited. Deviations are still present, particularly at intermediate SOC levels, where a wider spread of predictions can be observed. However, the magnitude of these deviations is generally smaller than in the SOH case. The residual plot further supports this observation, with residuals exhibiting a reduced spread.

This improved behavior can be attributed, at least in part, to the wider range of SOC values in the dataset (5 % to 95 %), which provides greater variability for the model to learn from. In contrast to SOH, where the values are concentrated within a narrower interval, the broader SOC distribution facilitates a more accurate approximation of the underlying relationships.

This ANN provides a more reliable estimation of SOC compared to SOH, both in terms of average performance and stability across runs. While some variability remains, the tighter clustering and reduced residual magnitude indicate that the SOC estimation problem is better captured by the current modeling approach.

V. CONCLUSIONS

This study analysed the use of vector fitting parameters derived from Electrochemical Impedance Spectroscopy (EIS) data as input features for the estimation of battery state variables. A rational approximation of the impedance response was obtained using the vector fitting method, and the resulting parameters were used to construct feature vectors for machine learning models.

Two Artificial Neural Network (ANN) models were developed to estimate the State Of Health (SOH) and State Of Charge (SOC), respectively. The models were evaluated using repeated random train-validation-test splits to ensure a robust performance assessment.

The results demonstrate that the proposed approach provides strong predictive capability for both targets. In particular, SOC estimation achieved consistently high performance, with average coefficients of determination exceeding $R^2 = 0.91$, indicating an excellent agreement between predicted and true values. SOH estimation also yielded solid results, with an average $R^2 \approx 0.73$, showing that a substantial proportion of the variability is captured despite increased dispersion.

The difference in performance between the two tasks can be largely attributed to the characteristics of the dataset. The SOC values span a wide range, providing sufficient variability for the model to learn the underlying relationships, whereas the SOH values are concentrated within a narrower interval, limiting the available information for accurate prediction. Although it is not possible to compare our results with other Deep Learning based methods, the presented results were trained and compared with the SOH and SOC from the used dataset.

The small size of the used dataset limits the accuracy of the results, but overall the findings indicate that vector fitting parameters constitute a meaningful and effective feature representation for battery state estimation, particularly when combined with non-linear models such as ANNs.

A database of SOH and SOC is being created through a measurement campaign of different batteries over multiple charge/discharge cycles. Future work will focus on extending the analysis to larger and more diverse datasets, particularly with a broader range of SOH values, as well as validating the proposed approach on different battery types with varying nominal capacities to assess its generalization capability.

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Measurement of the characteristics of EDL and hybrid supercapacitors versus temperature

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Abstract – This study presents an experimental characterization of supercapacitors (SCs) of different sizes and technologies under varying operating temperatures. The identification of SCs main parameters as a function of temperature primarily involves the measurement of capacitance (C) and equivalent series resistance (ESR). To assess C and ESR, the main characterization techniques employed were cyclic voltammetry and galvanostatic charge/discharge measurements. The latter was performed including a constant-voltage charging phase, in accordance with standard procedures.

Experimental results reveal markedly different behaviors between electric double-layer SCs and hybrid SCs. In both technologies, C decreases as temperature declines; however, this reduction is significantly more pronounced in hybrid supercapacitors. Conversely, the ESR increases as the temperature approaches sub-zero values.

Keywords: electric double-layer capacitors; hybrid supercapacitors; temperature effect.

I. INTRODUCTION

SC is a promising technology for enhancing battery performance in hybrid energy systems. They are particularly well suited to applications requiring high power delivery over short durations, such as uninterruptible power supplies (UPS), door actuation systems, and rapid movements mainly in aerospace contexts [1]. Their advantages stem from their intrinsic safety and robustness: SCs do not experience thermal runaway or abnormal heating, and they maintain reliable operation even at very low temperatures, down to $-40\text{ }^{\circ}\text{C}$. For this reason, understanding their temperature-dependent behavior is crucial in many applications [2].

The thermal behavior of SCs mostly depends on how temperature variations affect the SC parts, such as electrodes (especially the pore structure and pore volume of the electrode material [3],[4]), electrolyte materials [5],[6], and the separator [7].

A key distinction exists between electric double layer (EDL) SCs and hybrid SCs. EDL SCs store energy purely through electrostatic charge mechanisms, whereas hybrid SCs combine characteristics of both batteries and

capacitors. In hybrid devices, one electrode stores charge via ionic adsorption, while the other involves faradaic reactions [8]. The difference between EDL SCs and hybrid SCs comes from the different materials used in their main components [9] which leads to different behaviors in facing thermal variations. Studying the thermal behavior of these two types of SCs and observing how their characteristics change is very crucial in real applications, modeling and thermal management of these devices [10], [11] and it helps to choose the best technology for each application.

This study analyses the measurement setup and the methods for evaluating the significant parameters of these two types of SCs and how they differ in their thermal behavior.

II. MEASUREMENT SETUP

The analysis was performed using a commercial galvanostat-potentiostat (Biologic BCS-815) through which cyclic voltammetry (CV) and constant current – constant voltage (CCCV) tests were performed. CCCV specifically consists of a constant-current charging up to the SC rated voltage followed with 30 minutes of constant-voltage charging, then discharging at constant-current. CV test is described in the next section. Temperature variations were obtained by means of a climatic chamber (Binder MK 240) on three EDL SCs characterized from $-40\text{ }^{\circ}\text{C}$ to $65\text{ }^{\circ}\text{C}$ and two hybrid SCs characterized from $-25\text{ }^{\circ}\text{C}$ to $70\text{ }^{\circ}\text{C}$. For both types of SCs, the measurements are conducted at seven different temperatures within the specified temperature range, and $23\text{ }^{\circ}\text{C}$ is considered as the reference ambient temperature. Specifications of the devices under test (DUTs) can be found in Table 1.

Table 1. Specifications of the DUTs.

SC type	Rated C /F	Max. ESR / mΩ	Temp. Range / $^{\circ}\text{C}$	Min. voltage / V	Electrolyte
EDL	25	18	-40 to +65	0	non-aqueous organic
EDL	60	13	-40 to +65	0	As above
EDL	400	3.2	-40 to +65	0	As above
Hybrid	90	135	-40 to +85	2.4	Organic Li-based
Hybrid	400	80	-25 to +70	2.2	Organic Li-based

The temperature of each SC is monitored through a thermocouple and at every temperature variation, the SCs were left to reach the thermal equilibrium for at least eight hours.

All the measurements performed by authors are available in the public database platform Zenodo and are reported in detail in [12] - [46].

III. EVALUATION OF C AND ESR

The parameters evaluated for the proper characterization of SCs at different temperatures are C and ESR. The capacitance was computed from both the CV and the CCCV tests according to the energy method in agreement with IEC EN 62391-1:2022 [47], and IEC EN 62813:2025 [48] standards, which are guidelines for EDL SCs and hybrid SCs, respectively. The ESR was computed from the voltage drop at the beginning of the discharging phase of CCCV tests according to the same standards. The adopted methods are described as follows.

CCCV method is described in Section II, while the second method, CV, is performed by applying a linear voltage variation at the SCs terminals, following a triangular waveform. For EDL SCs, the voltage varies linearly from zero to the rated value and vice versa. For hybrid SCs, the voltage sweep is limited to a minimum and a maximum voltage value, since they cannot be fully discharged. For both types of SCs, C is calculated as:

$$C = \frac{1}{2 \cdot s \cdot \Delta V} \oint I(V) dV = \frac{Area}{2 \cdot s \cdot \Delta V} \quad (1)$$

where *Area* is the area enclosed by the I-V curve, *s* is the constant time derivative of the voltage, namely the scan rate having unit mV/s, and ΔV is the voltage range over which the voltage varies.

ESR and C are both determined during the discharge phase, while the SC is discharged with a constant current I_{dis} . ESR can be calculated by the sudden and nearly instantaneous voltage drop (ΔV_{cccv}) when the current stops in the beginning of the discharging phase of the CCCV test as follows:

$$ESR_{cccv} = R_{en} = \frac{\Delta V_{cccv}}{I_{dis}} \quad (2)$$

and C can be calculated according to the energy method as follows.

$$C_{en} = \frac{2 W}{U_1^2 - U_2^2} = \frac{2 \int_{t_1}^{t_2} I_{dis}(t) U(t) dt}{U_1^2 - U_2^2} \quad (3)$$

where $U(t)$ is the voltage time behavior during discharge, U_1 is 80% of the maximum voltage for EDL SCs and the voltage value reached after ΔV_{cccv} for hybrid SC, U_2 is the 40% of the maximum voltage for EDL SCs and the minimum voltage level for hybrid SCs. t_1 and t_2 are the times at which the terminal voltage reaches U_1 and U_2 , respectively.

IV. TEMPERATURE EFFECTS

The five SCs listed in Table 1 underwent thermal characterization across the specified ranges. For both categories, measurements were recorded at seven distinct temperatures. The divergence in the limits is due to the underlying cell chemistry:

- hybrid SCs: limited to -25 °C because their battery-like chemistry prevents reliable low-temperature operation, though they offer a higher upper limit.
- EDL SCs: capable of reaching -40 °C; however, they are restricted to 65 °C in this study to avoid the voltage derating required at higher temperatures.

Table 2 presents the results of the C evaluation obtained using the CV and CCCV methods. The first section of the table reports the results for the EDL SCs, whereas the second section summarizes the results for the hybrid SCs.

The capacitance relative variability was evaluated as the difference between the maximum and minimum C values measured during the thermal cycles, normalized with respect to the nominal C of the DUT. The highest variability was observed for the hybrid SCs, indicating a stronger sensitivity to temperature variations. For instance, the 90 F hybrid SC exhibited a capacitance variability of nearly 154% based on the CCCV measurements, going from 1.67 F registered at -25 °C to a maximum value of 139.81 F registered at 55 °C.

Table 2. Computed C obtained from CV and CCCV tests.

EDL SCs						
Temp /°C	400 F		60 F		25 F	
	C_{en}/F	C_{cv}/F	C_{en}/F	C_{cv}/F	C_{en}/F	C_{cv}/F
-40	336.69	324.55	53.79	51.26	23.42	20.85
-25	338.67	327.61	53.67	51.49	23.11	21.04
-10	339.18	330.08	53.63	51.55	23.01	21.15
5	341.96	332.20	53.62	51.66	22.98	21.23
23	347.01	334.89	57.14	53.55	23.57	21.63
40	350.24	337.31	56.87	53.61	23.60	21.67
65	358.07	341.75	54.51	52.62	23.65	21.71

Hybrid SCs				
Temp /°C	400 F		90 F	
	C_{en}/F	C_{cv}/F	C_{en}/F	C_{cv}/F
-25	230.30	90.53	1.67	3.267
-10	304.32	234.03	4.71	7.156
5	353.76	352.53	9.58	14.21
23	411.08	395.35	85.46	27.17
40	428.35	395.35	123.38	39.14
55	436.18	419.58	139.81	44.66
70	445.11	426.09	111.99	31.39

Figures 1 and 2 show the measurement results as a function of temperature for two SCs of the same 400 F rating, one hybrid and one EDL. The comparison reveals clear differences in both the temperature behavior and the shape of the voltammetric cycles. These latter, obtained for the 400 F EDL SC and the 400 F hybrid SC are presented in the insets of Fig. 1 and Fig. 2, respectively. The applied scan rates were 5 mV/s for the EDL SC and 1 mV/s for the hybrid SC. The EDL SC CV curves have the typical rectangular shape indicating the absence of faradaic processes in the storage mechanism. Instead, the hybrid SC CV curves exhibit the usual peaks representing the redox reactions involved in their storage mechanism [49].

For both DUTs, regardless the analyzed type of SC, the capacitance increases as the temperature rises. For the EDL SC the relative variability is lower than the one computed for the hybrid SC, which are almost 4% and 84%, respectively. This is consistent with what can be observed in the respective cyclic voltammograms: the EDL SC I-V curve remains relatively constant across the investigated temperature range, whereas the hybrid SC curve varies significantly as the temperature changes. It is also noted that the increase in capacitance for the EDL SC is almost linear, while the variation of C of the hybrid capacitor is strongly non-linear and has two zones: a region of low variability above 20 °C and a zone of strong, almost linear decrease towards negative temperatures, which makes this type of SC less suitable for applications below or near freezing temperatures.

The remaining SCs exhibit comparable trends. Generally, capacitance increases with temperature, reaching maximum values between 40 °C and 60 °C. However, for specific devices, such as the 60 F and 90 F SCs, a decrease in capacitance is observed at the highest temperatures, which is likely attributable to electrolyte degradation [7].

The CCCV tests were performed at the same temperatures used for the CV tests discussed earlier. The terminal voltages as a function of time for the 400 F EDL SC and the 400 F hybrid SC are shown in Fig. 3(a) and Fig. 3(b), respectively. The constant charging/discharging currents applied to the EDL SC and hybrid SC were 10 A and 2 A, respectively. The ESR and C obtained using the CCCV measurement technique, were determined according to equations (2) and (3). The resulting C values are reported in Table 2 as C_{en} , while the computed ESR values are presented in Table 3 for our five DUTs.

Fig. 4 illustrates an increasing trend in capacitance that is in strong agreement with the measurements obtained via the CV method. However, the absolute values differ, indicating an underestimation by the CV technique. This discrepancy arises partly from inherent differences between the measurement methodologies. Furthermore, the CV method only converges with the CCCV approach at very low scan rates s . Such low rates necessitate extended measurement times because, as observed,

capacitance C decreases as s increases [49]. For example, for the 400 F hybrid SC the used s is suitable to provide C values close to the CCCV method, while for the 90 F hybrid SC it is evident that the value of the applied s , although already very low with very long measurement times, is insufficient to provide reliable C values.

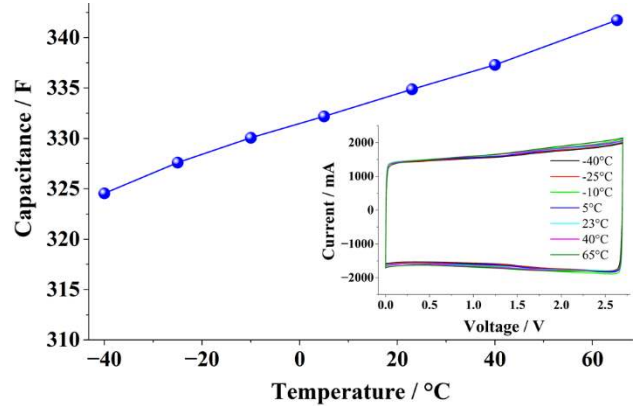


Fig. 1. The computed C of 400 F EDL SC obtained from CV tests at seven different temperatures. In-set: representative CV cycles.

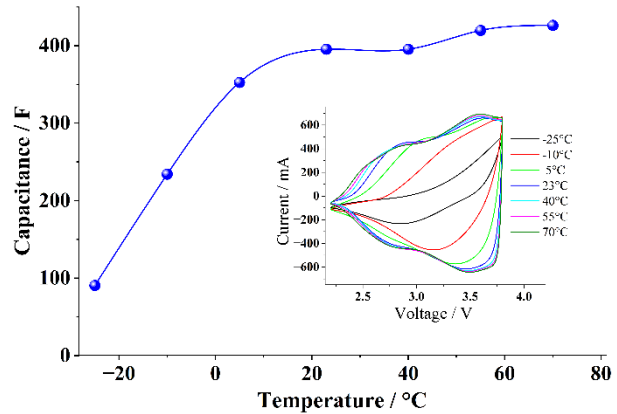


Fig.2. The computed C of the hybrid 400 F SC obtained from CV tests at seven different temperatures. In-set: representative CV cycles.

It can be highlighted from Fig. 3 that the charging time of the 400 F EDL SC remains relatively constant across the tested temperature range (85 s at -40 °C and 89 s at 65 °C), whereas the charging time of the 400 F hybrid SC varies significantly with temperature variation (0.2 s at -25 °C and 317 s at 70 °C). Furthermore, for the 400 F hybrid SC, the initial voltage drop during both charge and discharge phases increases significantly as the temperature decreases. This effect is particularly pronounced during discharge (Fig. 3(b)). This behavior corresponds to a decrease in ESR as temperature rises, as illustrated in Fig. 4(b). While the 400 F EDL SC is expected to exhibit a similar trend, Fig. 4(a) reveals an anomaly: its ESR decreases only up to room temperature, after which it unexpectedly increases.

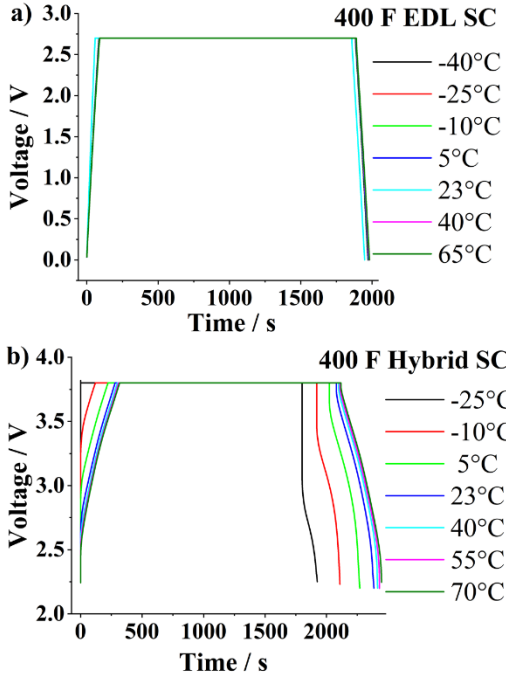


Fig. 3. CCCV tests at seven different temperatures of a) 400 F EDL SC with a charging and discharging current of 10 A; b) 400 F hybrid SC with a charging and discharging current of 2 A.

Table 3. Computed ESR obtained from CCCV tests.

EDL SCs			
Temp / °C	400 F	60 F	25 F
	ESR/mΩ	ESR/mΩ	ESR/mΩ
-40	3.35	12.26	24.10
-25	3.1	9.92	19.59
-10	3.07	9.94	17.95
5	3.06	9.35	16.02
23	3.08	8.77	14.93
40	3.13	8.63	14.65
65	3.27	8.69	14.49

Hybrid SCs		
Temp / °C	400 F	90 F
	ESR/mΩ	ESR/mΩ
-25	170.45	242.53
-10	122.35	166.68
5	65.48	118.55
23	30.41	85.71
40	19.92	71.47
55	16.28	64.79
70	14.64	69.23

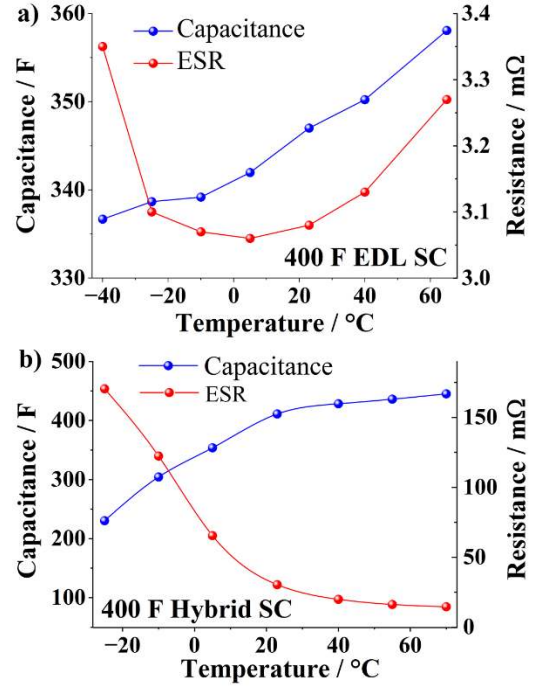


Fig. 4. Computed C and ESR from CCCV tests at seven different temperatures of a) 400 F EDL SC according to [47]; b) 400 F hybrid SC according to [48].

Notably, the hybrid 400 F SC exhibits a substantial ESR variation, with its maximum value reaching approximately eight times its minimum. In contrast, the variation for the 400 F EDL SC is limited to a few percent; thus, while this anomalous trend warrants further investigation, its overall magnitude is minor. Conversely, all other tested SCs follow the expected trend, with the ESR decreasing consistently as temperature rises, as detailed in Table 3.

V. DISCUSSION AND CONCLUSION

This study highlights that variations in capacitance and ESR are much more pronounced in hybrid SCs in comparison to EDL SCs, which, as expected due to their battery-like nature, exhibit a significantly stronger temperature dependence. In contrast, EDL SCs show only limited variations in both ESR and C, remaining within a few percent over a temperature range exceeding 100 °C. Consequently, greater measurement accuracy is required to reliably estimate parameter variations in EDL SCs.

For the 25 F EDL SC, the capacitance variation between the minimum and maximum values is approximately 0.64 F, corresponding to about 2.5% of the rated capacitance. Regarding ESR, larger SCs exhibit lower values. For the 400 F EDL SC, the ESR variation is 0.29 mΩ out of a maximum value of 3.35 mΩ, corresponding to 8.7%.

A clearly evident phenomenon, already highlighted, is the significant increase in ESR as the temperature decreases into the negative range that is particularly

pronounced in hybrid SCs. The difference can be attributed to the distinct charge-storage mechanisms of the two technologies.

In EDL SCs, charge storage is based on electrostatic ion adsorption at the electrode–electrolyte interface, a process characterized by relatively weak temperature dependence. In contrast, hybrid SCs involve not only electrostatic charge storage but also faradaic reactions and ion diffusion processes. These electrochemical mechanisms are strongly temperature dependent: at low temperatures, the electrolyte viscosity increases, ionic conductivity decreases, ion diffusion within the particles slows down, and the charge-transfer resistance rises sharply [3]. This behavior explains the remarkable increase in the ΔV_{CCV} value, and so in ESR, observed at negative temperatures in Fig. 3(b).

Another important remark is that C generally increases with temperature. This trend is noticed for all SCs investigated in this study. The only exception occurs at the highest temperatures, where in some cases (e.g., 60 F and 90 F Hybrid) a reduction in capacitance was observed. This behavior is likely attributable to electrolyte degradation. Moreover, the capacitance variation trends obtained using the CV and CCCV measurement techniques are consistent. However, the capacitance values provided by the CV technique are slightly lower. This discrepancy depends on the scan rate, which, as discussed in Section IV, may lead to significant capacitance underestimation if it is not sufficiently low.

Finally, commercial SC DUTs may exhibit capacitance values lower than their nominal ratings. This behavior may result not only from manufacturing variability, but also from prolonged storage periods or significant temperature fluctuations during storage. It is worth noting that, for hybrid SCs, the rated C is often exceeded when the temperature rises above room temperature, whereas for EDL SCs the increase is minimal and, even at elevated temperatures, for capacitance derated SCs, the nominal C is not always reached.

VI. ACKNOWLEDGEMENT

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Measurements of Wetsuits Thermal Properties at 100 kPa and 600 kPa Absolute Pressure

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Abstract – To assess the thermal properties of neoprene scuba-diving wetsuits for water immersion at different depths, it is fundamental to create a measurement setup able to characterize their thermal resistance in an environment that can replicate the real working conditions of the material. In this paper, neoprene wetsuits of different thicknesses, producers, and material composition, were tested in cold water conditions at two different working pressures. The wetsuit thermal resistance was measured under quasi-steady-state temperature conditions, recorded and studied as a function of time. Measurements of the heat transfer across the wetsuit sample were performed by means of an electrical-to-thermal watt balance in a *de facto* quasi-stationary regime of the thermal power flux. Experimental results show a significant degradation of the wetsuit insulation property at increasing water pressures, but also as a function of the wetting conditions and wetting time. Results in terms of reduced wetsuit insulation were empirically analyzed and collected, for the first time to our knowledge, in a comprehensive public report.

Keywords: wetsuit thermal resistance, pressure dependance, temporal degradation, thermal watt balance.

I. INTRODUCTION

The characterization of thermal properties of scuba diving wetsuits [1, 2] is important to assess the thermal insulation or resistance that these sport clothes can offer to the human body during water immersion at different pressures [3, 4].

A novel measurement system [5] for testing neoprene wetsuits in accordance with [6] was developed, improved [7], and used to test neoprene wetsuits at different pressures. Using neoprene wetsuit samples with different thickness, the wetsuit thermal resistance was measured at 100 kPa and 600 kPa pressures. Results show

a decrease of thermal resistance by a factor of around 4 when passing from the low to the high pressure, and a dispersion of the resistance values within a few percent among wetsuit samples of the same thickness.

During the experiments, a significant dependence of the wetsuit resistance on the wetting time at high water pressure was observed. This suggests adopting additional definitions of the testing ‘stationary’ condition, also to consider the time duration and the pressure of the test.

II. MEASUREMENT SYSTEM

The measurement system for the characterization of wetsuits thermal insulation was described in more detail in [5]. The basic principle underlying the measurement is an electrical-to-thermal watt balance [6] between the heat per unit time injected inside a phantom, enclosed by the wetsuit, and the same amount of heat transferred through the wetsuit sample to an external temperature controlled large water tank [7], that also allows changing the water pressure (from 100 kPa to 600 kPa absolute pressure values) during the experiments. The injected electrical power P comes from a low-voltage DC power supply, driving with current I (0-30 A) and voltage U (0-30 V) a stainless steel incapsulated resistive heater, with an electrical resistance $R \sim 3.4 \Omega$. The inner volume of the phantom (a cylinder with base diameter 16 cm and height 21 cm) is filled with water at an initial temperature T_{IN} , heating up when the heater is switched on. The outer volume, inside the water tank and outside of the phantom, is kept at constant cold temperature T_{OUT} , in the range 10-15 °C [6], by means of a commercial water chiller with 1.8 kW maximum cooling power. Automated temperature measurements are performed by multiple calibrated Pt-100 sensors, placed both inside and outside the cylindrical phantom enclosed by the wetsuit under test. A custom LabVIEWTM program (Virtual Instrument, VI) is used for the data acquisition, graphs, and continuous preliminary data analysis and display during the tests. Driving the heater at constant power $P=U \cdot I$, the inner water temperature T_{IN} will initially rise and then stabilize its value

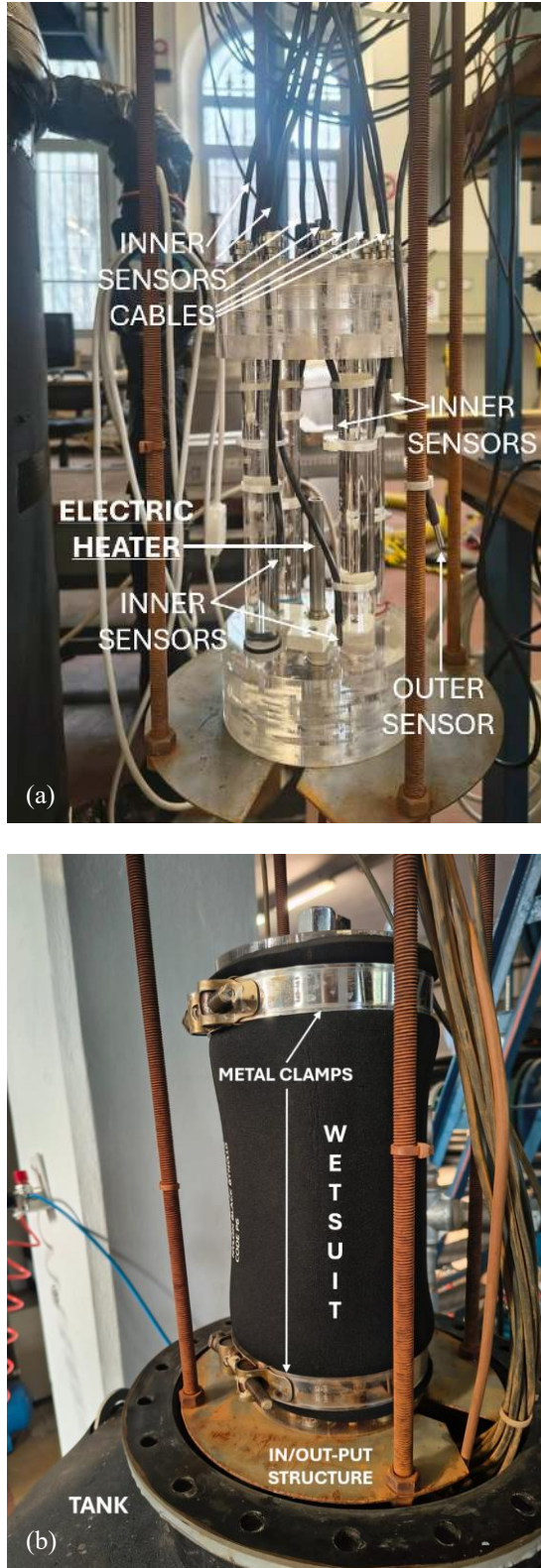


Fig. 1. (a) Phantom with 10 temperature sensors inside and 2 outside (not all visible). (b) Phantom enclosed by the wetsuit cylindrical sleeve and ready for insertion in the water tank .

to a steady-state value, undergoing a first-order transient [5]. The transient time constant depends on the phantom's thermal resistance, R_T , which is assumed equal to the wetsuit thermal resistance, under the assumption that all, or the greatest majority, of the thermal flow is across the wetsuit sample. In the steady state condition, by measuring the water temperatures T_{IN} (within the phantom) and T_{OUT} (outside the phantom), and knowing the electrical parameters driving the heater, one can indirectly measure the wetsuit resistance [6] as

$$R_T = \frac{T_{IN} - T_{OUT}}{UI} \quad [\text{K/W}] \quad (1)$$

Since in the steady-state conditions some temperature stratification in water occurs, both inside and outside of the phantom, more temperature sensors -as shown in Fig. 1- are used to assess T_{IN} and T_{OUT} as average temperatures of the water in contact with the inner surface and the outer surface, respectively, of the wetsuit cylindrical sample. As shown in Fig. 1, a total of 10 temperature sensors ($T_1 \dots T_{10}$) are used inside the phantom [6], placed at different heights in proximity to the wetsuit inner surface, while 2 more sensors (T_{11} and T_{12}) are placed in the tank, in proximity to the outer wetsuit surface and at two different heights. The temperature stratification in the large water tank is significantly mitigated, as verified measuring T_{11} and T_{12} , by a slow-flow water recirculation forced by a pump and a water diffuser placed at the bottom of the tank.

The water tank is an iron cylinder, with 460 L internal volume, certified to operate at pressures up to 800 kPa (see Fig. 2). The water pressure in the tank and in the phantom (thus acting on the wetsuit as well), is controlled by an external compressor and a regulating valve, while being continuously monitored by a calibrated analog manometer (see Fig. 2). In the experiments, absolute pressure values up to 600 kPa were achieved and maintained for several hours, without water leakage from the tank cover nor from the cables.

The LABVIEW™ software used for data acquisition is continuously saving each of the 12 water temperatures with a typical data acquisition rate of 1 Sa/s. To reduce electronic noise in the acquisition, each temperature value (from T_1 to T_{12}) is evaluated by averaging the digital output of the corresponding sensor over 13 samples in 1 s “integration time”. A first display panel of the VI can show all the 12 temperatures numerical values, with 1 s update time, drawing on one graph the plots of the different temperatures [7], also to allow a visual evaluation of the stratification effects in different experimental conditions. In particular, the VI continuously calculates the average internal temperature, as $T_{IN} = \text{AVE}(T_1, \dots, T_{10})$, and the average external temperature, as $T_{OUT} = \text{AVE}(T_{11}, T_{12})$, and the corresponding temperature gradient $\Delta T = T_{IN} - T_{OUT}$. These average values are available within the program as functions

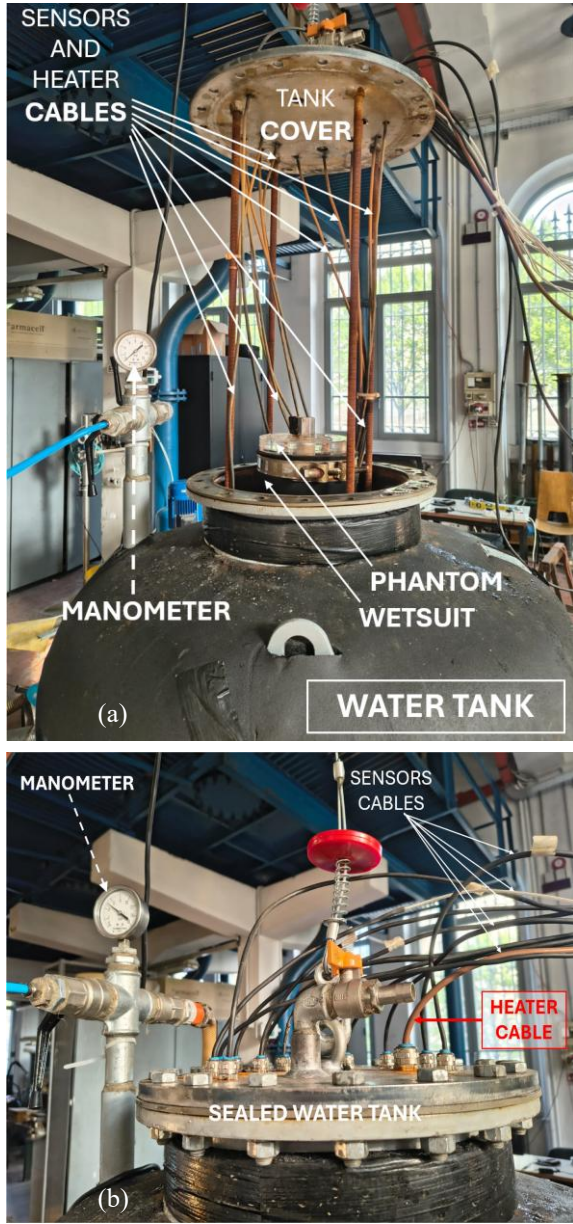


Fig. 2. (a) Photo of the externally thermally insulated high-pressure-resistant water tank with its manometer (relative pressure $p_R=0$ in open air) and the phantom almost completely vertically inserted in the tank. (b) Tank closed by its cover and sealed for high-pressure operation as indicated by the manometer ($p_R=5$ bar, i.e. $p=600$ kPa absolute water pressure).

of time, evaluated at 1 s time intervals, and they are graphically displayed in a second VI display panel as the plots of average temperatures of Fig. 3. Finally, upon inserting in the VI panel the electric power value (“ P [W]”) the software calculates and displays the current thermal resistance value (R_T), as evaluated by (1), using the last/current $\Delta T=T_{IN}-T_{OUT}$ value. Figure 3 shows the screenshot of the second VI display panel during one test,

when the stationary ΔT condition [6] is achieved. By visual inspection of the plots of T_{IN} and T_{OUT} , both temperatures are very stable, and, from the plot of ΔT over a time interval of 17 min (15 ± 1), one can observe a temperature gradient stability within ± 0.1 °C in a time interval of 15 minutes, as requested by the international regulations [6].

III. EXPERIMENTAL RESULTS

Neoprene wetsuits of different thickness were tested using the 2 pressure values prescribed by [6] to assess the wetsuit thermal resistance/resistivity. By applying the electric power, $P=U \cdot I$, to the heater within the phantom, after a time interval strongly depending on the pressure and wetsuit thickness, a quasi-steady-state condition [5] is obtained. When the temperature gradient $\Delta T=T_{IN}-T_{OUT}$ remains stable at ± 0.1 °C over 15 minutes, e.g. see Fig. 3, in accordance to the regulations [6] the wetsuit thermal resistance can be evaluated by (1). The results achieved during the tests at $p_{LOW}=100$ kPa (water surface level) and at $p_{HIGH}=600$ kPa (-50 m depth from surface) are reported in Table I. This is the first time that we can measure R_T at the high-pressure value, repeating the tests over 3 wetsuit samples of the same thickness and using 3 different neoprene thicknesses in the experiments, namely 5 mm, 7.5 mm, and 9 mm. The average inter-wetsuit repeatability is within 5%, being slightly better at 100 kPa (about 3% average standard deviation relative to the average value) than at 600 kPa (about 4% average deviation). The dependency of thermal resistance on the water pressure (p) is strong, with the average thermal resistance dropping by approximately a factor of 4 when p is increased from 100 kPa to 600 kPa. From Table I, one can see that when the pressure is increased from p_{LOW} to p_{HIGH} , i.e. by a factor of 6, the average thermal resistance of the wetsuits of the same thickness are decreasing by factors of 4.1, 4.1, and 4.7, for the 5 mm, 7.5 mm, and 9 mm thickness, respectively. It will be interesting, as a subject of further studies now possible with the improved measurement system, to characterize the variations of R_T with p , at different pressure values ranging more finely from p_{LOW} up to p_{HIGH} , e.g. in 5 steps of 100 kPa or 10 steps of 50 kPa.

The system can test the wetsuits thermal resistivity in accordance to [6]. In all our experiments, the wetsuit sample area is 0.1 m^2 and the thermal resistivity is obtained [7] by multiplying the measured resistance by this area. The next step to evaluate the new system accuracy and check for possible systematic errors is to establish intercomparisons with other metrological laboratories conducting similar tests. This is already agreed and is in progress between our laboratory and other laboratories operating in the thermal characterization of wetsuits. A precise evaluation of the thermal resistance of the top and bottom enclosures of the phantom will be needed [6] to confirm the assumption that this is much larger than the wetsuit resistance, thus assuring that the resistance of the wetsuit is approximately equal to the measured resistance of the phantom [5, 6, 7].

Table 1. Thermal resistance R_T [K/W] of the tested wetsuit samples at low and high pressures. Average values and relative standard deviations are indicated. The R_T average ratio is evaluated by dividing the average value by the one obtained at 100 kPa water pressure.

sample (mm #)	P_{LOW} 100 kPa	P_{HIGH} 600 kPa
5 mm #1	0.745	0.166
5 mm #2	0.738	0.178
5 mm #3	0.747	0.198
5 mm AVE	0.743	0.181
5 mm ST.D.	0.5%	7%
R_T AVE ratio	1	<u>4.1</u>
7.5 mm #1	1.014	0.236
7.5mm #2	0.991	0.226
7.5 mm #3	0.858	0.229
7.5 mm AVE	0.954	0.230
7.5 mm ST.D.	7%	2%
R_T AVE ratio	1	<u>4.1</u>
9 mm #1	1.213	0.261
9 mm #2	1.249	0.275
9 mm #3	1.250	0.257
9 mm AVE	1.237	0.264
9 mm ST.D.	1.4%	3%
R_T AVE ratio	1	<u>4.7</u>

IV. WETSUIT HUMIDITY AND WETTING TIME

As noticed and highlighted in [7], the wetsuit thermal resistance undergoes a degradation in time during the immersion. This happens since some of the water surrounding the wetsuit can enter the neoprene material and the sewing lines (wetsuit “wetting”) providing a better thermal path for the heat flow than the original one offered by the air bubbles and neoprene rubber of the dry wetsuit and dry sewing lines. This effect was initially observed also at the lower water pressure [7] but it is now more significant at the higher pressure used in the recent experiments. As one would expect, exposing the wetsuit to $p_{HIGH}=600$ kPa water pressure, provides significantly more rapid wetting times, and deeper humidity penetration within the wetsuit fabric, than the ones obtained at $p_{LOW}=100$ kPa. Therefore, in the case of the wetsuit operating at the higher water pressure, the thermal resistance worsens already in the first hours of immersion and by a significant percentage of its original/nominal value for the dry suit. Hours of immersion

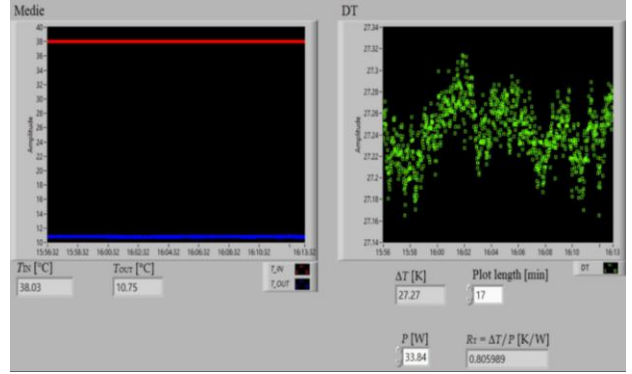


Fig. 3. Graphical interface of the data acquisition software, providing temperature graphs and R_T .

which are necessary in the laboratory test to achieve the ‘steady state’ condition required to perform the measurement of the wetsuit thermal resistivity.

In the practical experience of a scuba diver, after a rather long dive (e.g. 1 h), possibly at some depth and pressure (e.g. down to -50 m and up to 600 kPa), the wetsuit retains water within its neoprene structure and it will release this internal humidity in a drying process that can last up to a few days depending on the exposure to heat and dry air/wind or solar radiation. The wetting time of the wetsuit influences its internal humidity which in turn changes the thermal resistance. An increase in the wetsuit wetting time, as well as an increase in the original (pre-immersion) wetsuit internal humidity, will reduce the wetsuit thermal resistance in water immersion. Therefore, in the thermal tests, as much as in the practical dives, to experience the maximum thermal resistance for a given wetsuit it is important to start the immersion with a “perfectly dried” wetsuit. To quantify this behavior of scuba diving neoprene wetsuits, we plan to empirically study, with the availability of the present thermal resistance measurement system, the effect of the original dryness condition on the measured wetsuit thermal resistance. First, we have observed that all wetsuits coming out of the tests, even after an initial dripping-drying phase of some 0.5 h, still exhibit a significant residual internal humidity resulting in an increased value of the humid wetsuit mass with respect to the minimum mass of the completely dry suit. Starting after half an hour of vertical dripping and drying in the typical laboratory conditions (hanging the wetsuit cylindrical sleeve at temperature $T \sim 25$ °C in still air with humidity $RH \sim 40\%$), we can define the wetsuit sample residual humidity as the ratio of the mass of water still trapped in the suit (m_{H_2O}) to the mass of the perfectly dried suit (m_{DS}). Being the humid wetsuit mass $m_{HW} = m_{DS} + m_{H_2O}$ and in practical cases with $m_{H_2O} < m_{DS}$, the wetsuit residual humidity can be seen as a percentage value relative to the mass of the dry suit:

$$WS_{\%H} = \frac{m_{H_2O}}{m_{DS}} = \frac{m_{HS} - m_{DS}}{m_{DS}} = \frac{m_{HS}}{m_{DS}} - 1 \quad [\%] \quad (2)$$

Experimentally, the quantity of water trapped in the suit (m_{H_2O}) can be evaluated by the difference between the mass of the humid suit (m_{HS}) and the mass of the perfectly dried suit (m_{DS}), both measured by a precision scale. Equation (2) defines get a wetsuit humidity parameter ranging from high percentage values when the wetsuit is still retaining significant humidity, down to zero when the wetsuit is completely dry. A graph showing the measured wetsuit residual humidity vs. time, in our laboratory conditions, is given in Fig. 4 for a 5-mm wetsuit exiting a few hour test at 600 kPa. The measurements of the mass of the humid wetsuit were manually repeated over a few days. The lack of a larger number of measurement points in the first 2 days of drying is evident from Fig. 4. Therefore, this measurement will be repeated when a digital scale with automated serial data acquisition will be available. In future experiments, the originally dripped wetsuit will be positioned over a precision digital scale, during the drying process, with continuous discrete data acquisition. During the drying, both the internal and external wetsuit surfaces will be exposed to still air (or a controlled air flow) in the laboratory environment. Measuring and using the wetsuit humidity parameter defined by (2), we shall evaluate the performance of the wetsuit thermal resistance as a function of the wetsuit's original humidity.

We also investigated the dependence of R_T on the wetting time in prolonged tests, as originally observed in [7], now considering this effect at the high pressure ($p_{HIGH}=600$ kPa) used in the tests. Figure 5a shows the temperature gradient ($\Delta T=T_{IN}-T_{OUT}$) as a function of time from the beginning of the test, for the 7.5-mm wetsuit sample #2 at 600 kPa. The time t is now counted in hours with starting point ($t=0$) when the heater is switched on inside the phantom, placed into the tank rapidly brought at the high pressure. Keeping the electrical and thermal power $P=U \cdot I$ constant throughout the experiment (apart from two slight adjustments, in this case, during the exponential transient needed to achieve the required temperature gradient $\Delta T=25-30$ °C [6] in the 'steady state' condition), we prolonged the wetsuit thermal test for some time after the time interval Δt_{PEAK} , from $t=0$ to $t_{PEAK} \approx 3$ h in this case, needed to reach the thermal resistivity peak ($R_{T,MAX}$) in the 'steady state' conditions. Actually, the same recommended [6] 'steady state' temperature gradient (ΔT in the range 25-30 °C and with a ΔT stability of ± 0.1 °C over 15 min), that we typically observe around the peak value of ΔT , can be practically observed also at other times after the peak condition, where ΔT , and consequently R_T , are slowly decreasing with time (not too slowly at the 600 kPa pressure). We typically record the temperature gradient ΔT_{MAX} at the steady-state peak condition and then, using (1), we get the maximum value, $R_{T,MAX}$, of the measured thermal resistance during the test. After reaching its peak, the value of R_T is decreasing with time, due to the increasing wetting time, to which the wetsuit is exposed. The decrease of $R_T(t)$ is approximately linear for some hours after reaching its peak 'steady state' value. We can use the relative variation

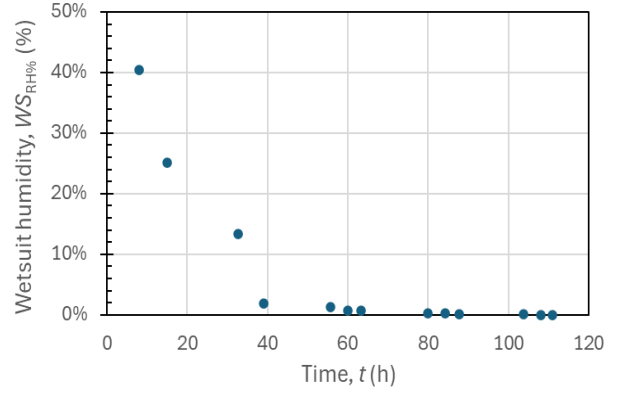


Fig. 4. Wetsuit residual humidity as a function of the drying time.

of R_T with respect to $R_{T,MAX}$, to define its relative slope $S\%=\Delta R_T(t)/R_{T,MAX}$ and quantify this linearly decreasing behavior with time, which is observed and measured after the peak. However, this linear trend of $R_T(t)$ is expected to be present also in the beginning of the test, where the linear negative trend is masked by the much stronger exponential transient of the warming up temperature T_{IN} and thus by the corresponding significant increase of ΔT during the initial transient. This is an important aspect to be quantitatively and precisely known for a given wetsuit test since the typical duration of the measurement to achieve the peak values, of both ΔT and R_T , can last from a few to several hours, depending on the wetsuit thickness and water pressure. Therefore, even when assessing the wetsuit thermal resistance value by the conventional value $R_{T,MAX}$ achieved in the in test, one is underestimating the wetsuit thermal resistance at the beginning of the immersion by a percentage value $\Delta R_{T,\%}=S\% \cdot \Delta t_{PEAK}$. Knowing the relative slope $S\%$, of the thermal resistance degrading with wetting time, one can evaluate the correct thermal resistance value, $R_{T,C}$, at any time t from the beginning of the test/immersion, by the linear correction formula

$$R_{T,C}(t) = R_{T,MAX} \{1 - S\% \cdot (t - \Delta t_{MAX})\} \quad [K/W] \quad (3)$$

Equation (3) is a general formula that can be used when the slope $S\%$ is well known for a given wetsuit family (with specific producer, fabric, thickness, ...). However, since we do not have a large experience and confidence in our tests yet, nor we do have many repeated measurements with the same wetsuit samples, we prefer to use the absolute slope $S=dR_T/dt$, evaluated as the slope of the absolute decrease of the thermal resistance with time after the peak (see Fig. 5b, where this slope is evaluated by the best fit straight line of $R_T(t)$ after the peak). As one can see from the fitting straight line of Fig. 5b, the intercept is at $R_{T,0}=R_{T,C}(0)=0.244$ K/W and this is theoretically the initial wetsuit thermal resistance at $t=0$, i.e., at the beginning of its immersion into water. In general, $R_{T,0} > R_T$ and, in the case of this wetsuit sample (see

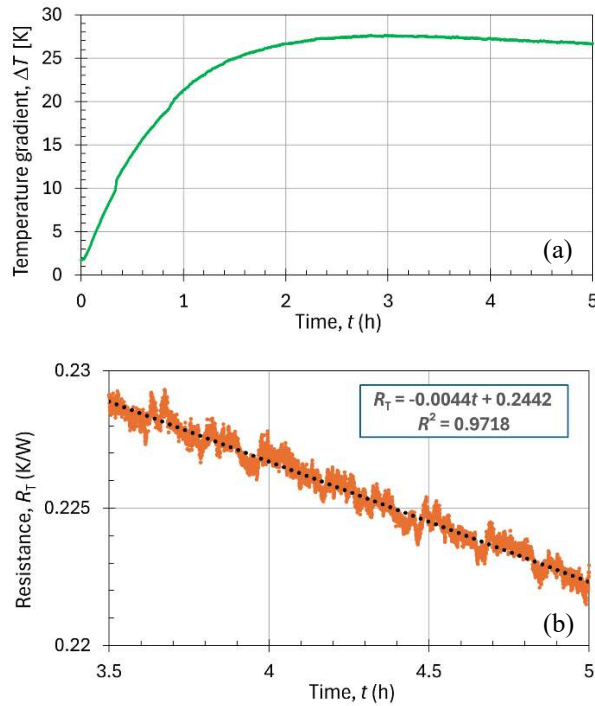


Fig. 5. (a) Temperature gradient from the beginning of the test until its peak value and after. (b) Decrease of the thermal resistance and its regression line after the temperature gradient peak.

Table I), $R_T=0.226$ K/W, which is significantly lower than $R_{T,0}$. By (3), we can evaluate $R_T(0.5\text{ h})=0.242$ K/W= $*R_T$ as the theoretical, or proper, value of the wetsuit thermal resistance at half-time duration of a typical scuba diving immersion (1 h). The measurement, performed in our test at the peak stationary condition of ΔT (in this case after 3 h of wetting time, as can be seen in Fig. 5a), provides an underestimated value of the proper thermal resistance $*R_T$ by $\sim 7\%$. If the stationary measurement is performed a few hours later or more after the peak condition, an underestimation of some 10% or greater can easily occur.

Since the wetsuit thermal resistance decreases with the wetting time, in particular at high water pressure, and since the wetting time which is needed to perform the test in accordance to [6] is rather longer than a typical scuba diving immersion, it is the Authors' opinion that (3) or some other correction model should be used to properly evaluate the wetsuit thermal resistivity in these tests. Simply stating that the steady state temperature gradient stability is sufficient to ensure a correct measurement of the thermal resistance and to assess the thermal class of a wetsuit, as indicated in [6], seems inadequate. In fact, the process is inherently non-stationary and the longer the time used to measure R_T the greater the error in the measured value, which can lead to

underestimate the correct thermal resistance at the beginning, or $\frac{1}{2}$ - or $\frac{3}{4}$ -duration, of the dive. With the phantom's size prescribed by [6] and the corresponding inner water volume and thermal capacitance [5], one cannot expect the 'quasi stationary' measurement time to be less than a few hours and, thus, some more indications about the correct measurement condition and time must be added to the current regulation requirements.

V. CONCLUSION

The thermal resistance of different neoprene wetsuits was tested at 100 kPa and 600 kPa water pressures, with a method and instrumentation compliant with the international regulations. The wetsuits resistance reduces roughly by a factor of 4 due to the pressure increase. During the tests, a significant thermal resistance degradation with the wetsuit wetting time was quantitatively evaluated. This suggests that the tests specifications might be improved by considering also the immersion time and water pressure, when assessing the wetsuit thermal resistance and resistivity.

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